

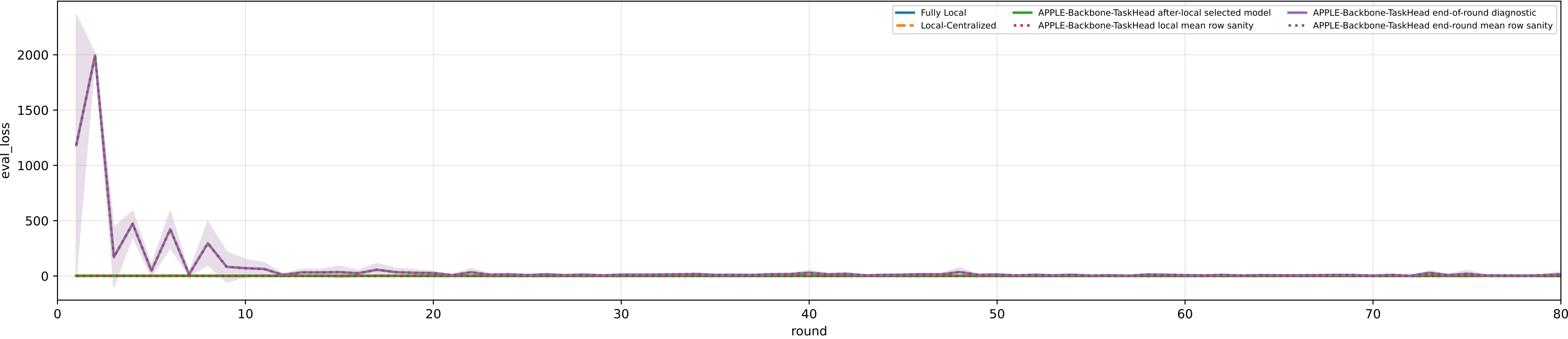
Controlled q-label heterogeneity: iid | **APPLE-Backbone-TaskHead** diagnostics | aggregated /  
pooled over 5 clients |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

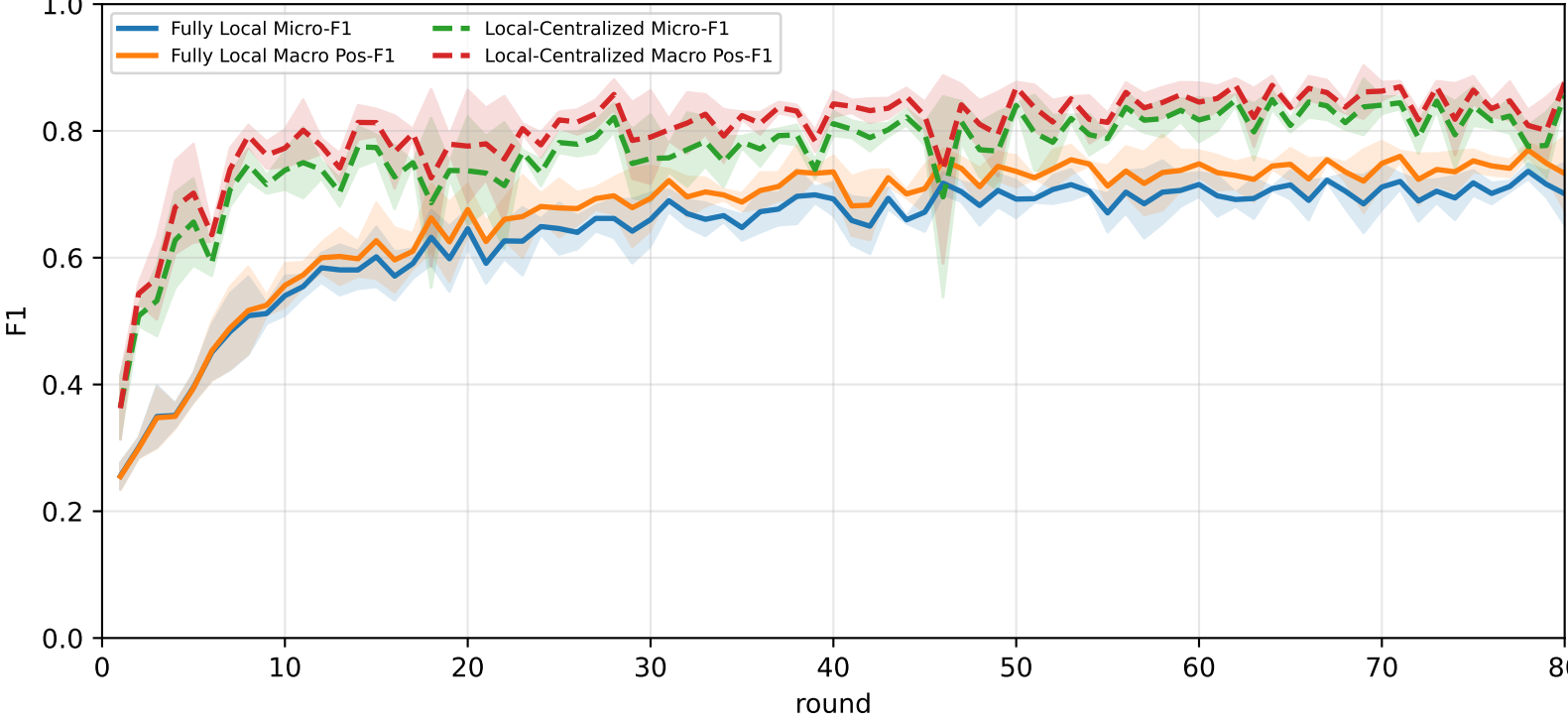
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | <u>67.9% ± 1.6%</u> | <u>72.3% ± 1.5%</u> | <u>90.0% ± 1.5%</u> | <u>85.0% ± 1.7%</u> | <u>67.5% ± 3.5%</u> | <u>66.8% ± 2.2%</u> | 50.9% ± 1.2%        |
| Local-Centralized       | 85.0% ± 0.7%        | 87.0% ± 0.6%        | 98.0% ± 1.1%        | 94.5% ± 1.4%        | 89.7% ± 1.8%        | 80.3% ± 0.5%        | 72.5% ± 1.3%        |
| APPLE-Backbone-TaskHead | 64.3% ± 3.1%        | 66.0% ± 3.6%        | 72.1% ± 10.7%       | 71.3% ± 3.2%        | <u>67.5% ± 3.3%</u> | 62.6% ± 3.7%        | <u>54.7% ± 1.5%</u> |

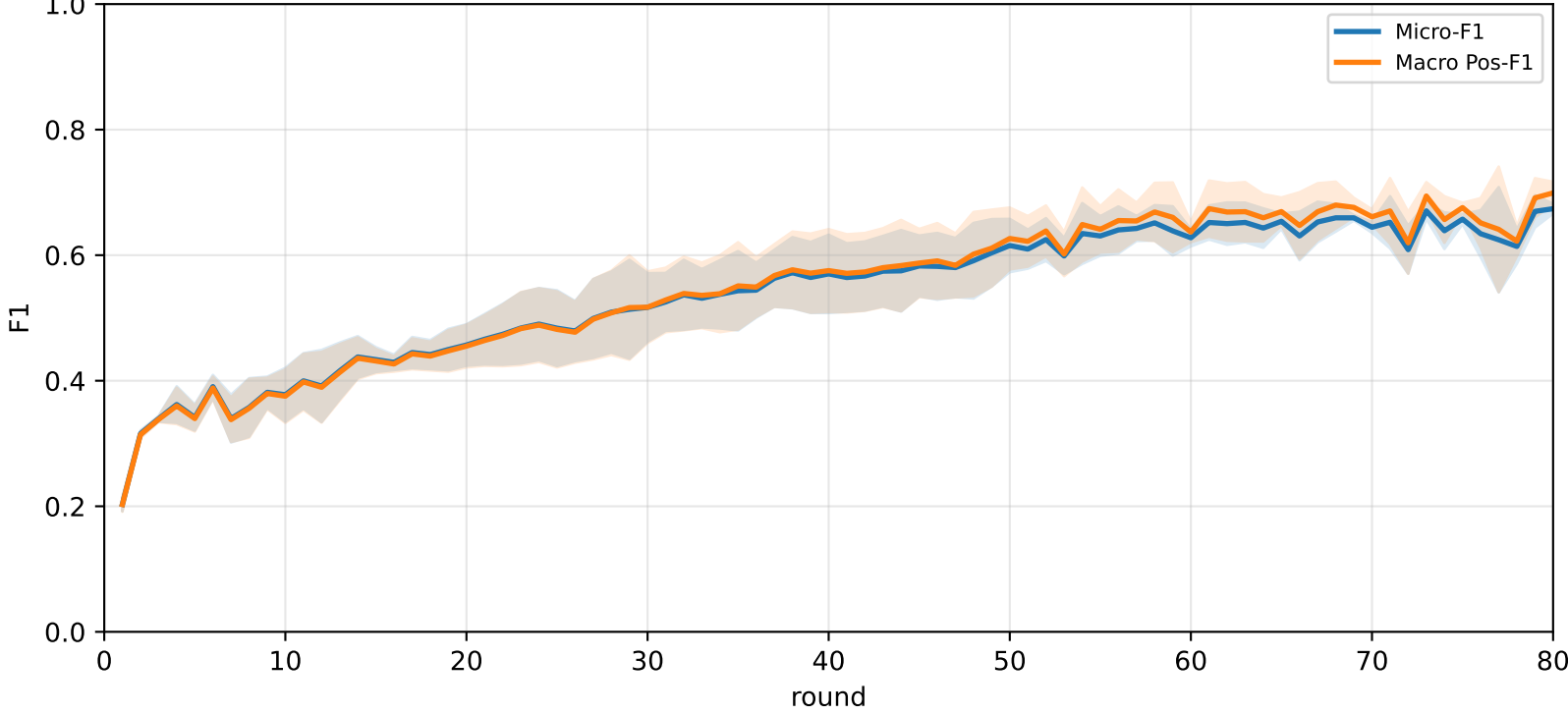
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



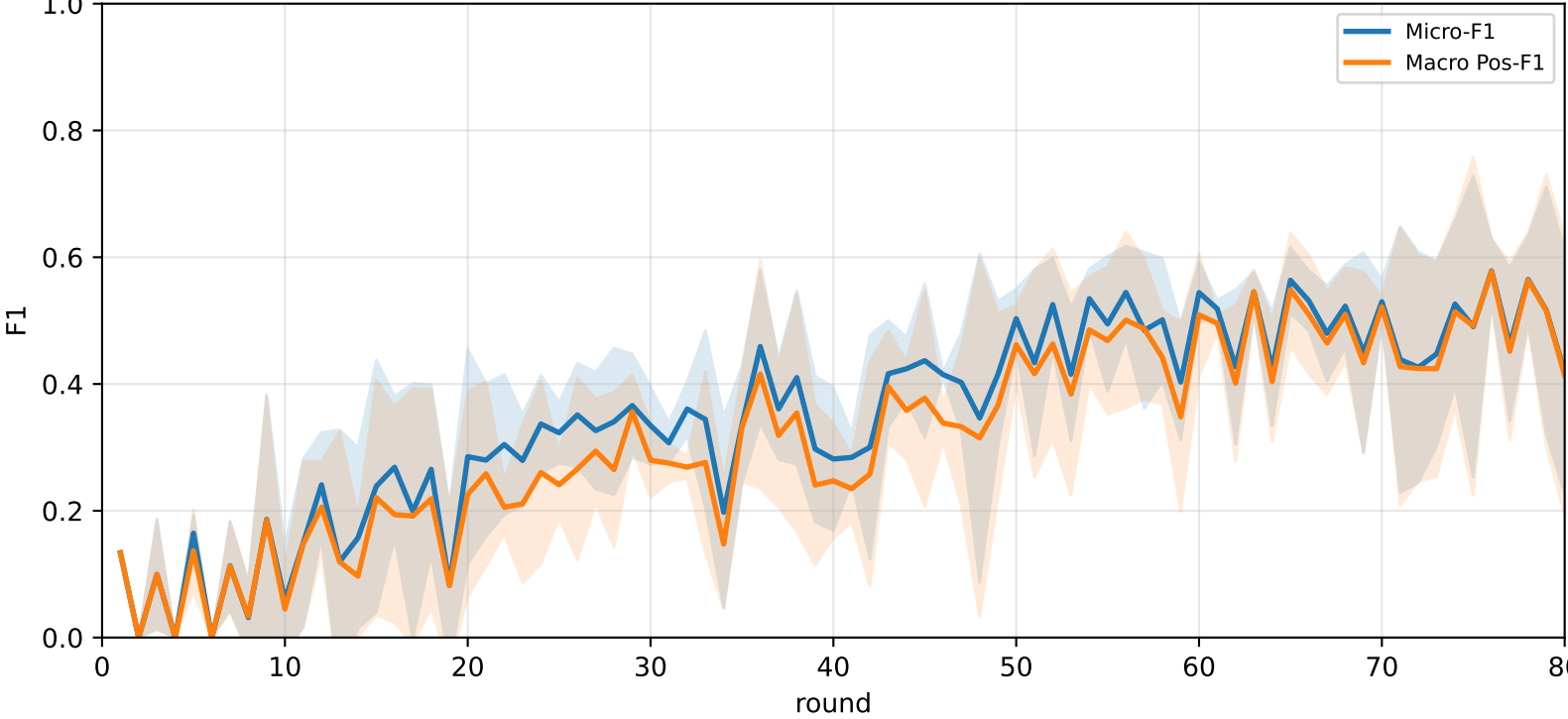
Pooled baseline micro / macro F1



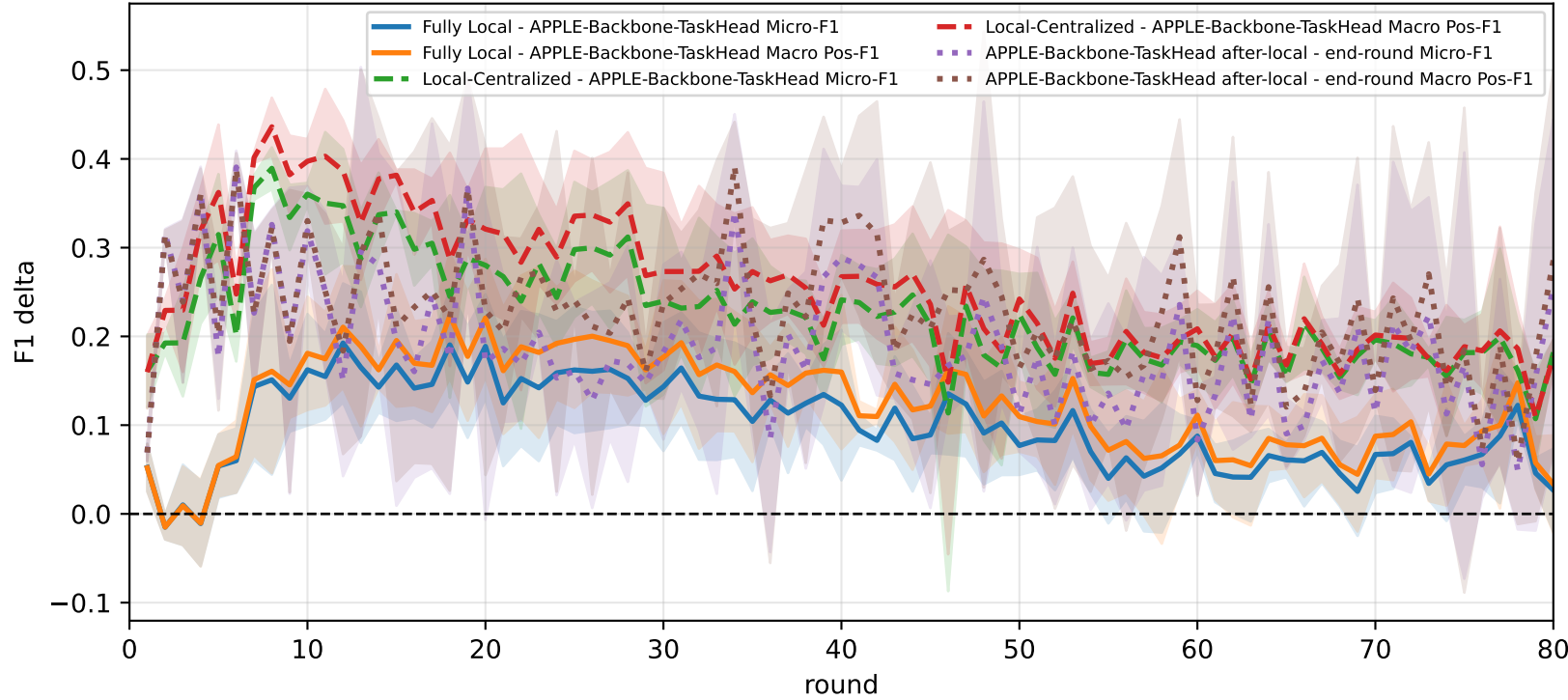
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap

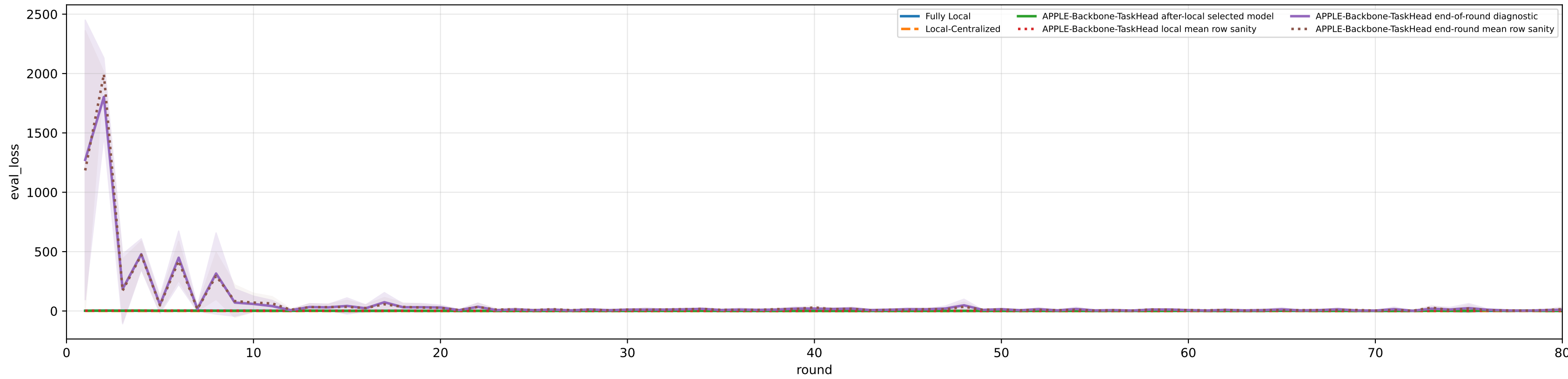


family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

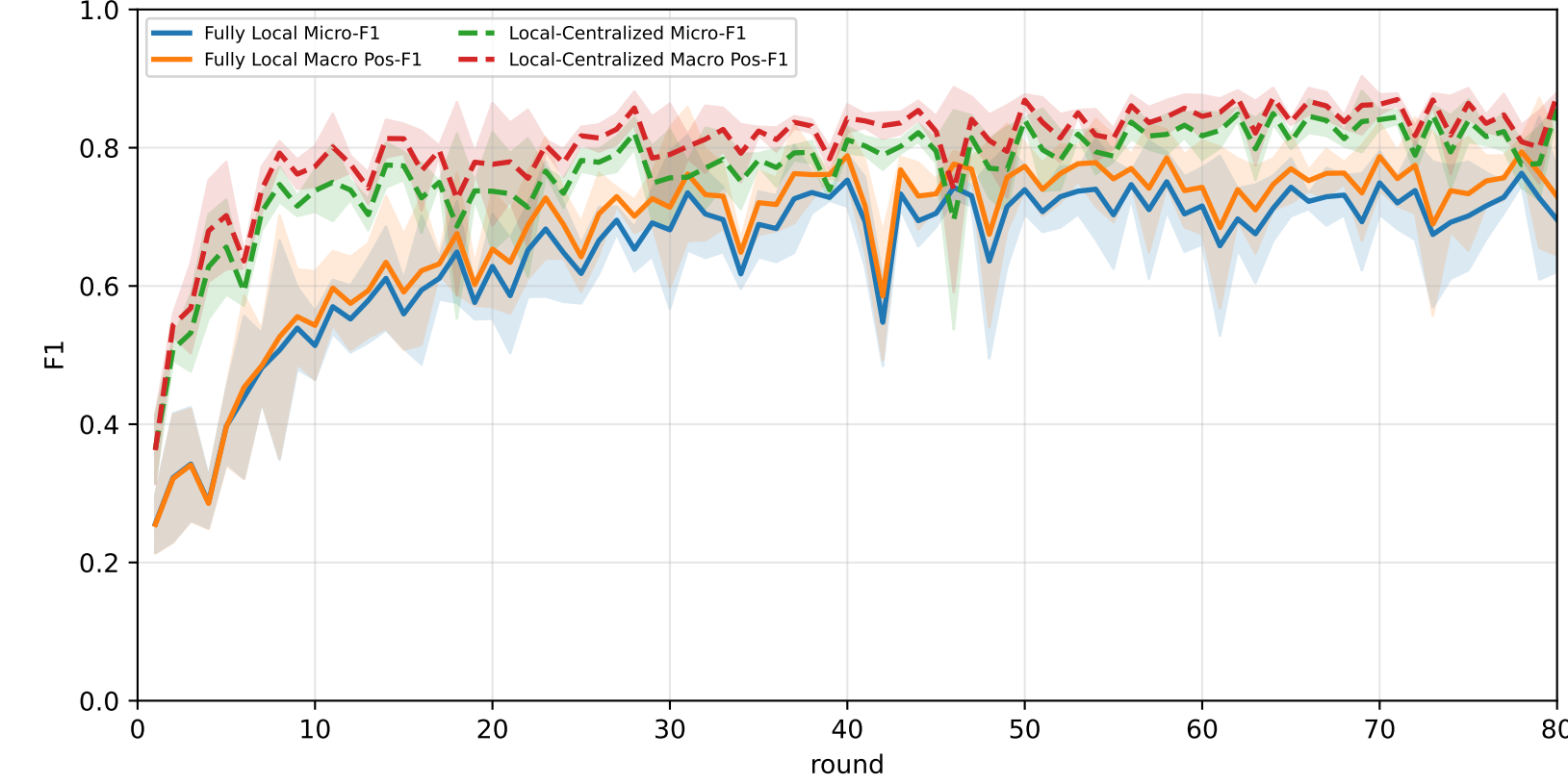
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | <u>70.1% ± 3.2%</u> | <u>74.3% ± 2.5%</u> | <u>93.1% ± 1.1%</u> | <u>86.7% ± 1.6%</u> | <u>70.3% ± 4.6%</u> | <u>67.5% ± 2.2%</u> | 53.8% ± 5.0%        |
| Local-Centralized       | 85.0% ± 0.7%        | 87.0% ± 0.6%        | 98.0% ± 1.1%        | 94.5% ± 1.4%        | 89.7% ± 1.8%        | 80.3% ± 0.5%        | 72.5% ± 1.3%        |
| APPLE-Backbone-TaskHead | 65.8% ± 4.0%        | 67.9% ± 4.7%        | 78.0% ± 13.8%       | 74.4% ± 7.5%        | 69.1% ± 3.6%        | 62.3% ± 3.5%        | <u>55.9% ± 2.6%</u> |

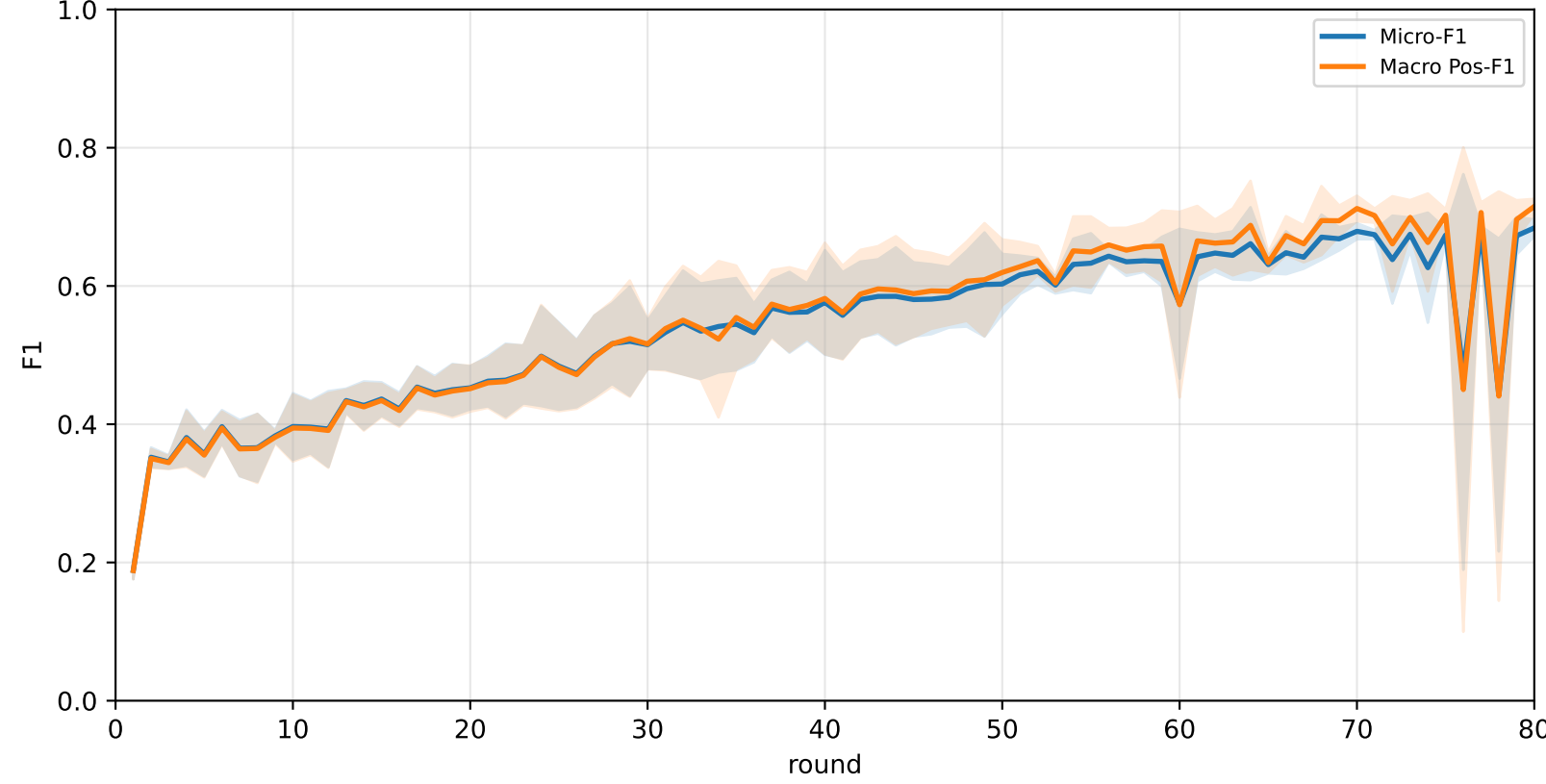
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



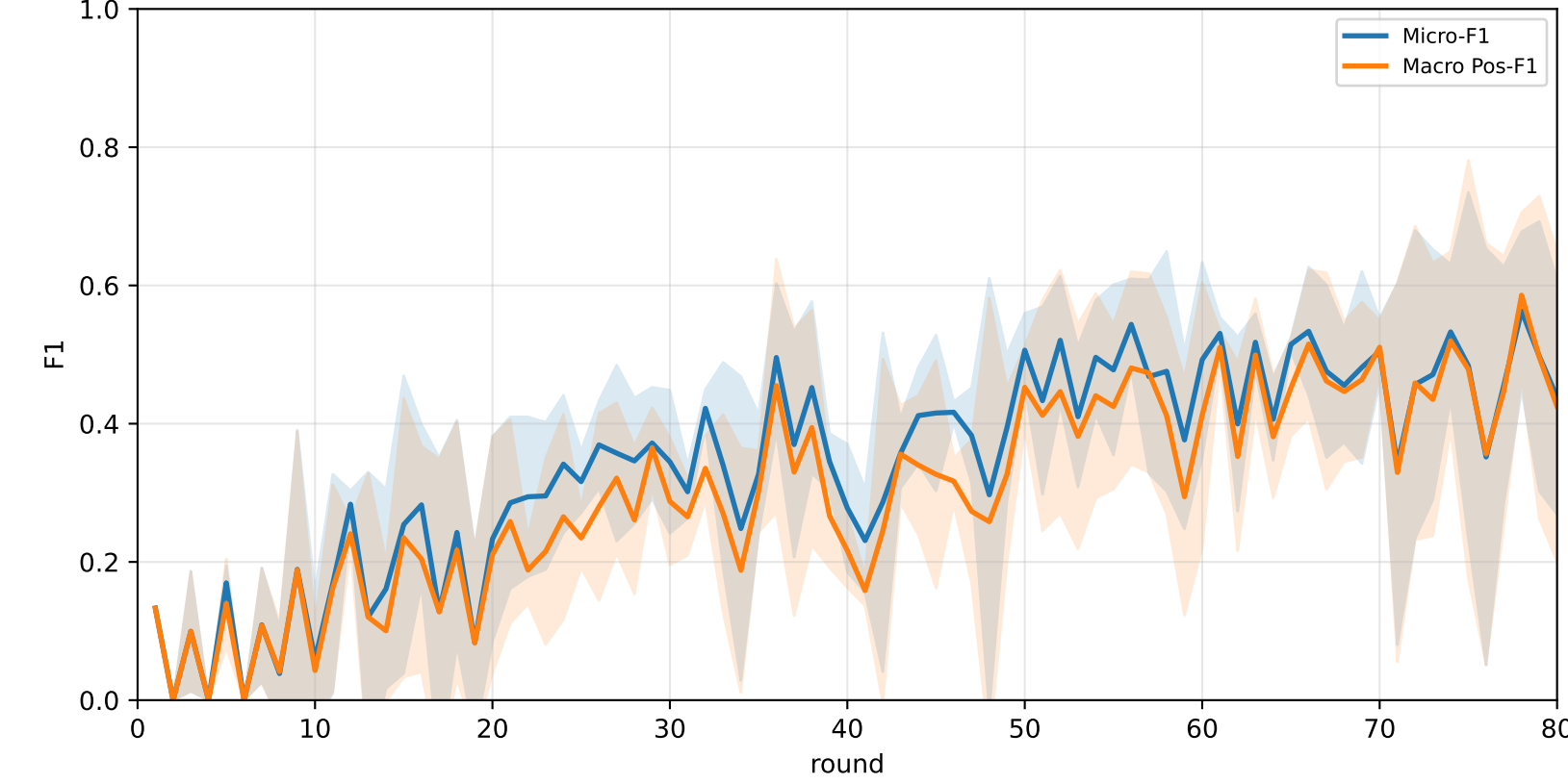
Pooled baseline micro / macro F1



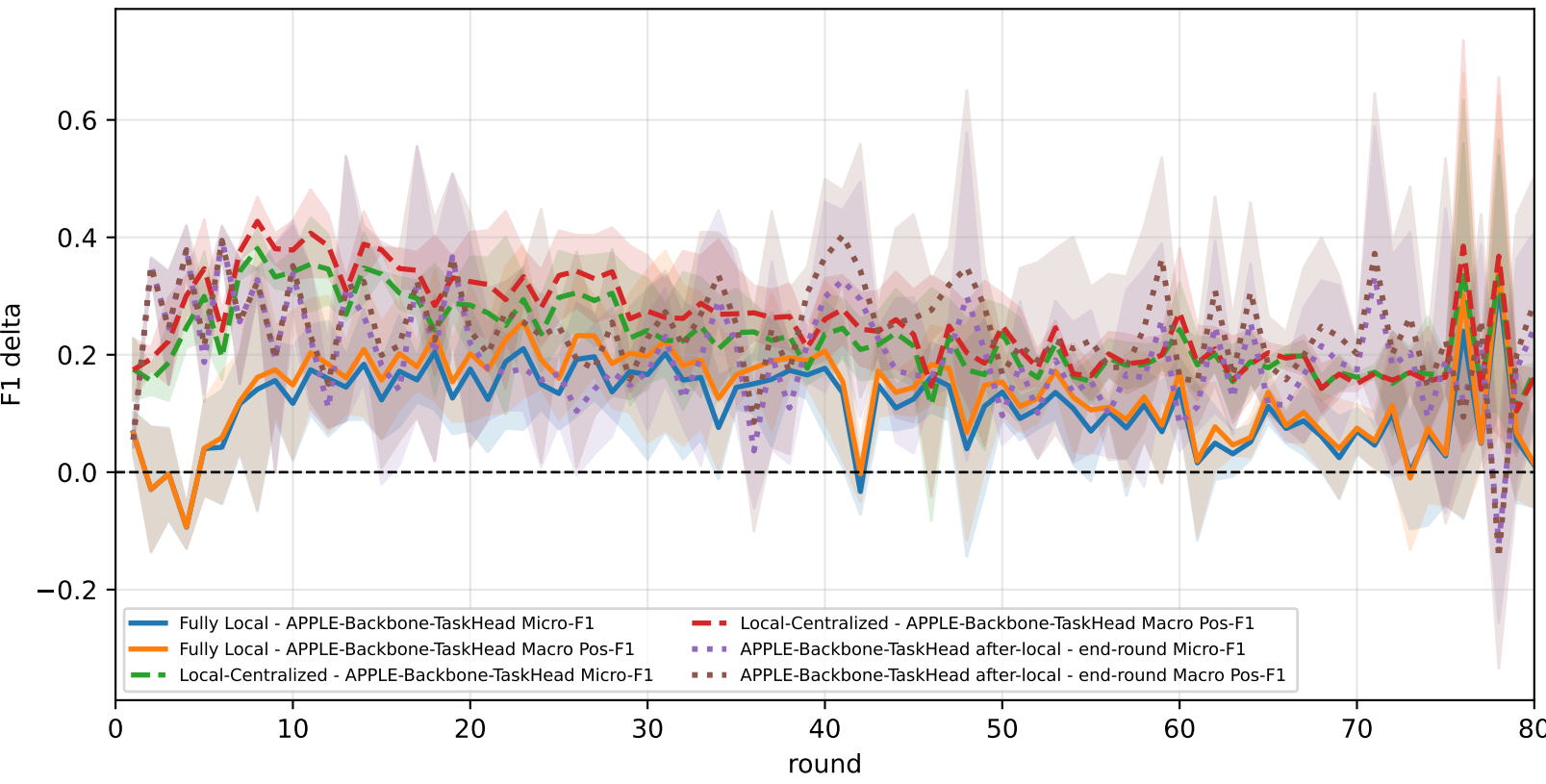
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap



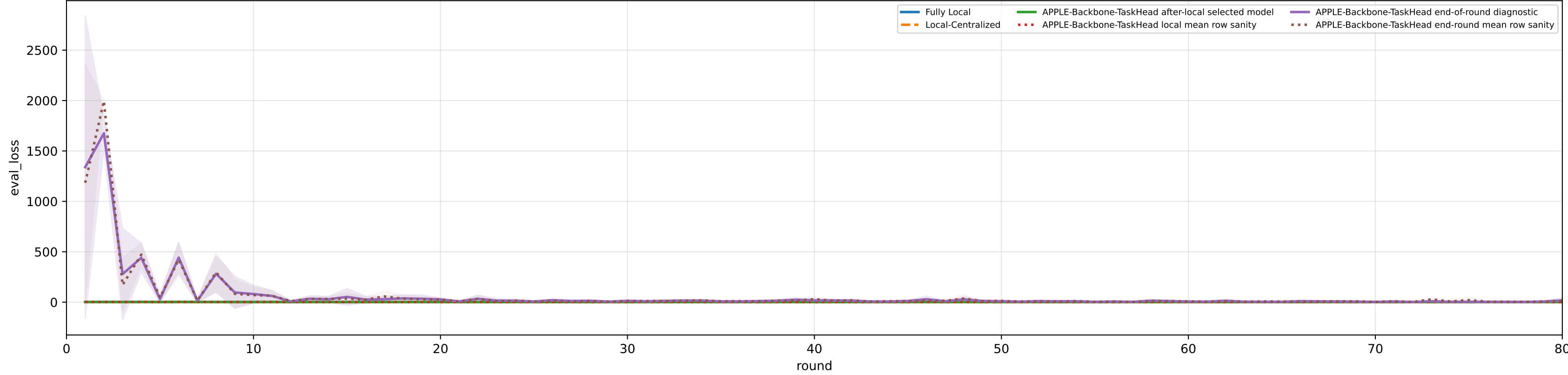


family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

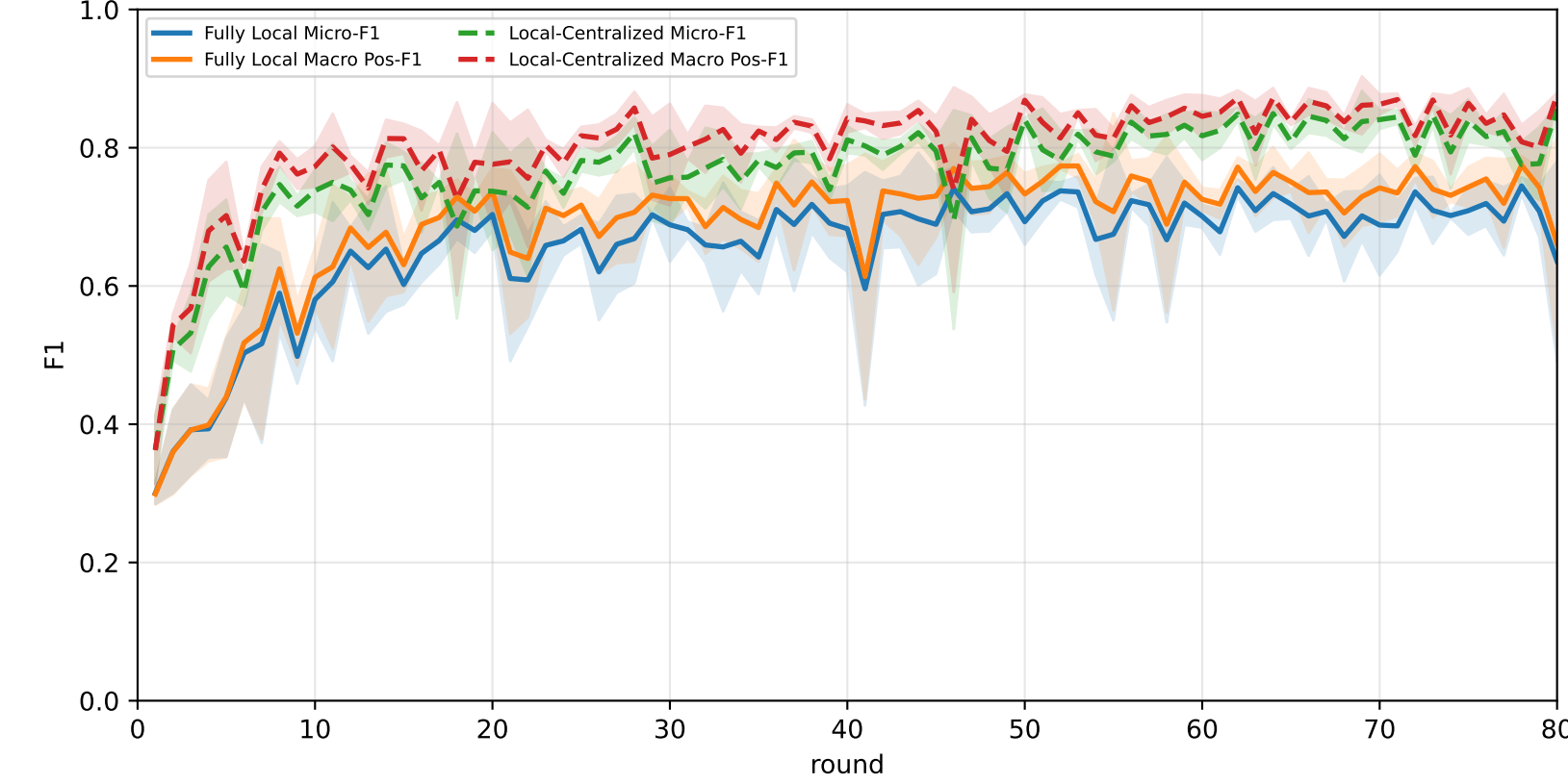
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | <u>69.5% ± 0.8%</u> | <u>73.5% ± 0.6%</u> | <u>90.4% ± 0.9%</u> | <u>85.2% ± 1.6%</u> | <u>67.7% ± 0.6%</u> | <u>71.2% ± 1.1%</u> | 52.9% ± 2.1%        |
| Local-Centralized       | 85.0% ± 0.7%        | 87.0% ± 0.6%        | 98.0% ± 1.1%        | 94.5% ± 1.4%        | 89.7% ± 1.8%        | 80.3% ± 0.5%        | 72.5% ± 1.3%        |
| APPLE-Backbone-TaskHead | 65.4% ± 2.5%        | 67.6% ± 3.2%        | 76.8% ± 15.1%       | 73.7% ± 6.1%        | 67.2% ± 2.9%        | 64.9% ± 1.7%        | <u>55.2% ± 1.3%</u> |

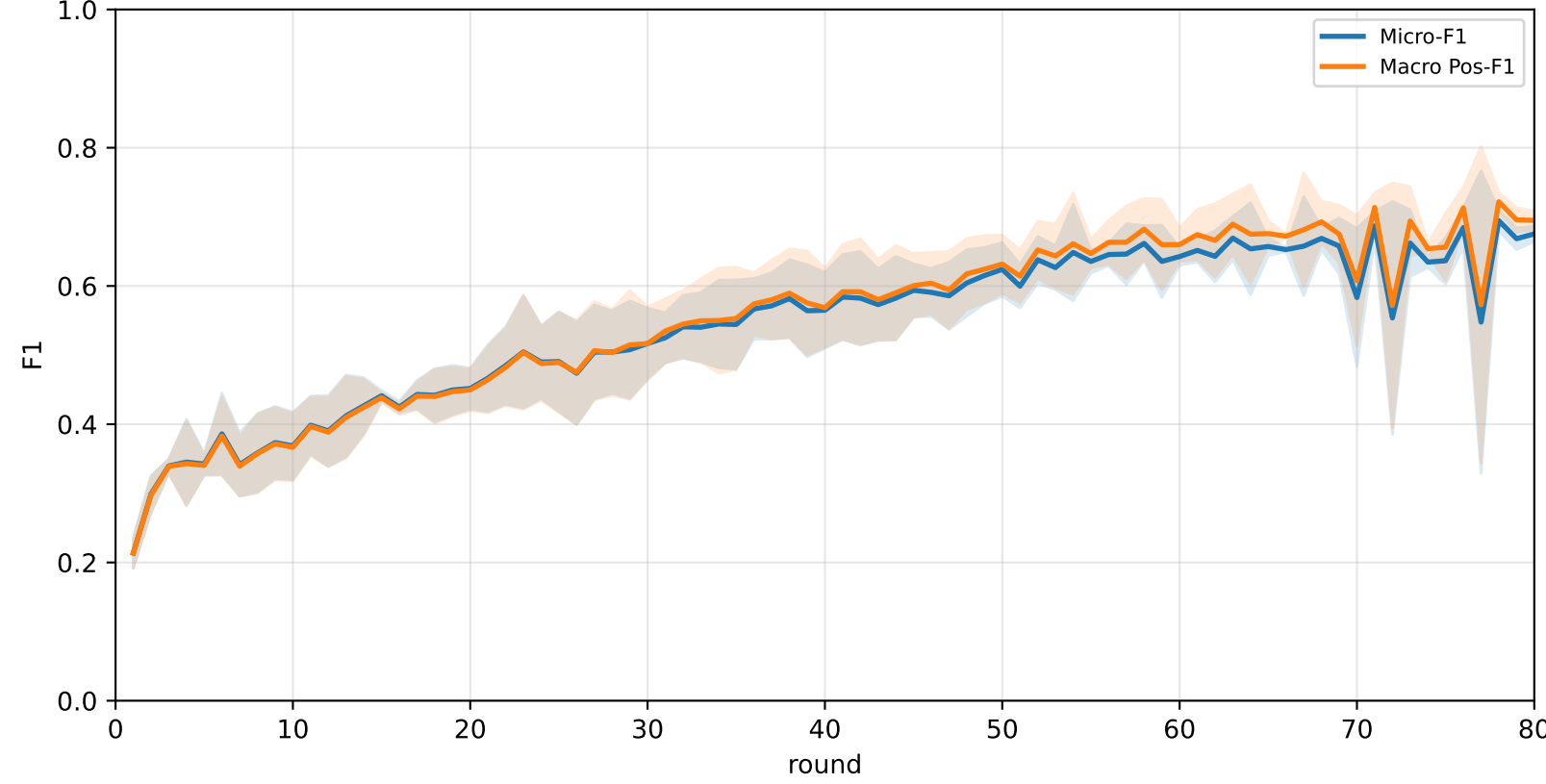
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



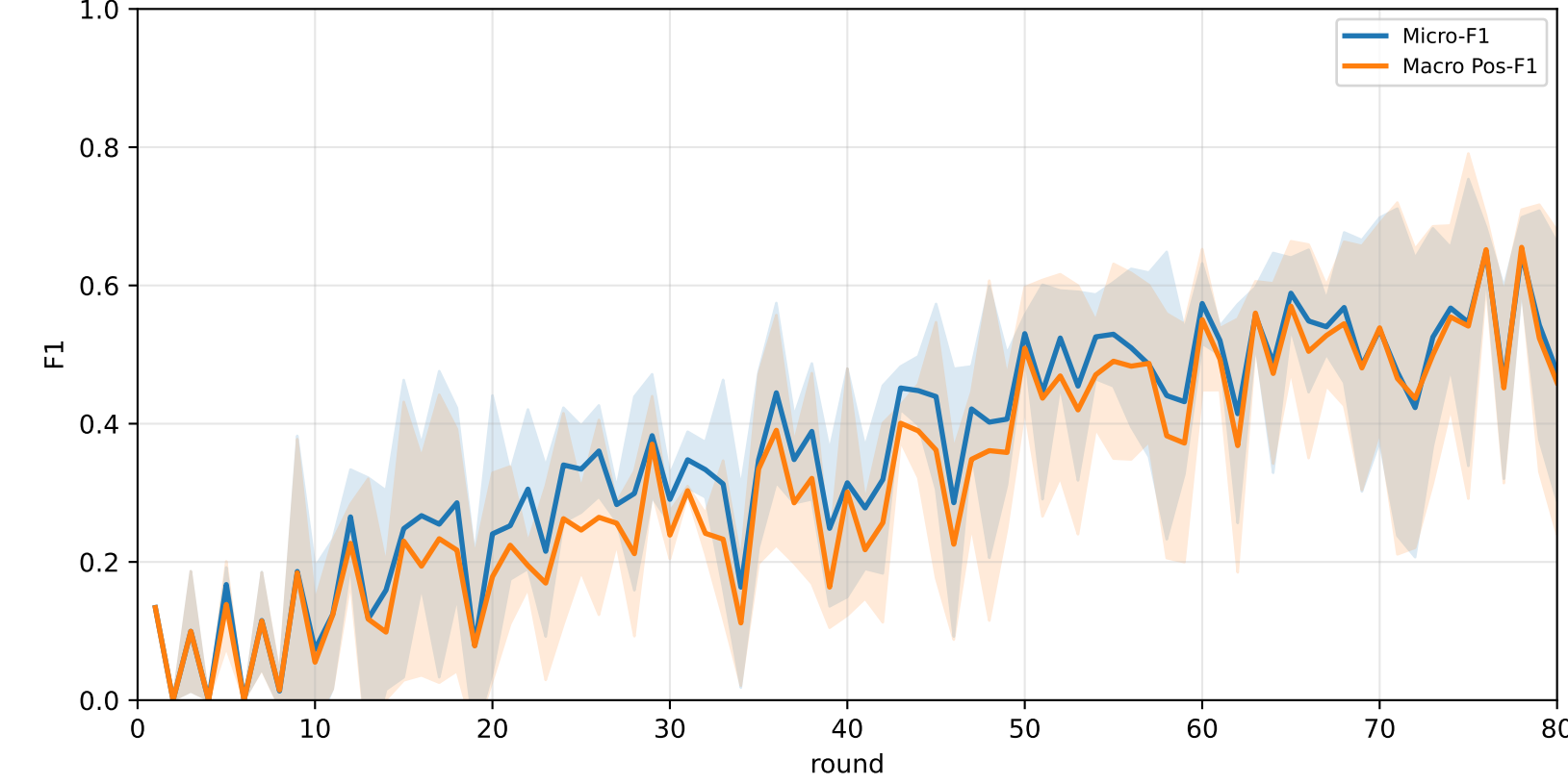
Pooled baseline micro / macro F1



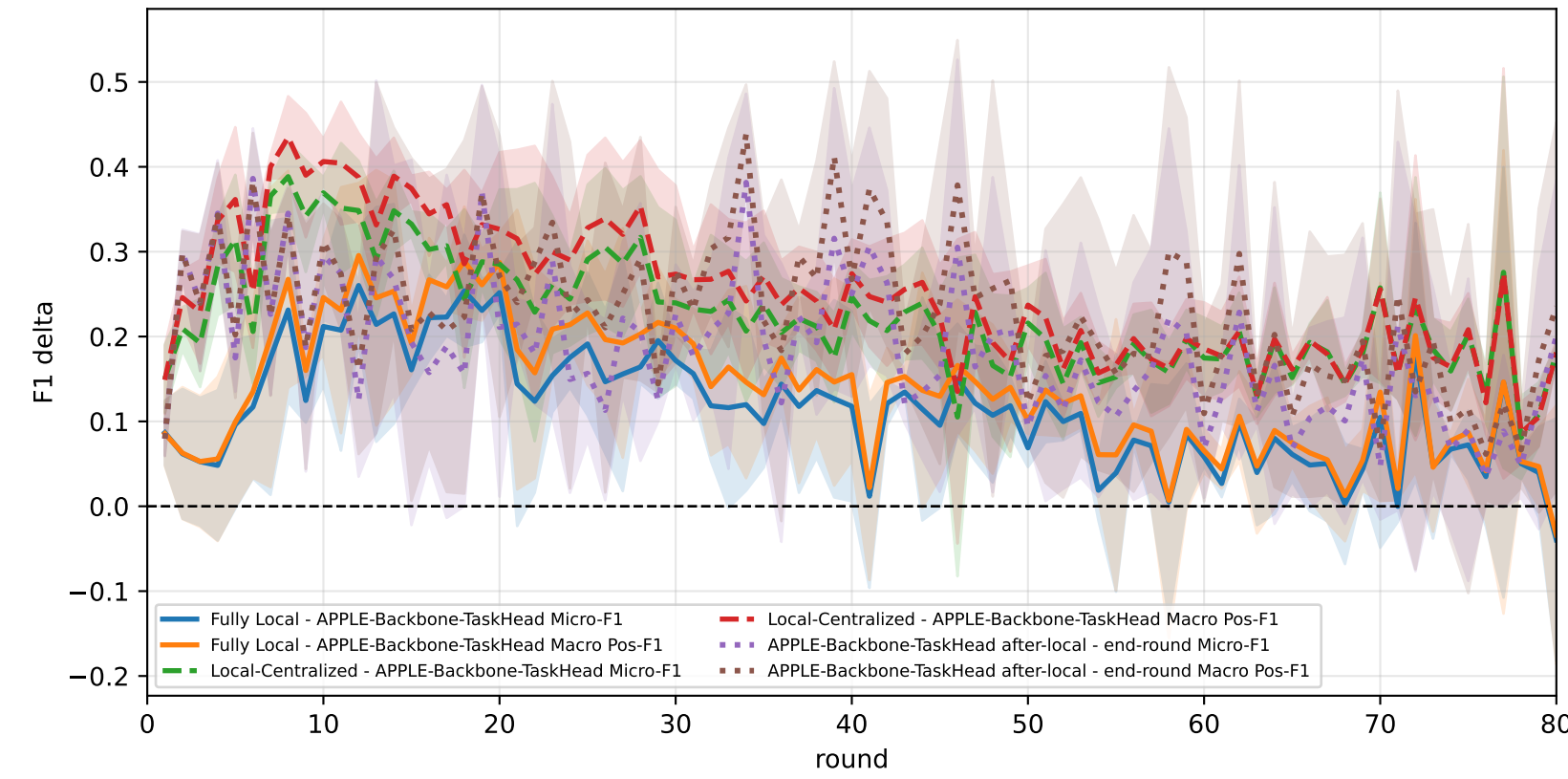
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap

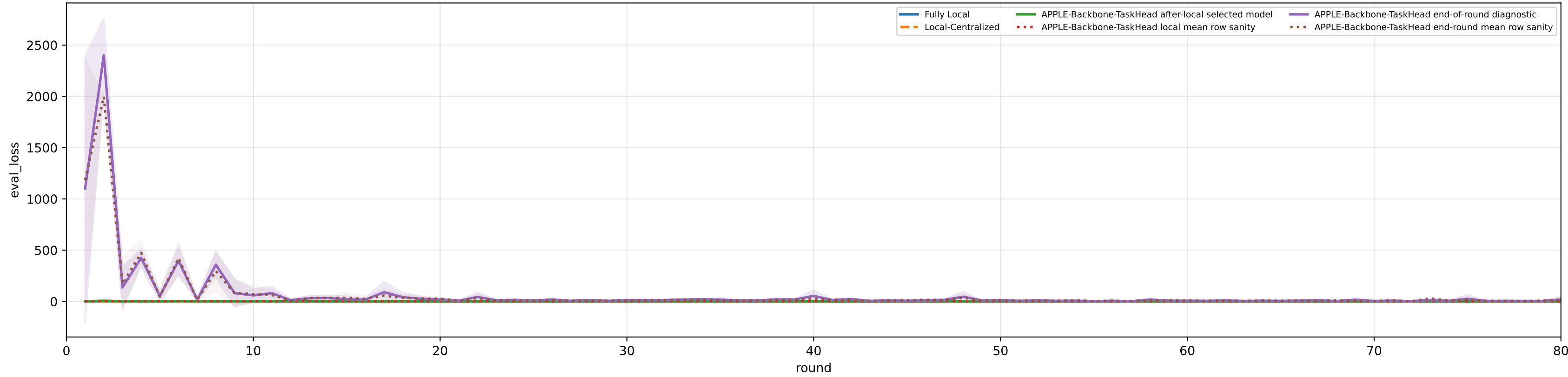


family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

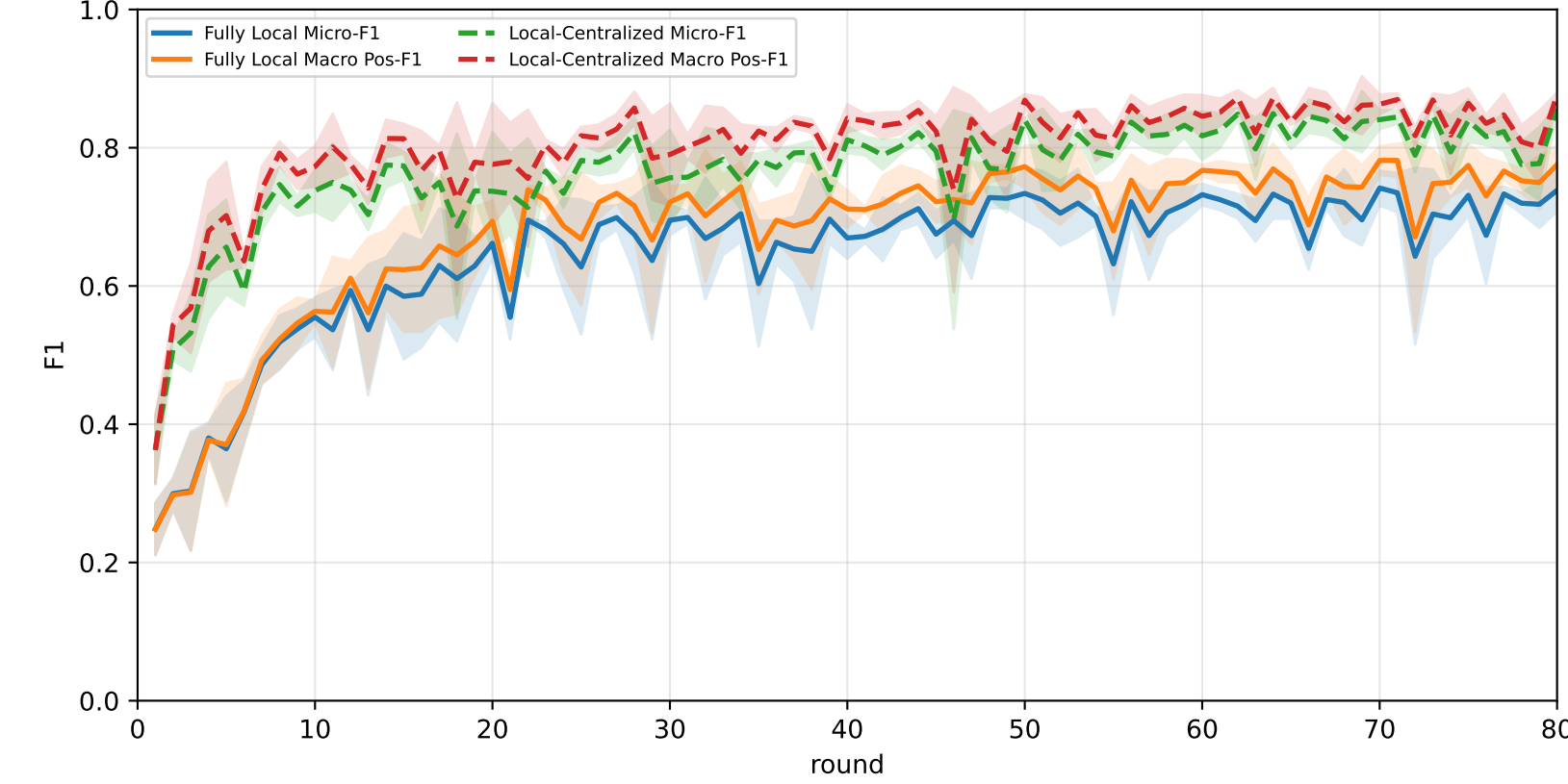
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | <u>64.8% ± 3.3%</u> | <u>70.1% ± 2.7%</u> | <u>92.2% ± 3.4%</u> | <u>82.7% ± 3.5%</u> | 65.7% ± 8.3%        | 60.9% ± 7.0%        | 48.9% ± 2.7%        |
| Local-Centralized       | 85.0% ± 0.7%        | 87.0% ± 0.6%        | 98.0% ± 1.1%        | 94.5% ± 1.4%        | 89.7% ± 1.8%        | 80.3% ± 0.5%        | 72.5% ± 1.3%        |
| APPLE-Backbone-TaskHead | 62.7% ± 3.8%        | 63.7% ± 4.1%        | 65.2% ± 12.0%       | 69.9% ± 0.9%        | <u>67.1% ± 3.2%</u> | <u>61.4% ± 3.5%</u> | <u>54.9% ± 3.5%</u> |

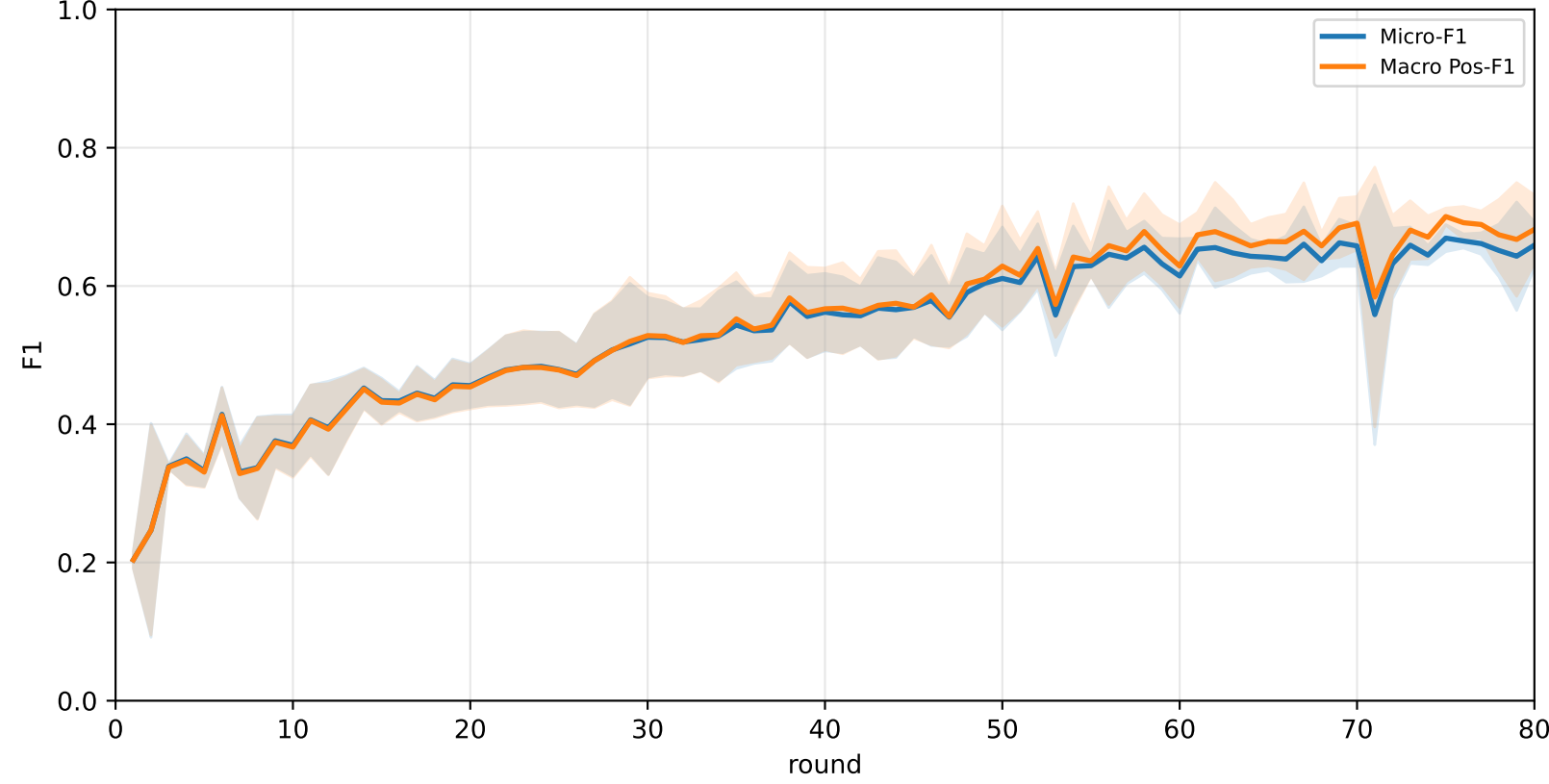
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



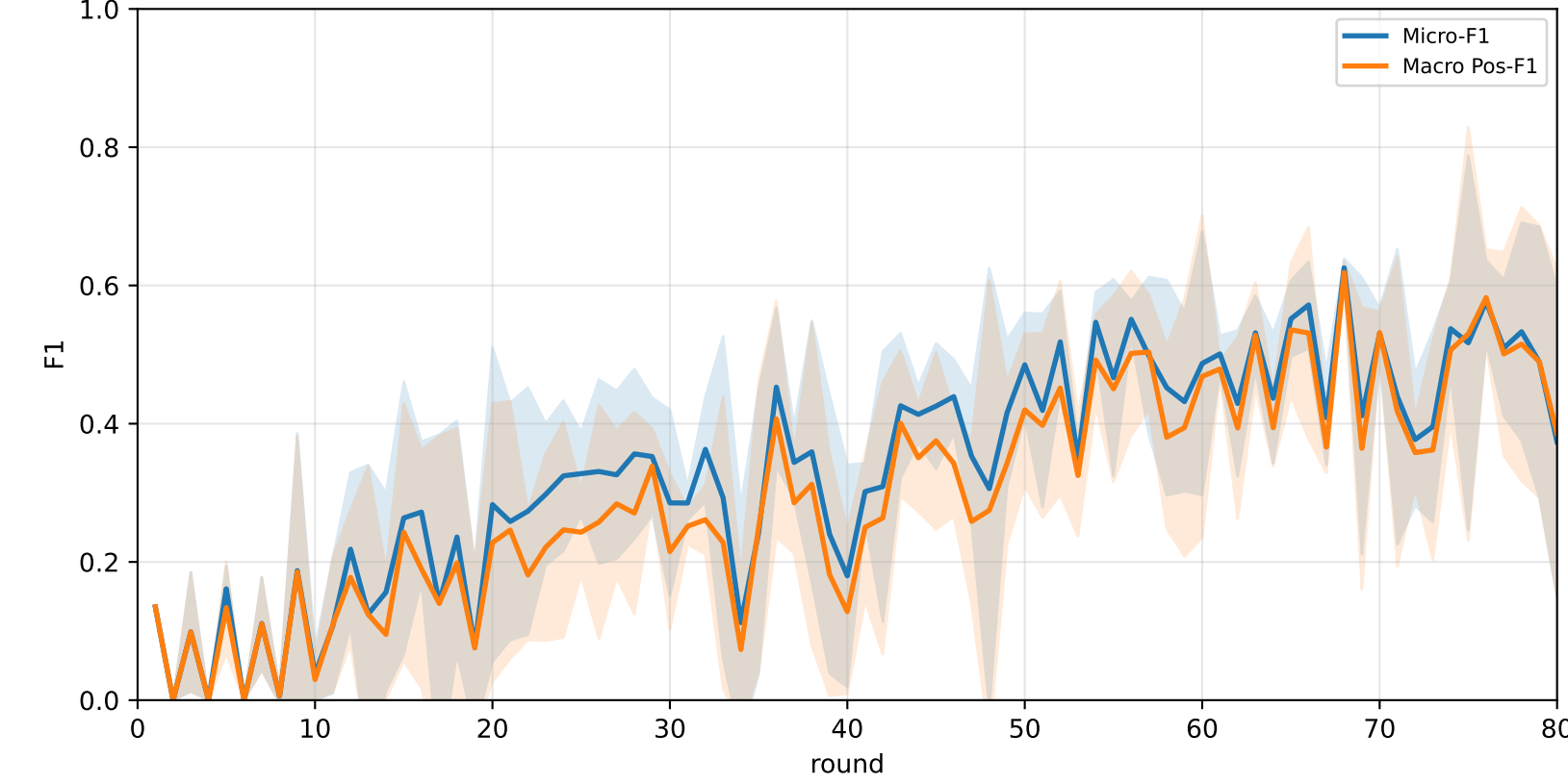
Pooled baseline micro / macro F1



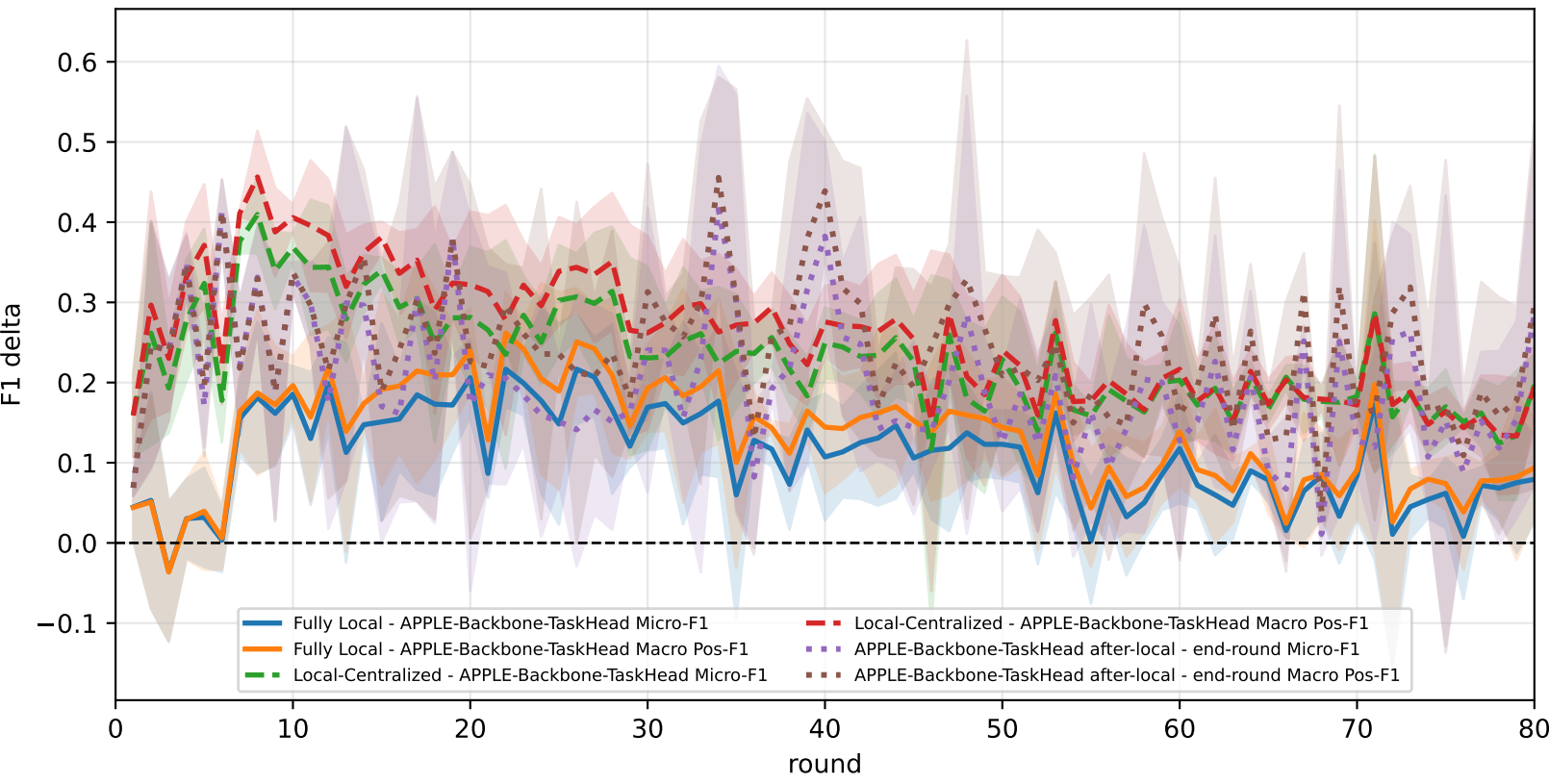
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap





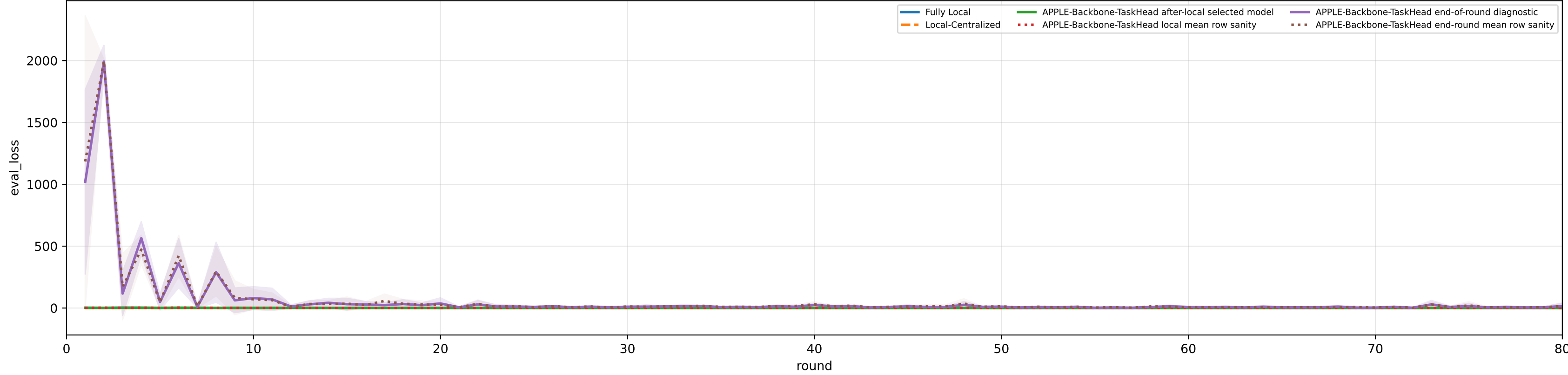
Controlled q-label heterogeneity: iid | APPLE-Backbone-TaskHead diagnostics |  
family=q\_task\_cycle5 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

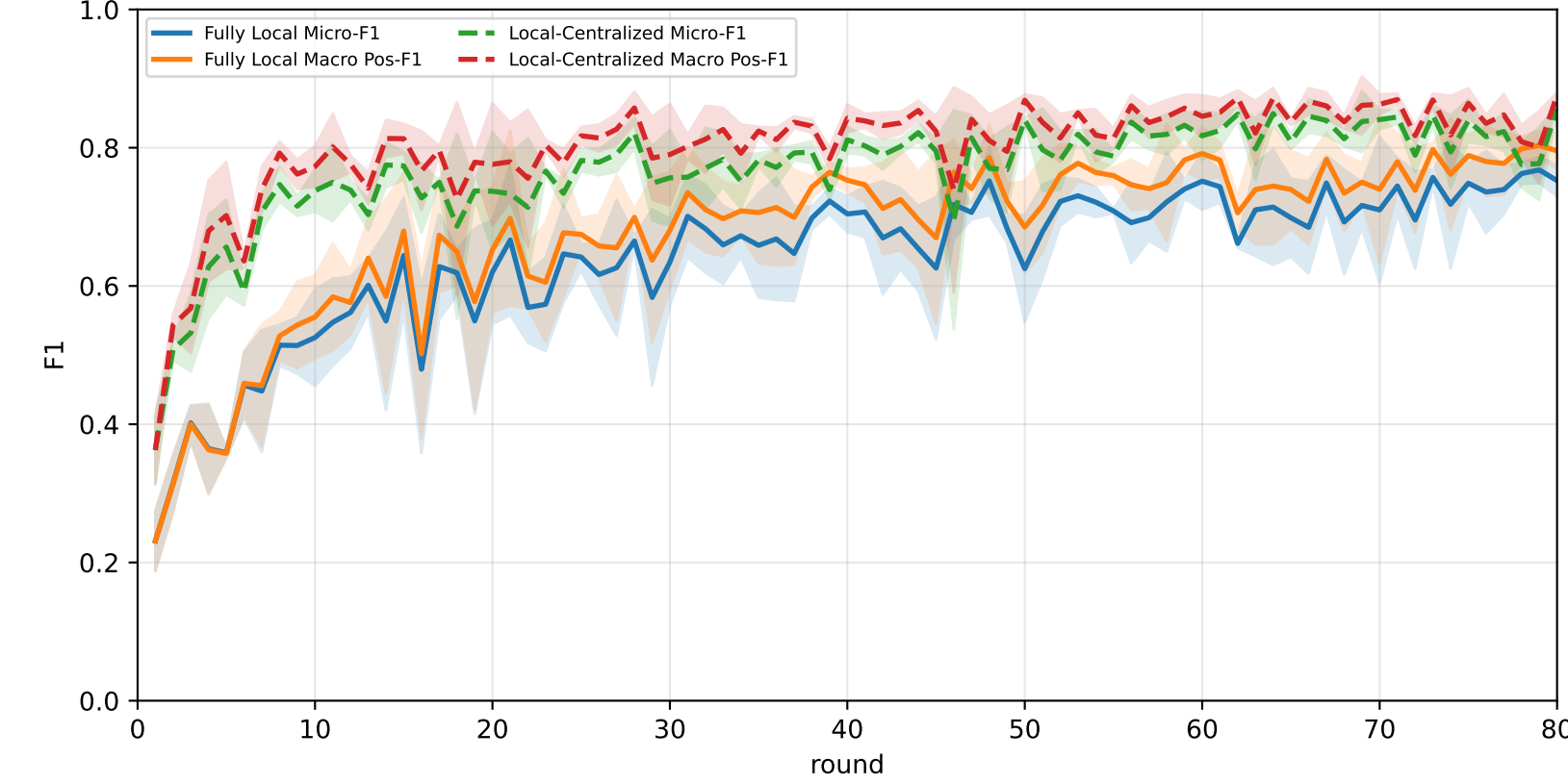
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | <u>70.5% ± 1.9%</u> | <u>74.2% ± 2.2%</u> | <u>87.8% ± 6.8%</u> | <u>86.4% ± 2.4%</u> | <u>74.0% ± 2.0%</u> | <u>69.7% ± 1.8%</u> | <u>53.3% ± 1.7%</u> |
| Local-Centralized       | 85.0% ± 0.7%        | 87.0% ± 0.6%        | 98.0% ± 1.1%        | 94.5% ± 1.4%        | 89.7% ± 1.8%        | 80.3% ± 0.5%        | 72.5% ± 1.3%        |
| APPLE-Backbone-TaskHead | 62.1% ± 3.7%        | 63.4% ± 4.0%        | 69.4% ± 6.6%        | 66.8% ± 1.0%        | 66.4% ± 4.9%        | 61.2% ± 5.5%        | 53.0% ± 1.9%        |

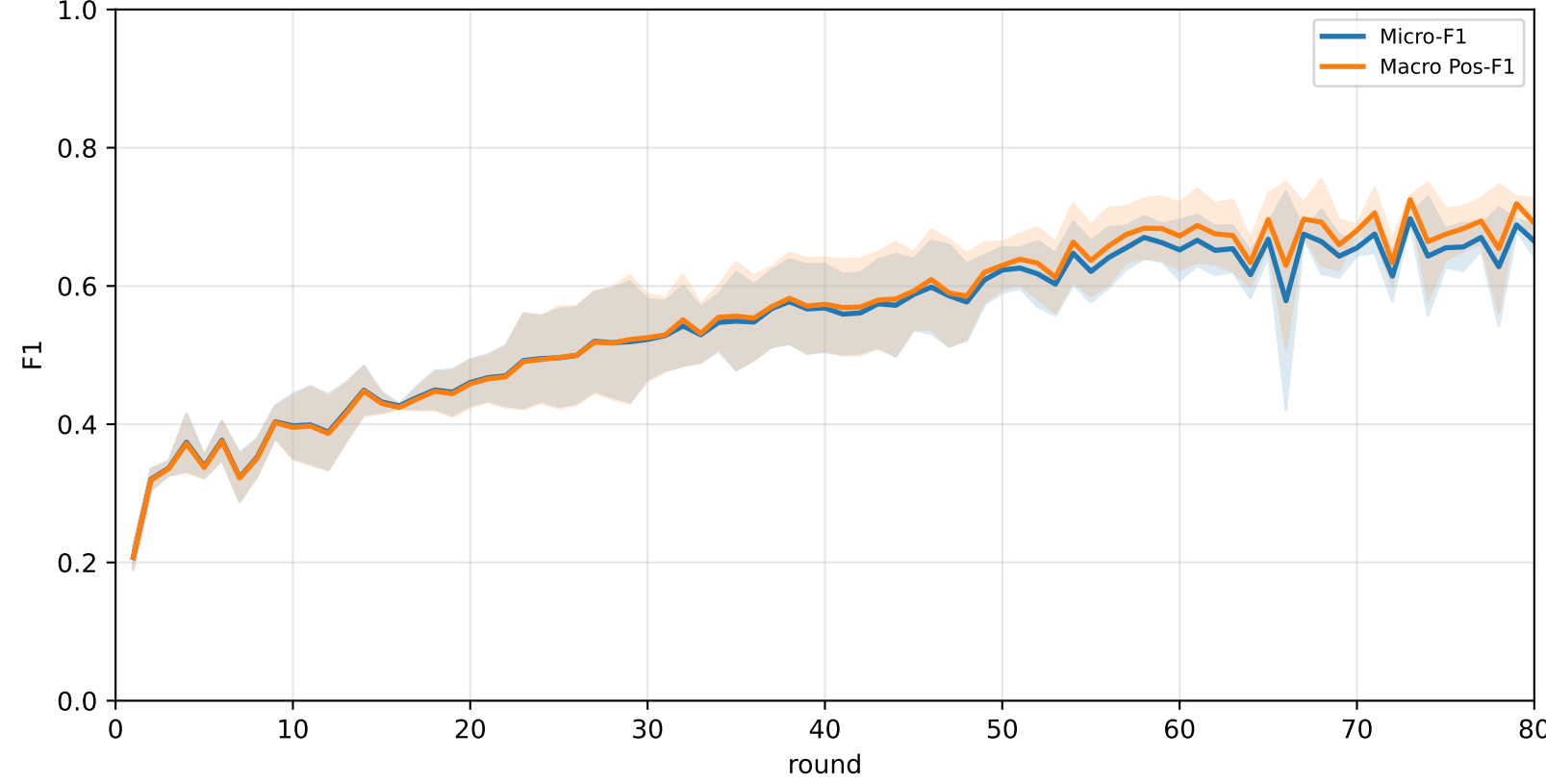
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



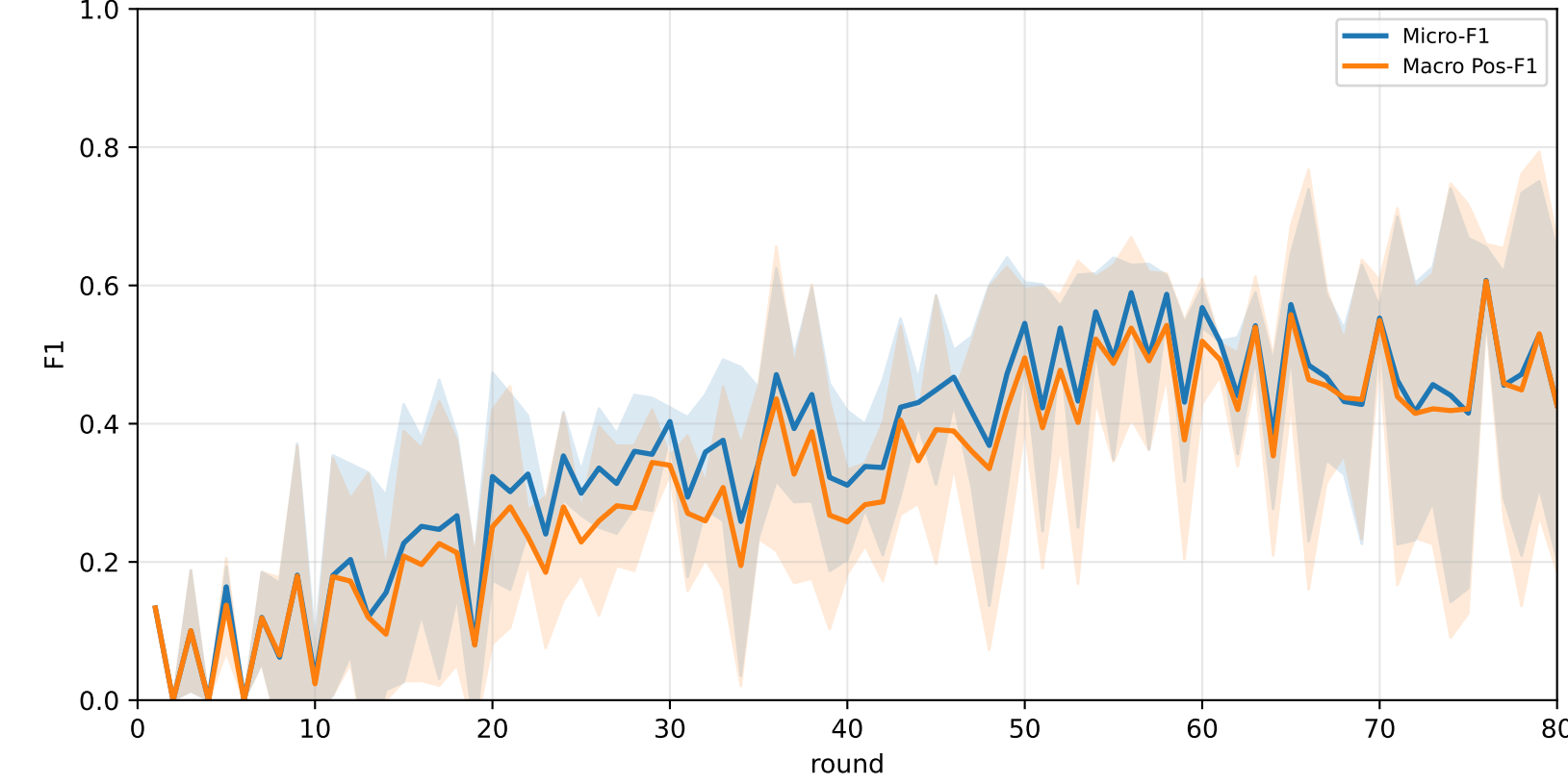
Pooled baseline micro / macro F1



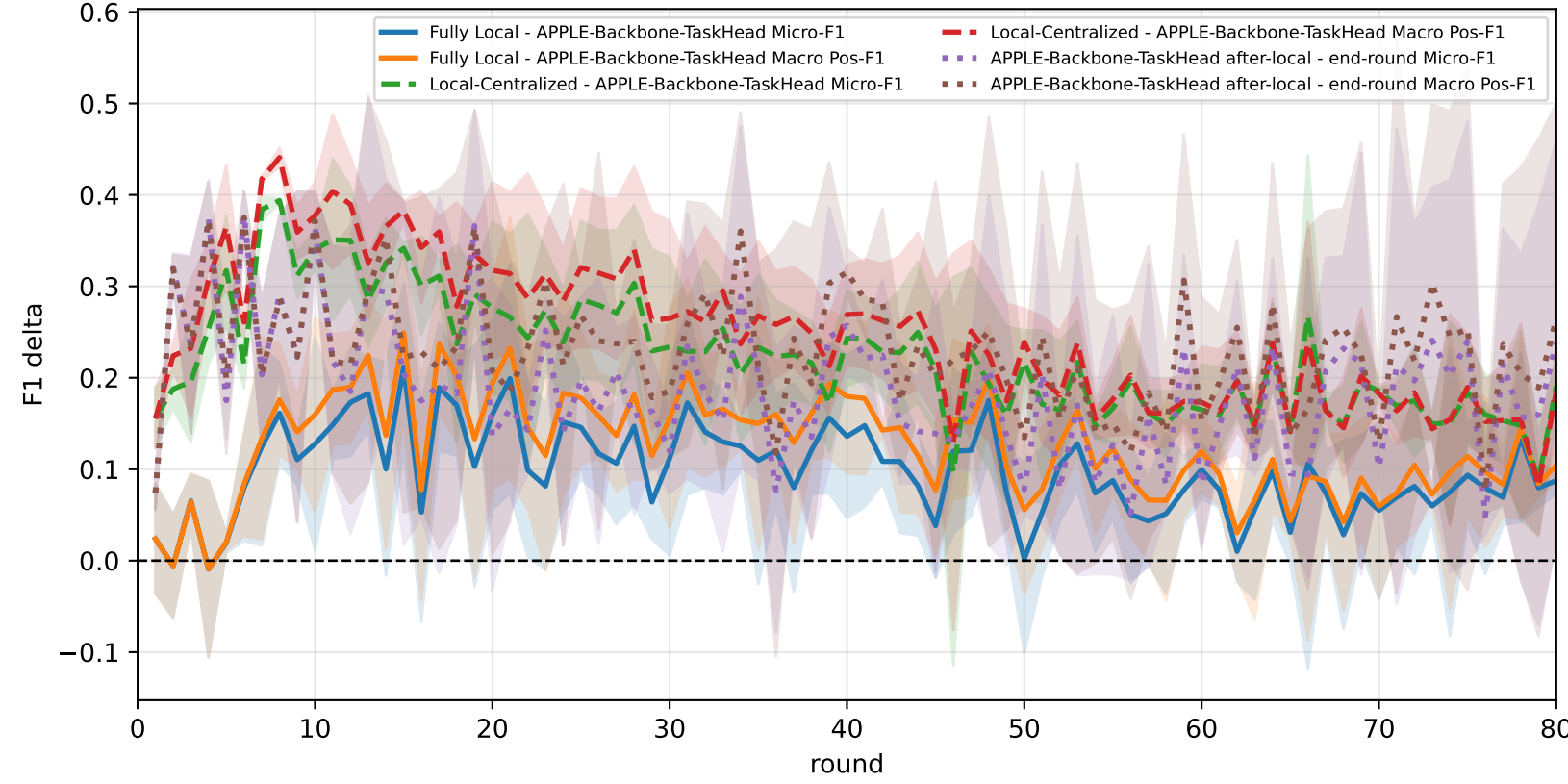
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap

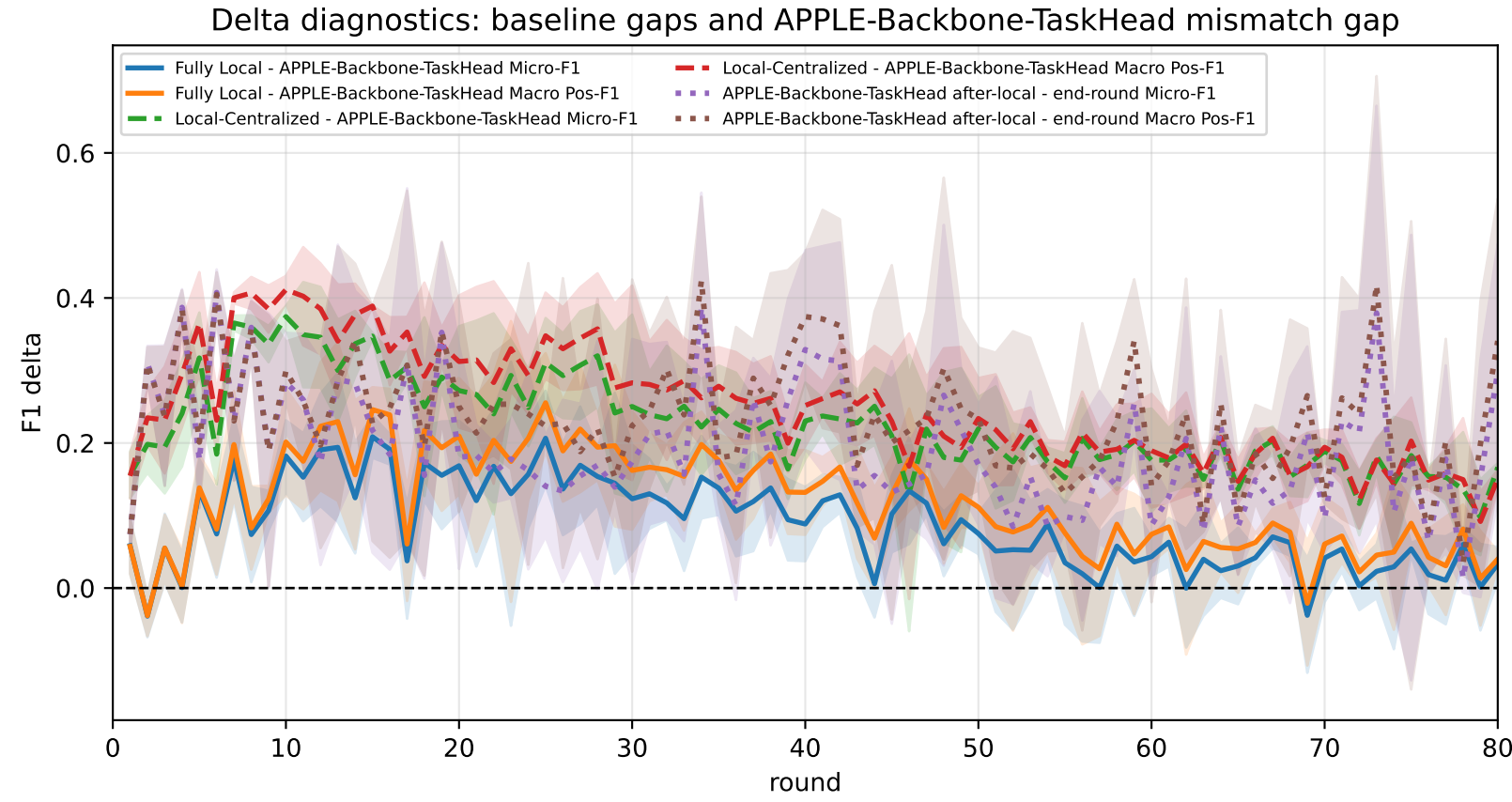
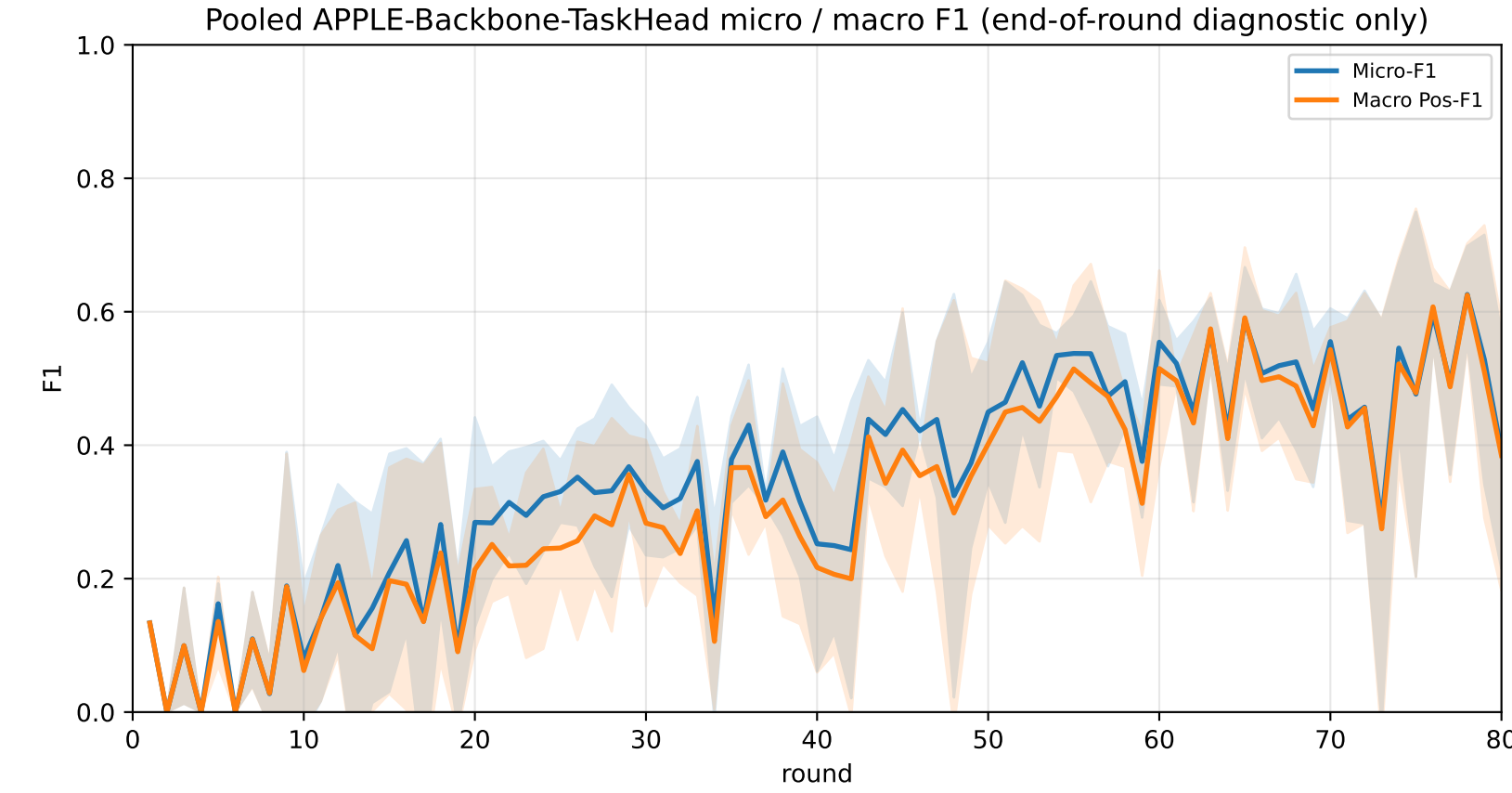
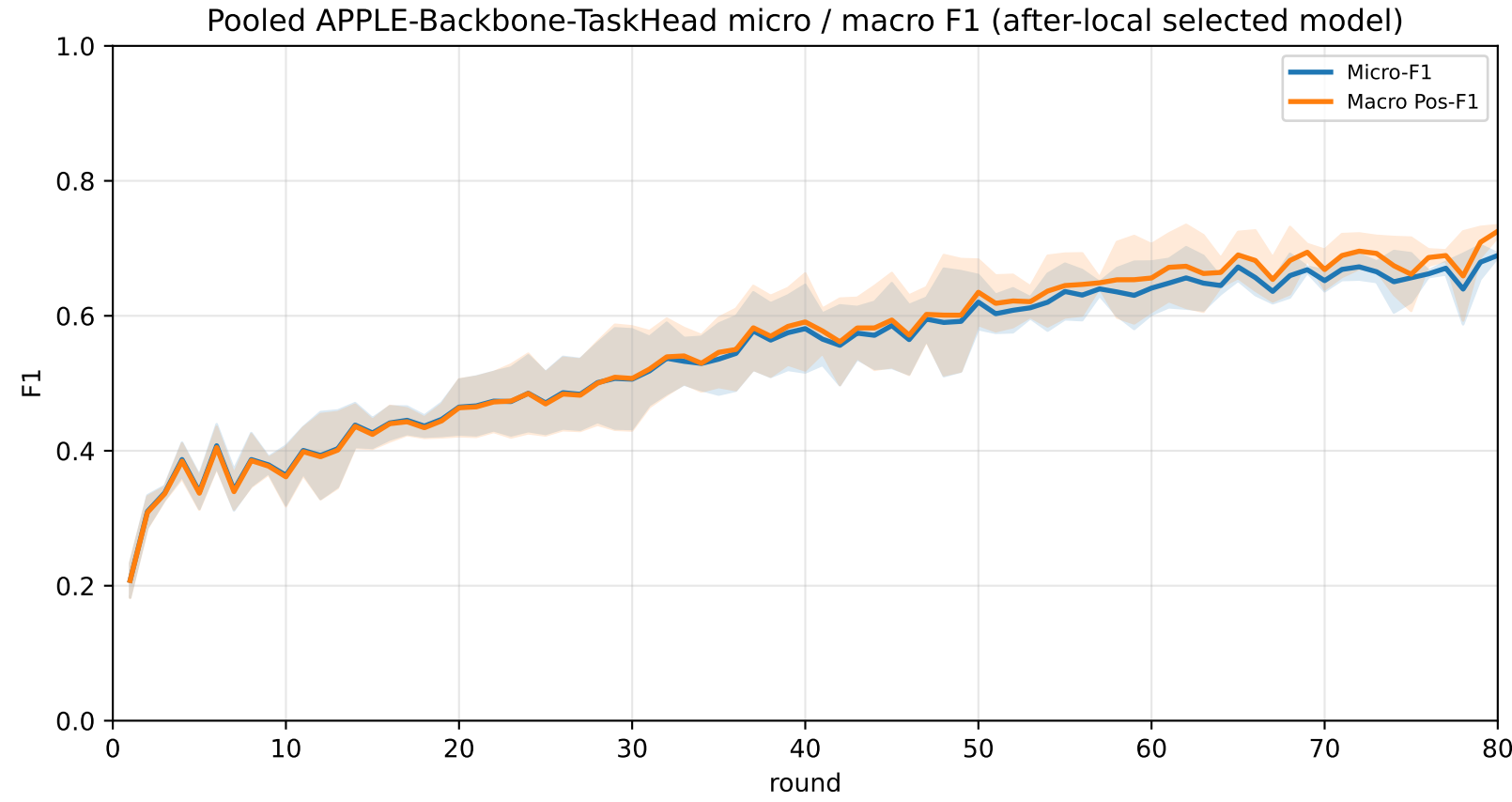
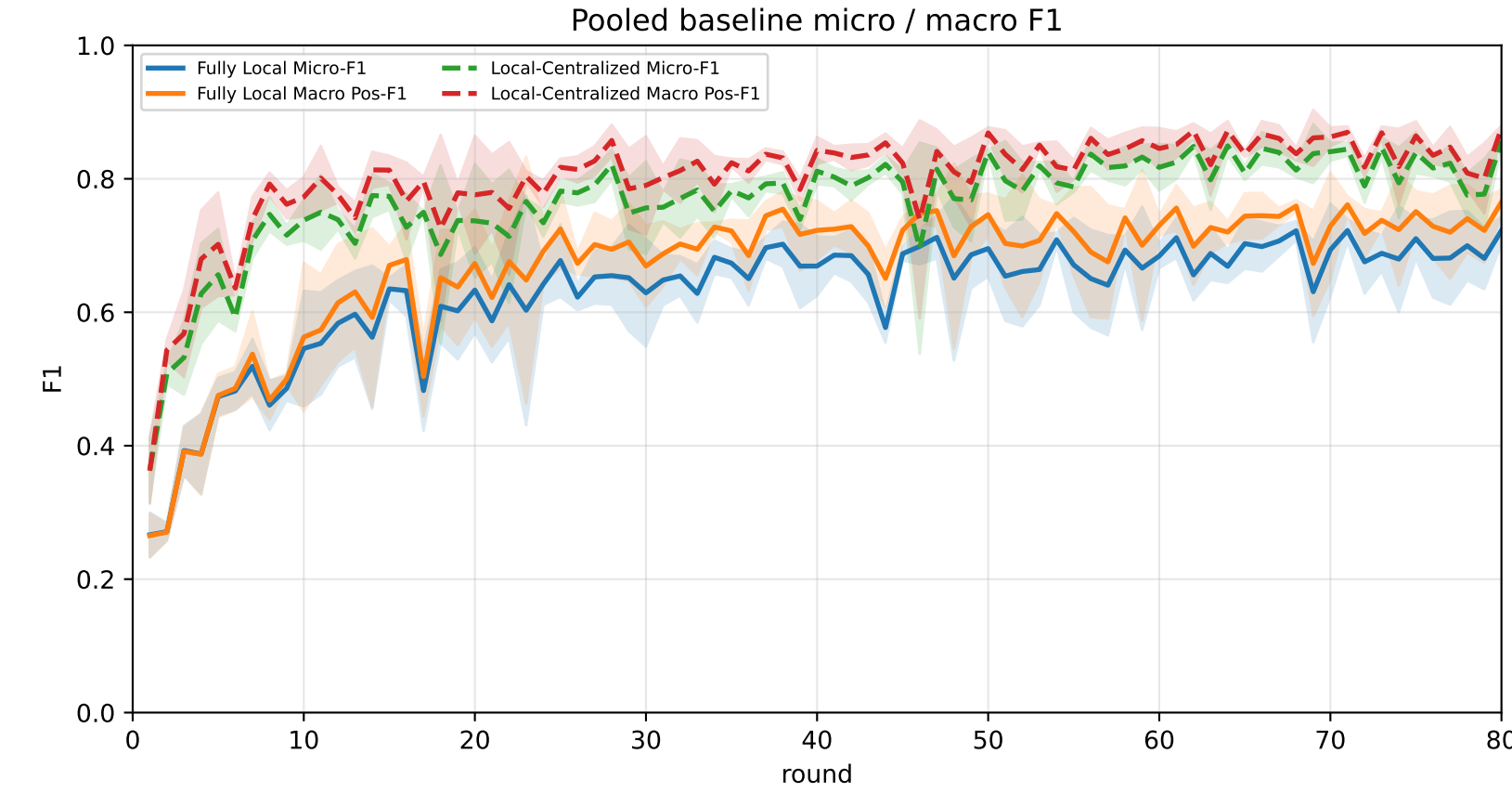
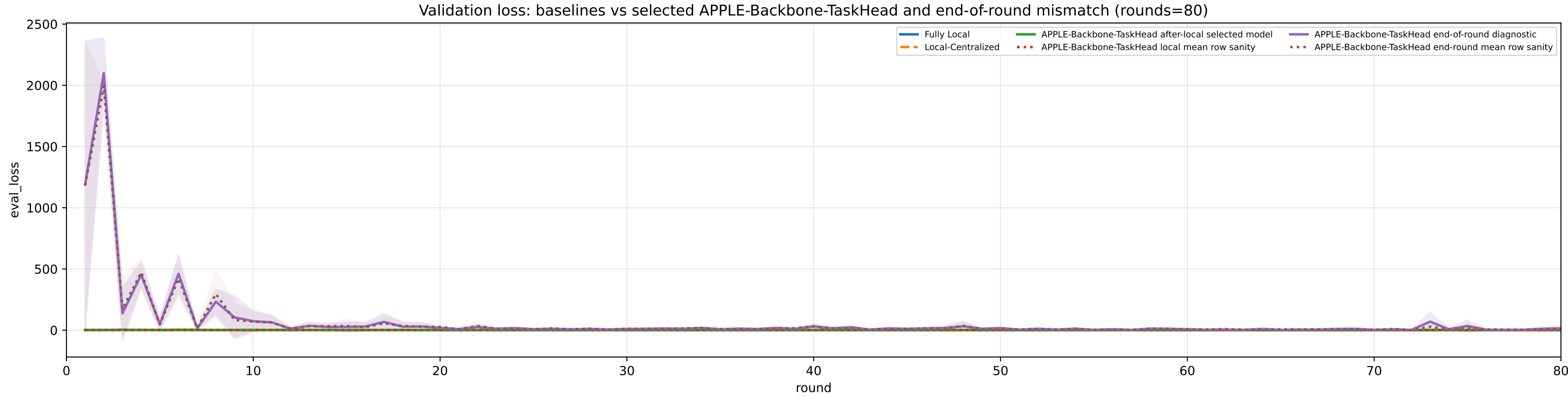


Controlled q-label heterogeneity: iid | APPLE-Backbone-TaskHead diagnostics |  
family=q\_task\_cycle6 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |  
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |  
heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 64.5% ± 3.5%        | <u>69.6% ± 3.3%</u> | <u>87.3% ± 2.7%</u> | <u>84.4% ± 3.4%</u> | 62.5% ± 7.2%        | <u>67.0% ± 5.0%</u> | 46.8% ± 1.7%        |
| Local-Centralized       | <b>85.0% ± 0.7%</b> | <b>87.0% ± 0.6%</b> | <b>98.0% ± 1.1%</b> | <b>94.5% ± 1.4%</b> | <b>89.7% ± 1.8%</b> | <b>80.3% ± 0.5%</b> | <b>72.5% ± 1.3%</b> |
| APPLE-Backbone-TaskHead | <u>65.3% ± 3.9%</u> | 67.2% ± 4.6%        | 76.0% ± 13.9%       | 73.0% ± 4.0%        | <u>68.0% ± 3.9%</u> | 63.8% ± 3.9%        | <u>55.2% ± 2.4%</u> |

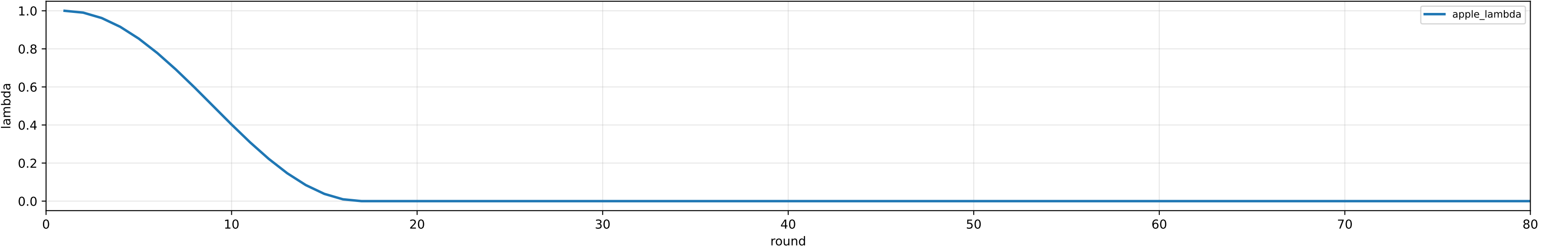




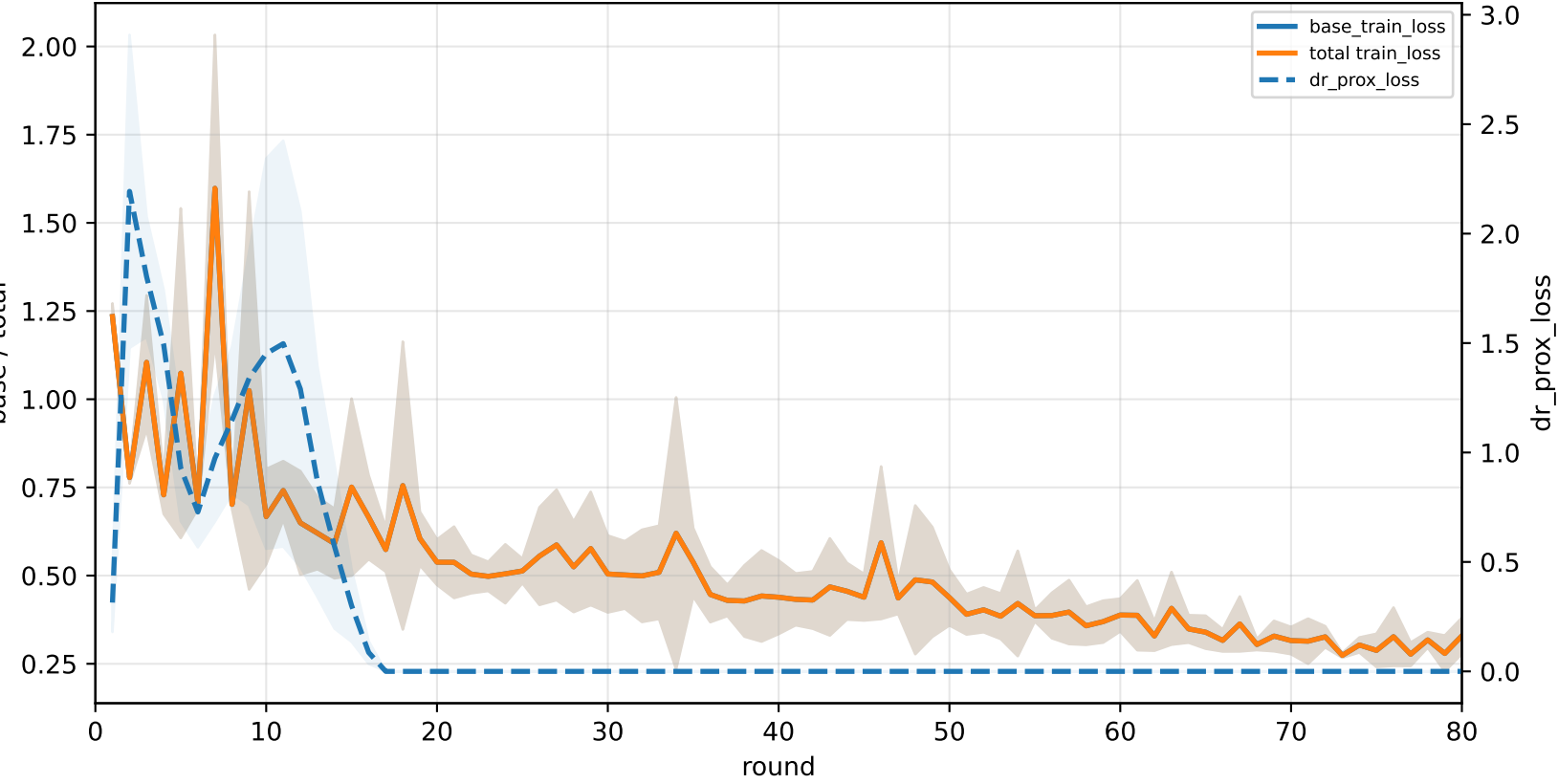
APPLE-Backbone-TaskHead training-loss decomposition | lid heterogeneity

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

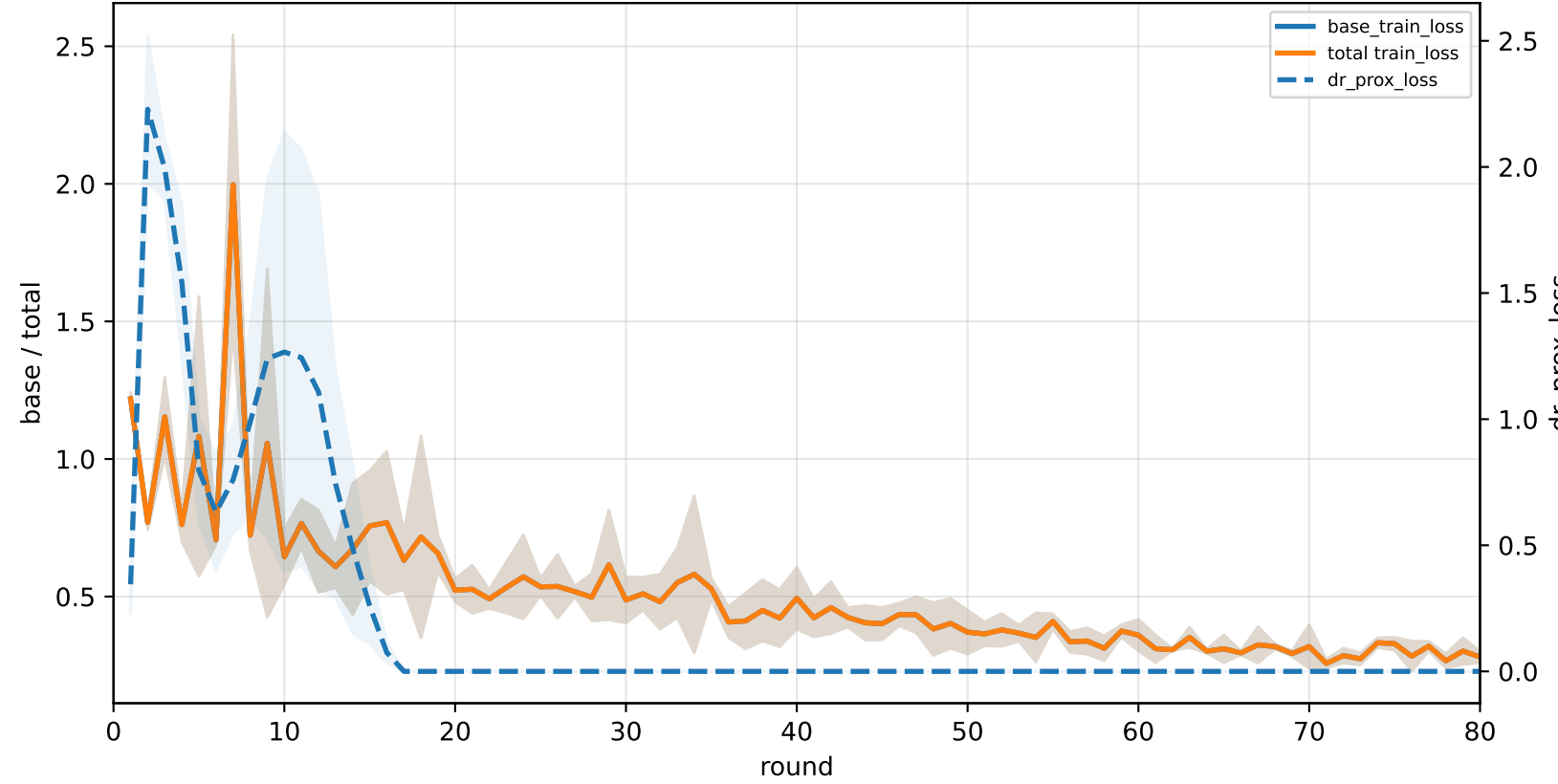
APPLE lambda scheduler | apple\_mu=0.001



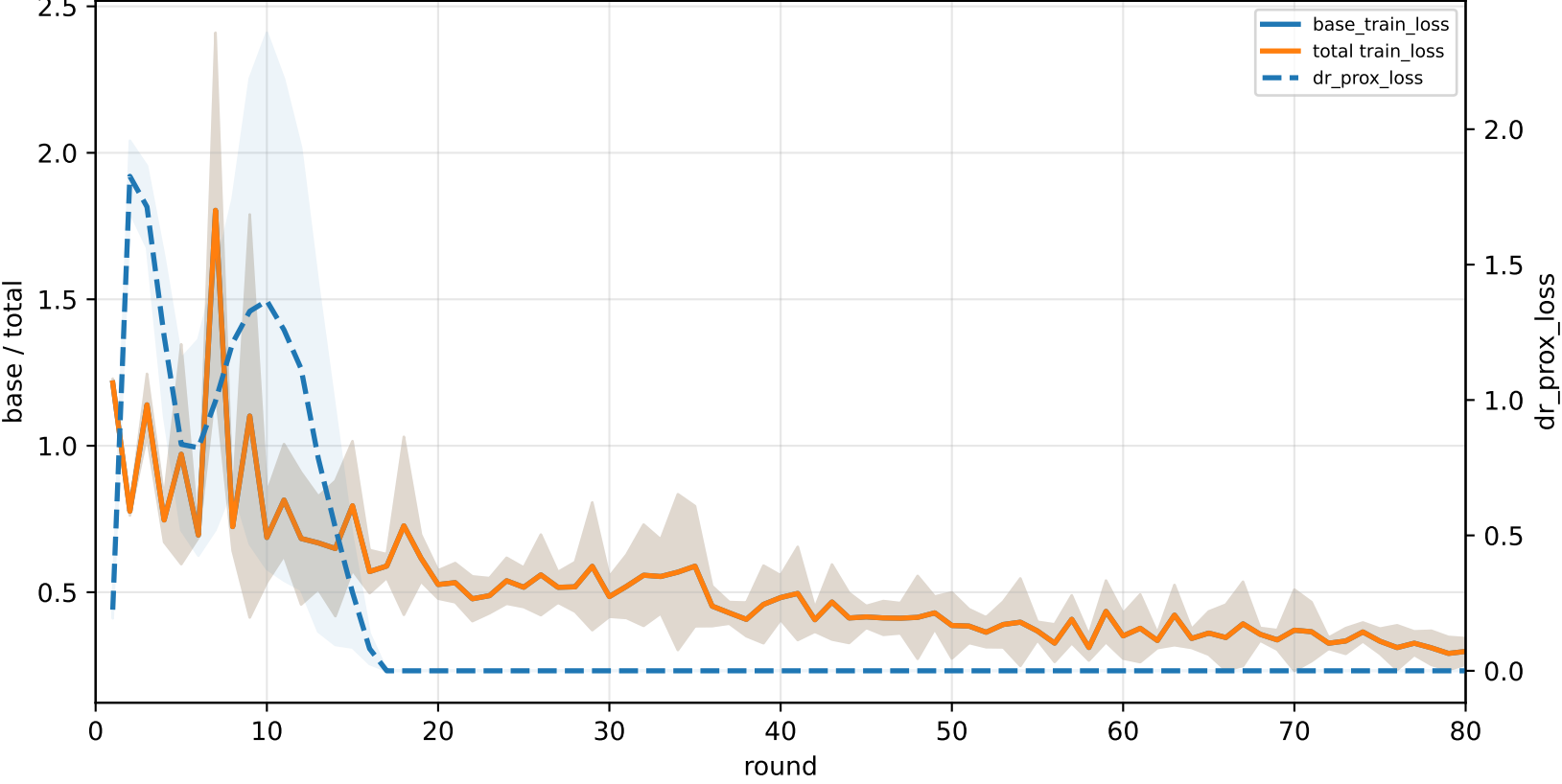
C2-specialist | graph\_id=3000000



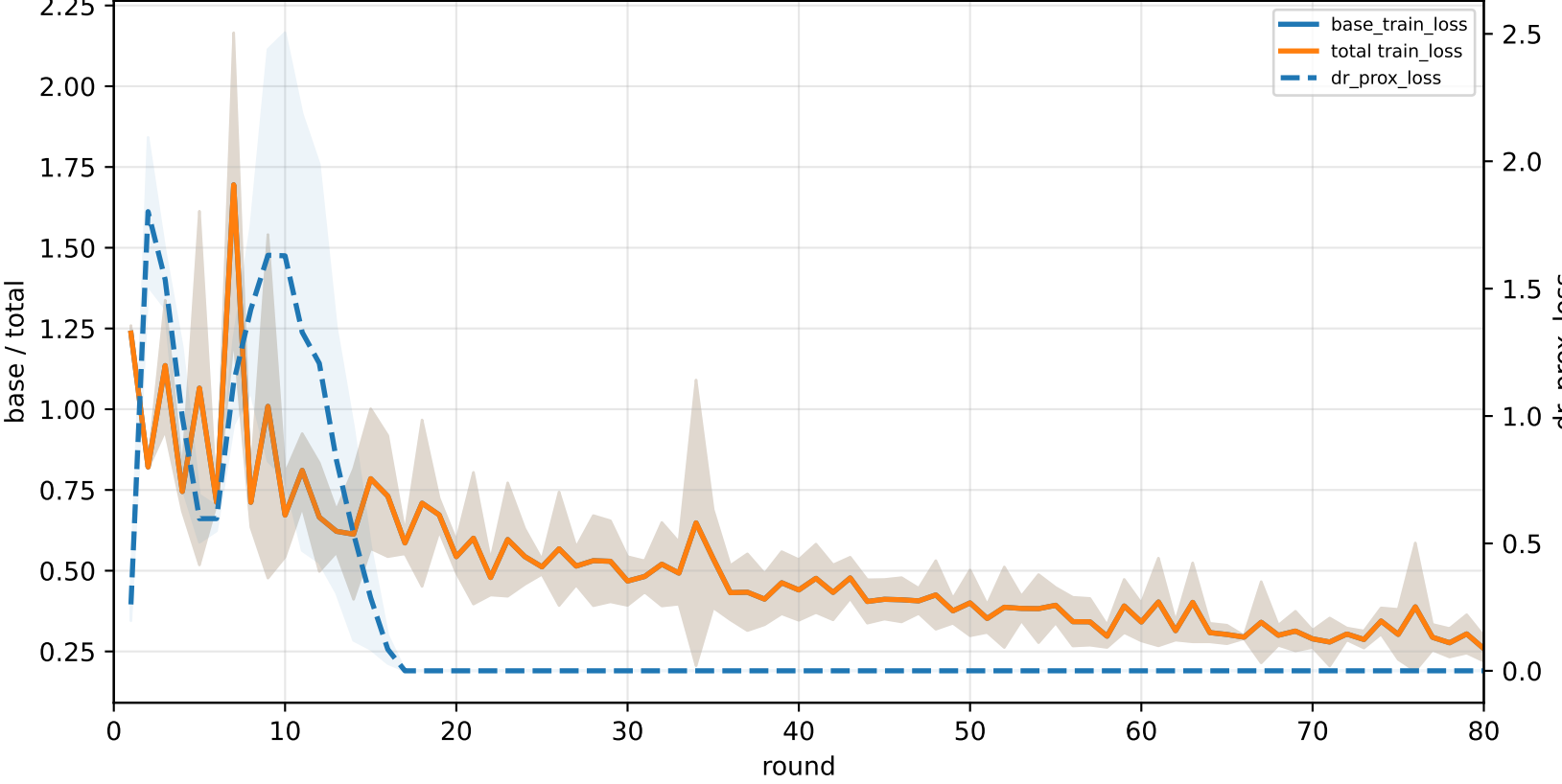
C3-specialist | graph\_id=3000001



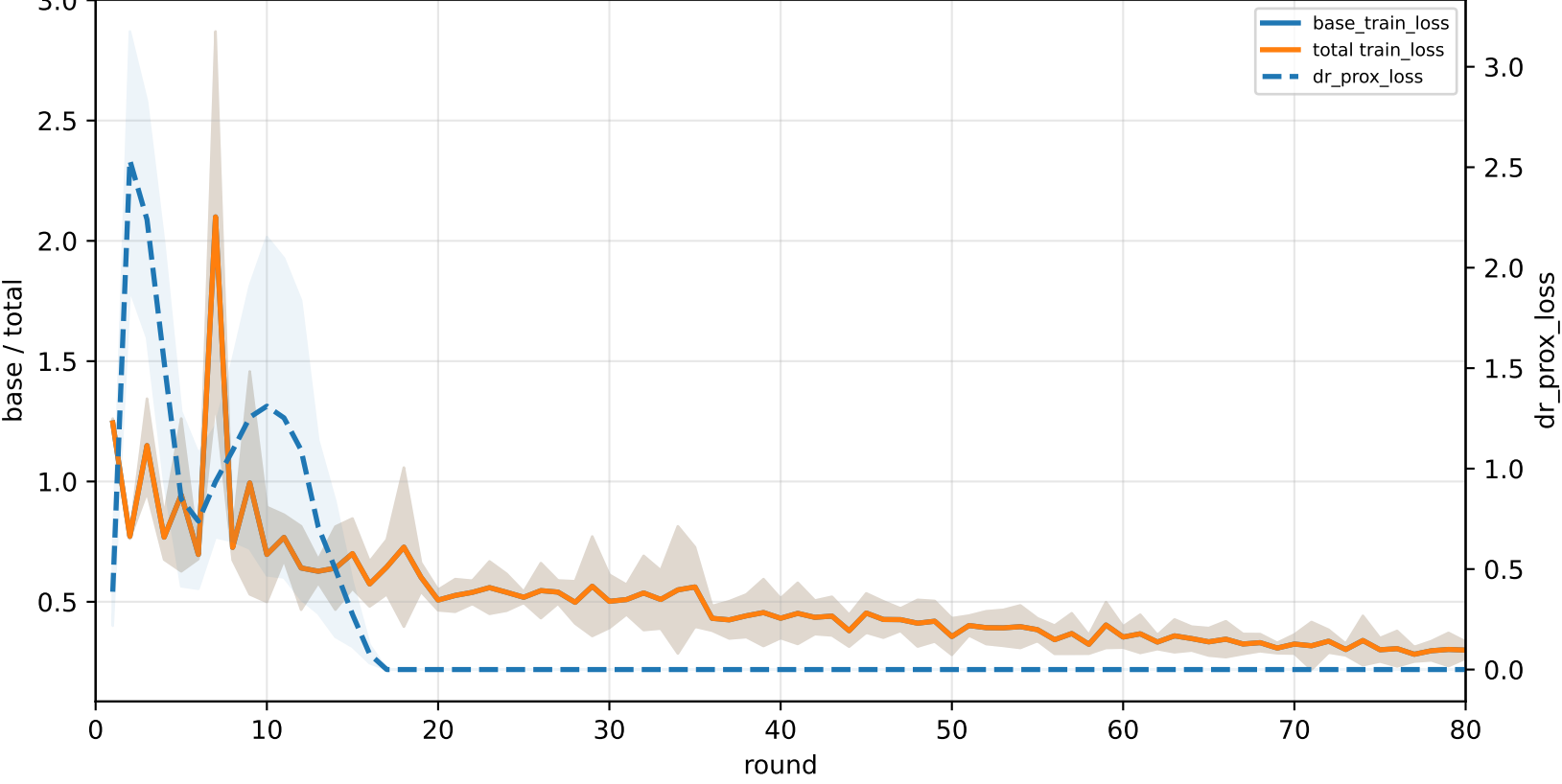
C4-specialist | graph\_id=3000002



C5-specialist | graph\_id=3000003



C6-specialist | graph\_id=3000004

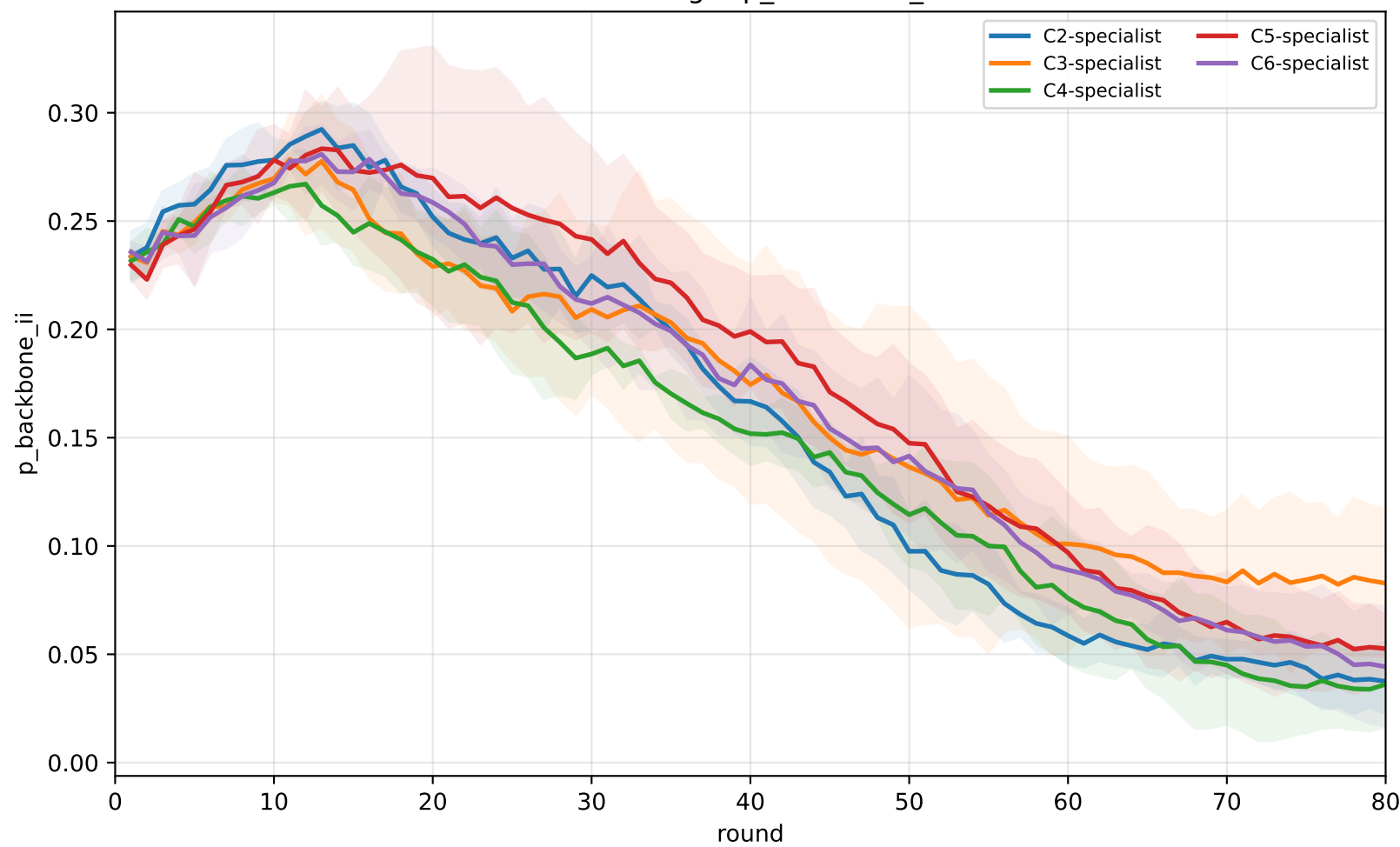




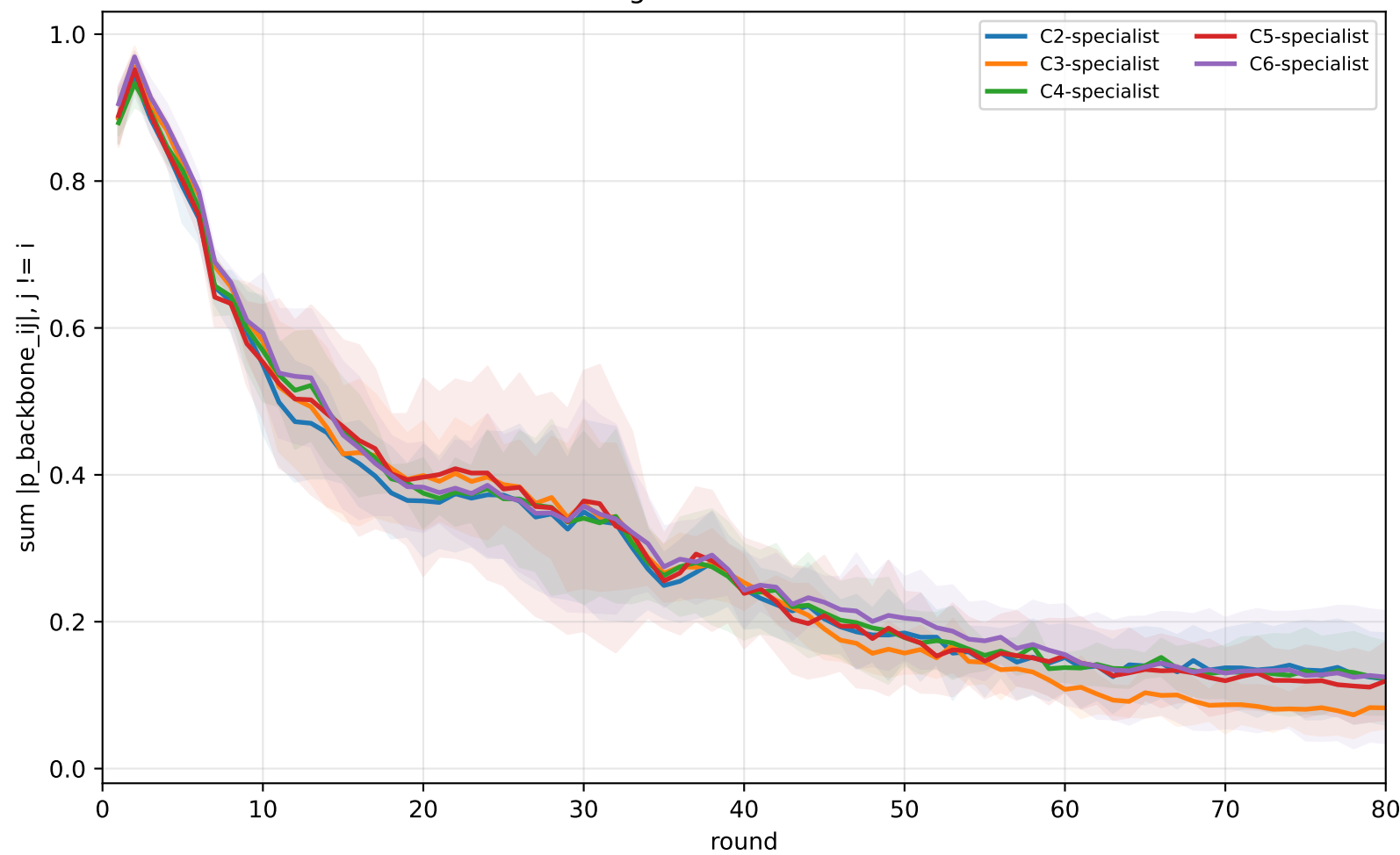
APPLE-Backbone-TaskHead backbone DR behavior | iid heterogeneity

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

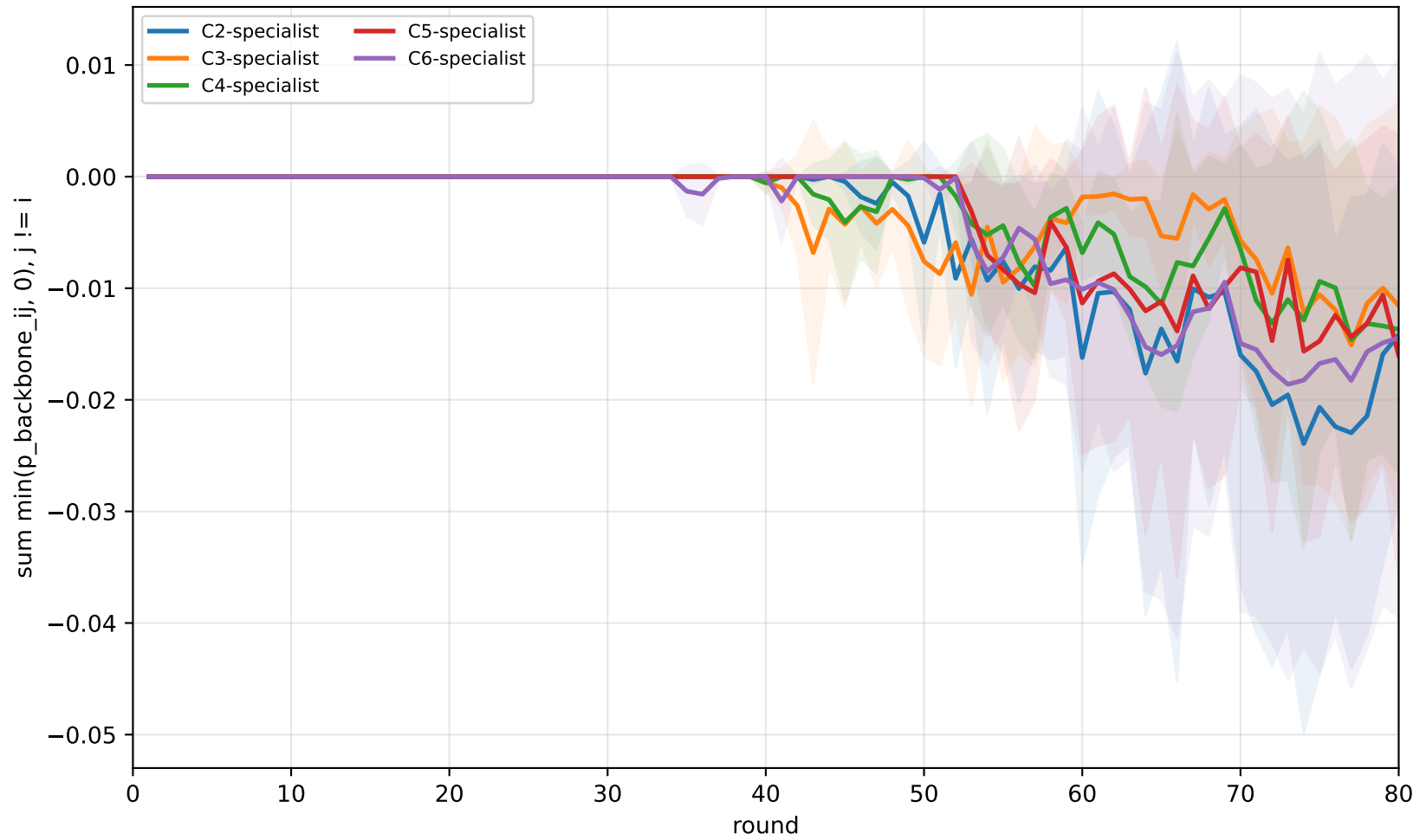
Backbone DR self-weight  $p_{\text{backbone\_ii}}$  over rounds



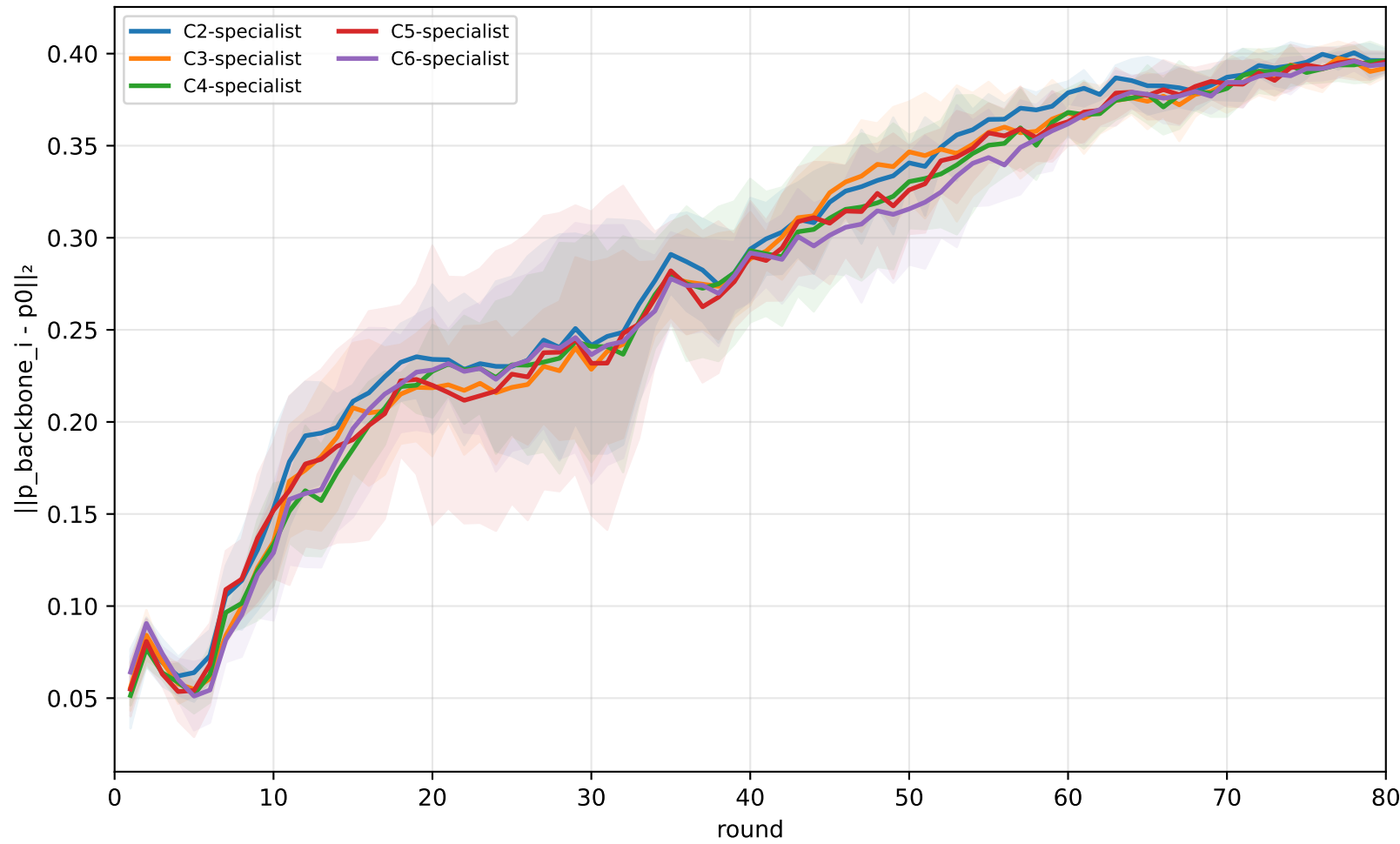
Backbone off-diagonal absolute mass over rounds



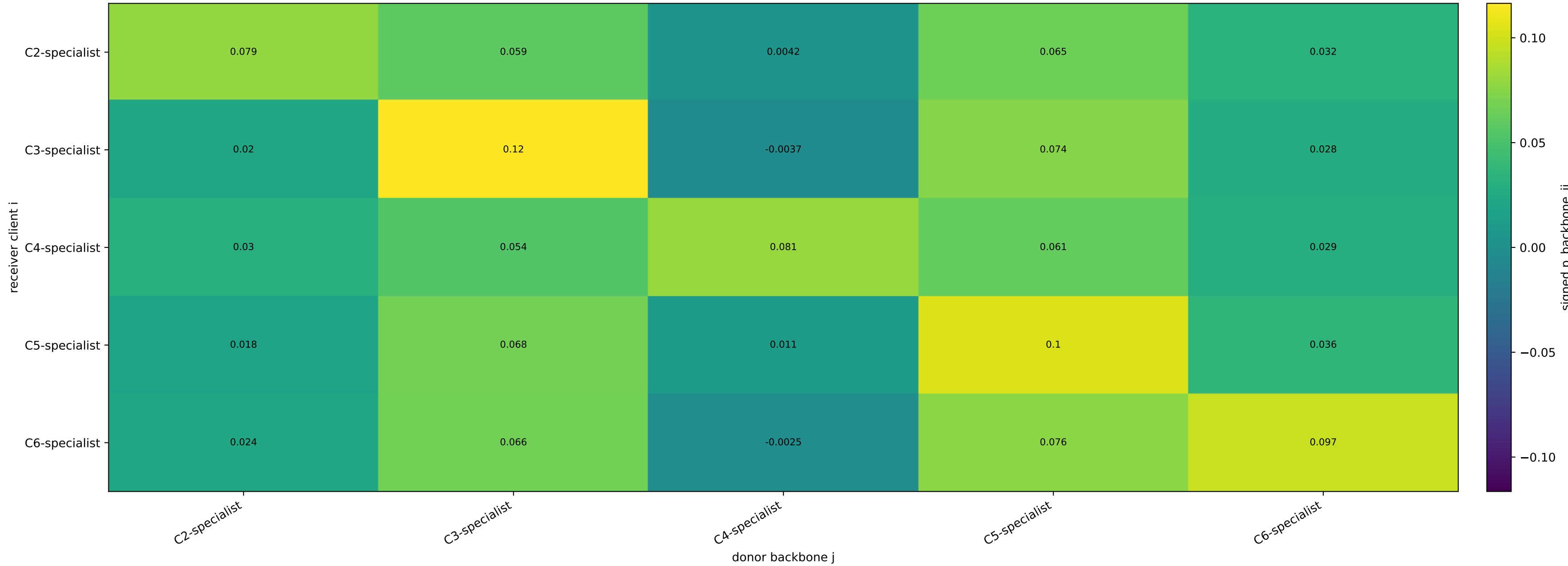
Backbone off-diagonal negative mass over rounds



Backbone DR movement away from  $p_0$



Selected-best-round backbone DR coefficients, mean over seeds

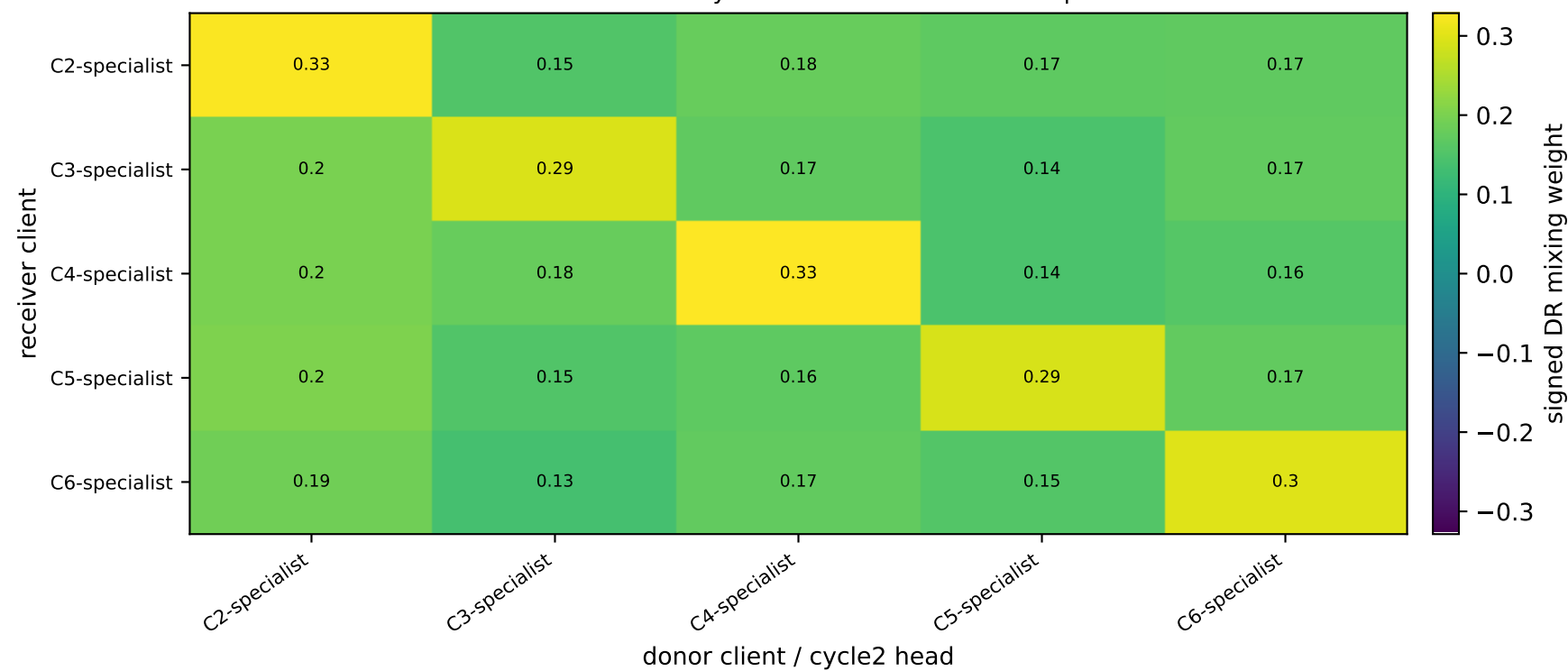




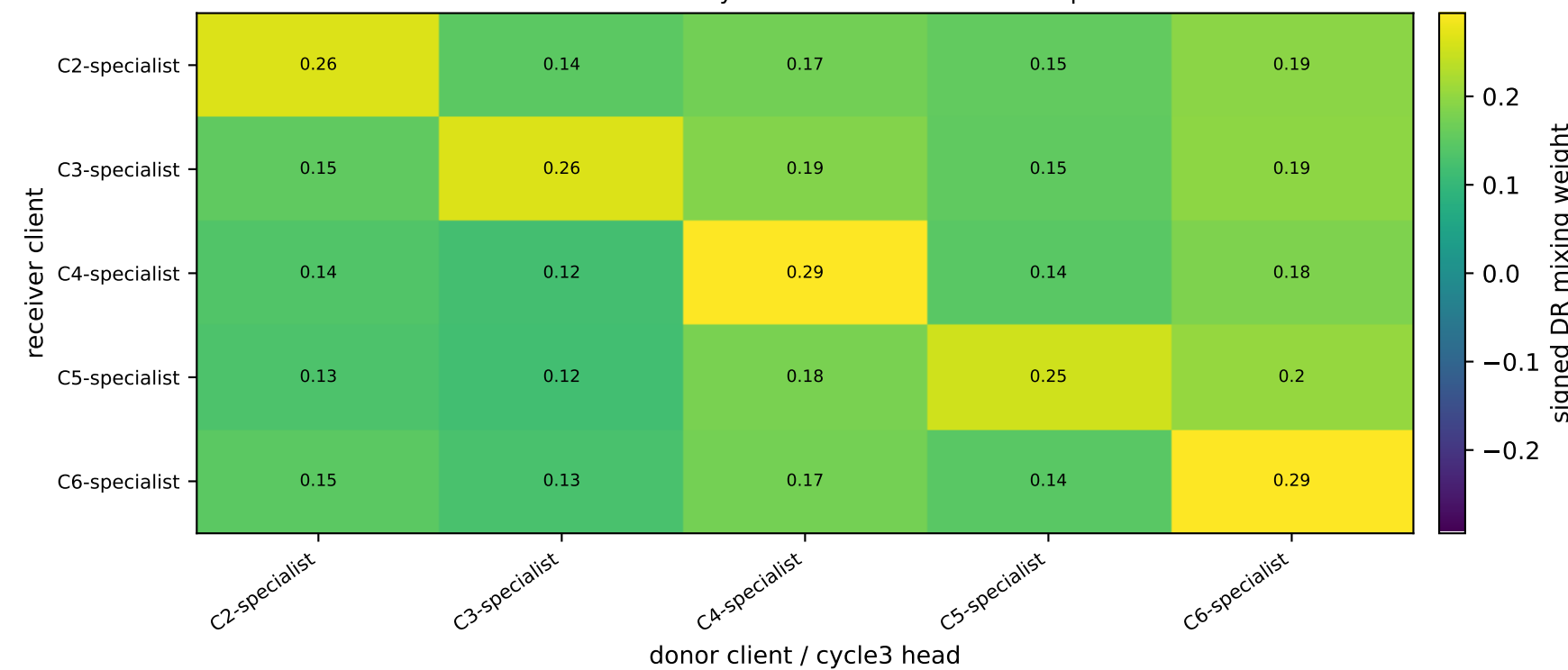
# APPLE-Backbone-TaskHead task-wise DR matrices: receiver clients × donor task-heads | lid

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

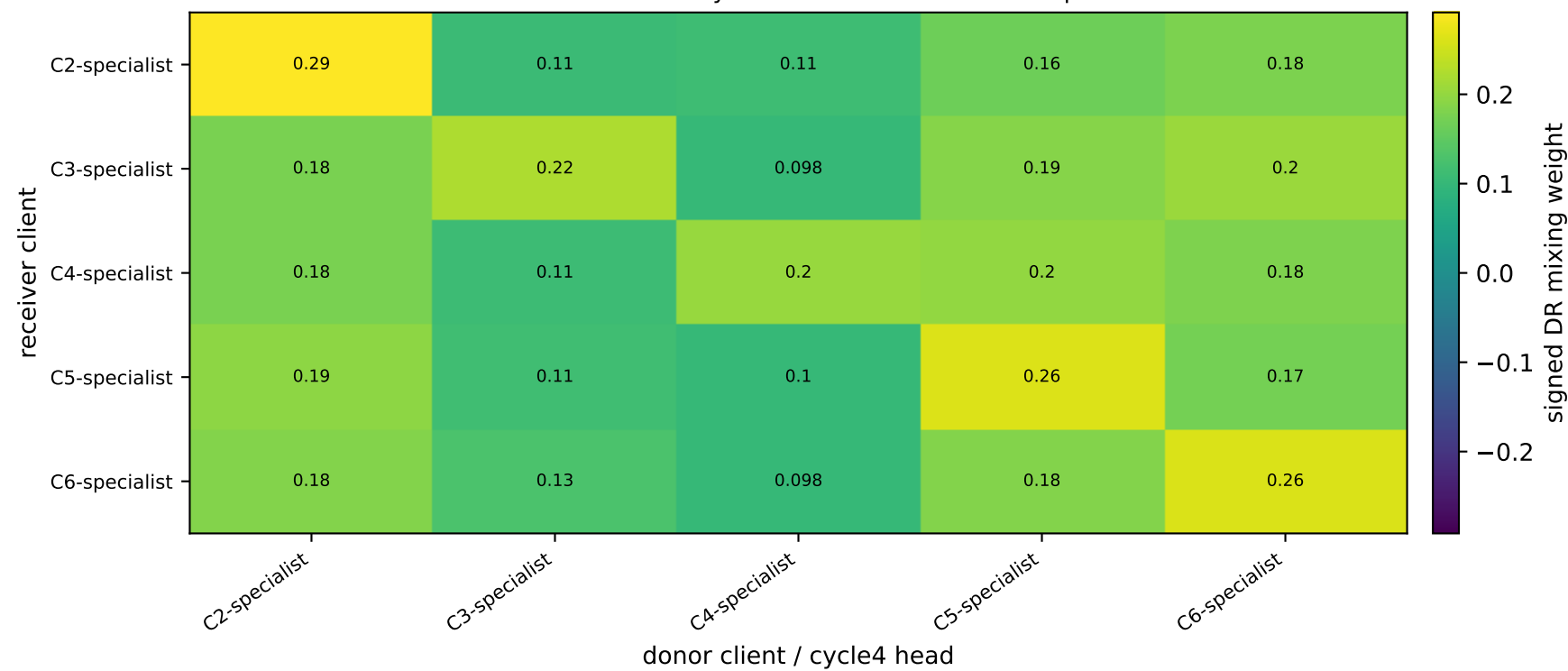
DR matrix for task cycle2: at best selected checkpoint



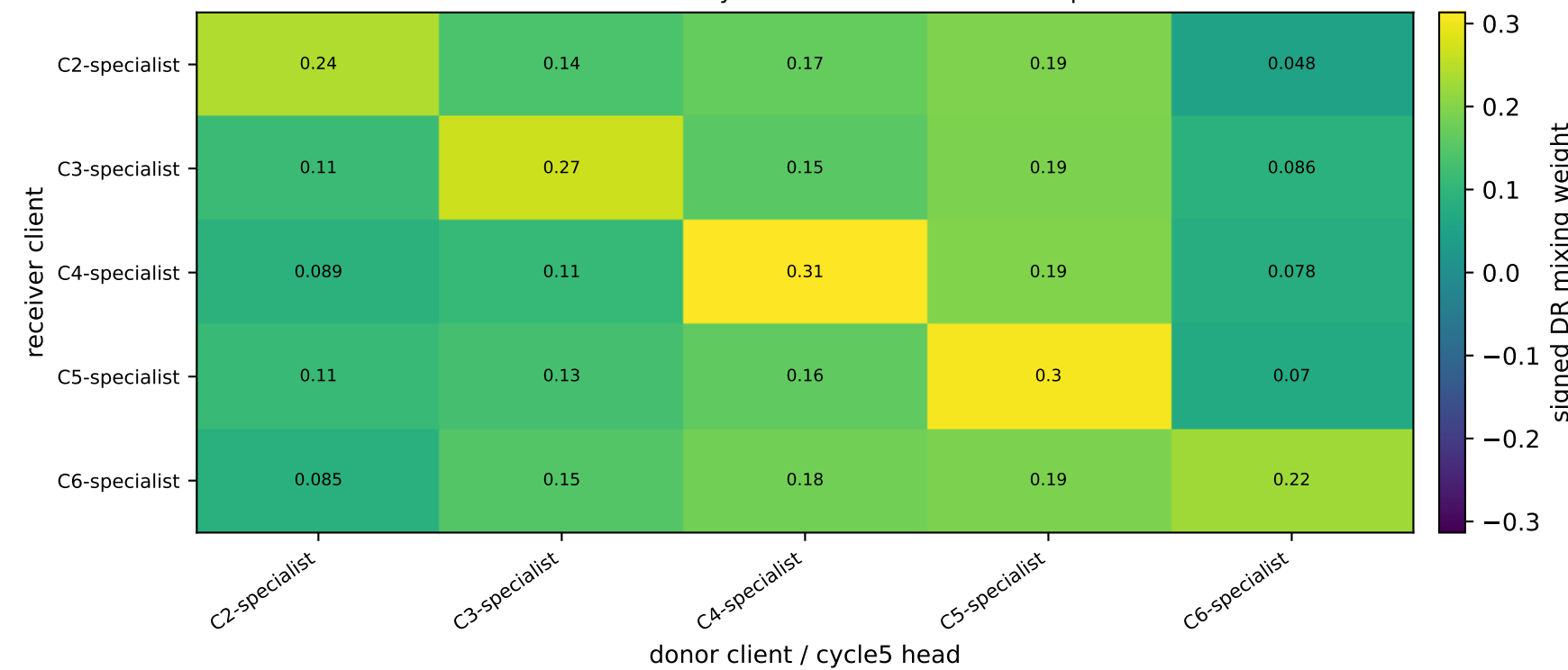
DR matrix for task cycle3: at best selected checkpoint



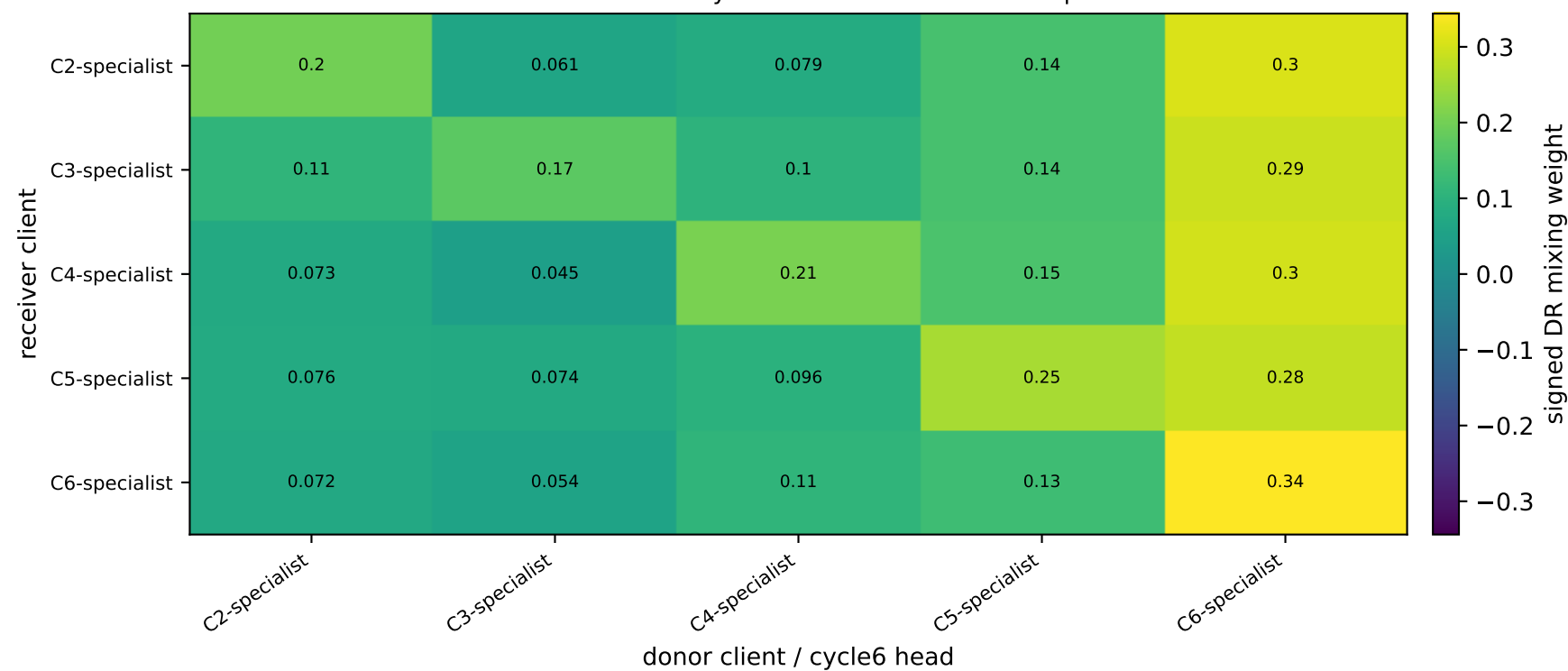
DR matrix for task cycle4: at best selected checkpoint



DR matrix for task cycle5: at best selected checkpoint



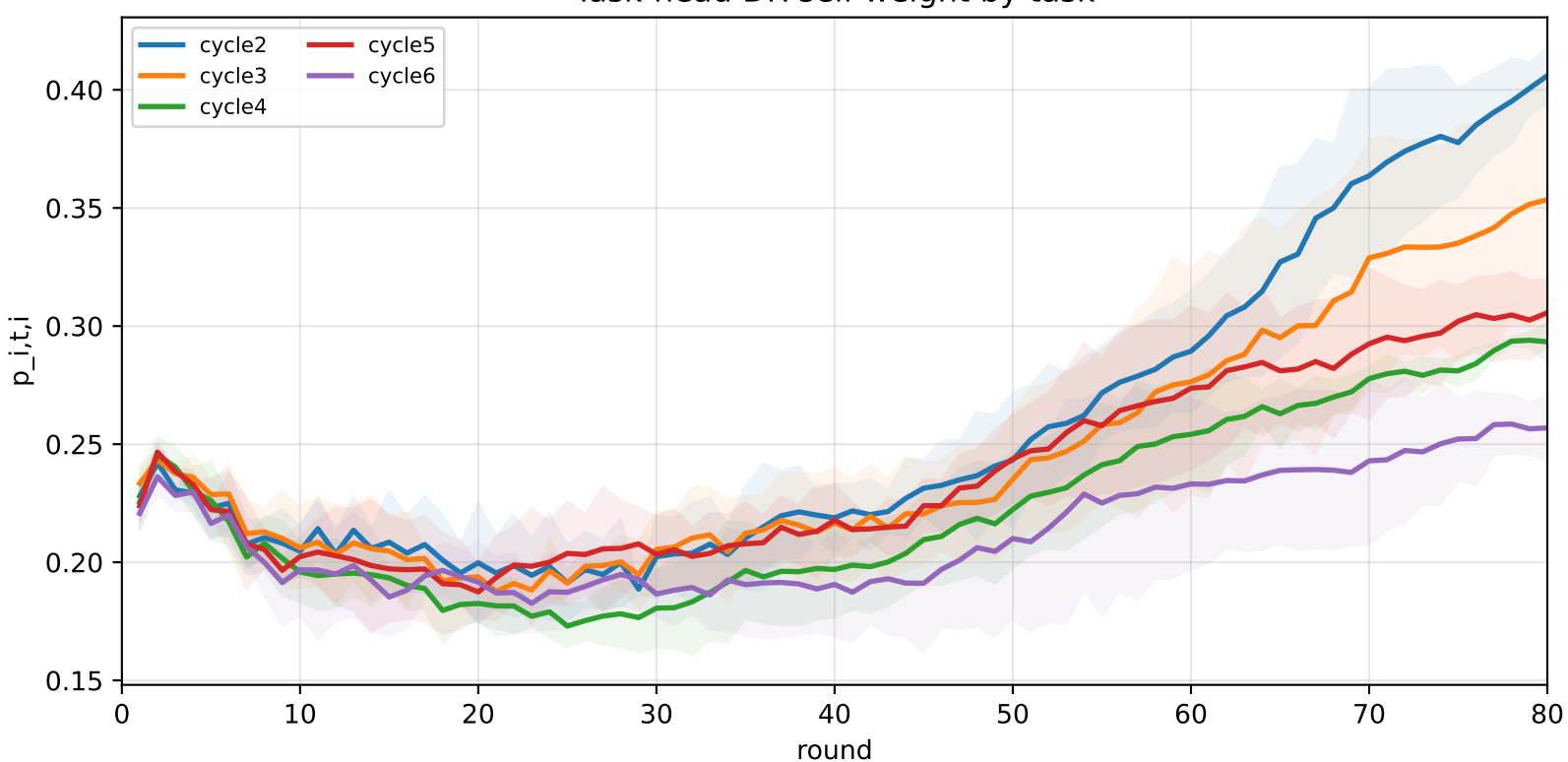
DR matrix for task cycle6: at best selected checkpoint



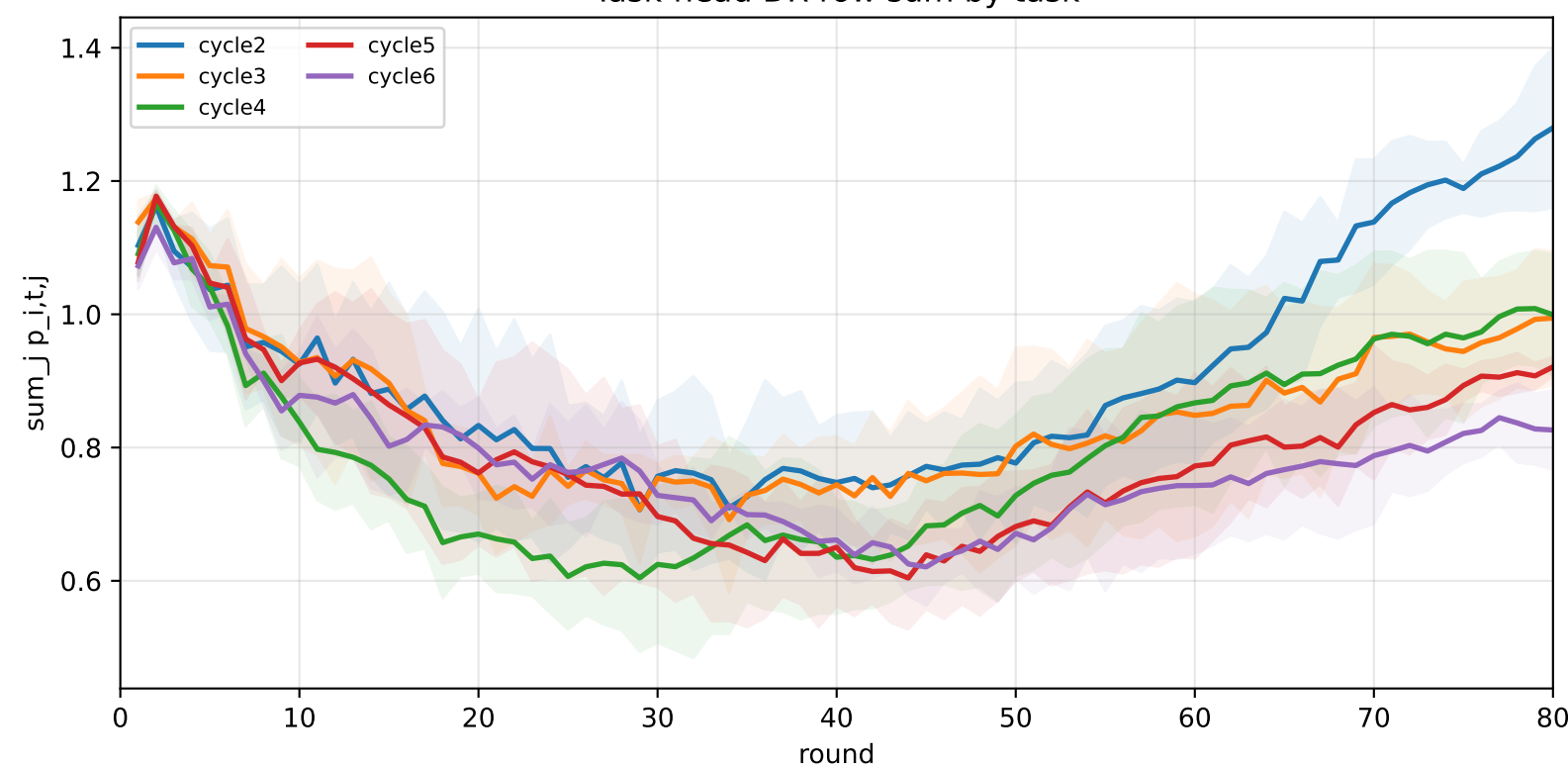
# APPLE-Backbone-TaskHead task-wise DR dynamics | iid

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%  
 heterogeneity=iid | gamma=0.20 | task\_jsd\_mean=0.000 | task\_jsd\_median=0.000 | task\_jsd\_max=0.000 | family\_centroid\_jsd\_mean=0.000

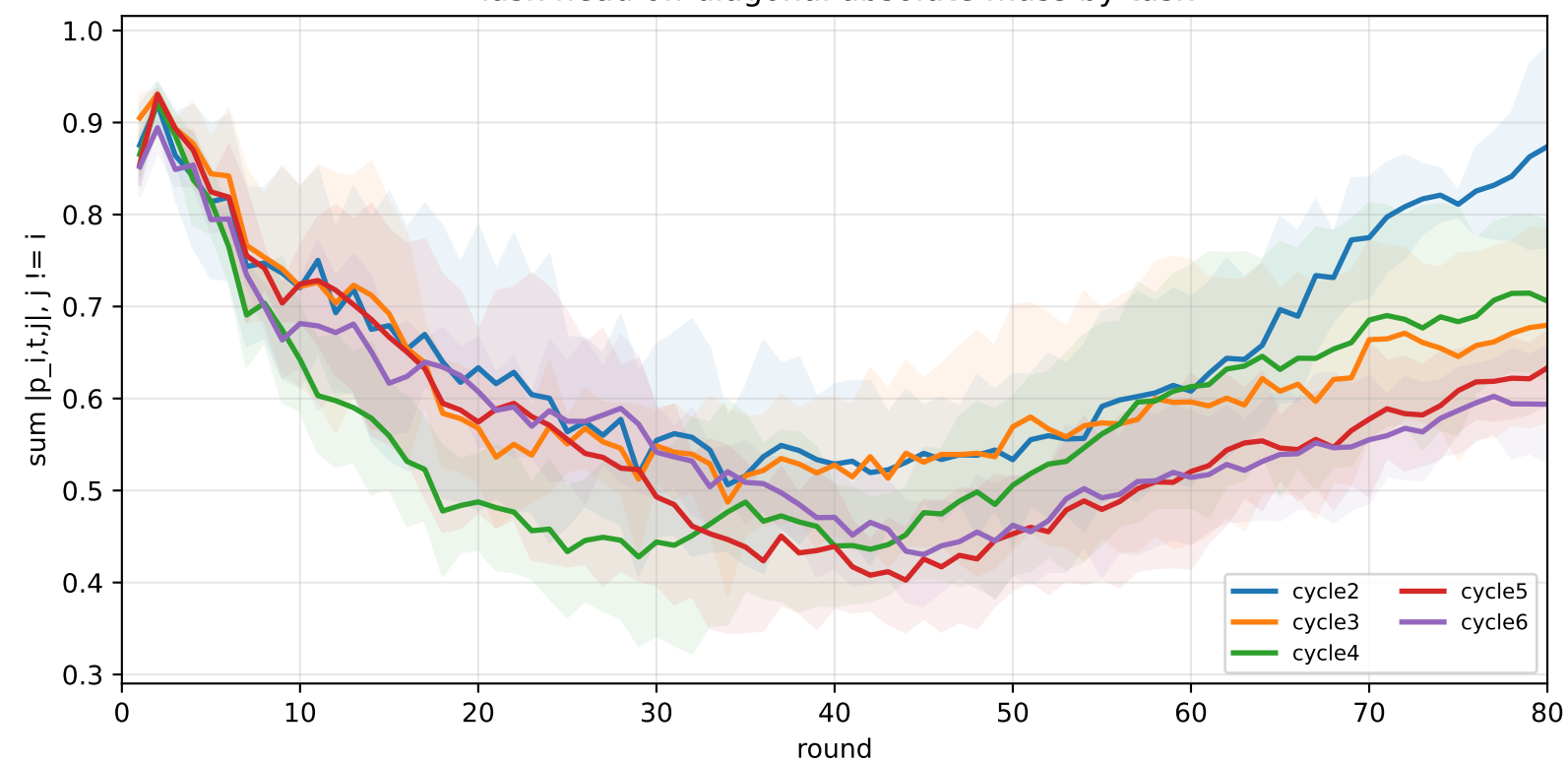
Task-head DR self-weight by task



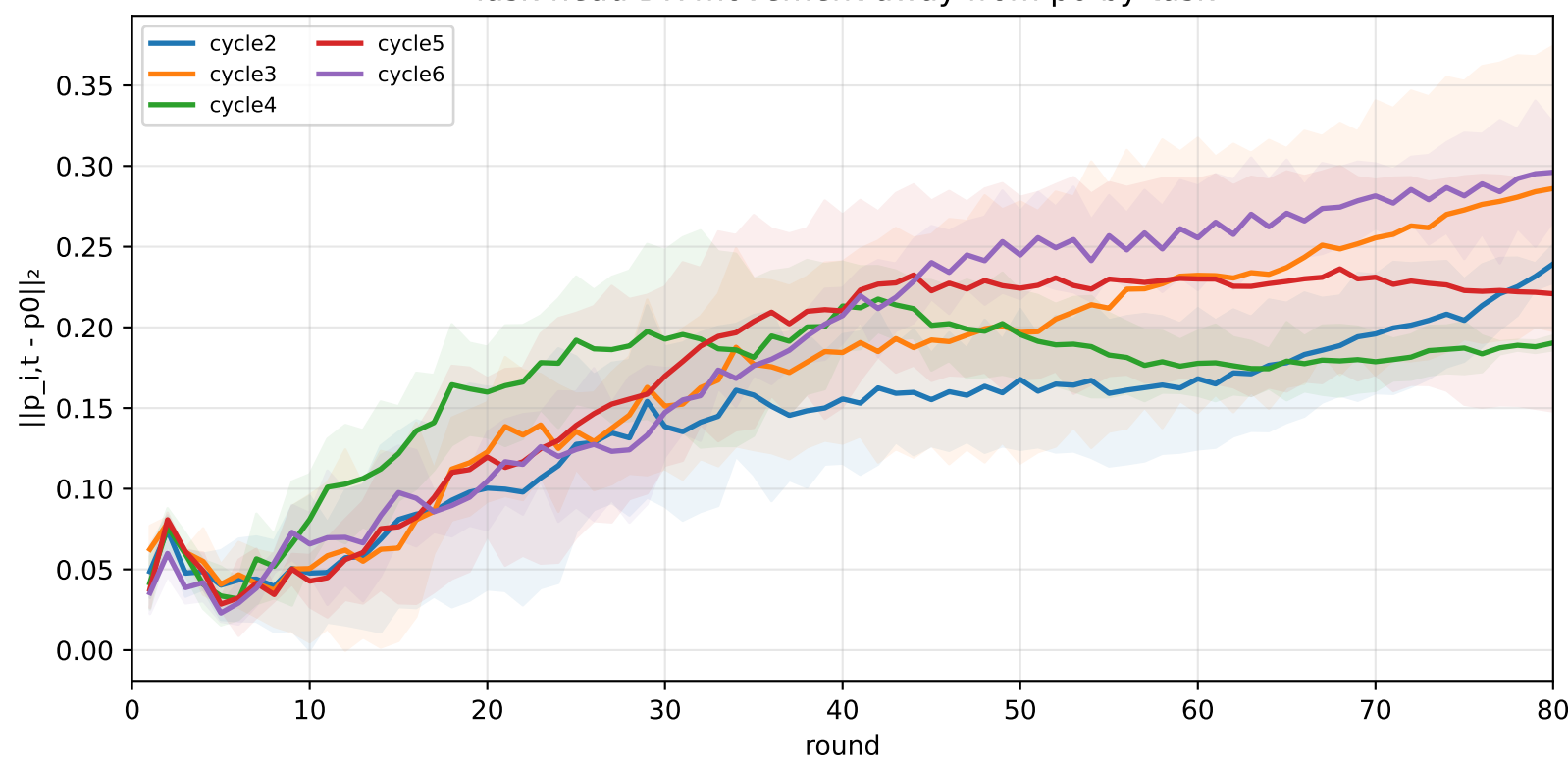
Task-head DR row sum by task



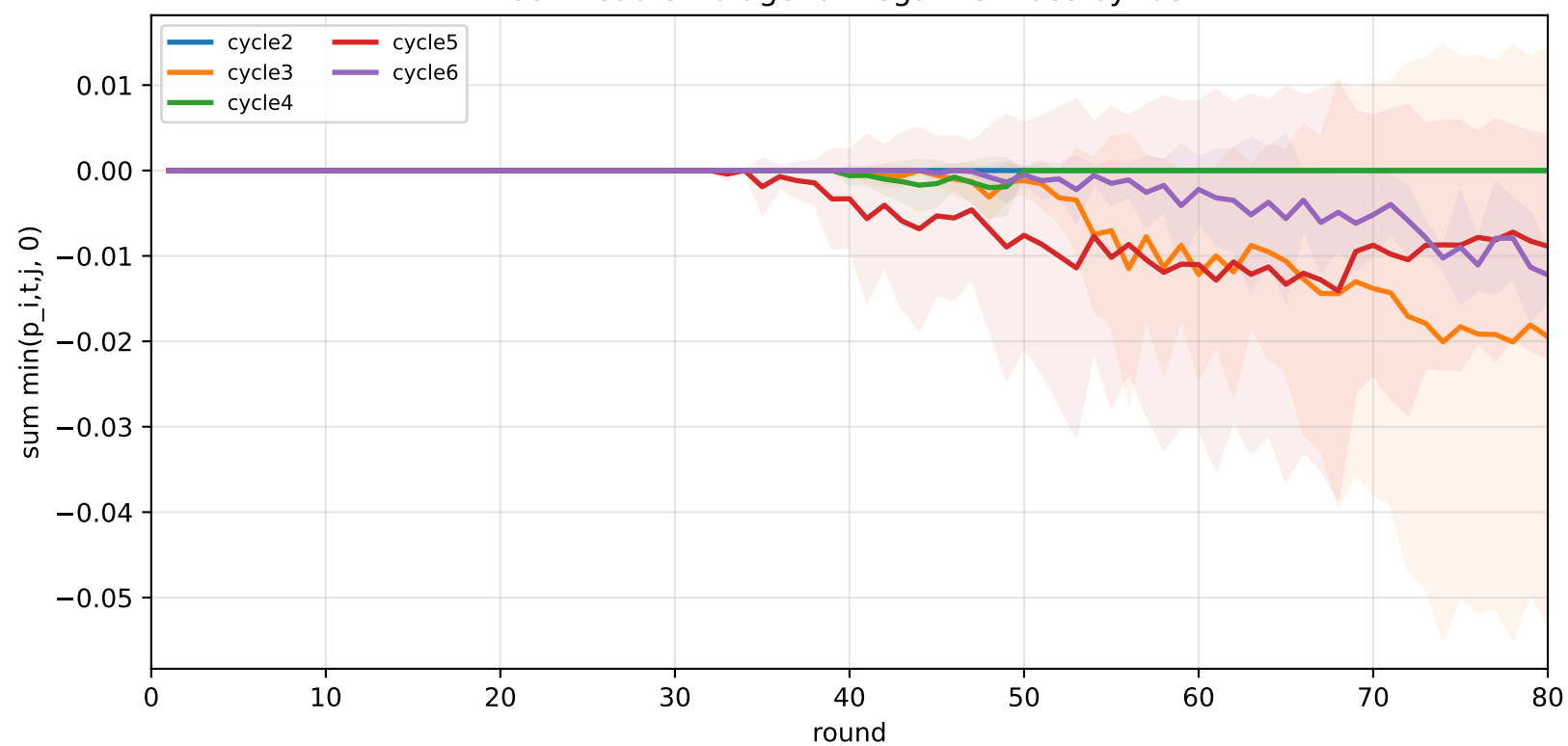
Task-head off-diagonal absolute mass by task



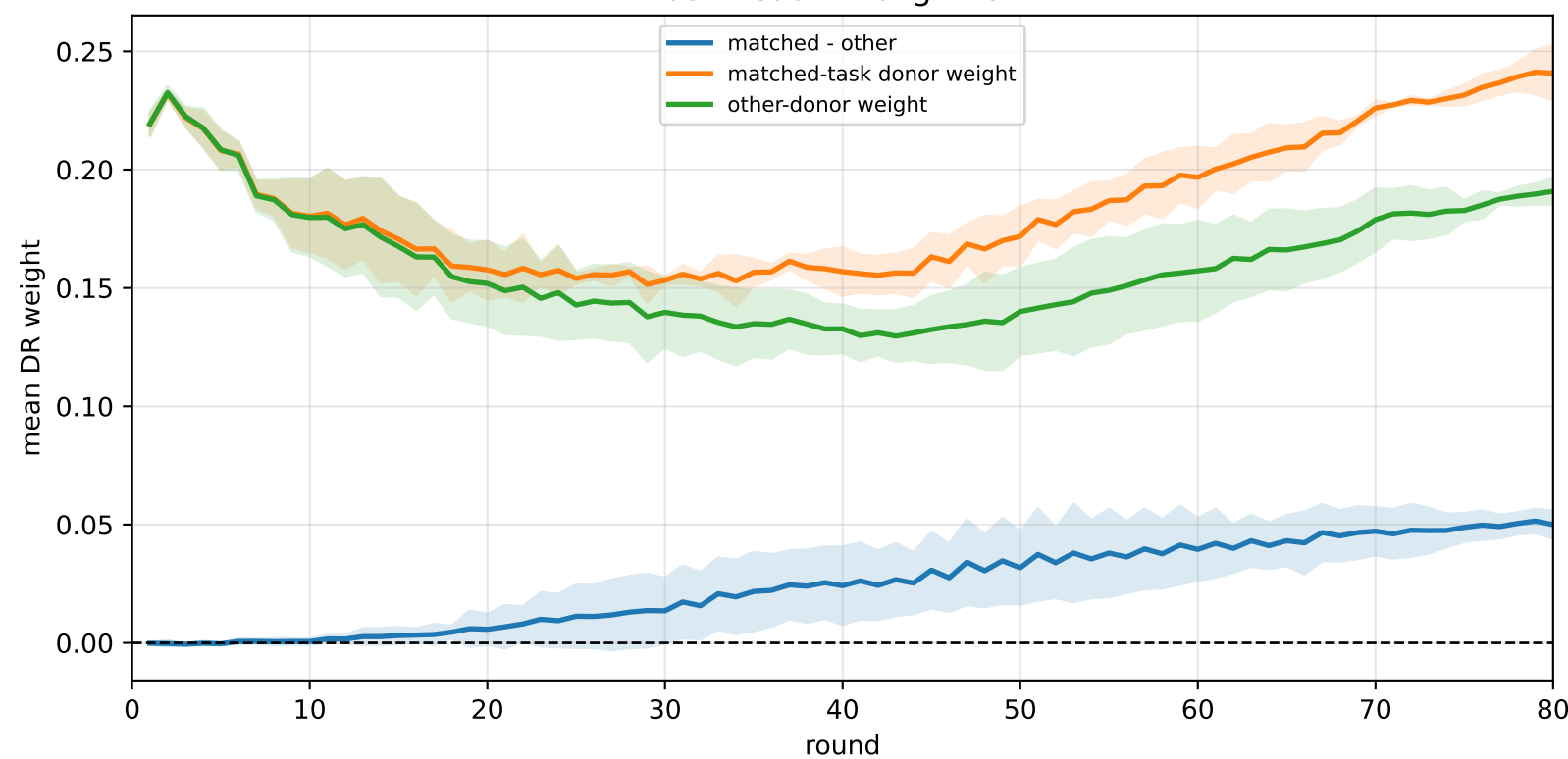
Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment





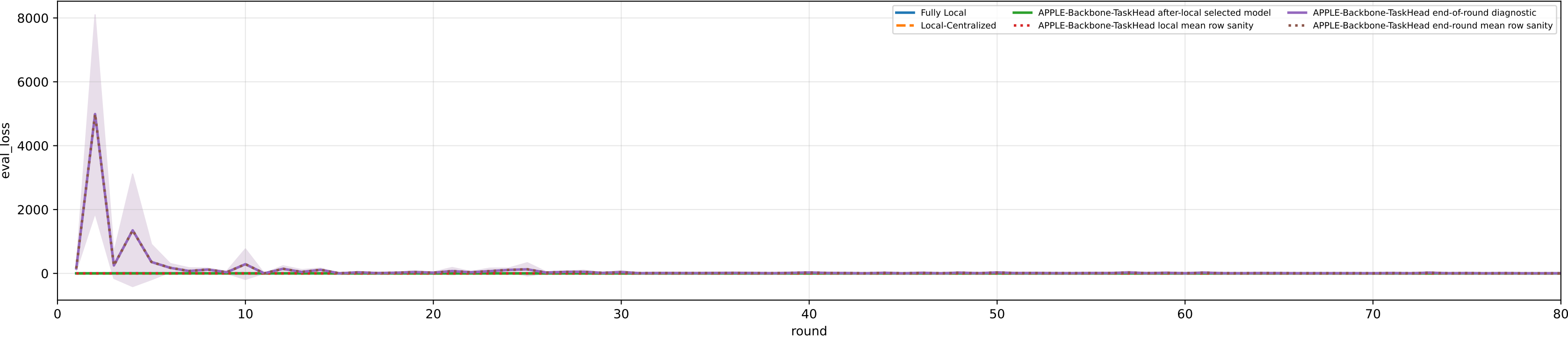
Controlled q-label heterogeneity: Mid | APPLE-Backbone-TaskHead diagnostics | aggregated /  
pooled over 5 clients |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

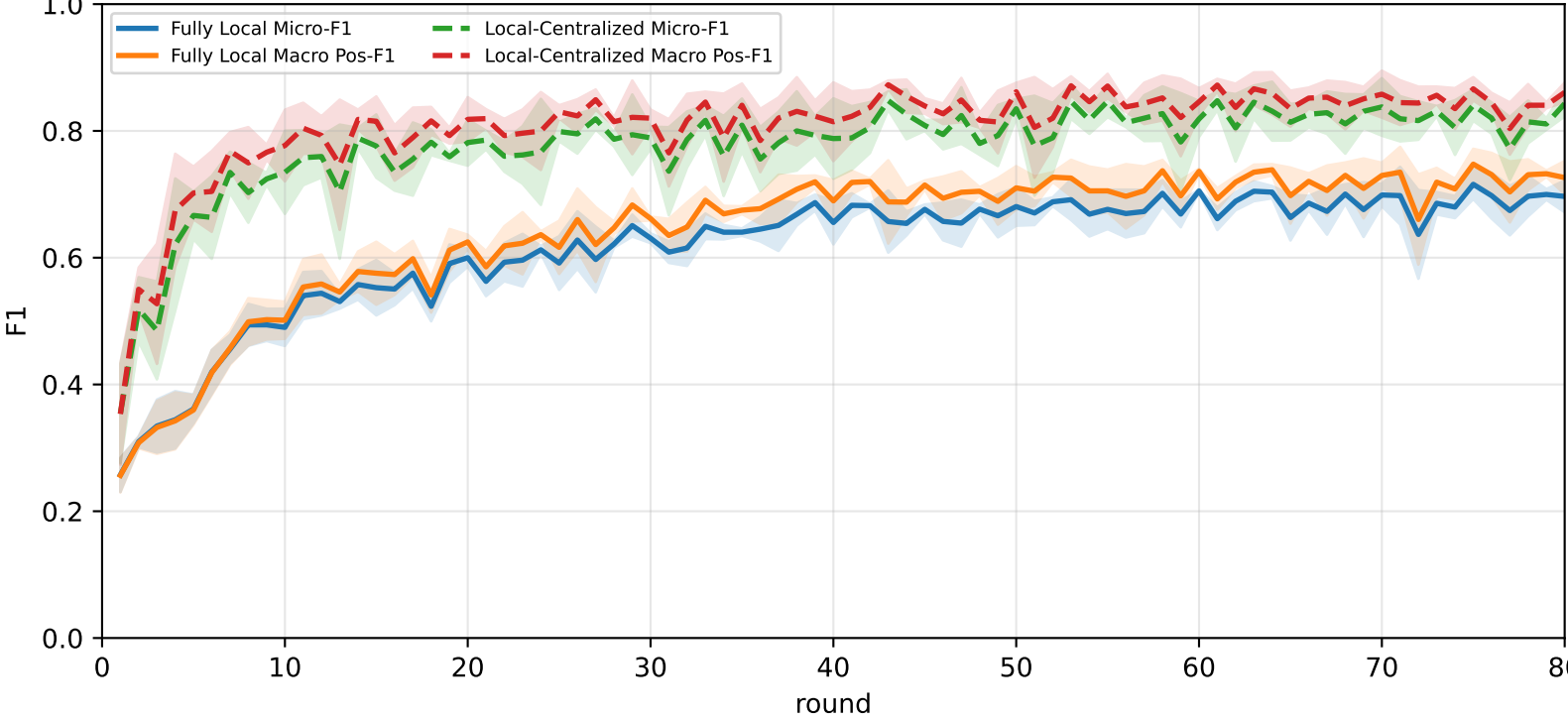
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 61.6% ± 4.0%        | 66.0% ± 4.3%        | <u>81.6% ± 7.0%</u> | <u>78.2% ± 5.1%</u> | 64.0% ± 3.6%        | 58.6% ± 5.2%        | 44.4% ± 3.0%        |
| Local-Centralized       | <b>84.3% ± 1.6%</b> | <b>86.3% ± 1.4%</b> | <b>96.7% ± 1.1%</b> | <b>94.7% ± 0.6%</b> | <b>87.4% ± 4.2%</b> | <b>80.9% ± 2.2%</b> | <b>71.8% ± 3.9%</b> |
| APPLE-Backbone-TaskHead | <u>65.2% ± 1.9%</u> | <u>66.6% ± 2.2%</u> | 71.8% ± 7.2%        | 73.4% ± 5.1%        | <u>64.9% ± 0.3%</u> | <u>66.3% ± 5.4%</u> | <u>56.0% ± 2.2%</u> |

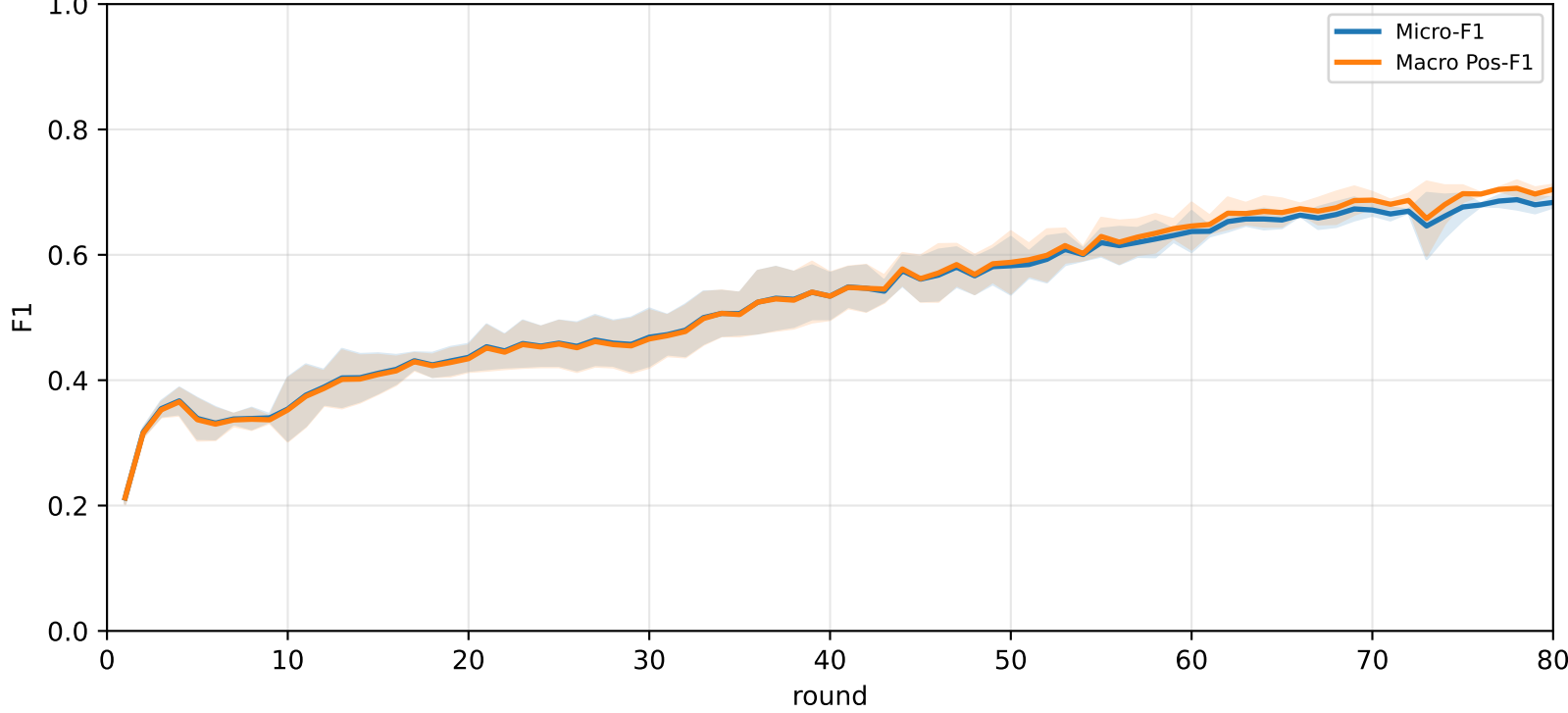
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



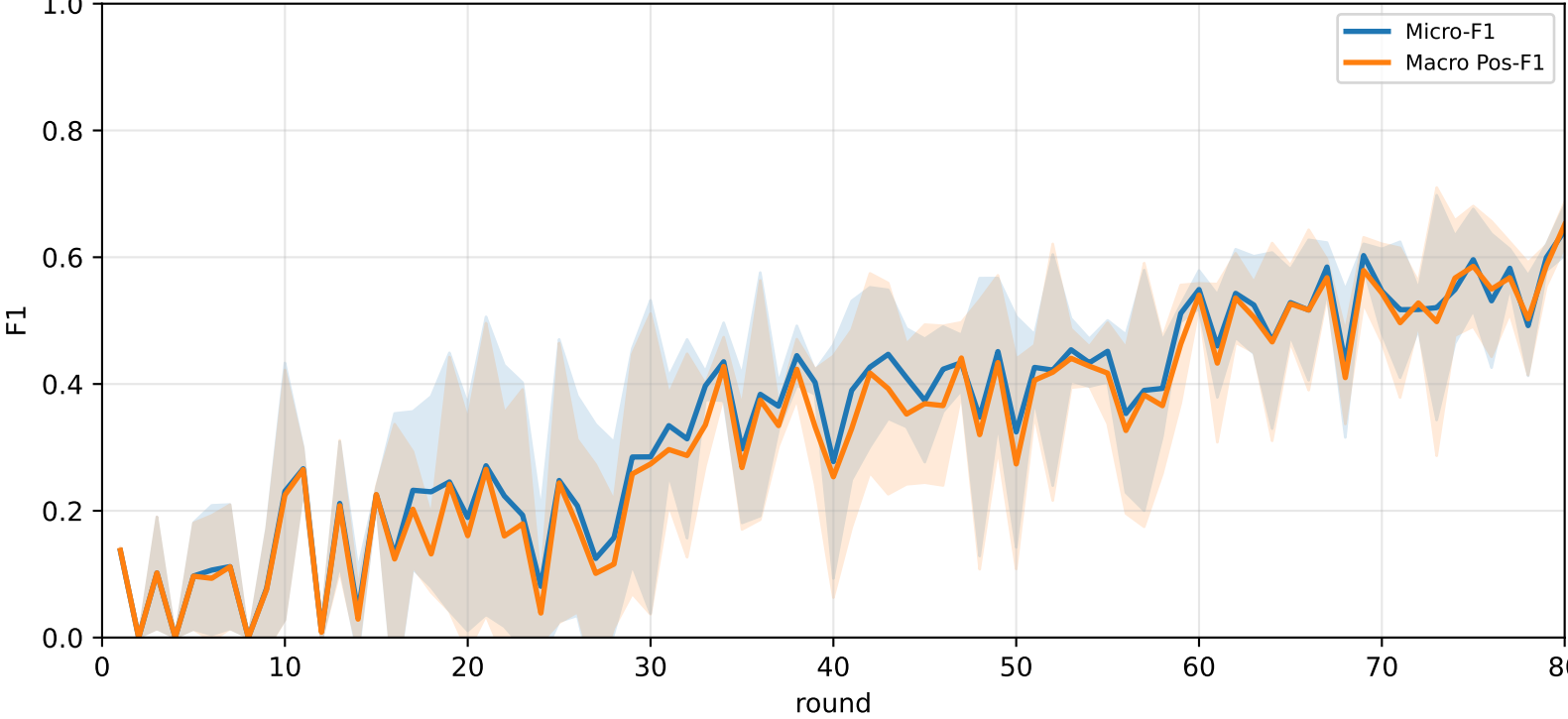
Pooled baseline micro / macro F1



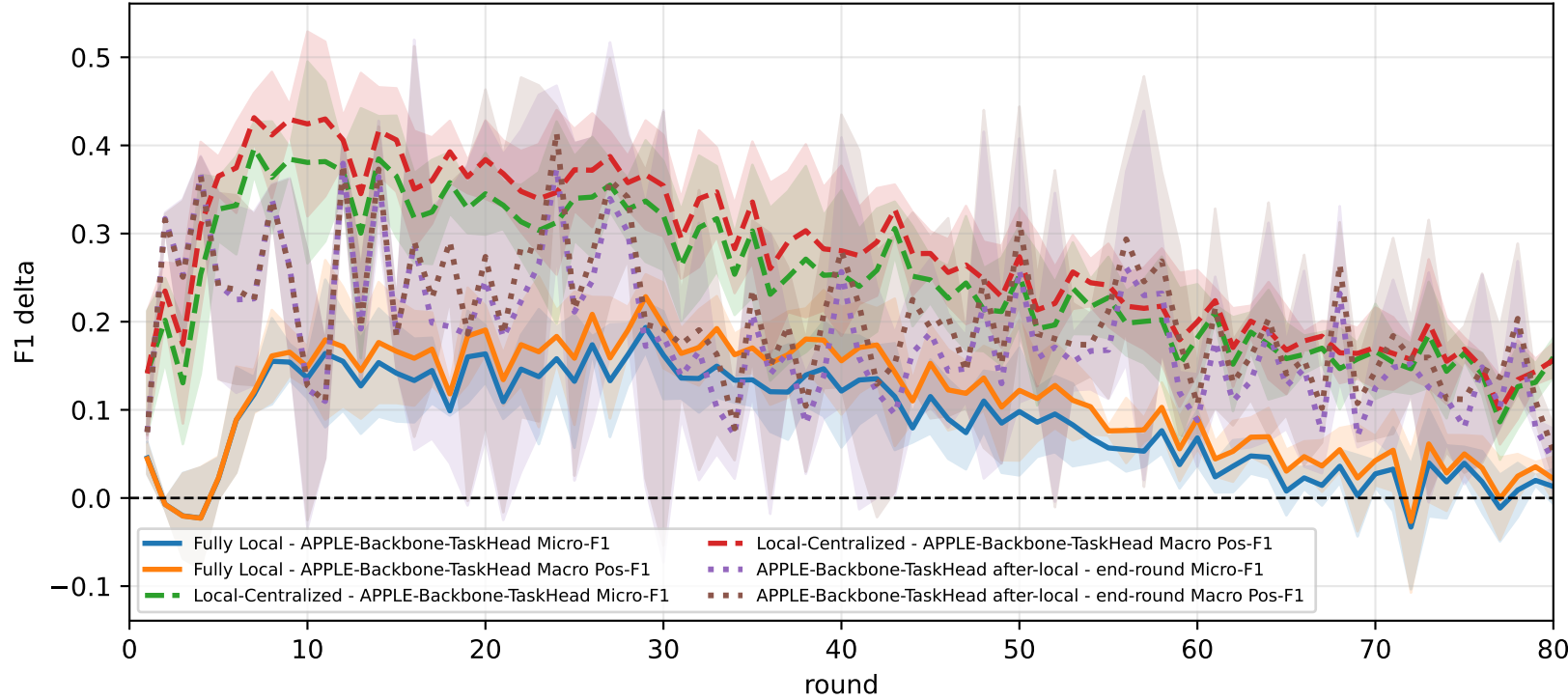
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap

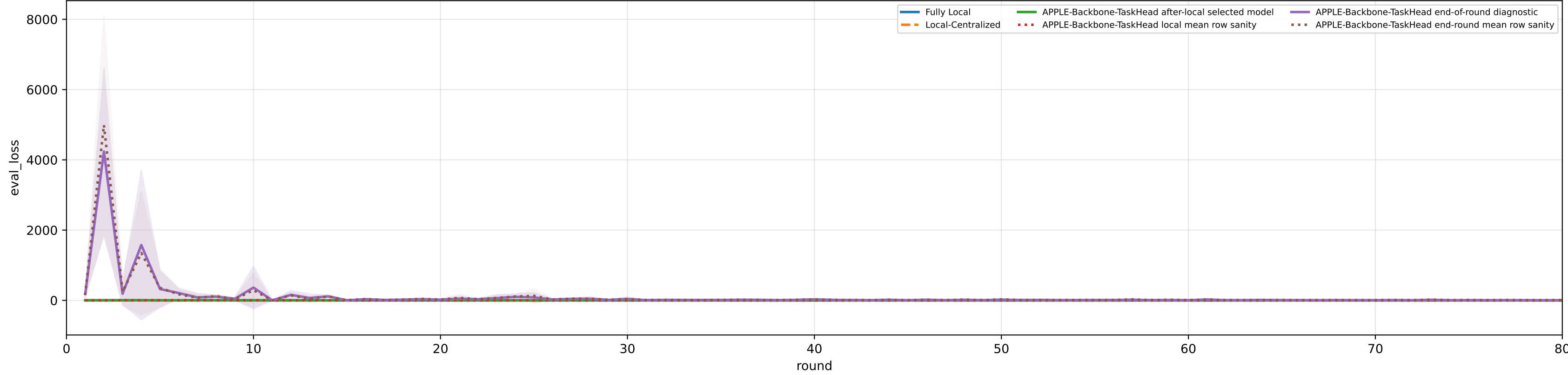


family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |  
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% |  
heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

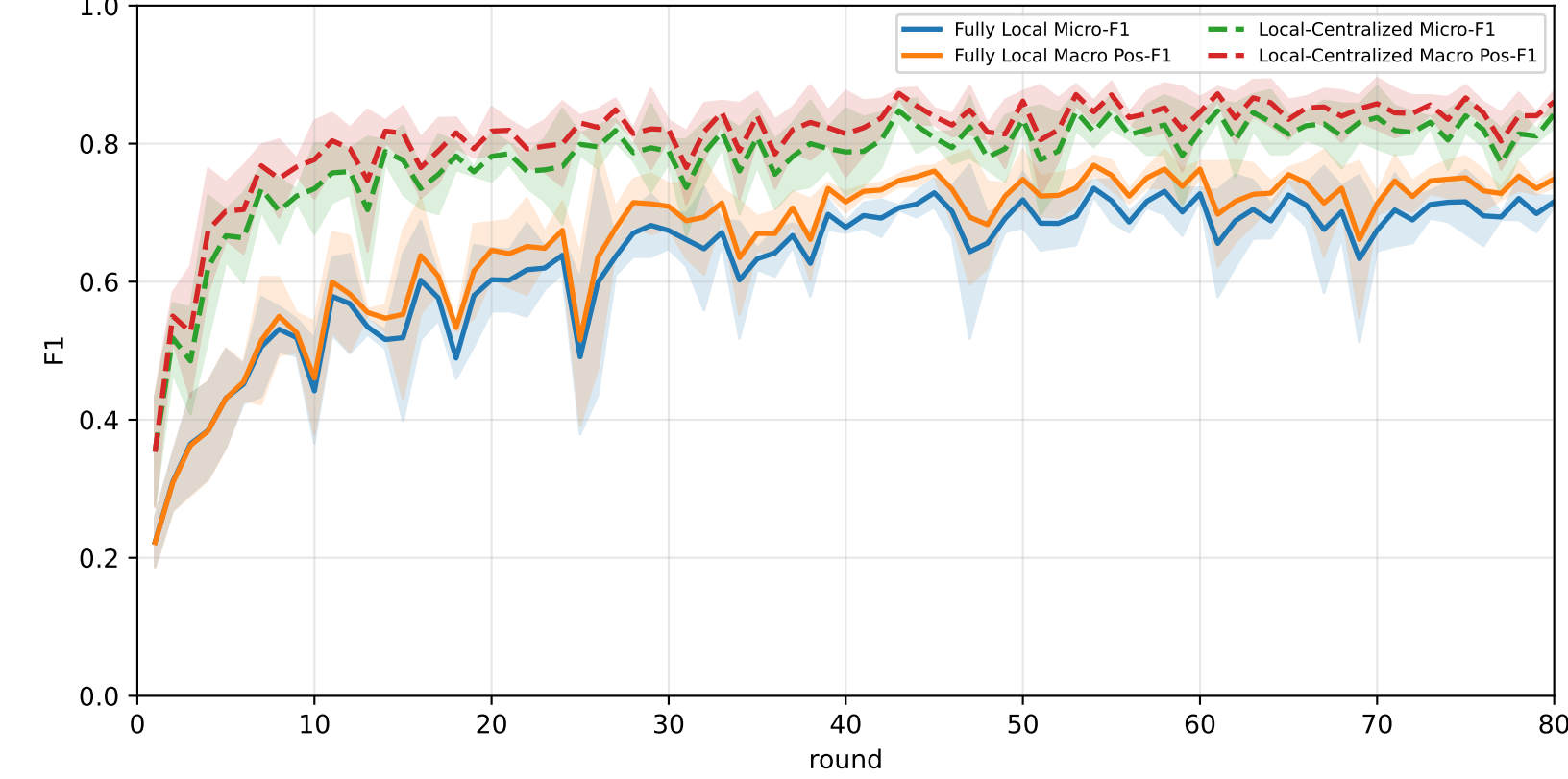
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 63.8% ± 5.4%        | 68.0% ± 5.8%        | <u>85.6% ± 8.0%</u> | <u>80.6% ± 4.3%</u> | 65.4% ± 6.2%        | 59.3% ± 9.8%        | 49.2% ± 2.7%        |
| Local-Centralized       | <b>84.3% ± 1.6%</b> | <b>86.3% ± 1.4%</b> | <b>96.7% ± 1.1%</b> | <b>94.7% ± 0.6%</b> | <b>87.4% ± 4.2%</b> | <b>80.9% ± 2.2%</b> | <b>71.8% ± 3.9%</b> |
| APPLE-Backbone-TaskHead | <u>67.5% ± 1.0%</u> | <u>69.5% ± 0.1%</u> | 77.9% ± 8.9%        | 78.1% ± 2.0%        | <u>67.0% ± 2.8%</u> | <u>67.4% ± 5.9%</u> | <u>57.2% ± 3.1%</u> |

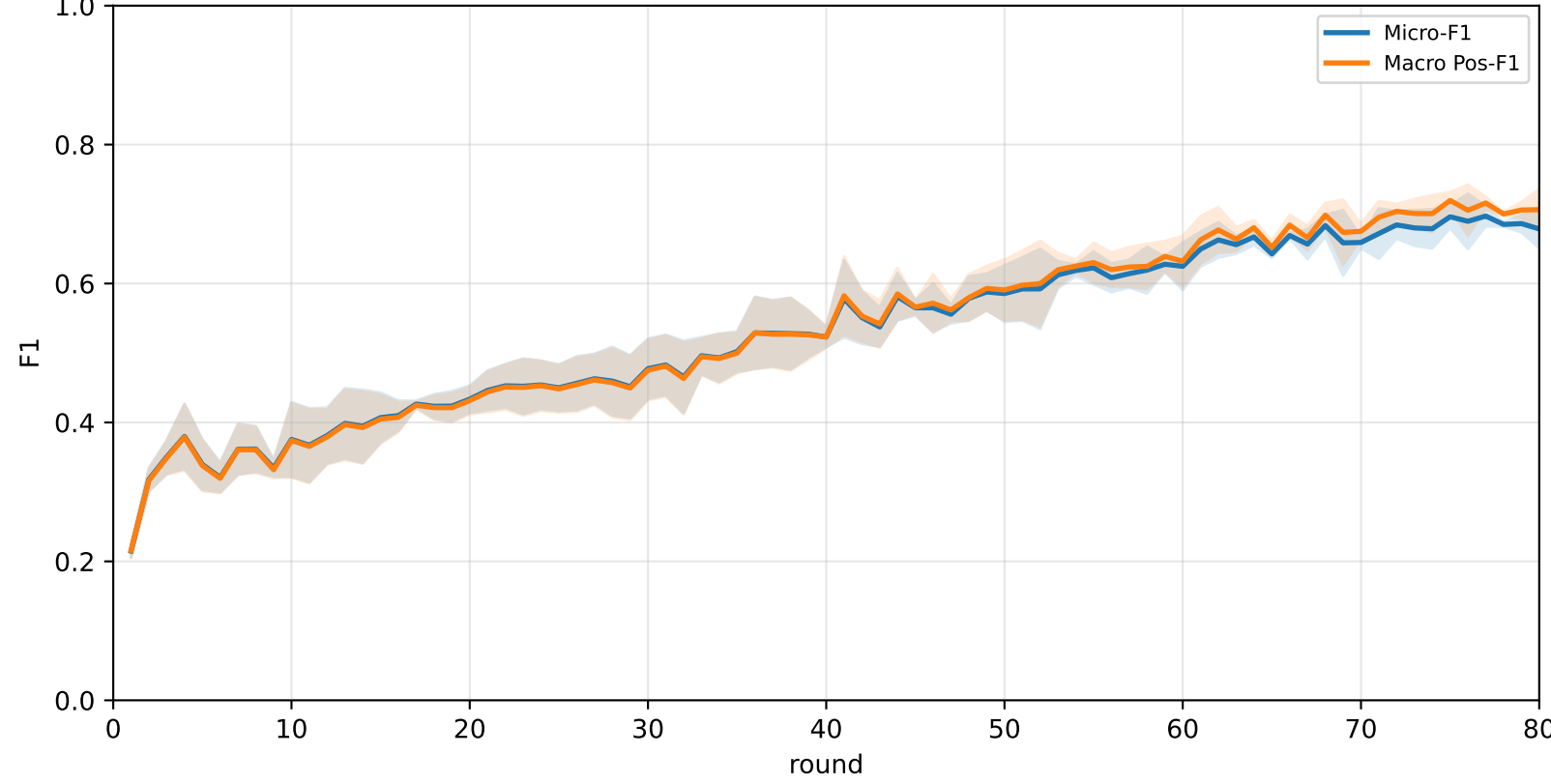
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



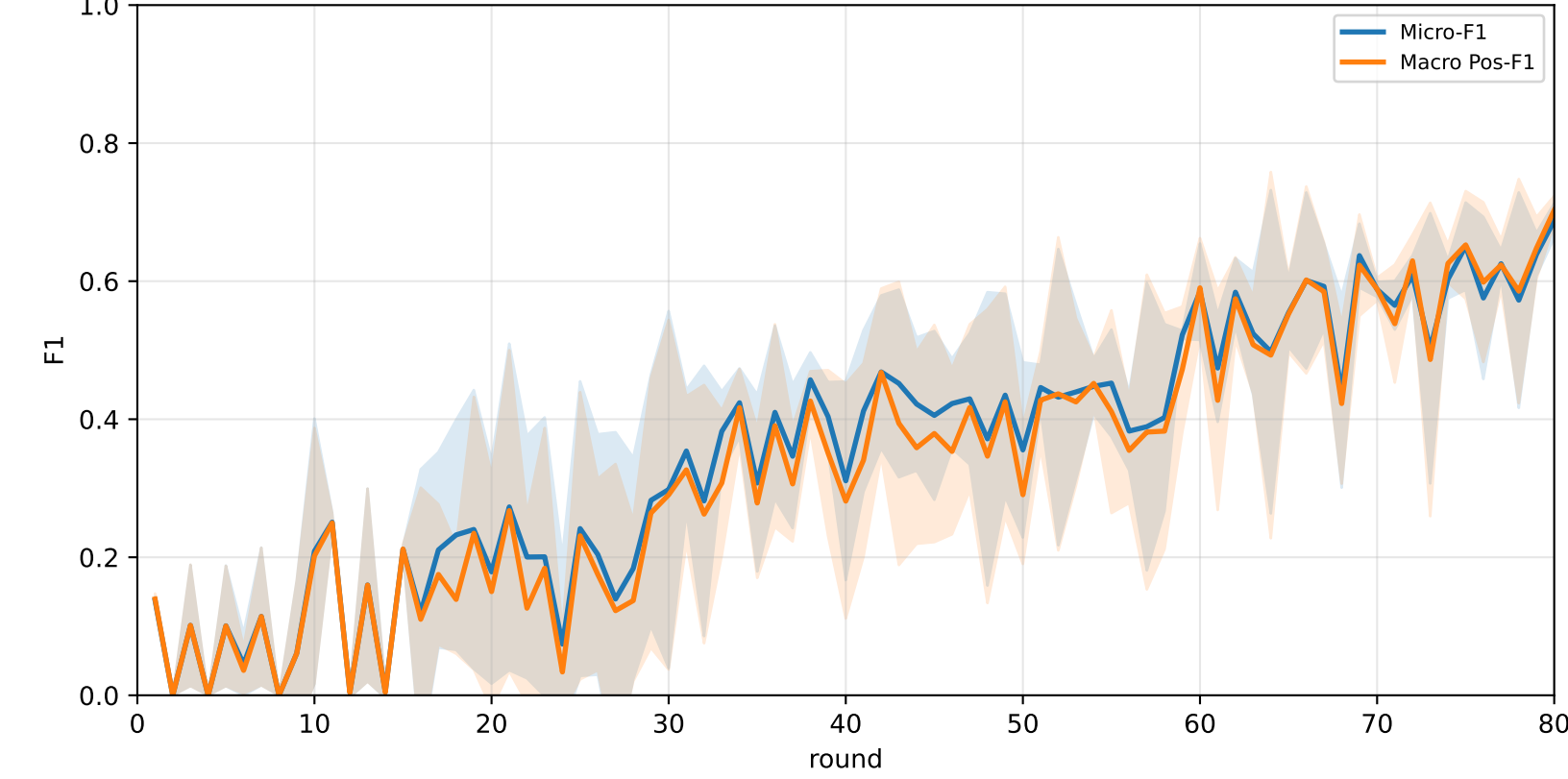
Pooled baseline micro / macro F1



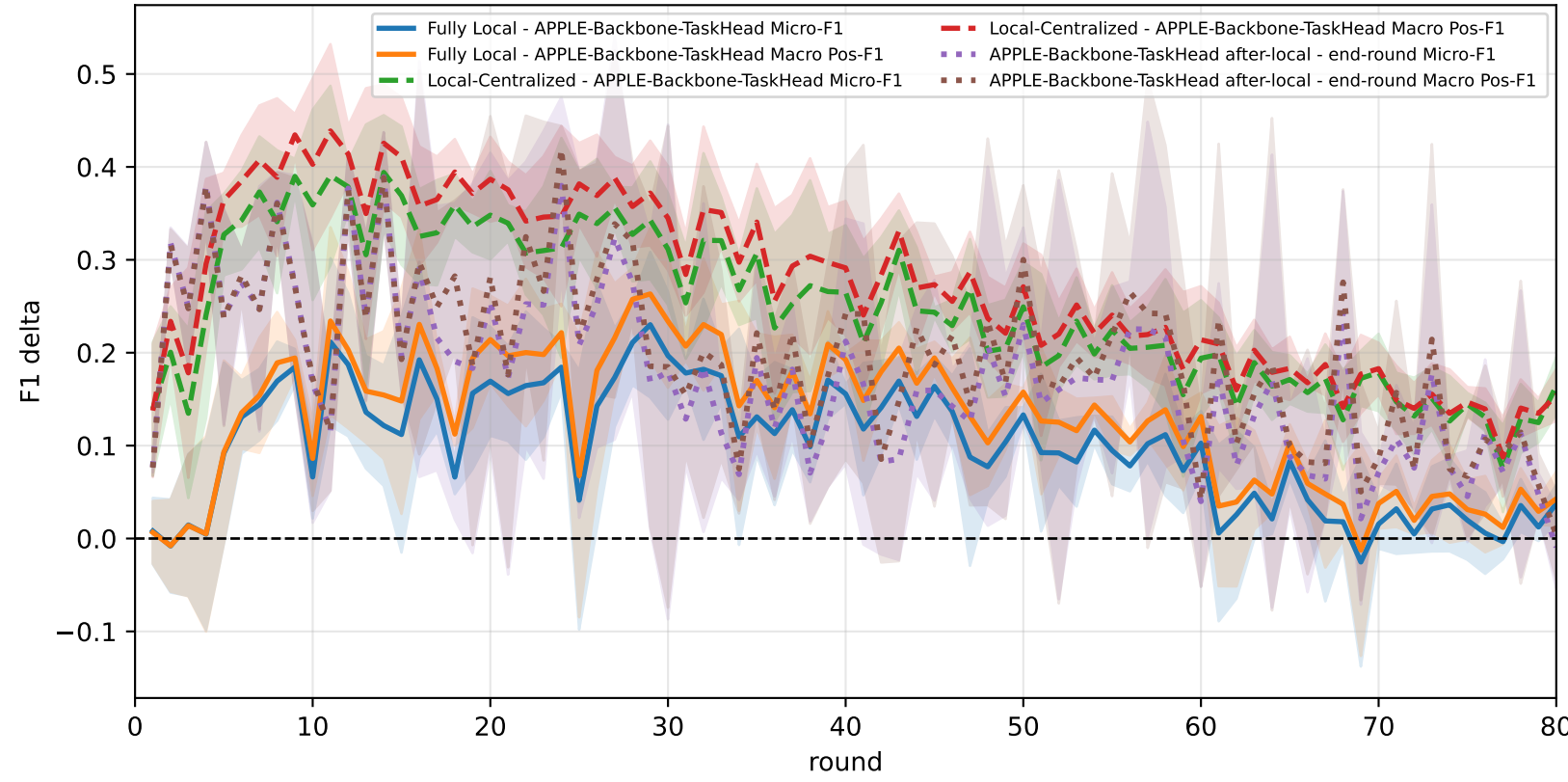
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap



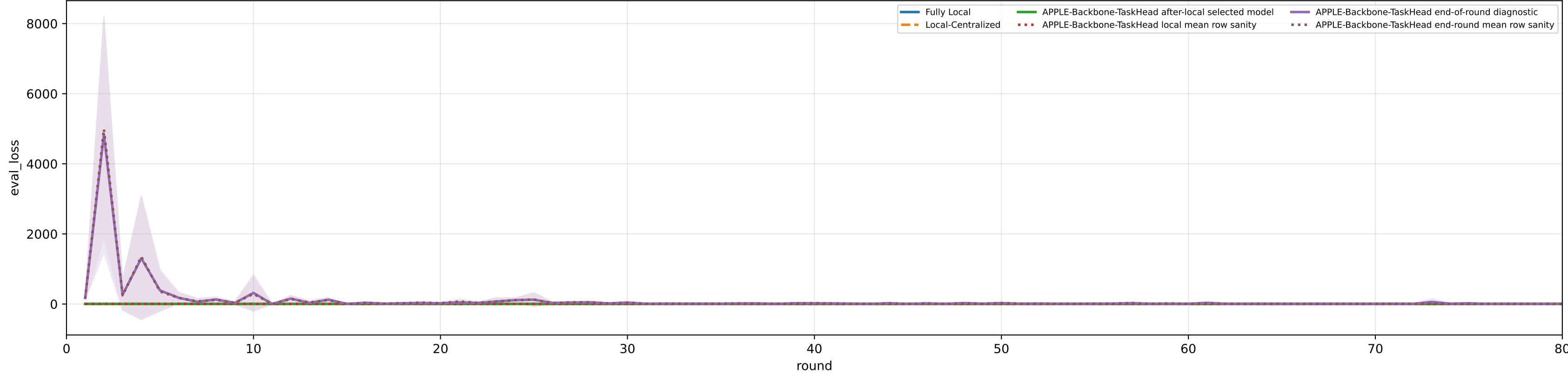


family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |  
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% |  
heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

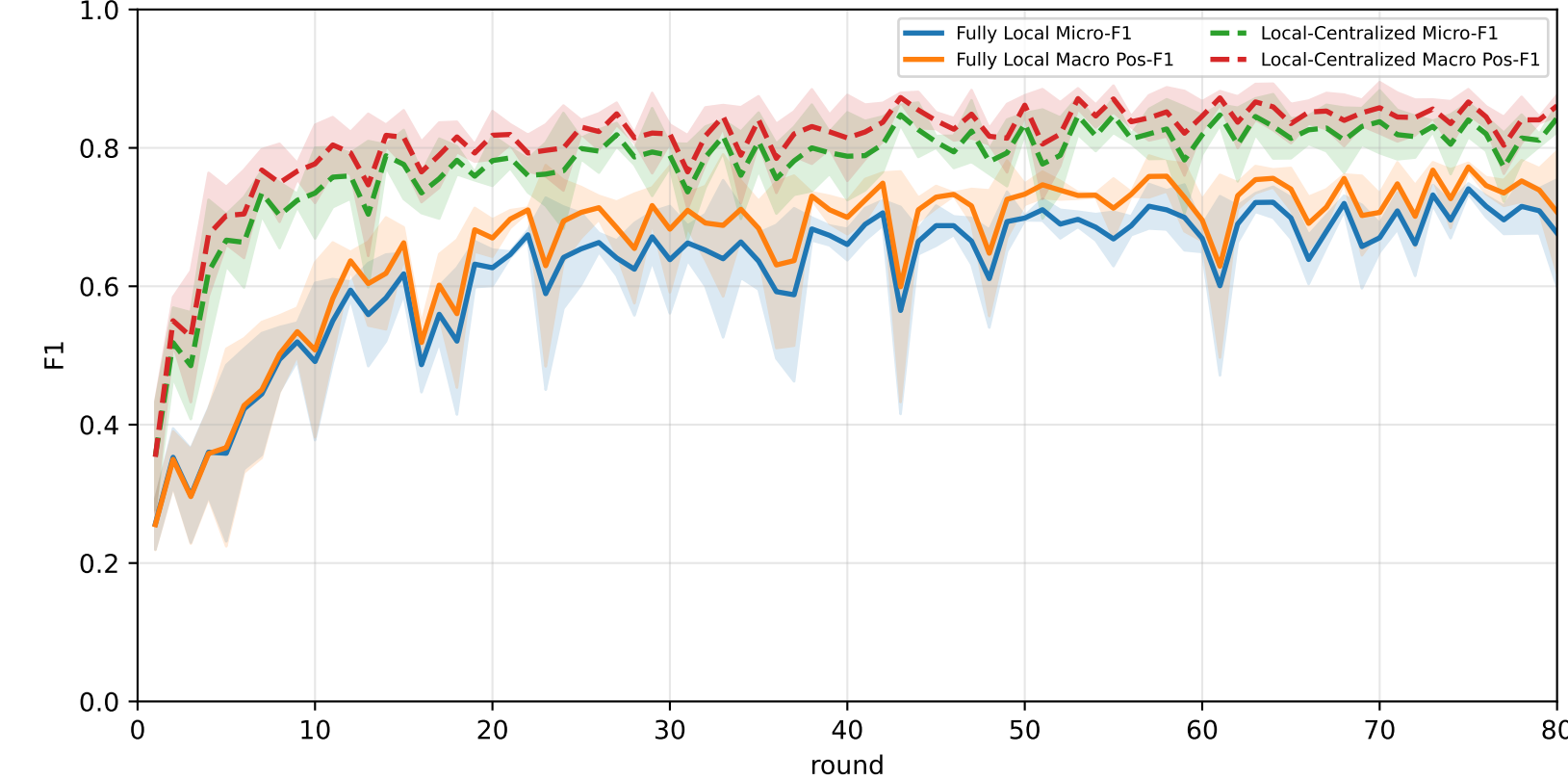
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1        | cycle3 pos-F1        | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
| Fully Local             | 60.1% ± 11.6%       | 64.2% ± 10.9%       | <u>75.3% ± 14.5%</u> | <u>77.9% ± 12.0%</u> | 63.8% ± 6.1%        | 60.7% ± 11.5%       | 43.1% ± 11.1%       |
| Local-Centralized       | <b>84.3% ± 1.6%</b> | <b>86.3% ± 1.4%</b> | <b>96.7% ± 1.1%</b>  | <b>94.7% ± 0.6%</b>  | <b>87.4% ± 4.2%</b> | <b>80.9% ± 2.2%</b> | <b>71.8% ± 3.9%</b> |
| APPLE-Backbone-TaskHead | <u>65.6% ± 2.2%</u> | <u>67.1% ± 2.4%</u> | 73.3% ± 7.7%         | 71.4% ± 5.7%         | <u>66.5% ± 1.2%</u> | <u>68.7% ± 5.6%</u> | <u>55.9% ± 2.8%</u> |

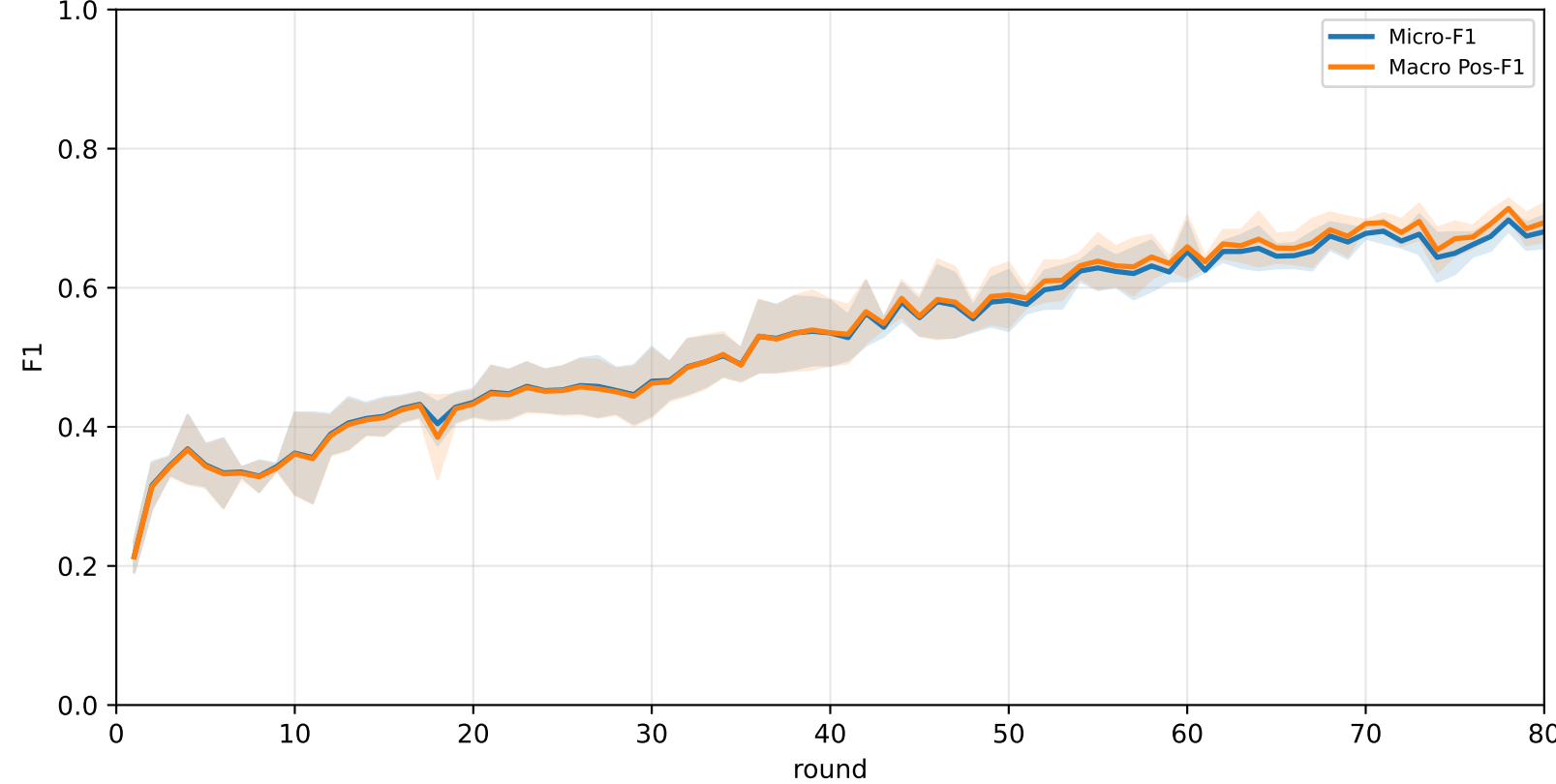
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



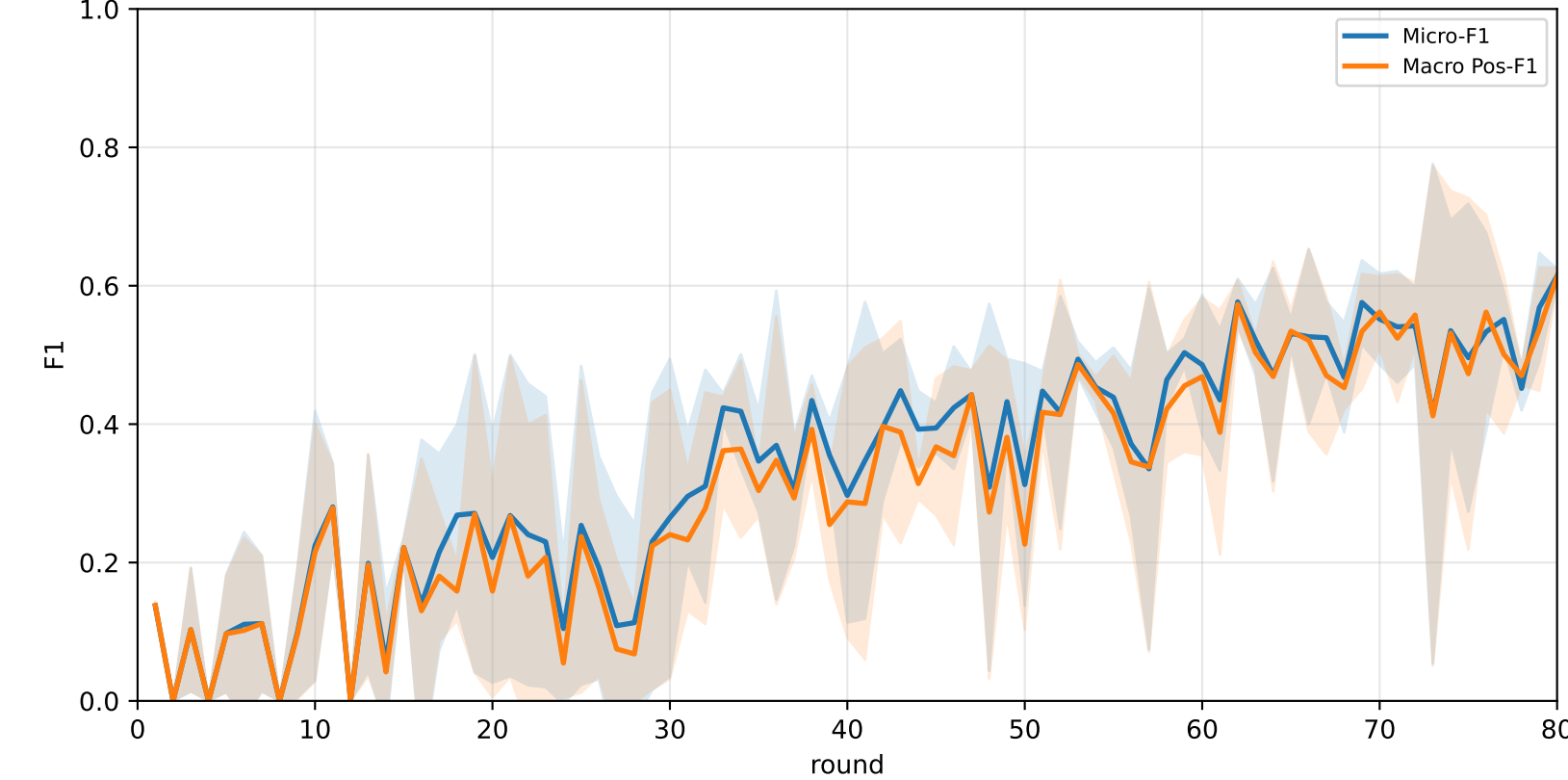
Pooled baseline micro / macro F1



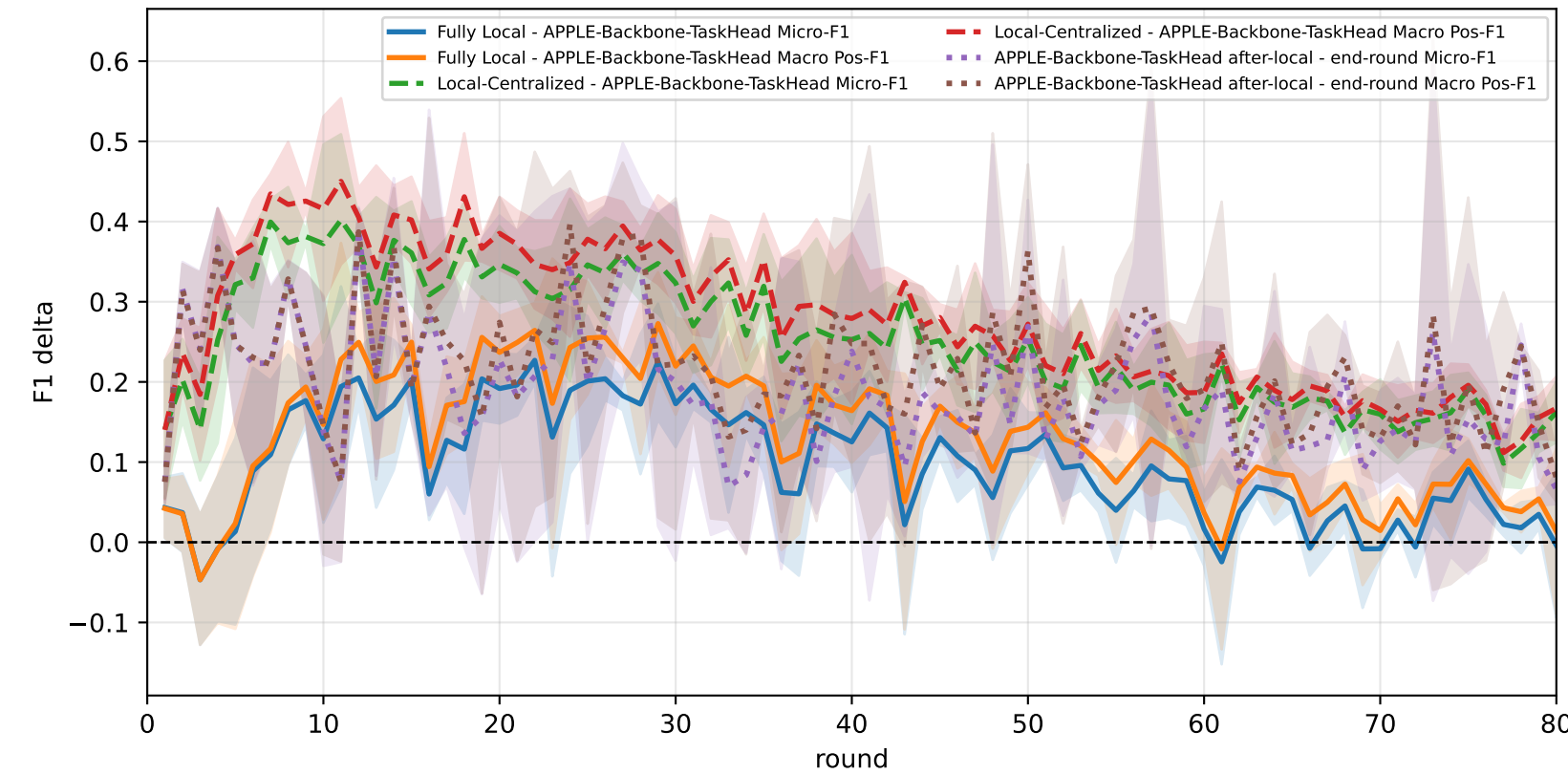
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap





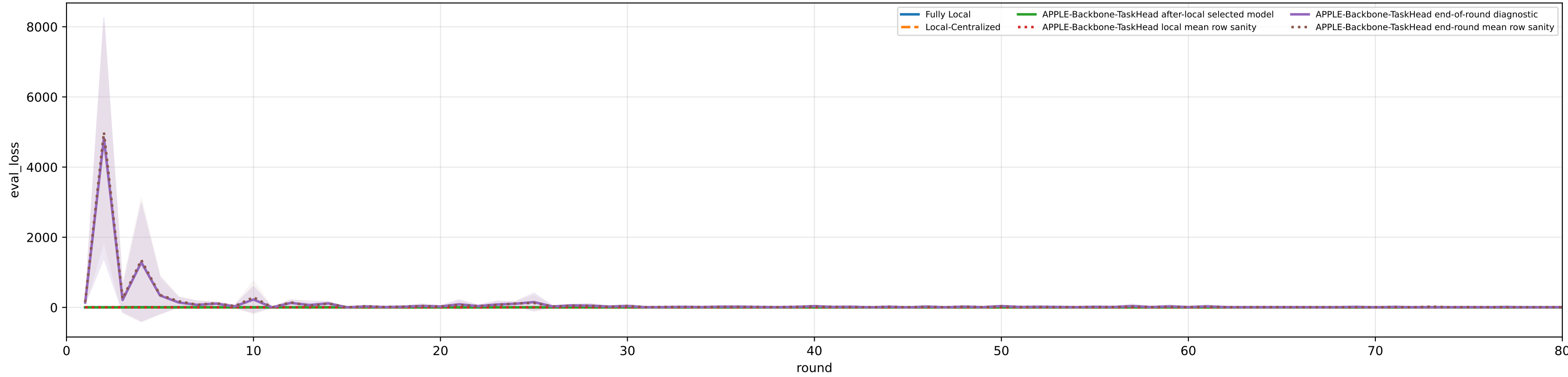
Controlled q-label heterogeneity: Mid | APPLE-Backbone-TaskHead diagnostics |  
family=q\_task\_cycle4 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |  
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% |  
heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

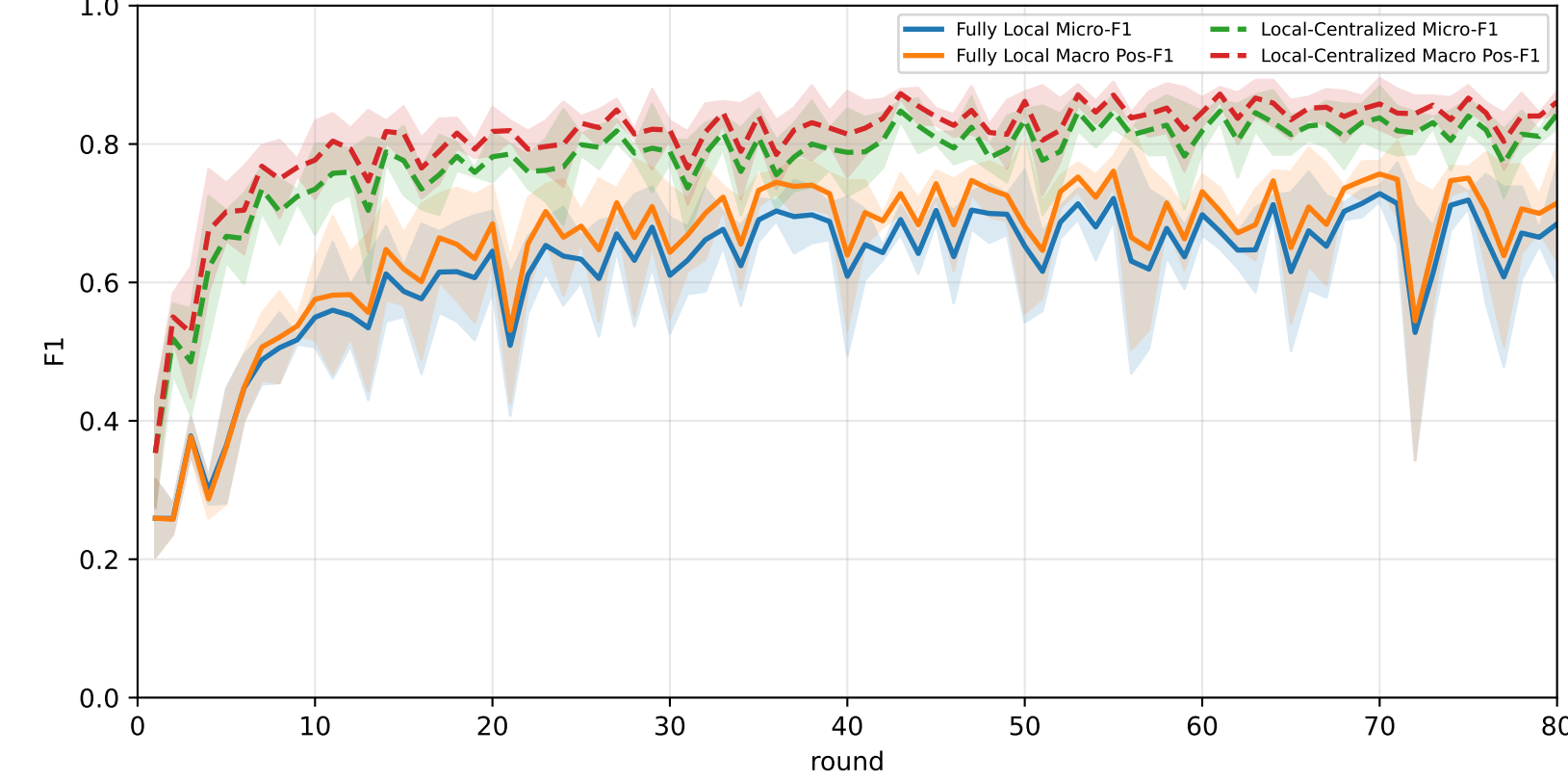
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 58.7% ± 5.5%        | 63.8% ± 5.1%        | <u>82.7% ± 2.7%</u> | <u>76.5% ± 5.8%</u> | <u>63.3% ± 5.5%</u> | 54.2% ± 7.8%        | 42.1% ± 3.8%        |
| Local-Centralized       | <b>84.3% ± 1.6%</b> | <b>86.3% ± 1.4%</b> | <b>96.7% ± 1.1%</b> | <b>94.7% ± 0.6%</b> | <b>87.4% ± 4.2%</b> | <b>80.9% ± 2.2%</b> | <b>71.8% ± 3.9%</b> |
| APPLE-Backbone-TaskHead | <u>62.9% ± 3.3%</u> | <u>63.9% ± 3.5%</u> | 65.9% ± 4.9%        | 71.9% ± 5.7%        | 62.1% ± 2.3%        | <u>64.4% ± 6.3%</u> | <u>55.2% ± 2.9%</u> |

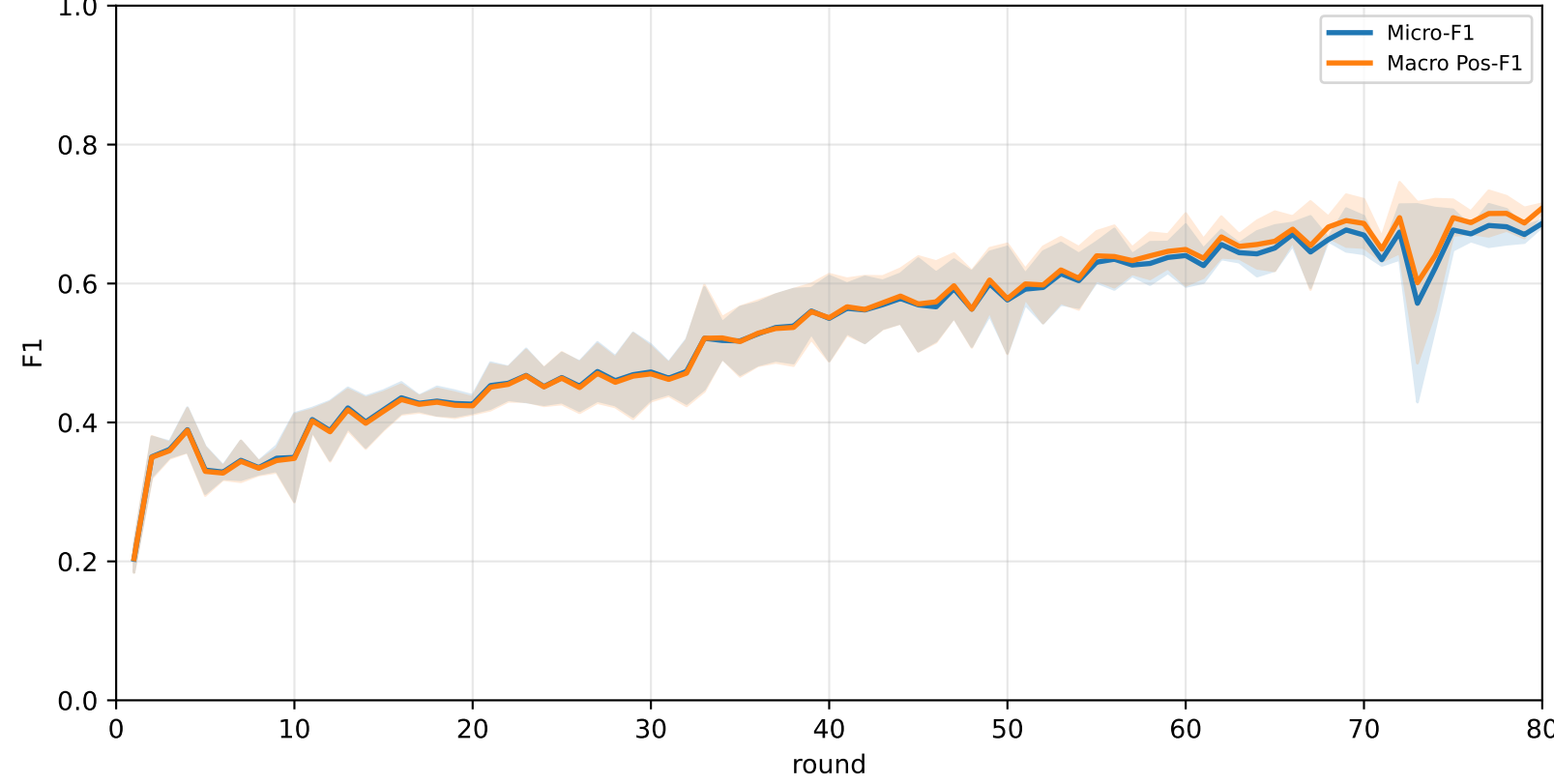
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



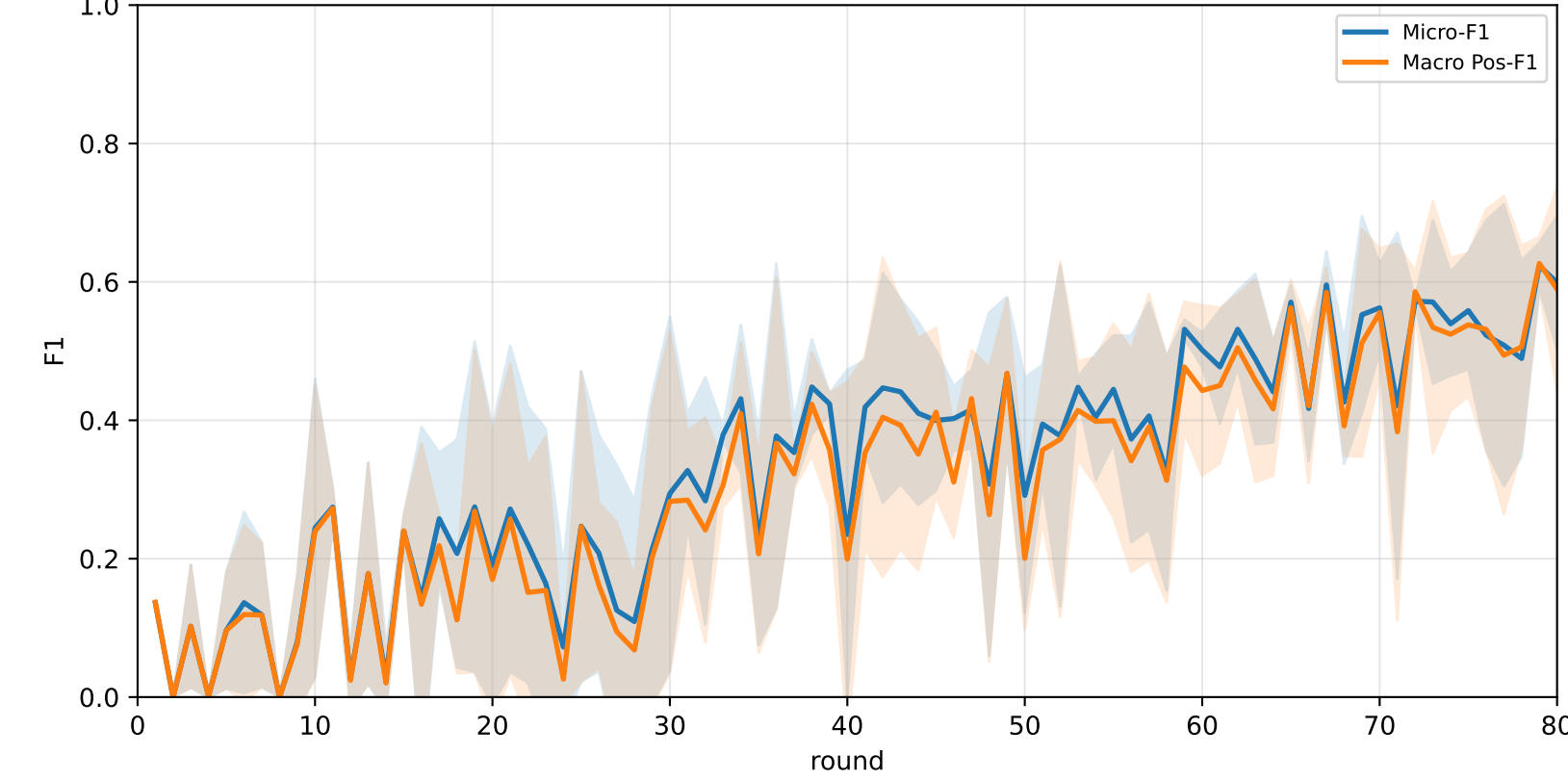
Pooled baseline micro / macro F1



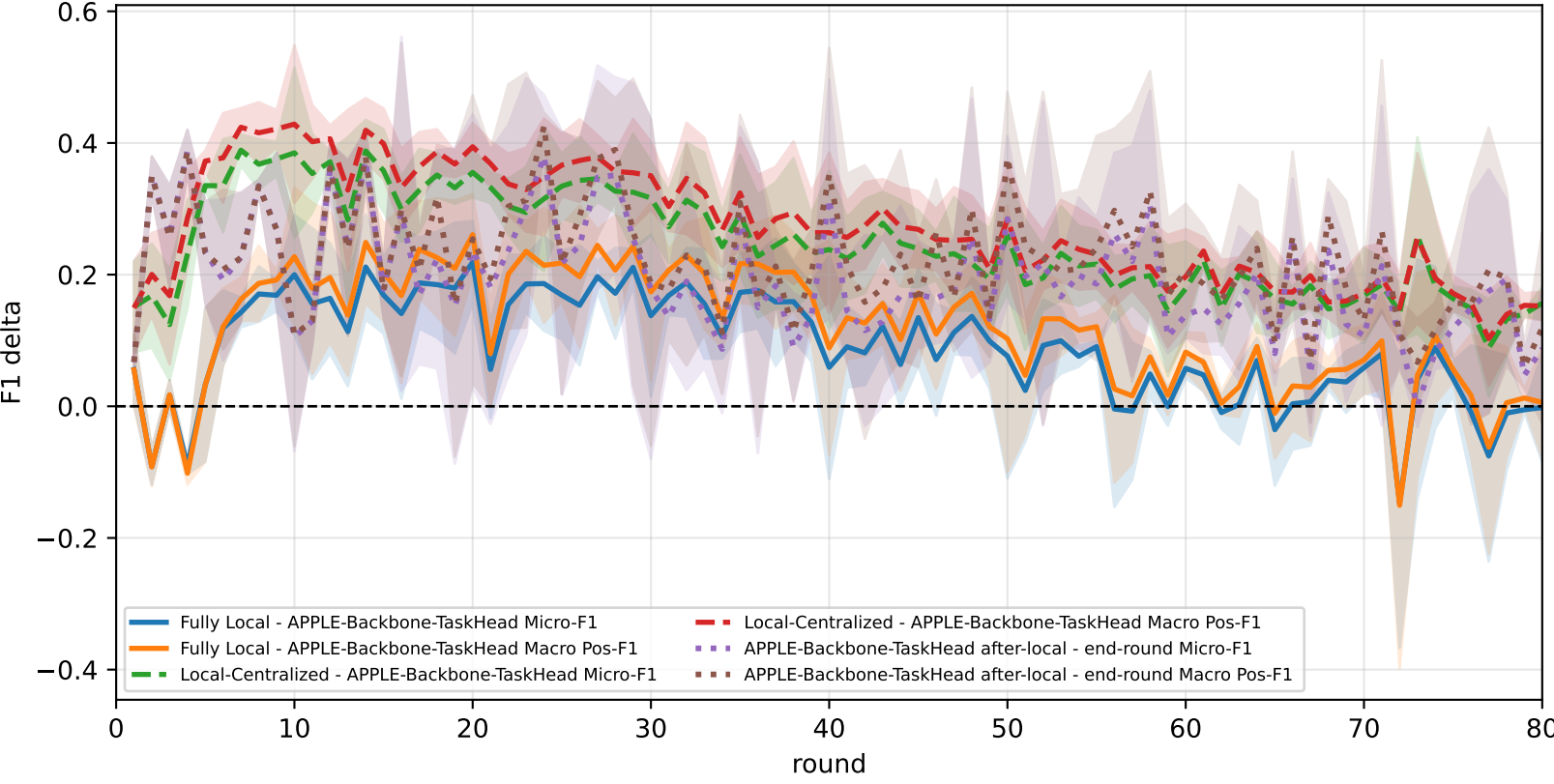
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



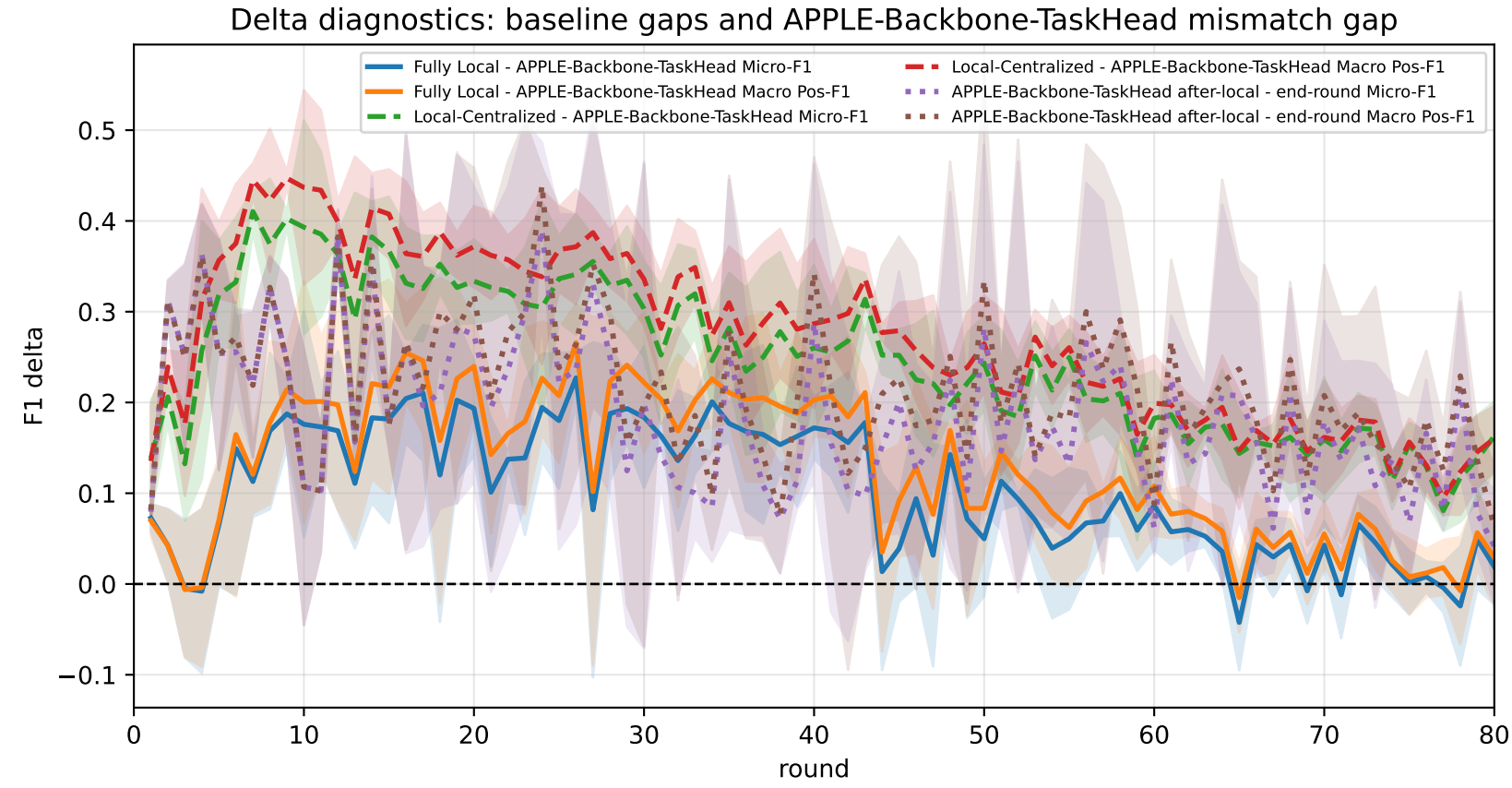
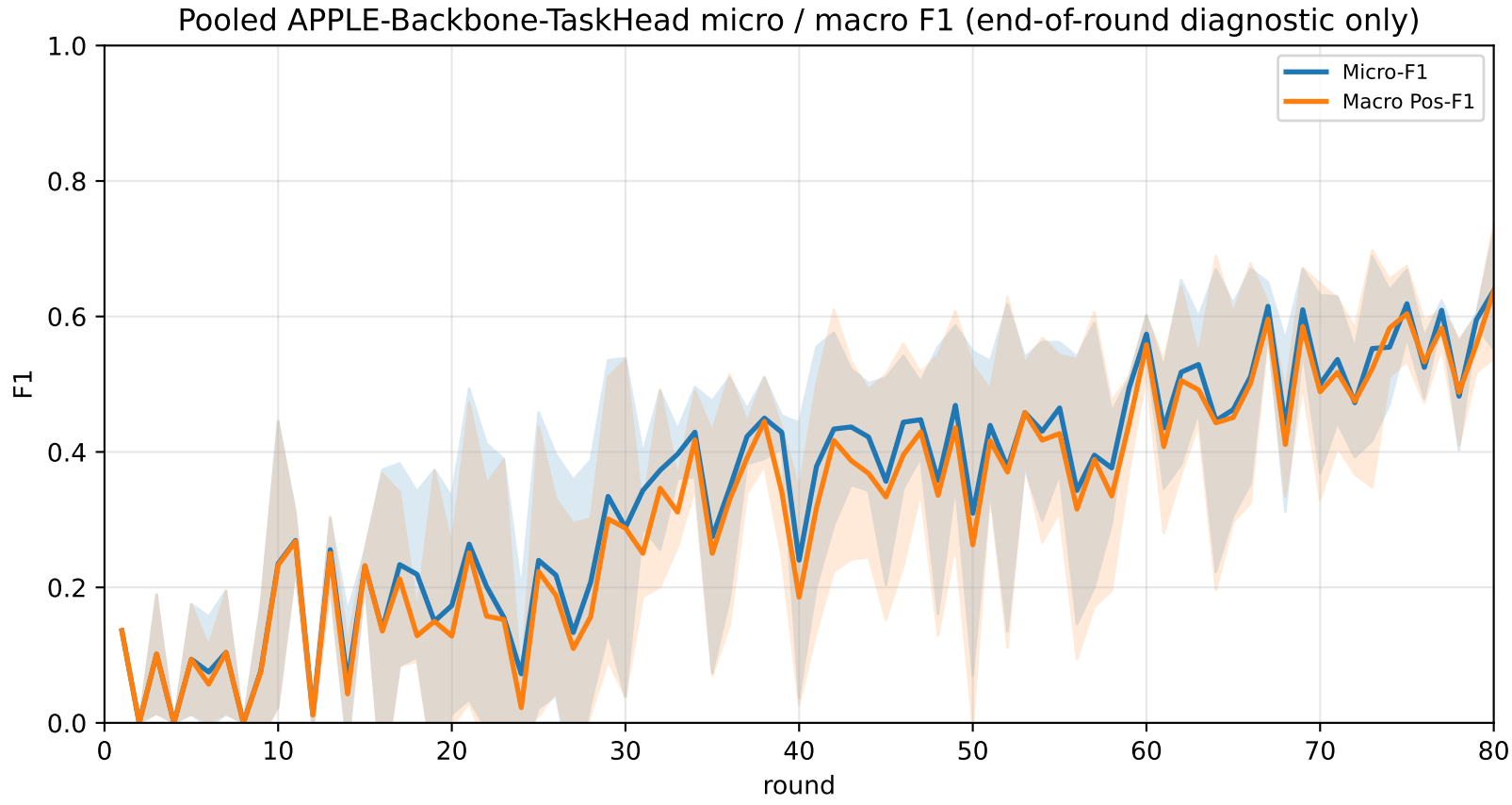
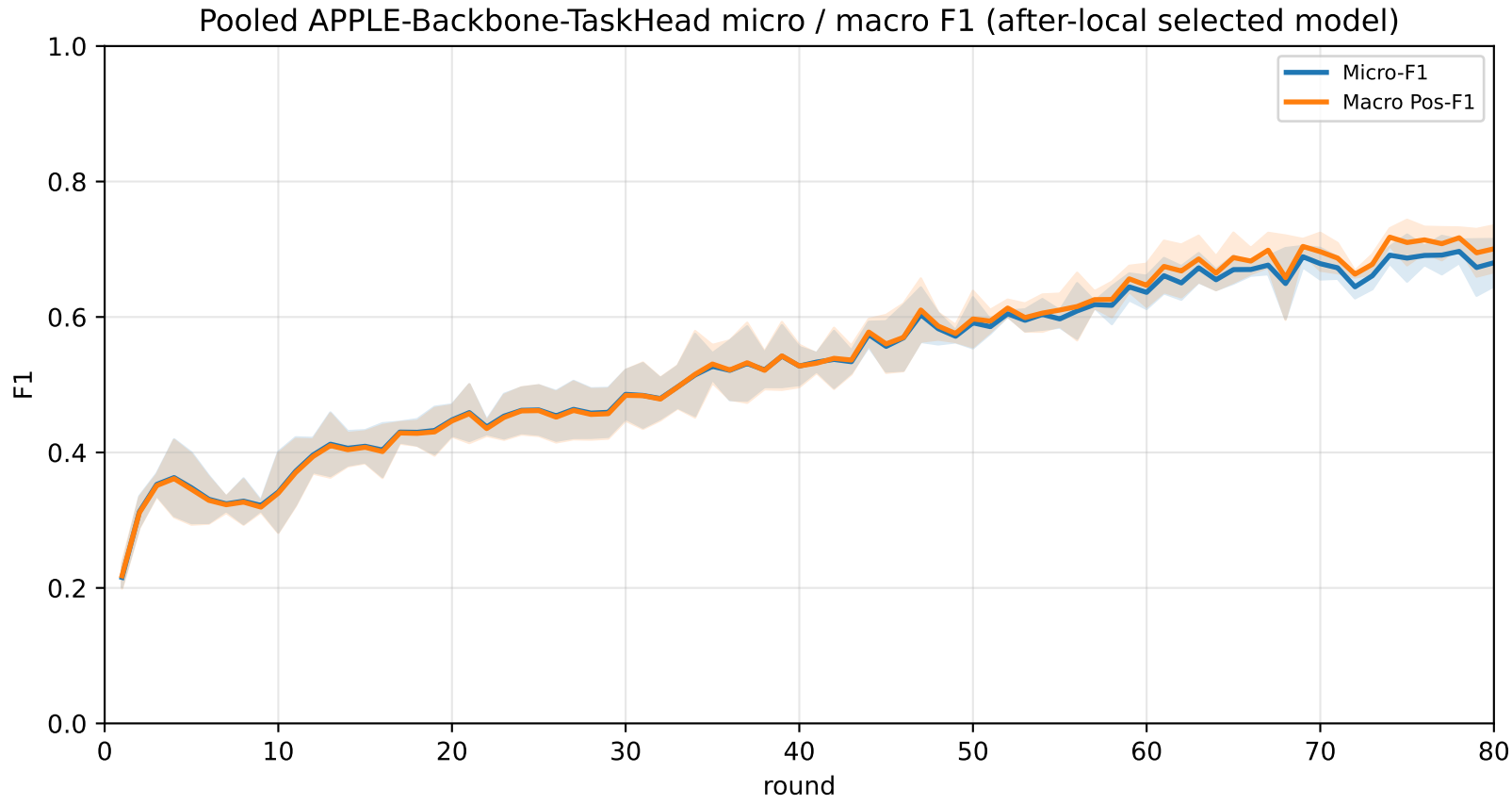
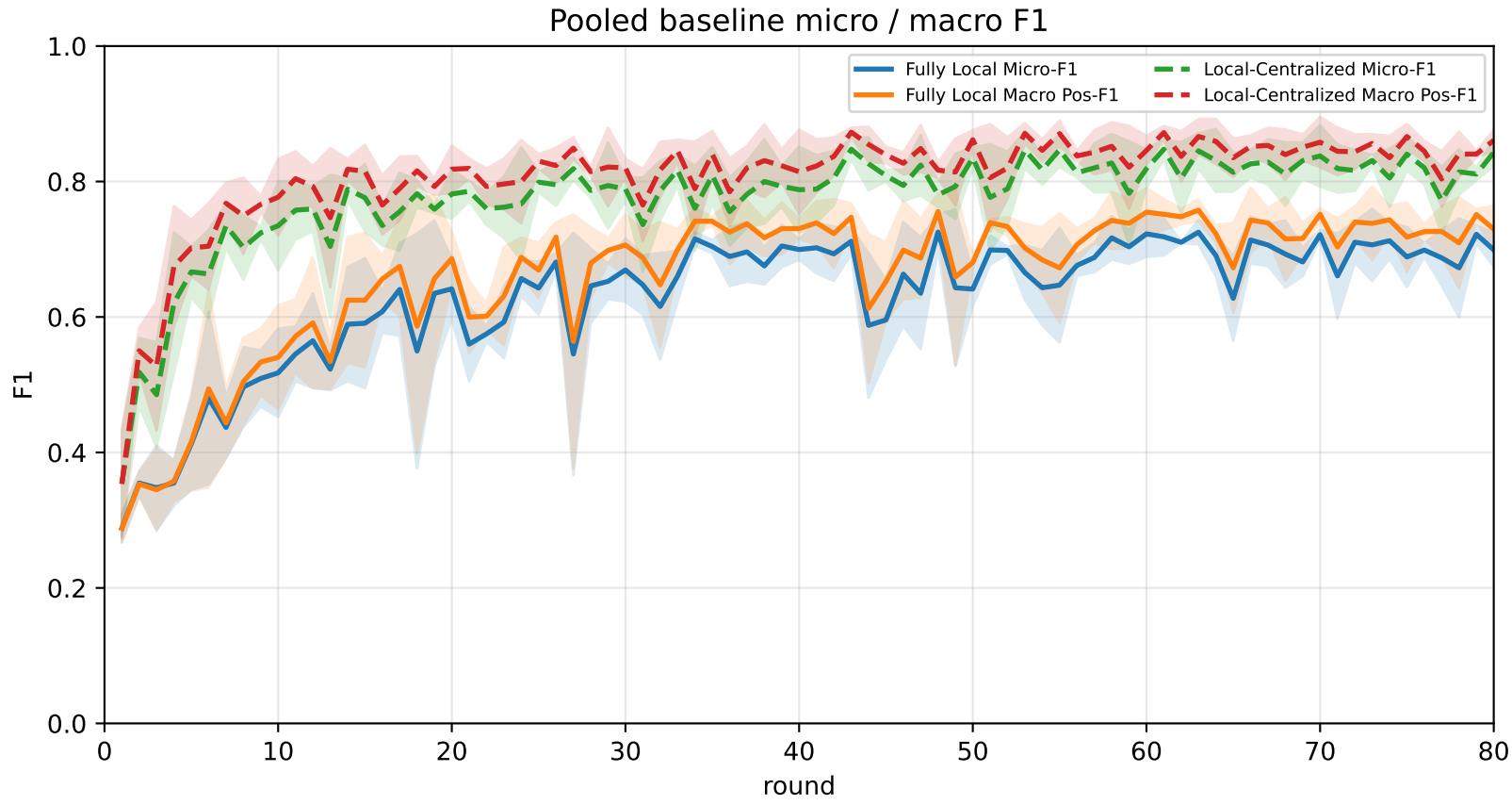
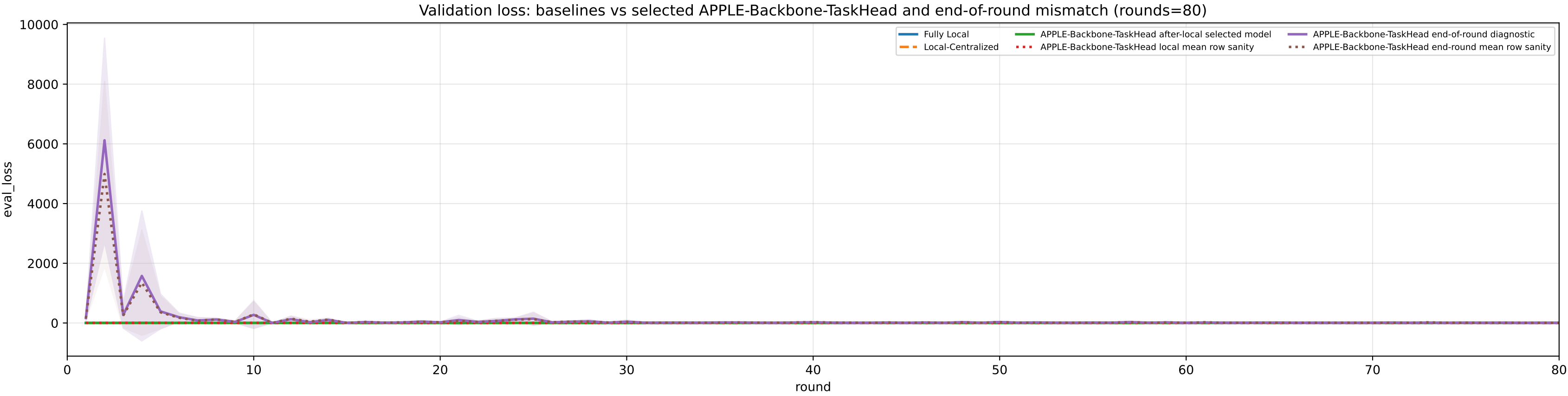
Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap



family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |  
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% |  
heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 61.2% ± 0.9%        | 66.3% ± 1.4%        | <u>86.3% ± 4.7%</u> | <u>79.8% ± 5.8%</u> | 63.6% ± 1.5%        | 58.1% ± 1.4%        | 43.8% ± 3.4%        |
| Local-Centralized       | <b>84.3% ± 1.6%</b> | <b>86.3% ± 1.4%</b> | <b>96.7% ± 1.1%</b> | <b>94.7% ± 0.6%</b> | <b>87.4% ± 4.2%</b> | <b>80.9% ± 2.2%</b> | <b>71.8% ± 3.9%</b> |
| APPLE-Backbone-TaskHead | <u>65.0% ± 1.8%</u> | <u>66.4% ± 2.5%</u> | 71.8% ± 9.0%        | 74.6% ± 5.3%        | <u>64.9% ± 0.6%</u> | <u>64.7% ± 4.5%</u> | <u>56.1% ± 1.8%</u> |





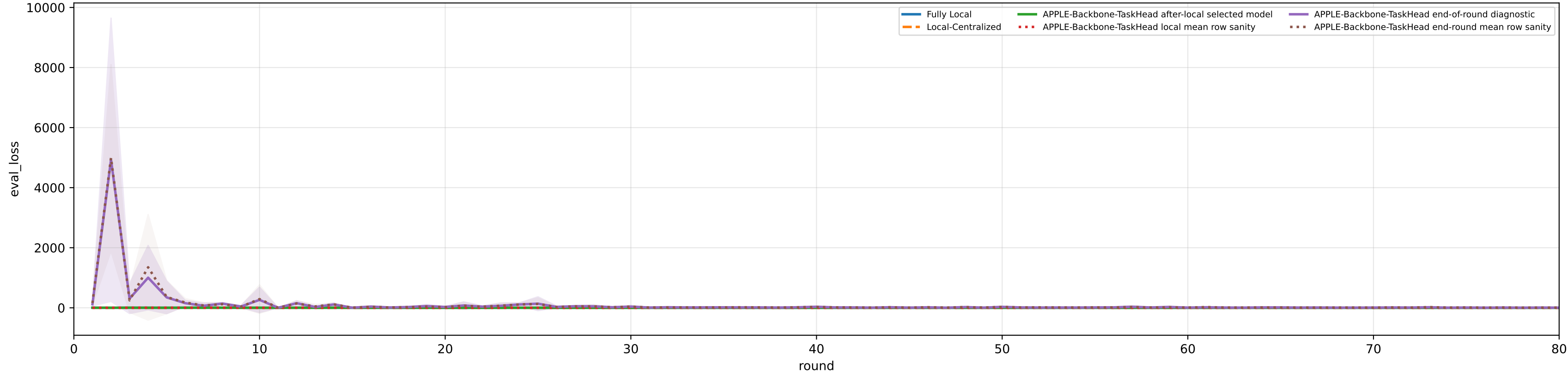
Controlled q-label heterogeneity: Mid | APPLE-Backbone-TaskHead diagnostics |  
family=q task\_cycle6 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |  
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% |  
heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

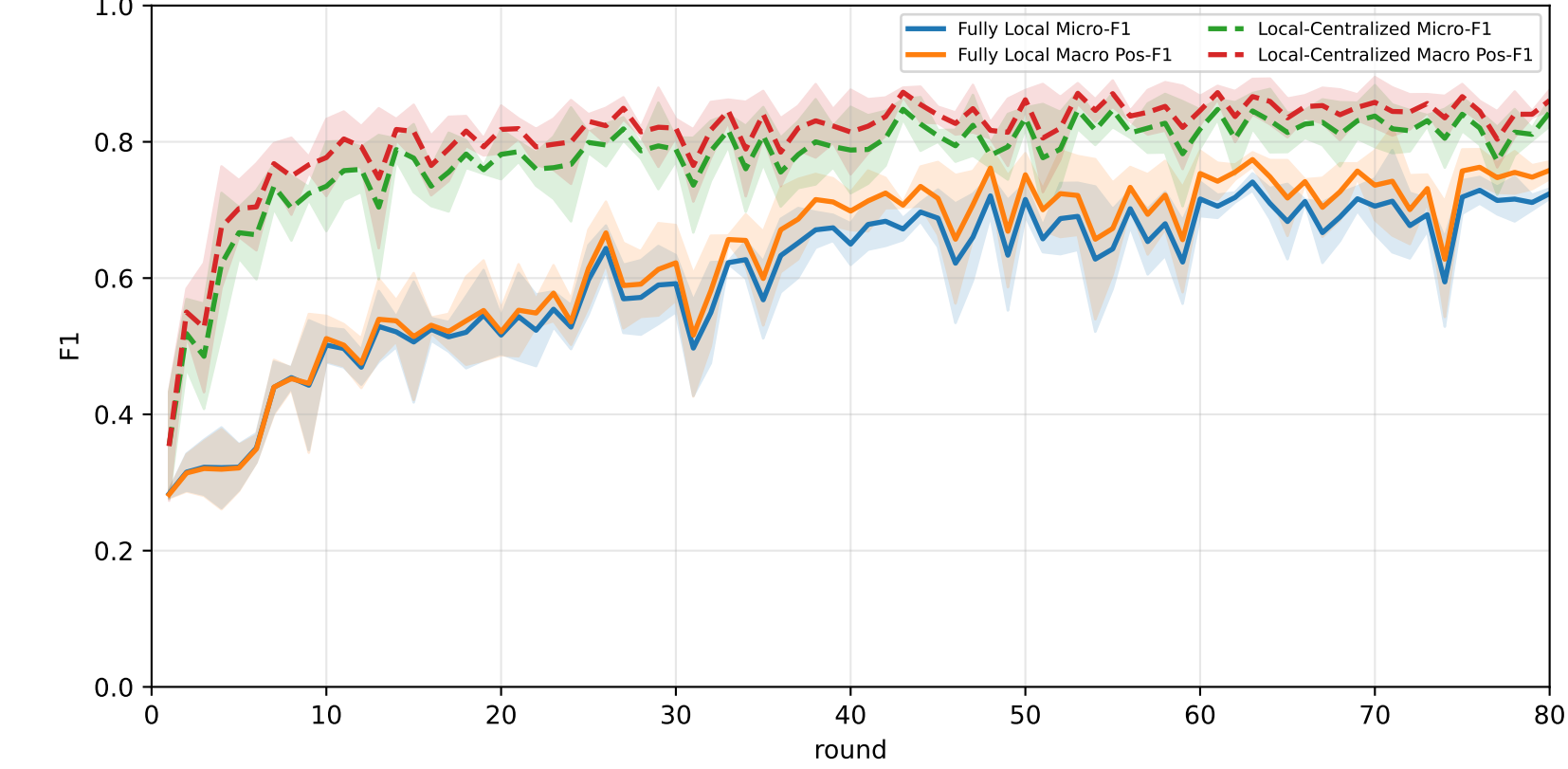
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 64.1% ± 1.8%        | <u>67.8% ± 2.4%</u> | <u>81.8% ± 8.2%</u> | <u>78.3% ± 6.8%</u> | <u>64.9% ± 1.4%</u> | 64.9% ± 4.4%        | 49.0% ± 3.2%        |
| Local-Centralized       | <b>84.3% ± 1.6%</b> | <b>86.3% ± 1.4%</b> | <b>96.7% ± 1.1%</b> | <b>94.7% ± 0.6%</b> | <b>87.4% ± 4.2%</b> | <b>80.9% ± 2.2%</b> | <b>71.8% ± 3.9%</b> |
| APPLE-Backbone-TaskHead | <u>64.7% ± 2.0%</u> | 66.2% ± 2.8%        | 71.9% ± 8.9%        | 71.9% ± 6.5%        | 64.5% ± 1.8%        | <u>66.6% ± 5.1%</u> | <u>55.9% ± 1.3%</u> |

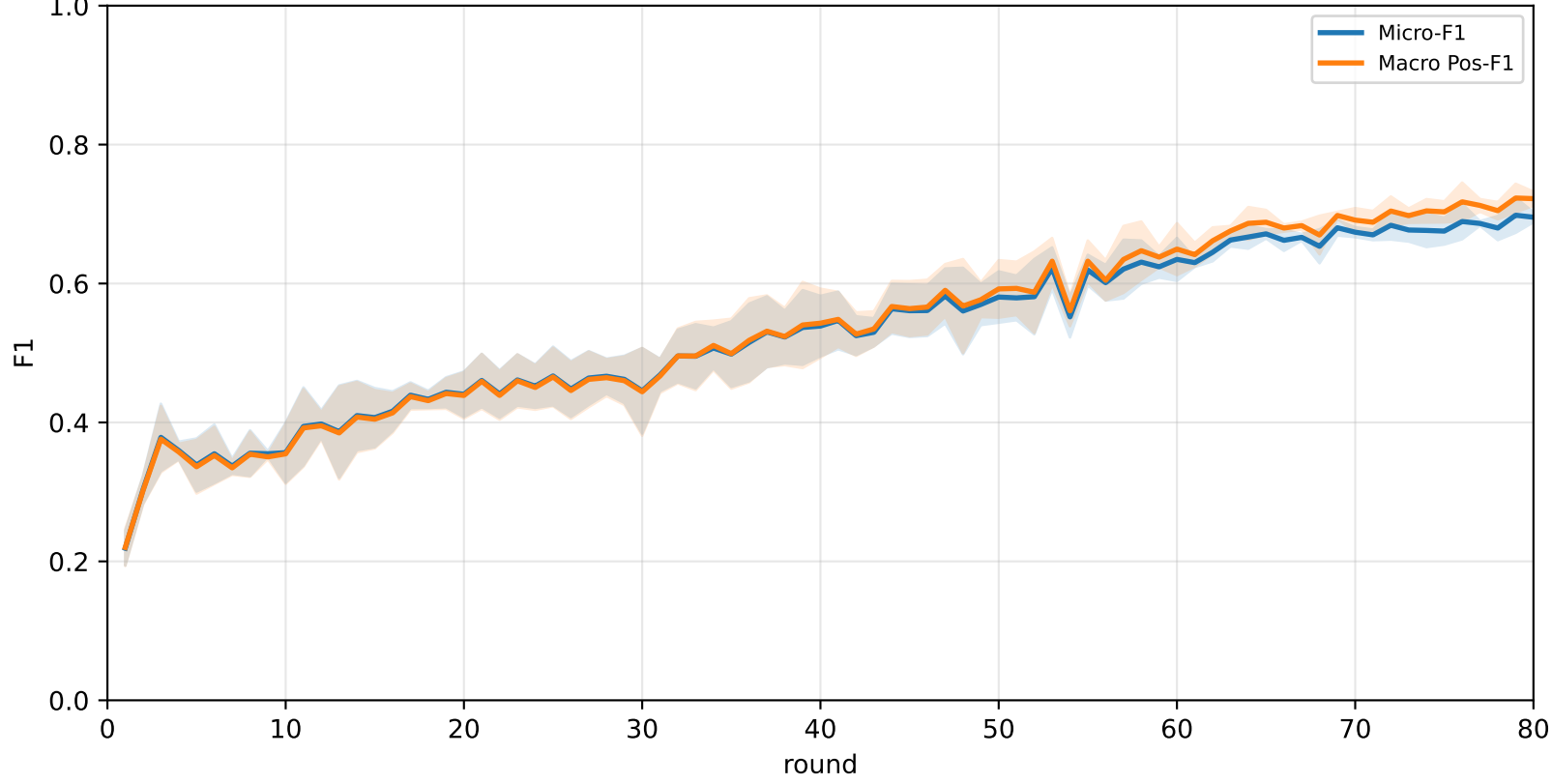
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



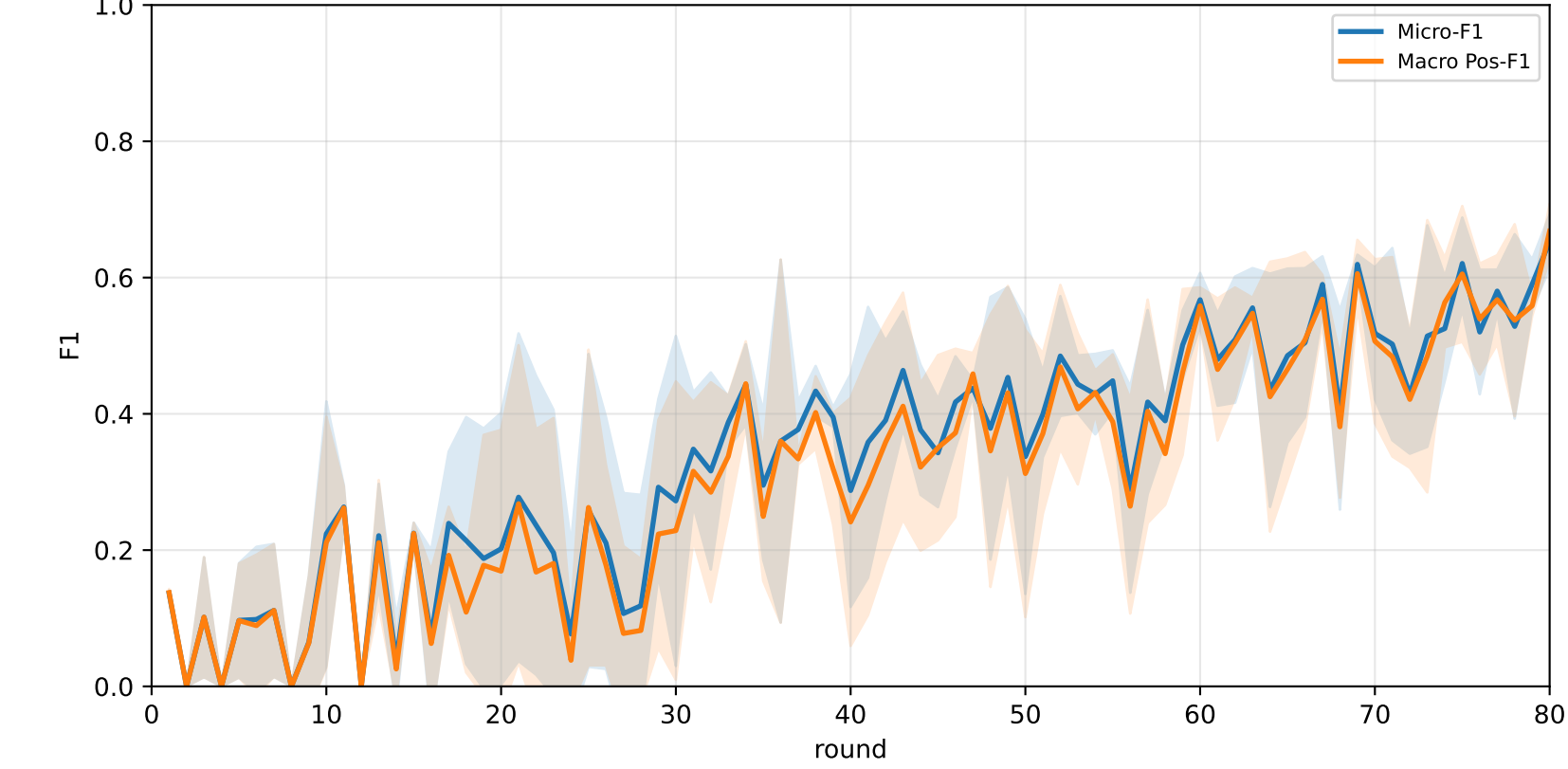
Pooled baseline micro / macro F1



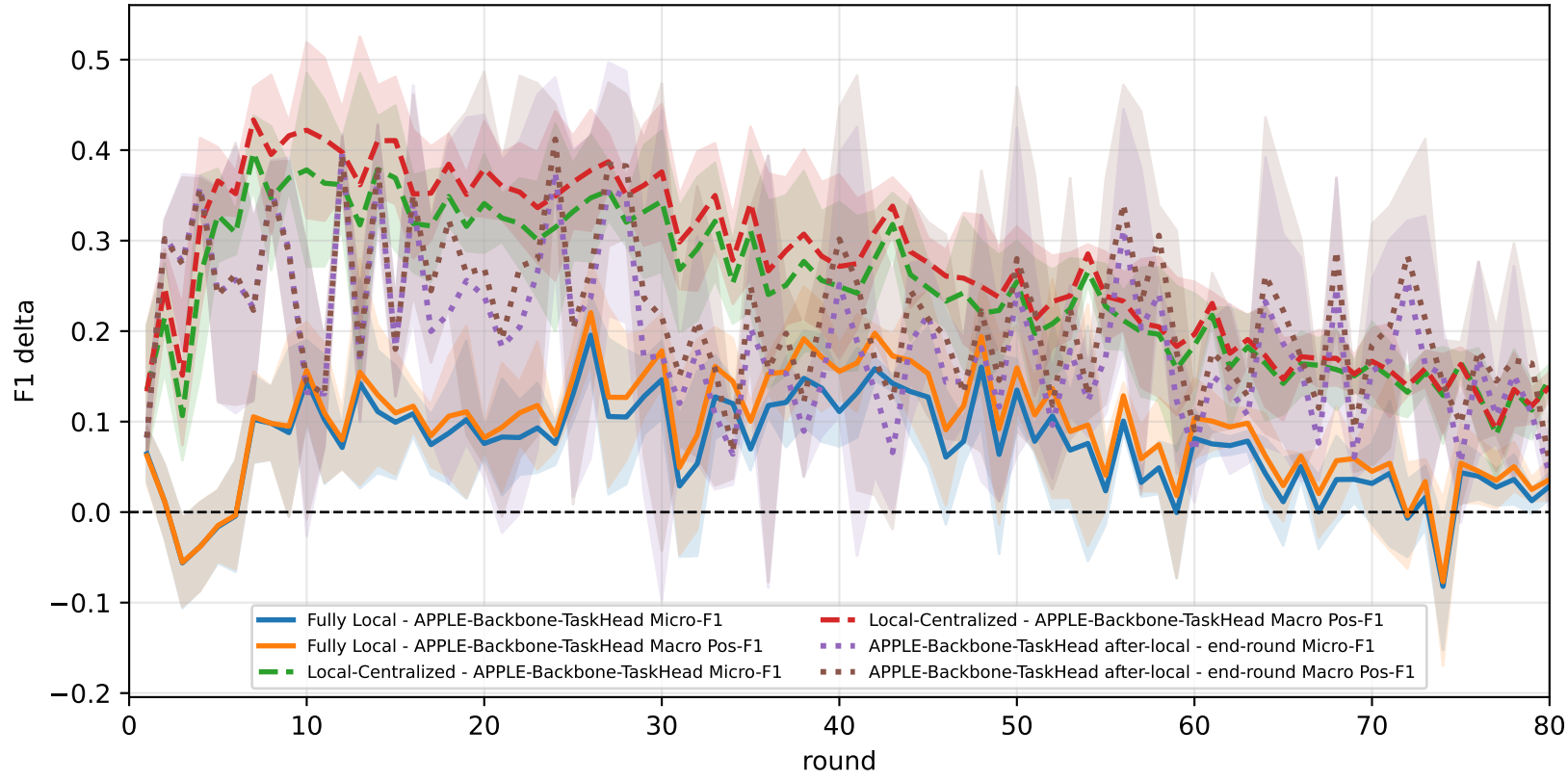
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



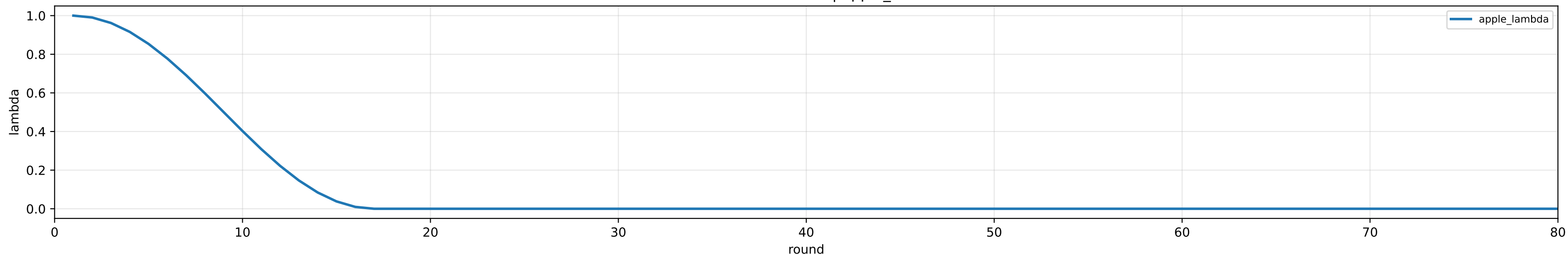
Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap



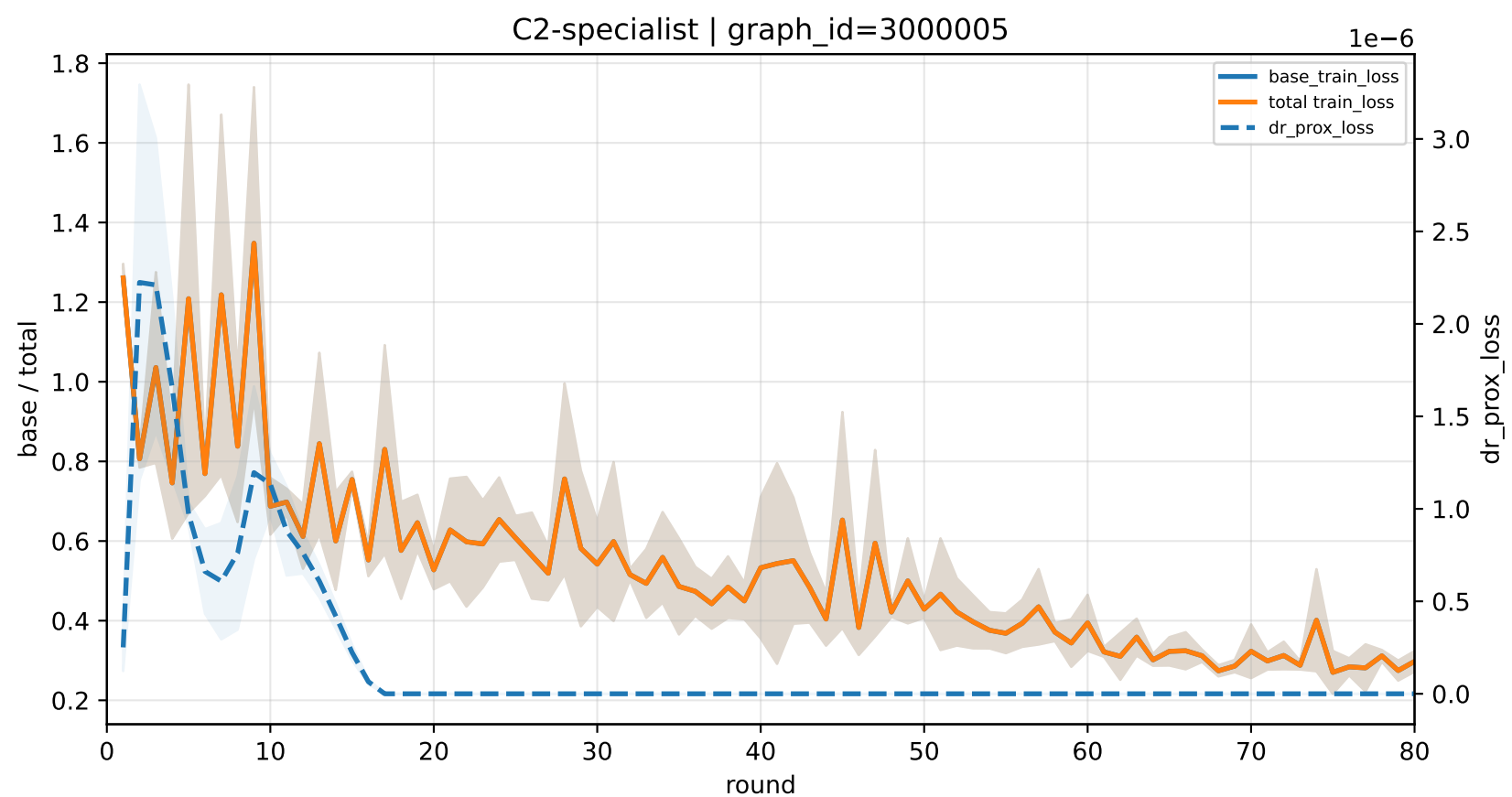
APPLE-Backbone-TaskHead training-loss decomposition | Mid heterogeneity

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

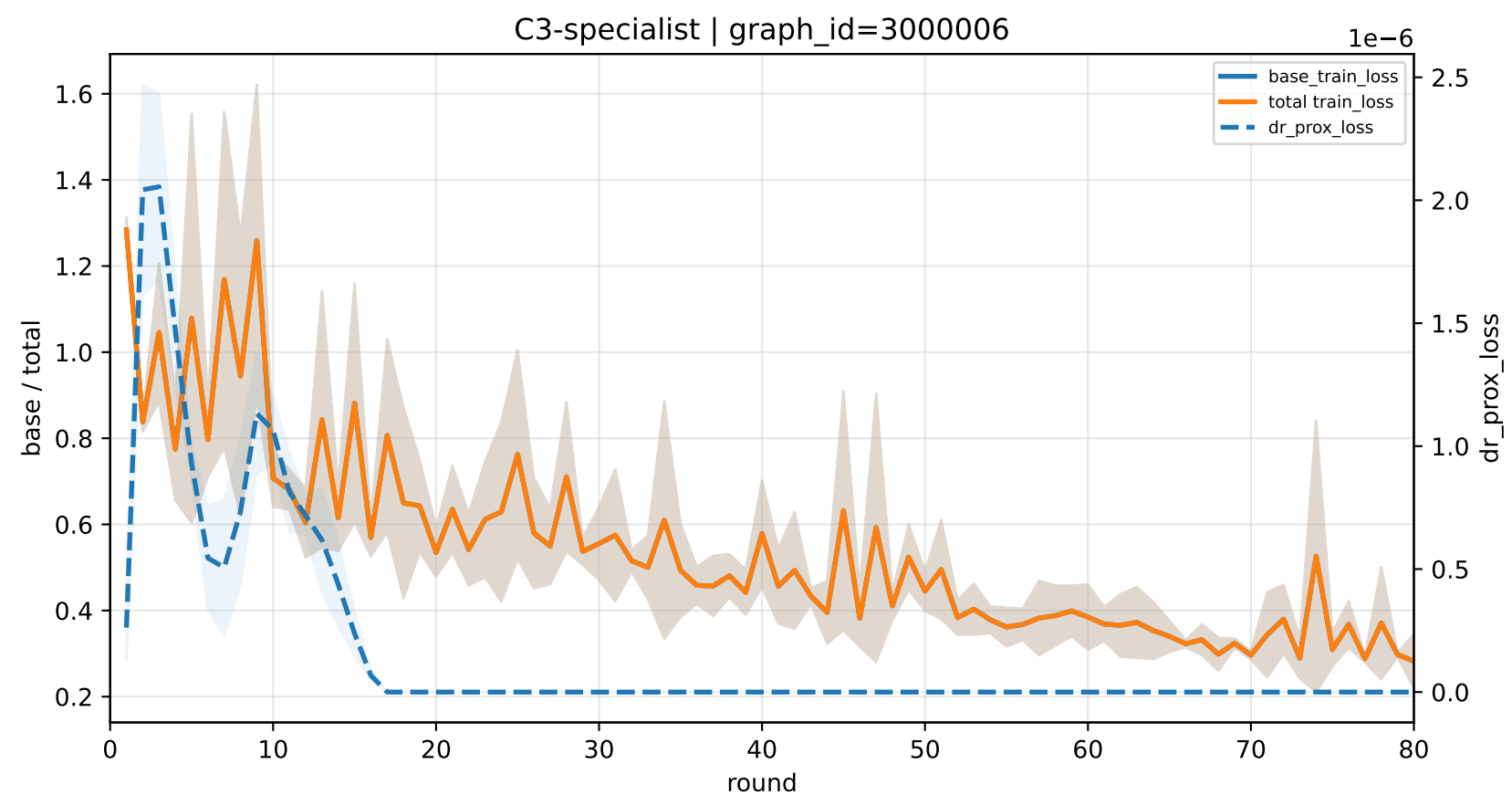
APPLE lambda scheduler | apple\_mu=0.001



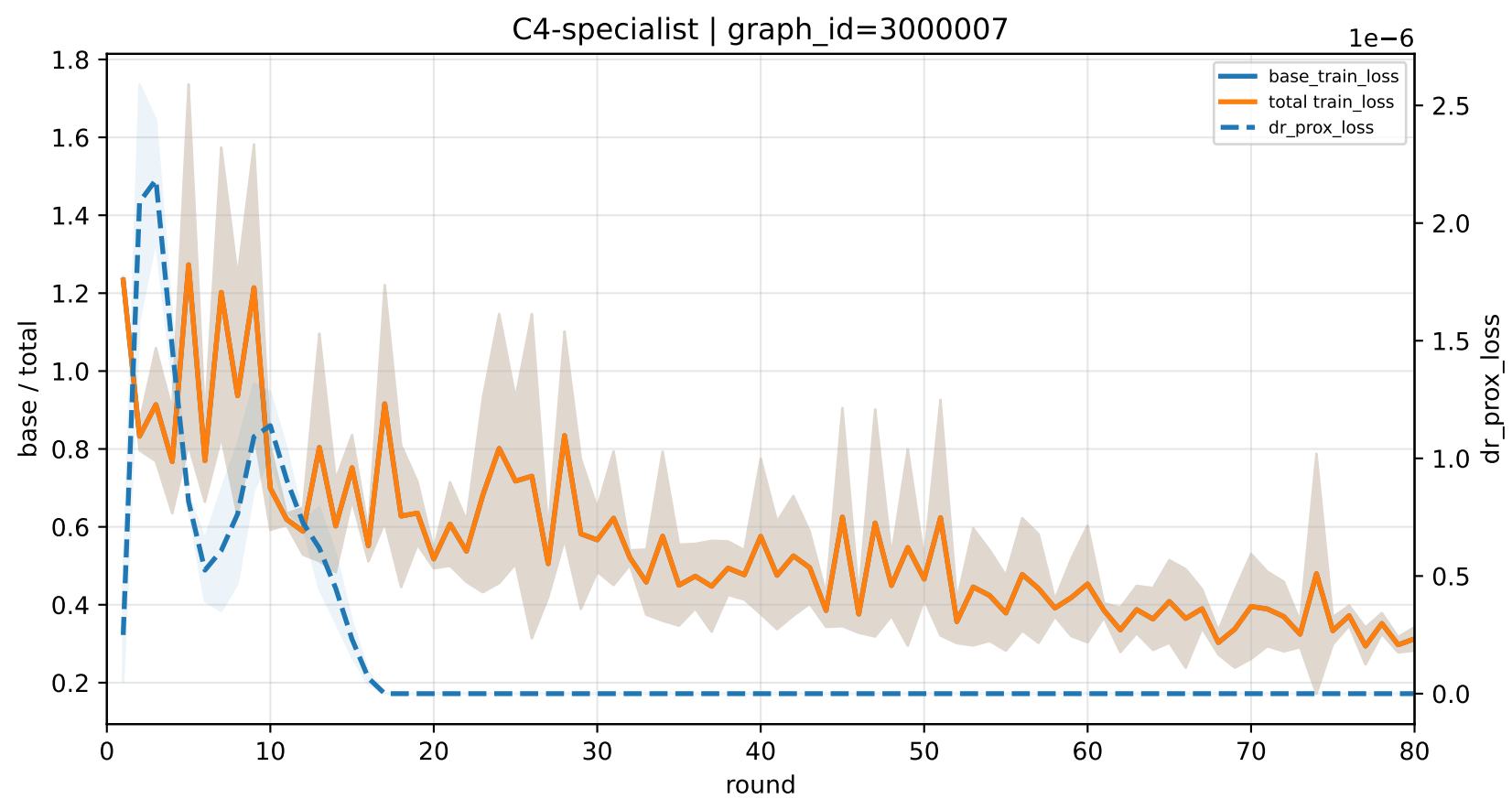
C2-specialist | graph\_id=3000005



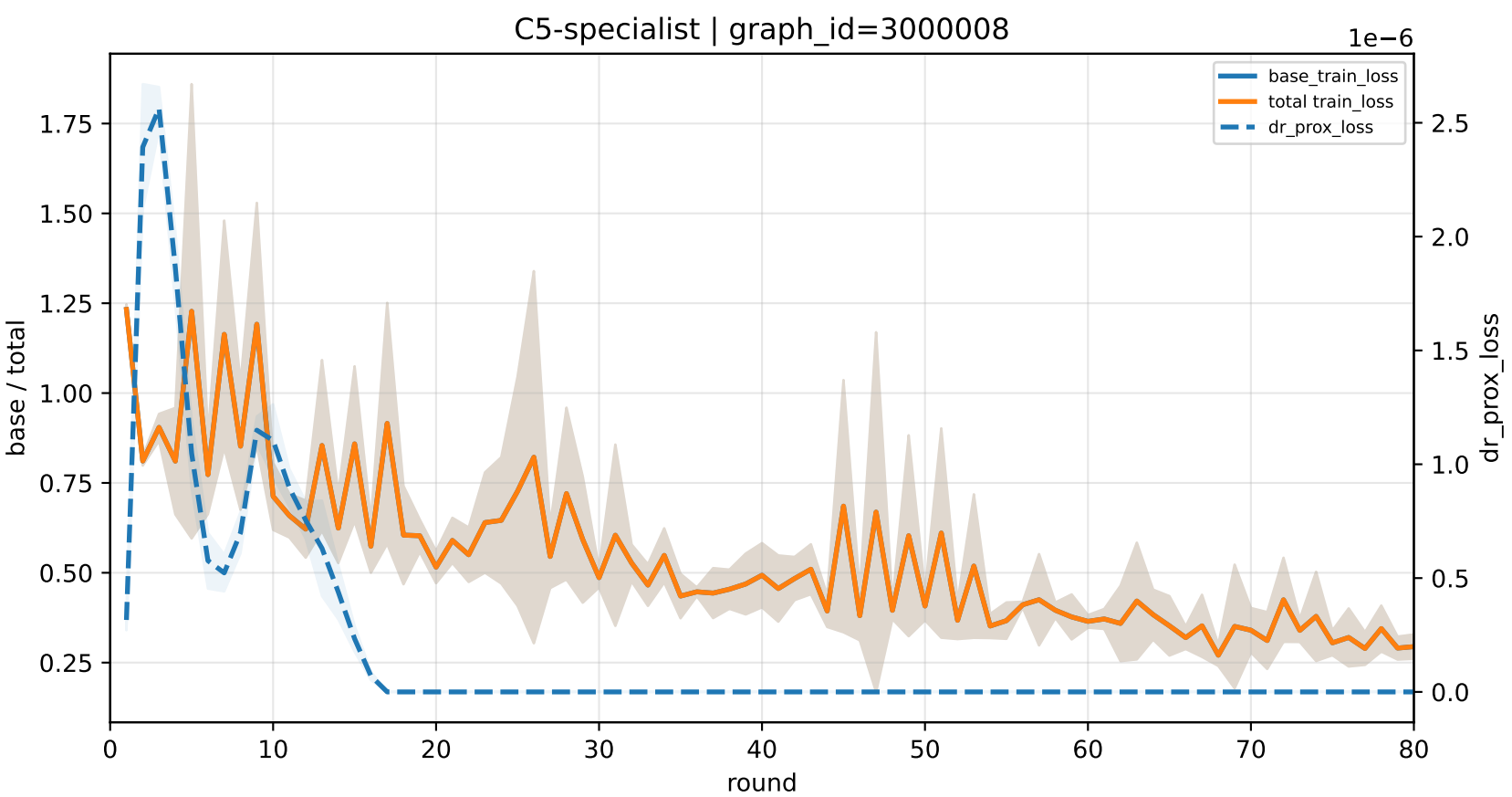
C3-specialist | graph\_id=3000006



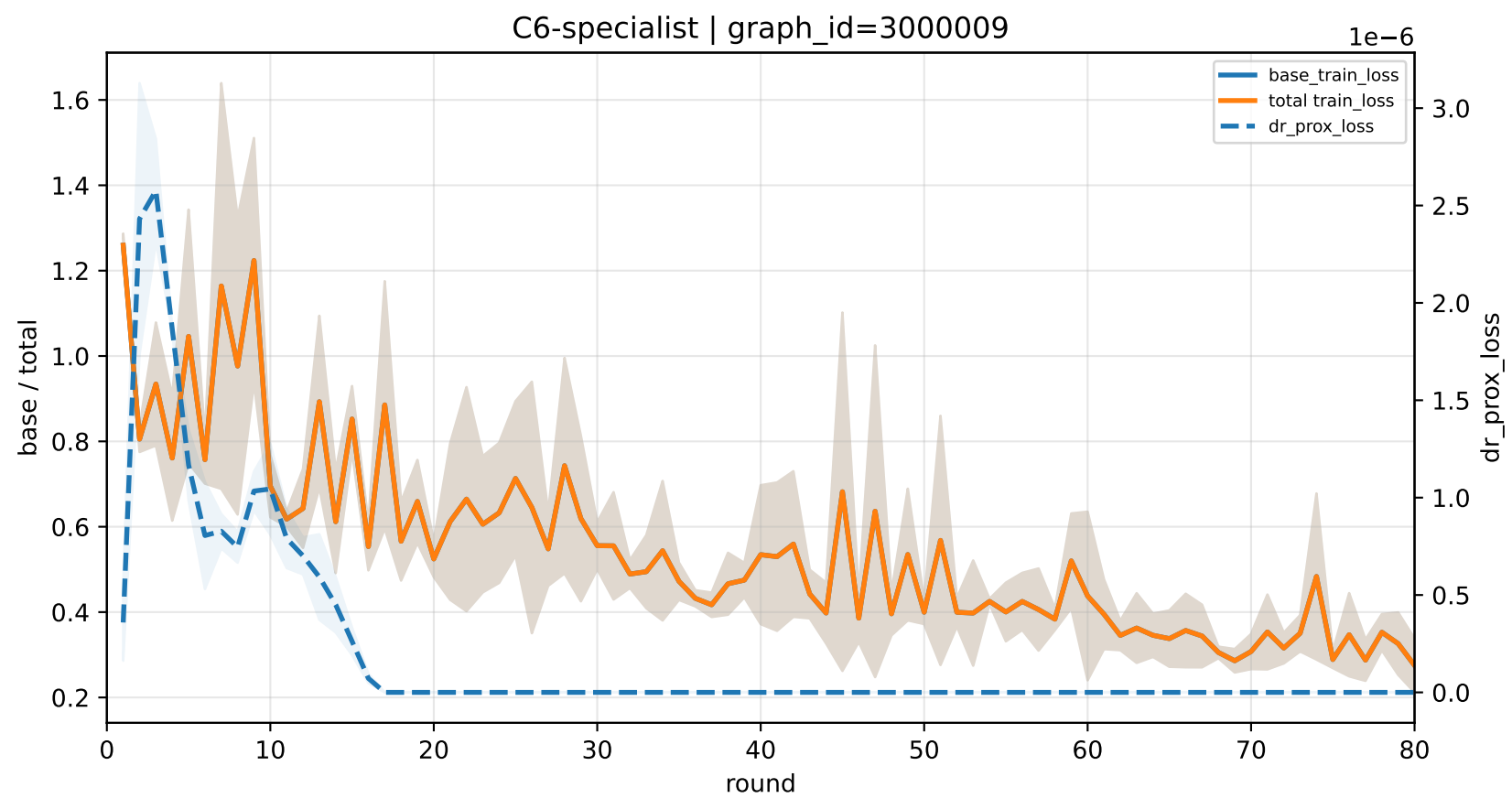
C4-specialist | graph\_id=3000007



C5-specialist | graph\_id=3000008



C6-specialist | graph\_id=3000009

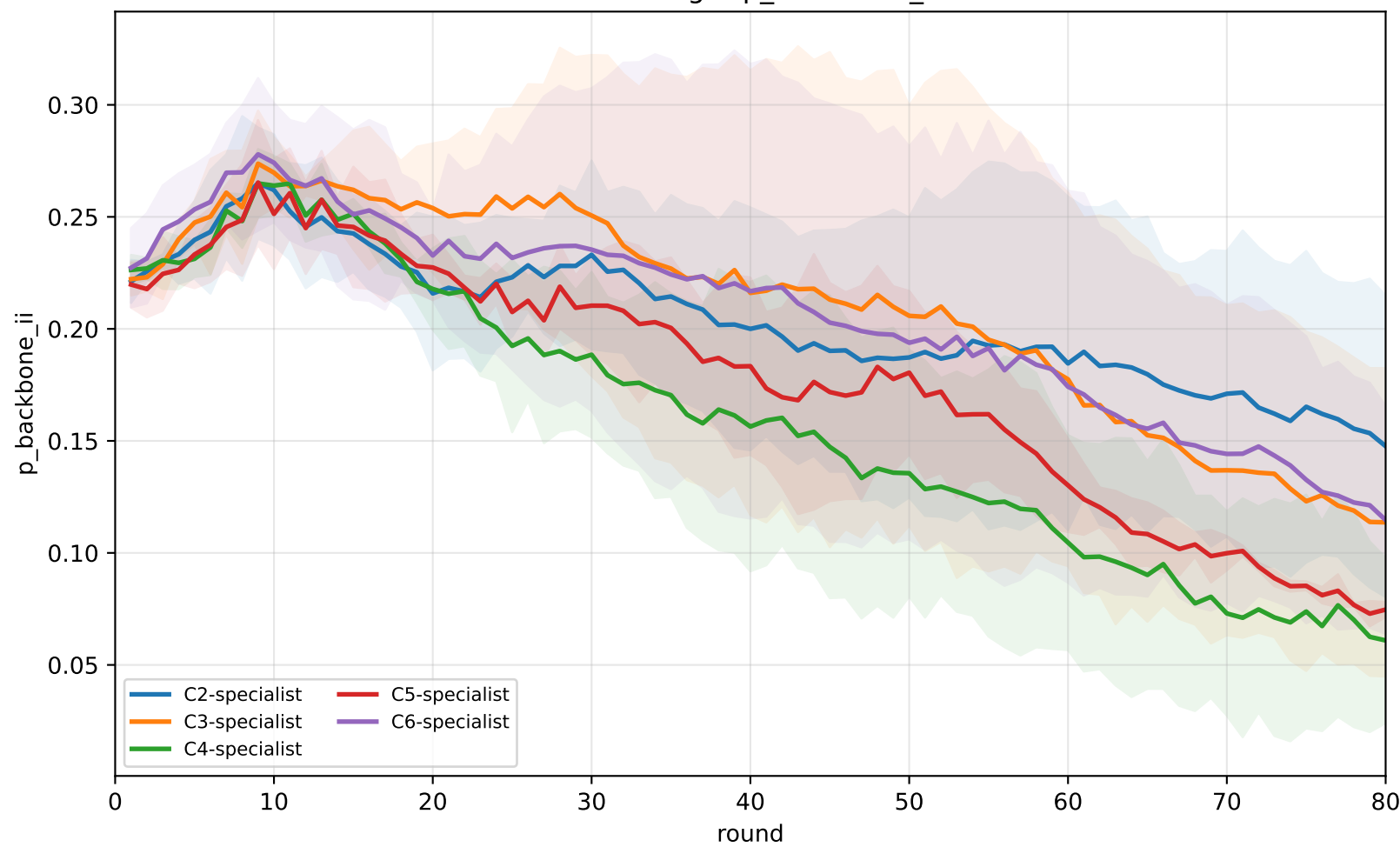




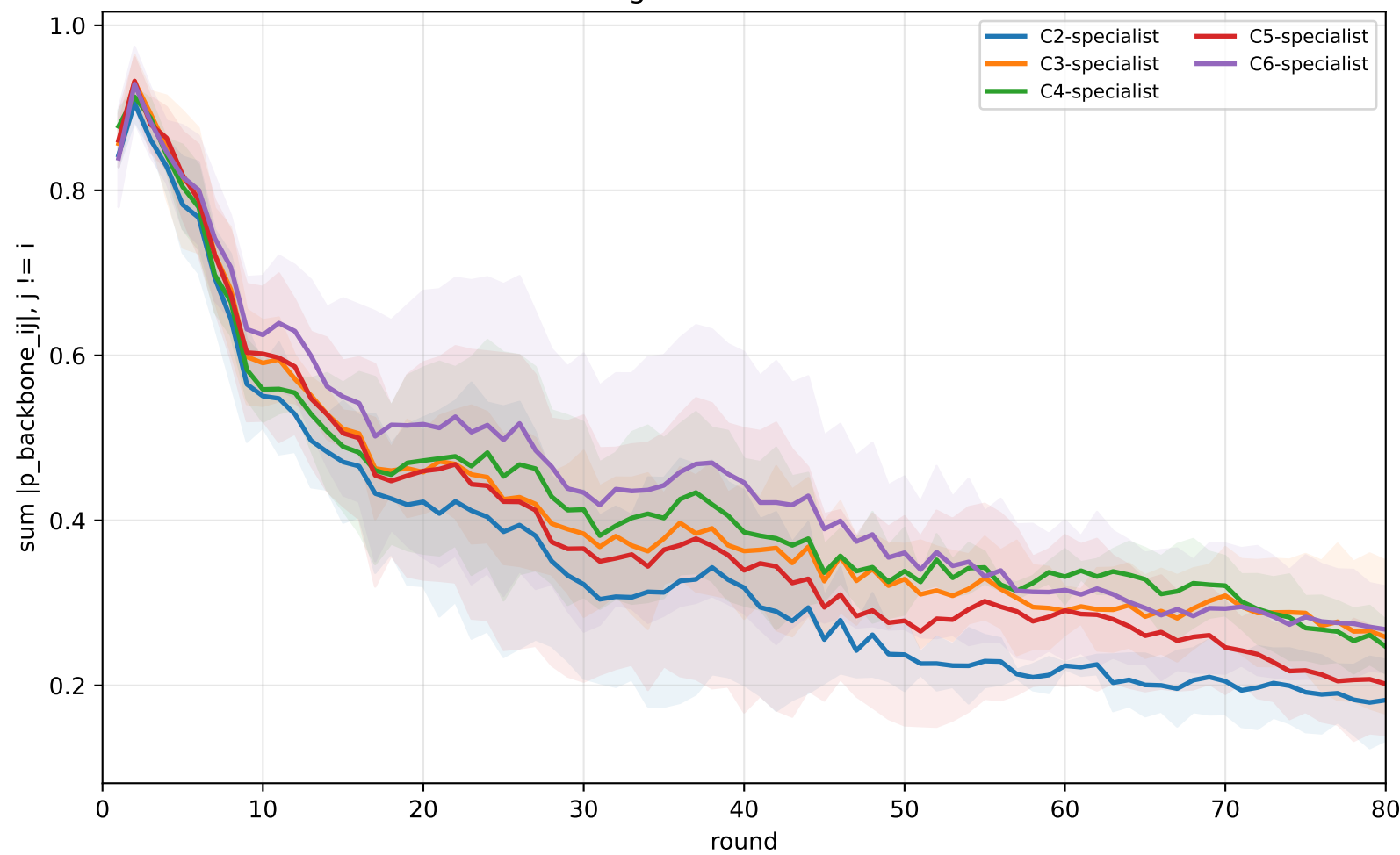
APPLE-Backbone-TaskHead backbone DR behavior | Mid heterogeneity

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

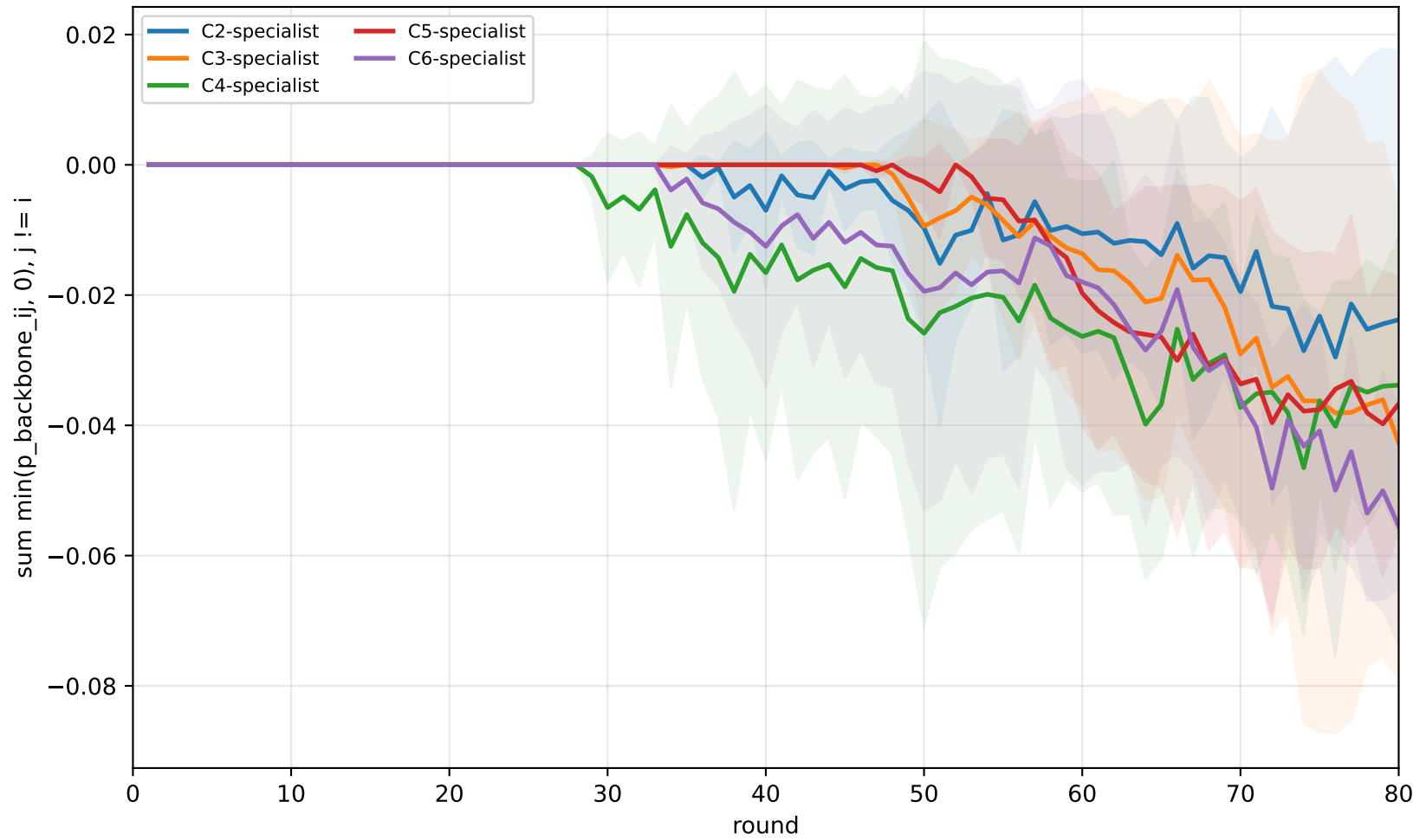
Backbone DR self-weight  $p_{\text{backbone\_ii}}$  over rounds



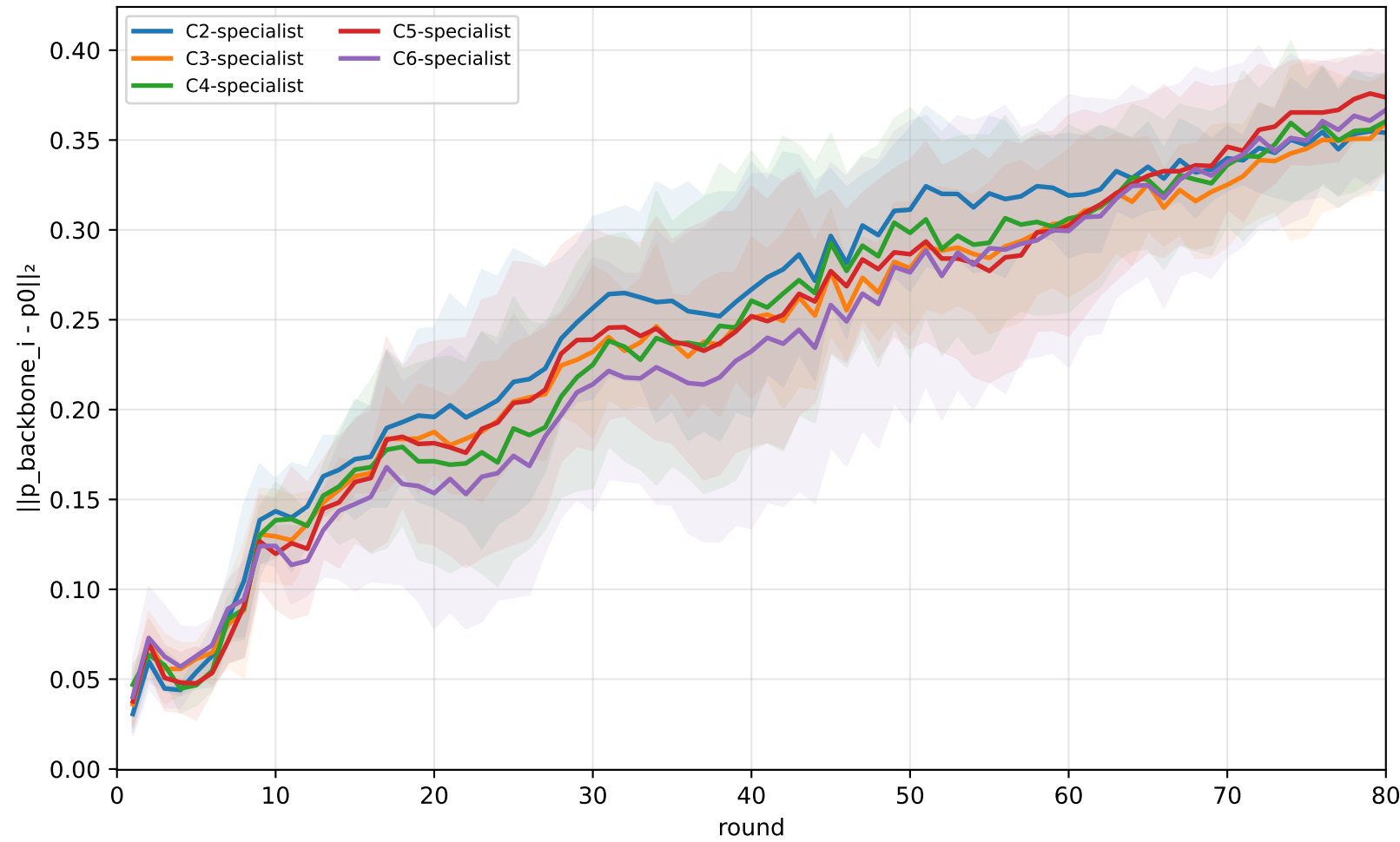
Backbone off-diagonal absolute mass over rounds



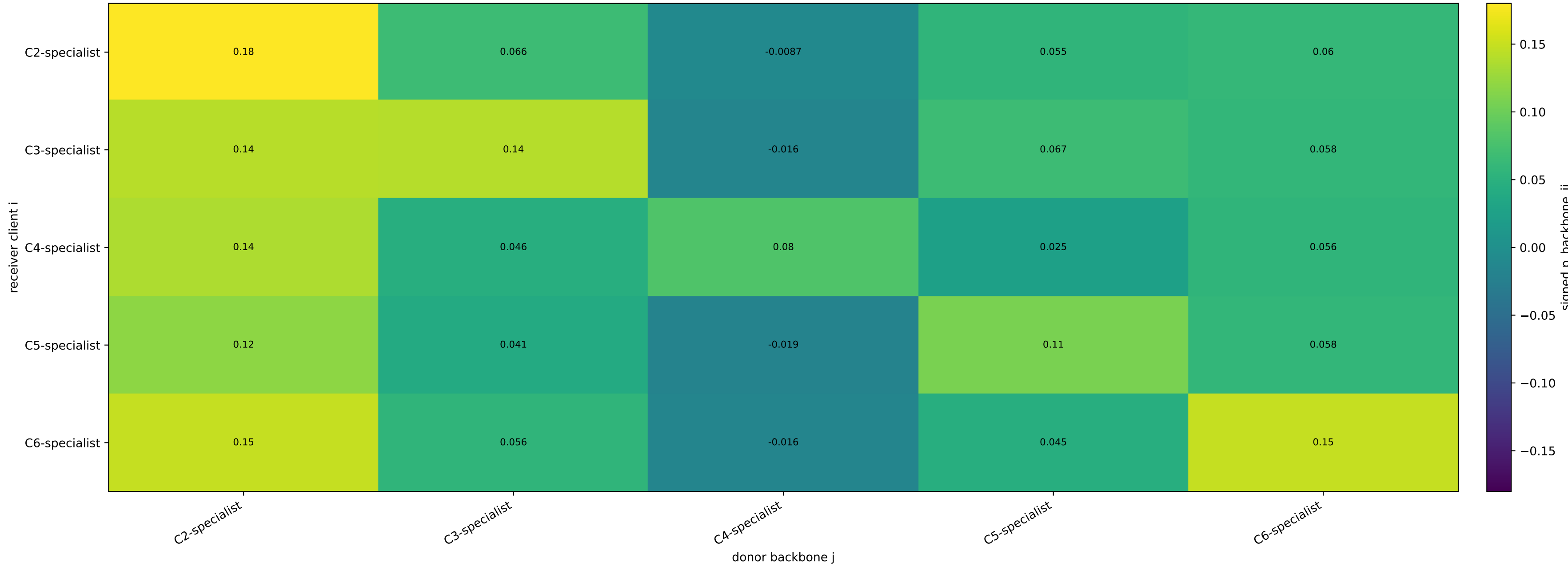
Backbone off-diagonal negative mass over rounds



Backbone DR movement away from  $p_0$



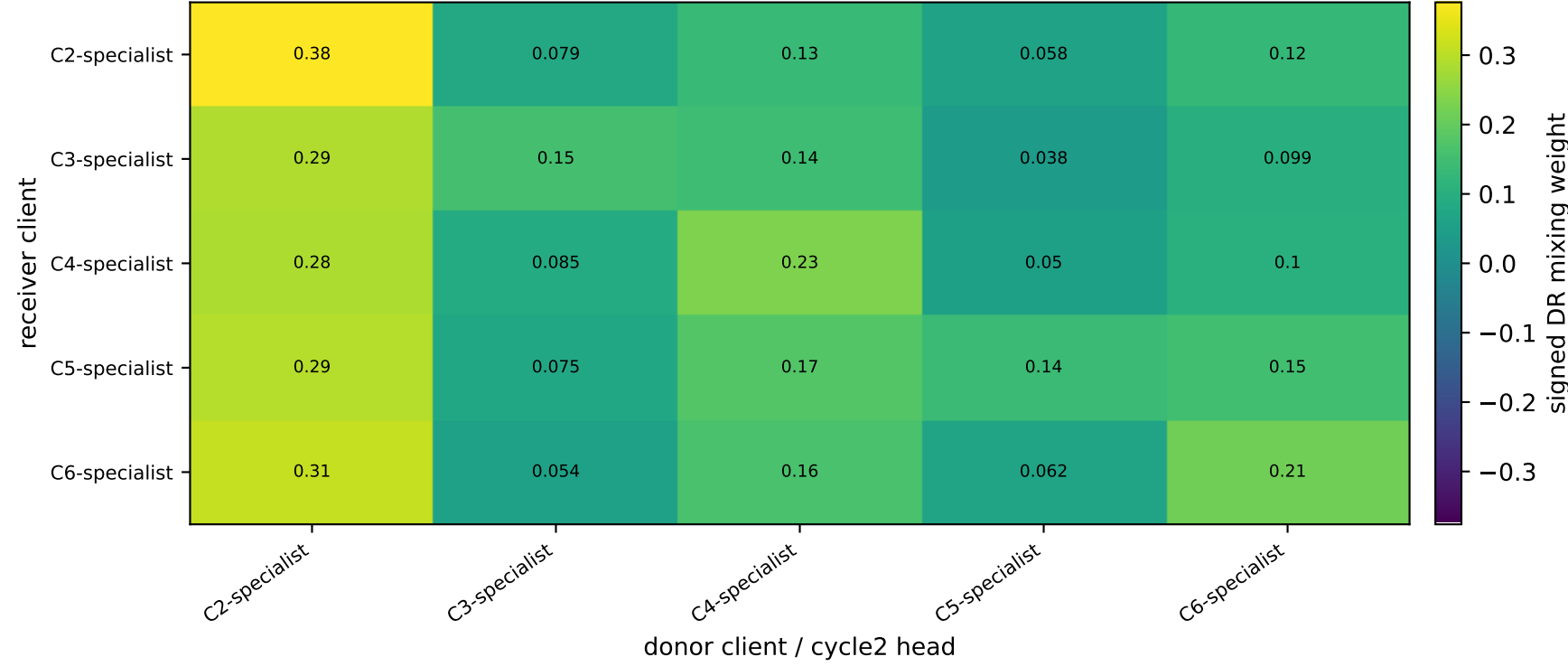
Selected-best-round backbone DR coefficients, mean over seeds



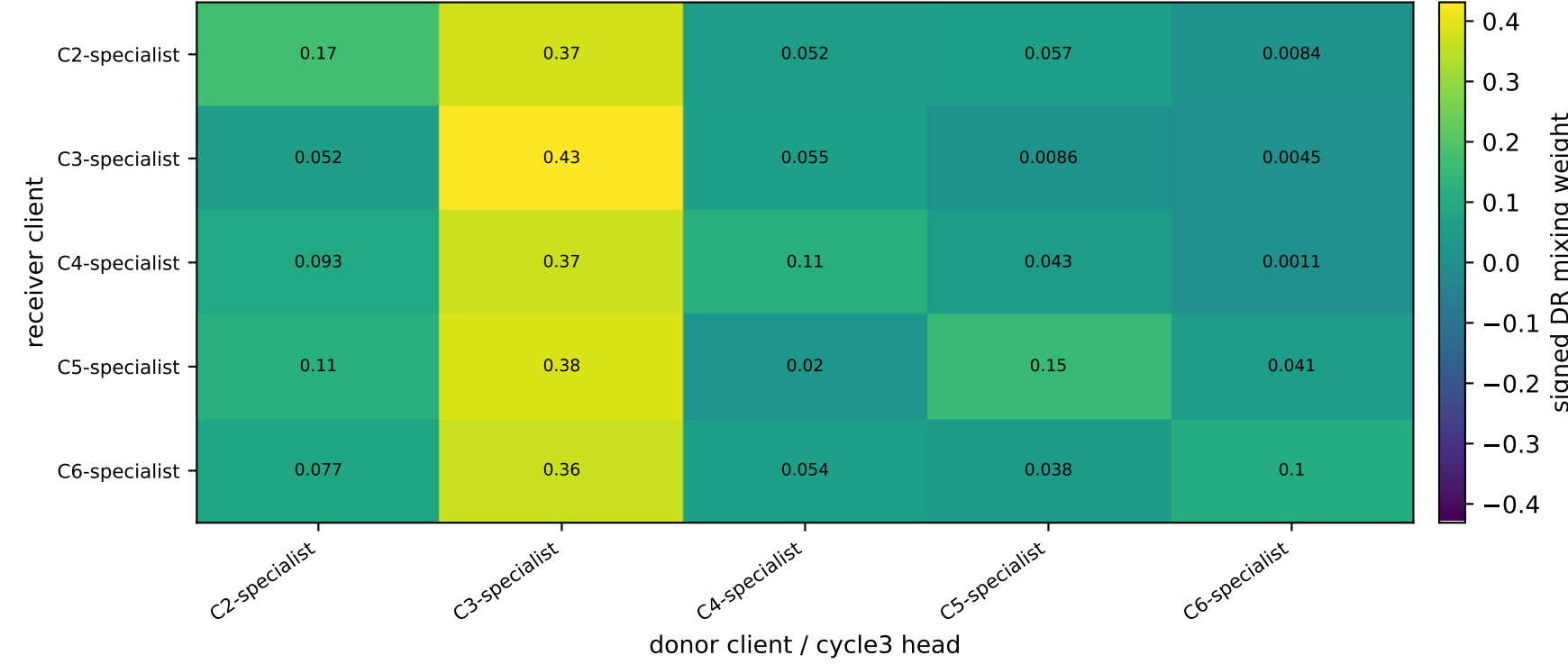
# APPLE-Backbone-TaskHead task-wise DR matrices: receiver clients × donor task-heads | Mid

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
 family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
 family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
 family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
 family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
 heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

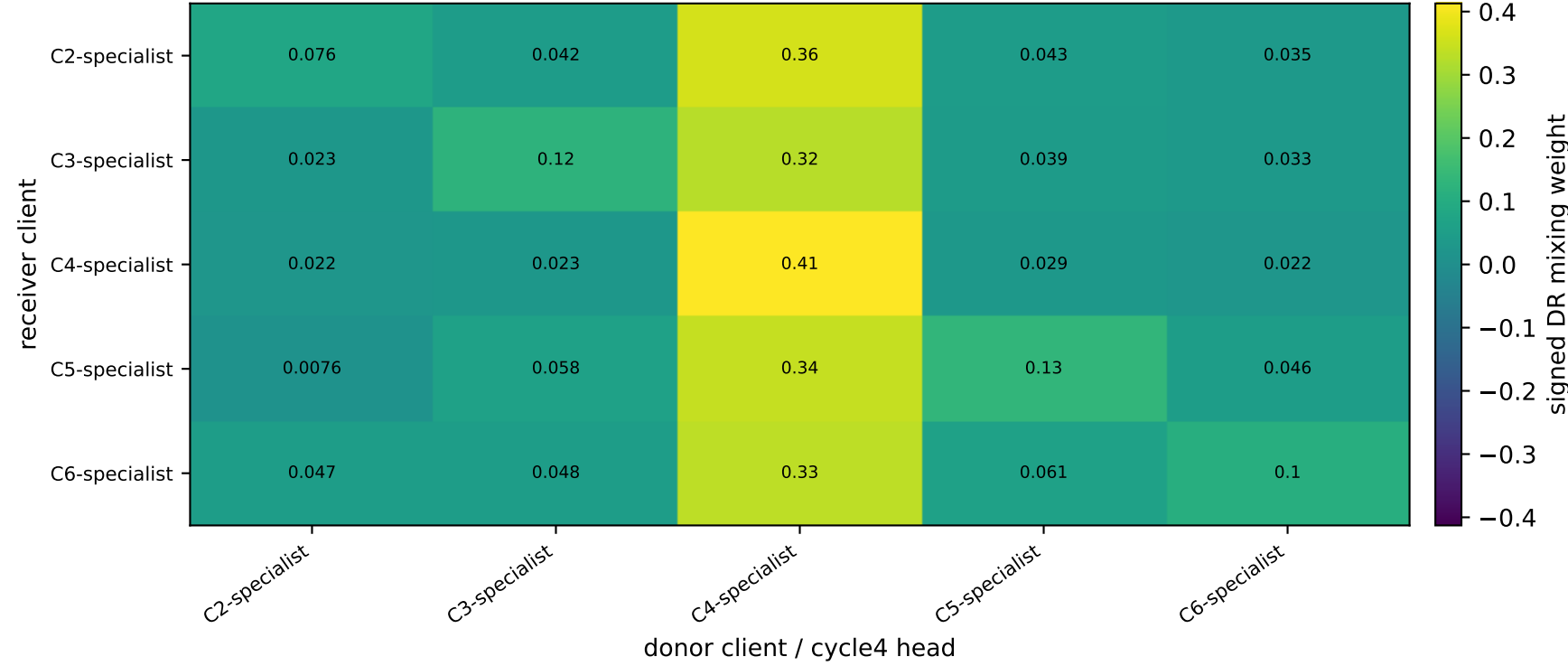
DR matrix for task cycle2: at best selected checkpoint



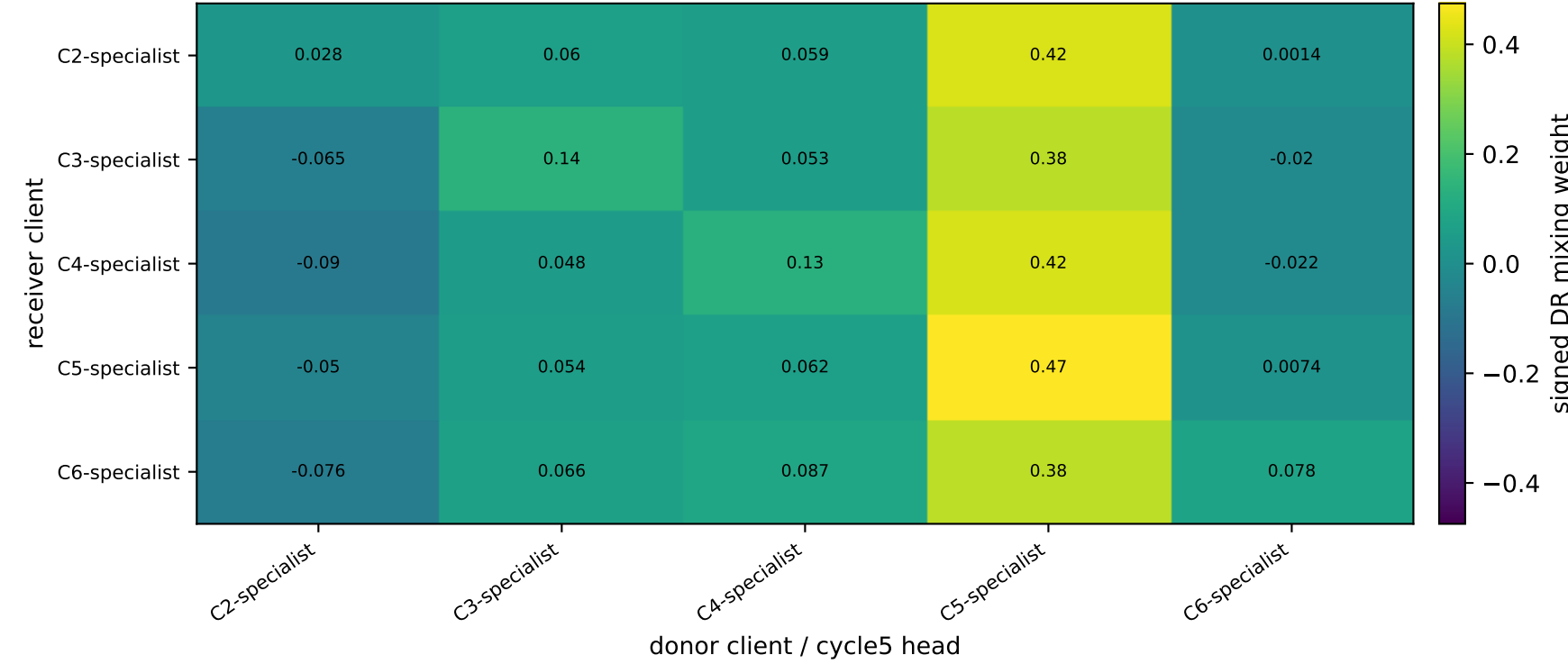
DR matrix for task cycle3: at best selected checkpoint



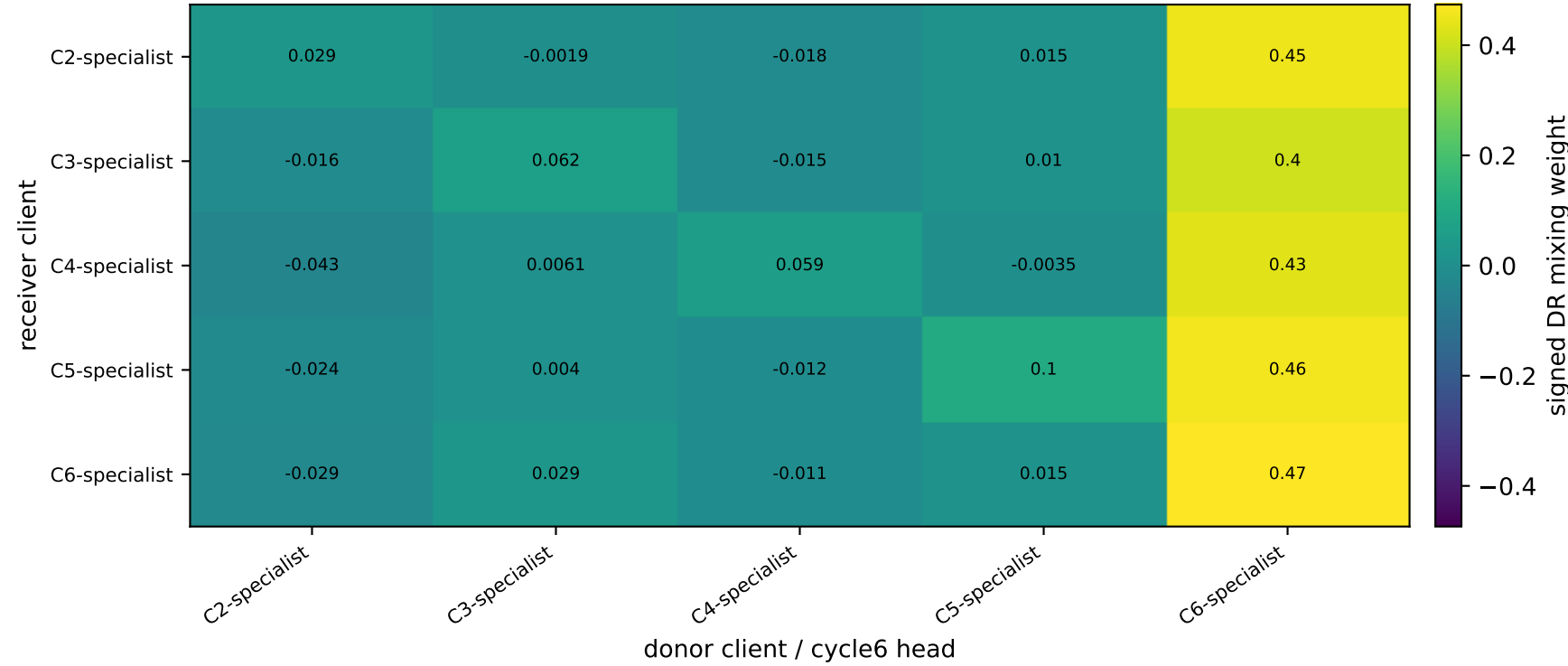
DR matrix for task cycle4: at best selected checkpoint



DR matrix for task cycle5: at best selected checkpoint



DR matrix for task cycle6: at best selected checkpoint

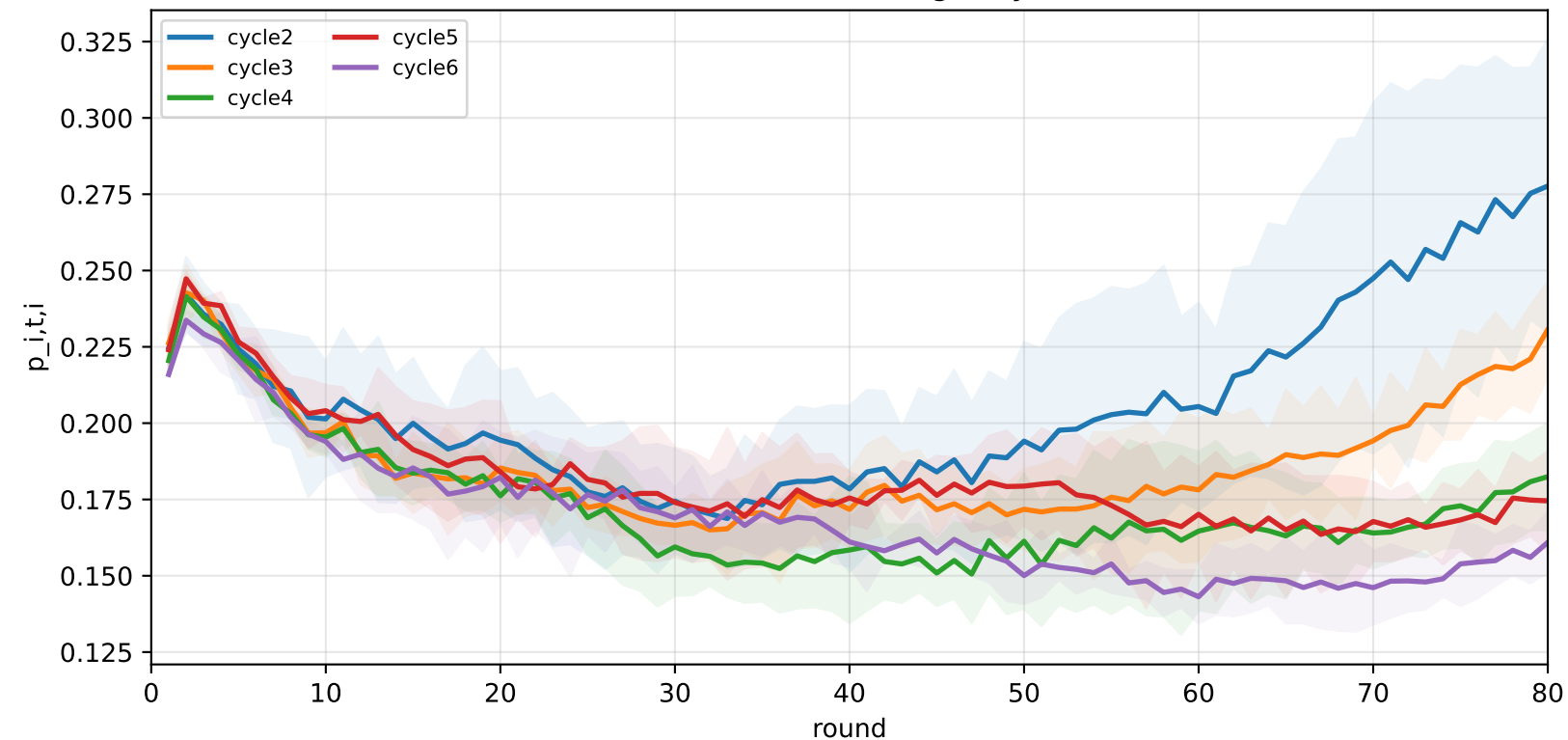




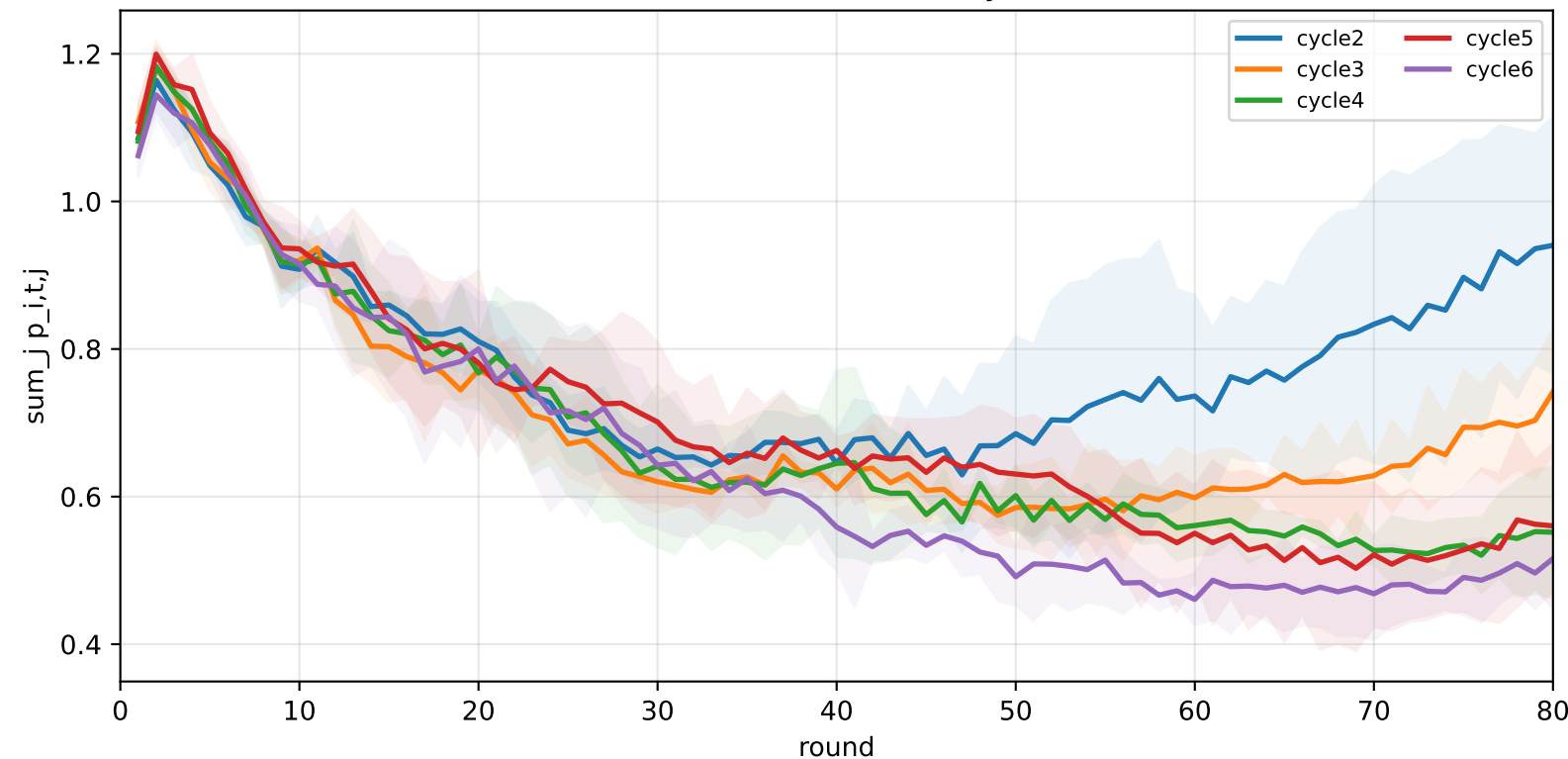
# APPLE-Backbone-TaskHead task-wise DR dynamics | Mid

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
 family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
 family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
 family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%  
 family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%  
 heterogeneity=mid | gamma=0.50 | task\_jsd\_mean=0.120 | task\_jsd\_median=0.121 | task\_jsd\_max=0.123 | family\_centroid\_jsd\_mean=0.120

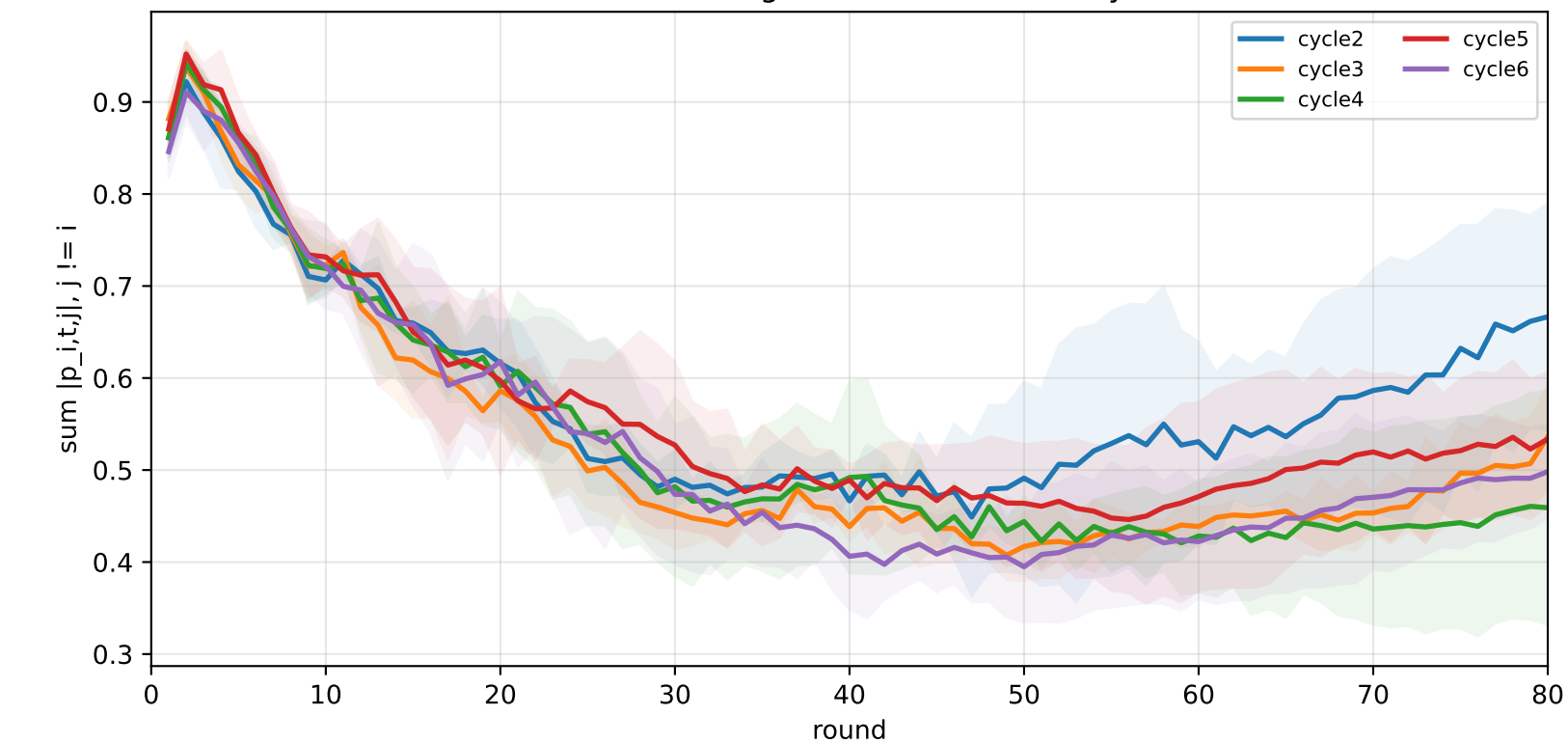
Task-head DR self-weight by task



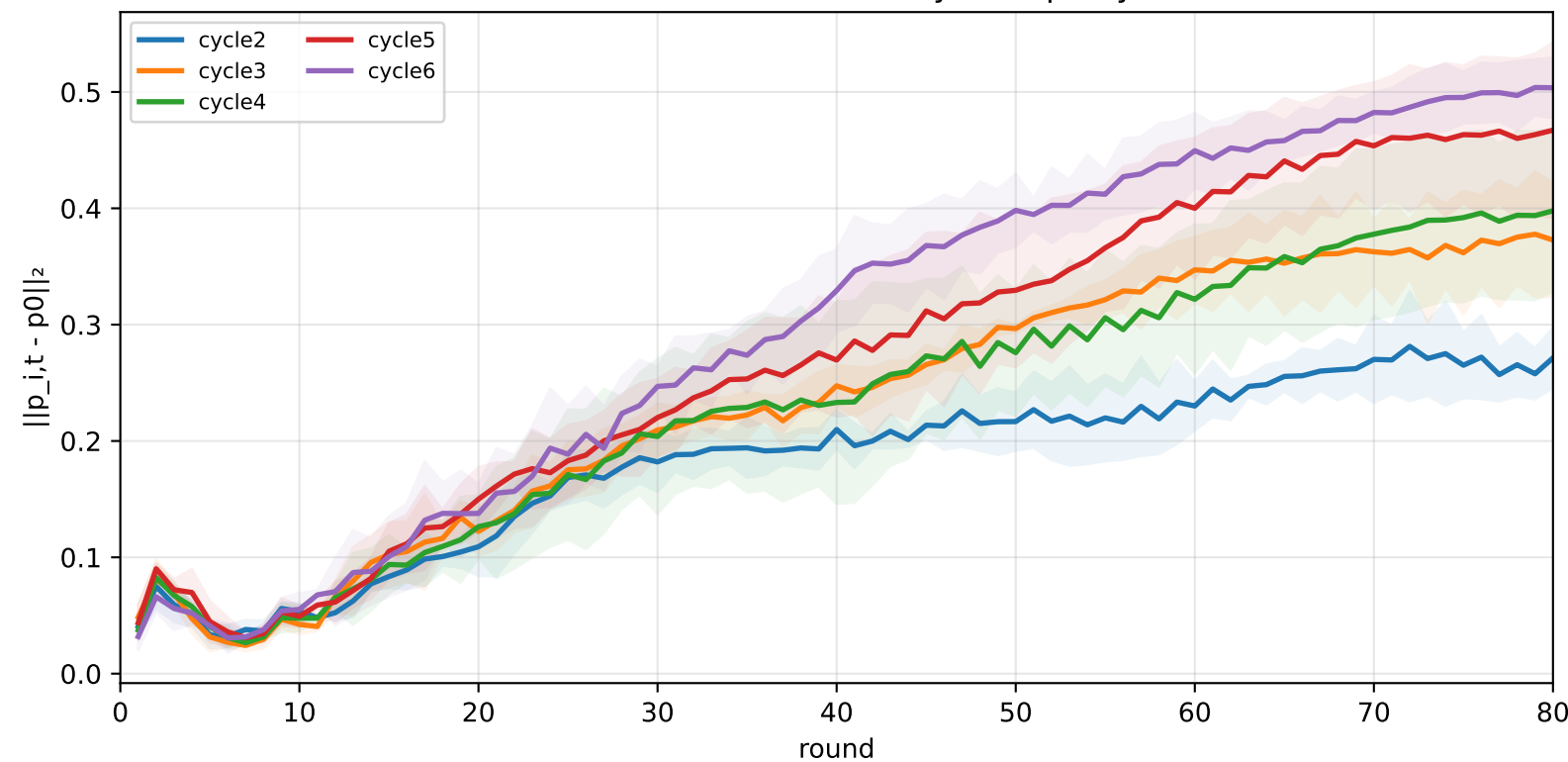
Task-head DR row sum by task



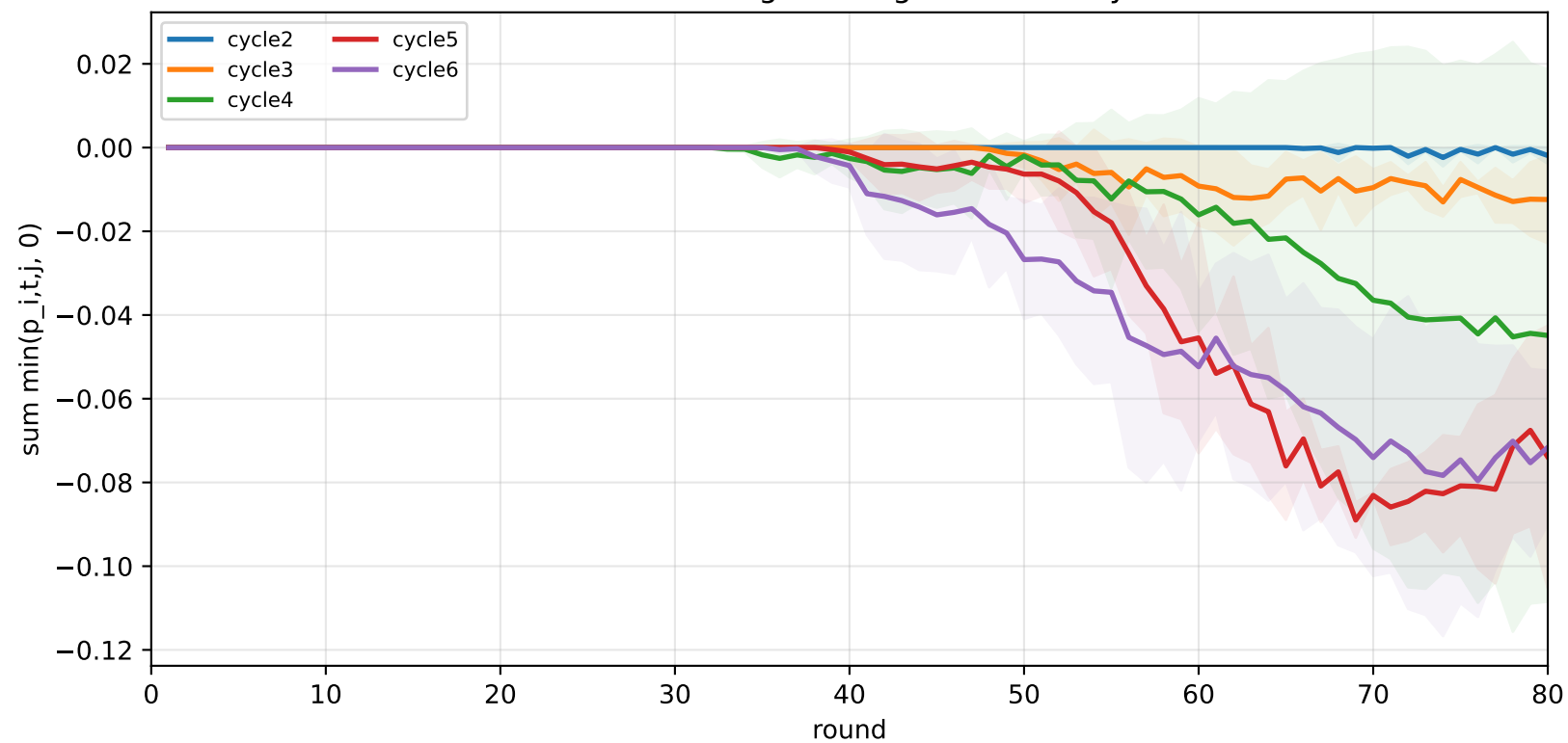
Task-head off-diagonal absolute mass by task



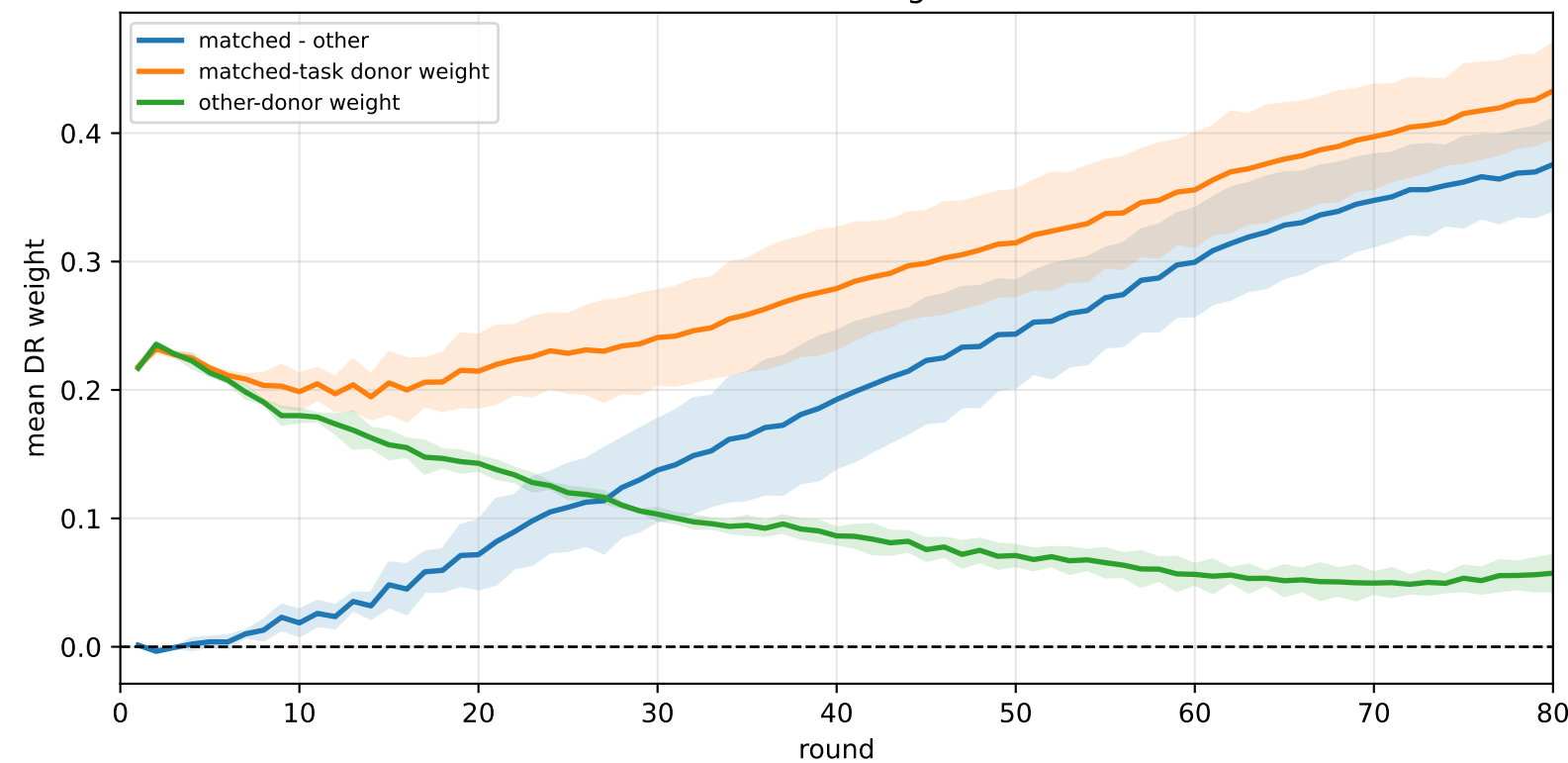
Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment

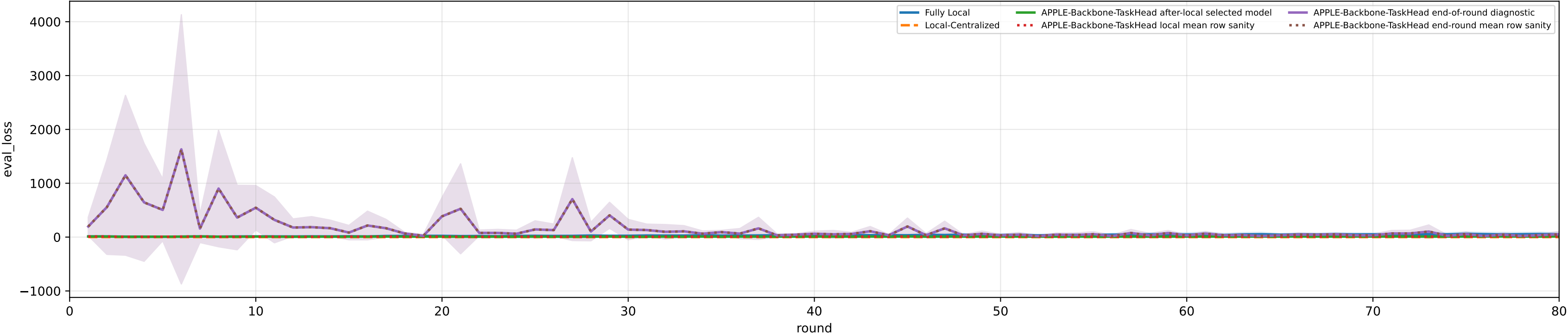


family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |  
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |  
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.2%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |  
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.2%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |  
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |  
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

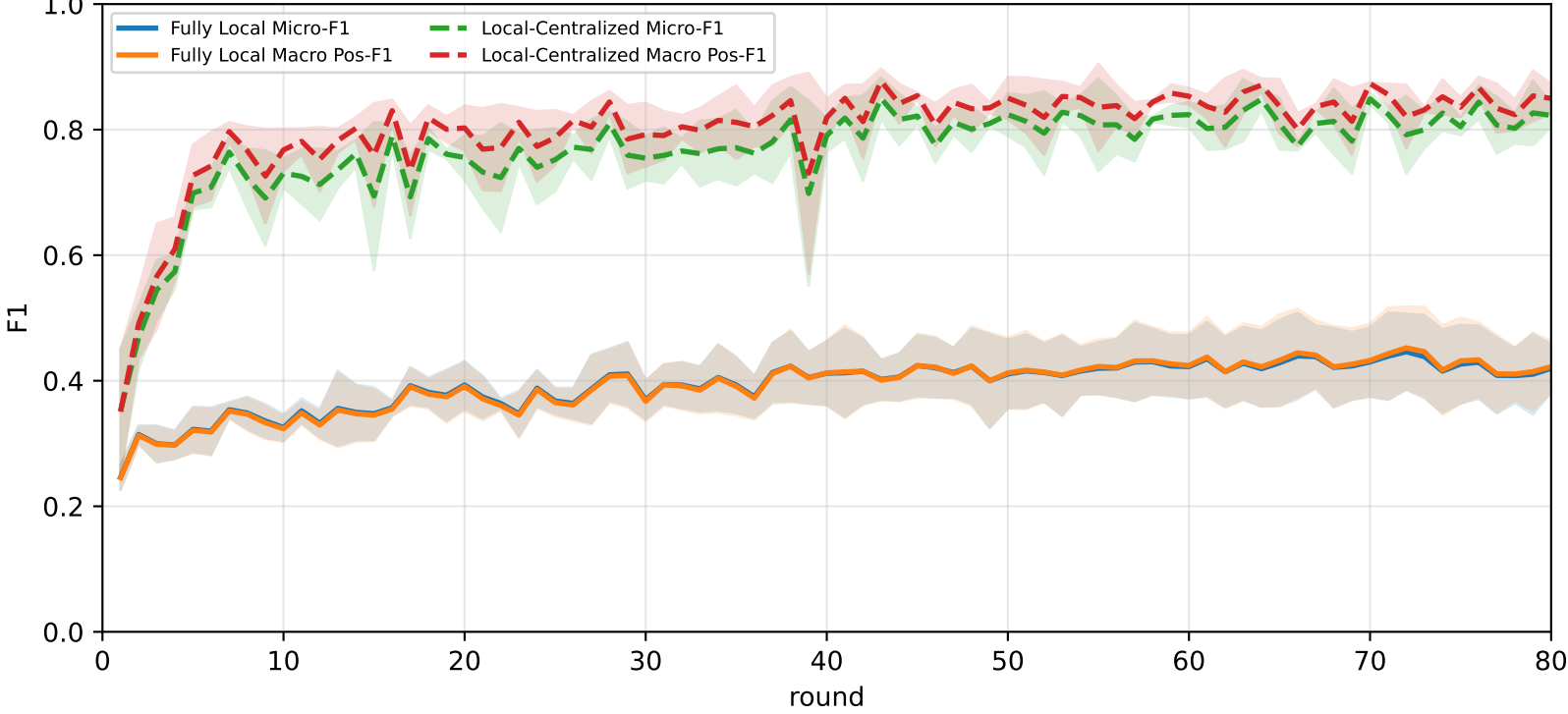
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 27.6% ± 1.8%        | 27.6% ± 1.8%        | 22.4% ± 1.5%        | 27.5% ± 2.1%        | 27.3% ± 1.2%        | 26.7% ± 1.2%        | 30.2% ± 1.5%        |
| Local-Centralized       | 84.9% ± 1.3%        | 86.7% ± 0.8%        | 96.8% ± 1.6%        | 95.1% ± 1.4%        | 86.6% ± 2.5%        | 81.4% ± 1.3%        | 73.7% ± 5.8%        |
| APPLE-Backbone-TaskHead | <u>36.0% ± 6.5%</u> | <u>36.0% ± 6.5%</u> | <u>30.7% ± 7.3%</u> | <u>36.7% ± 7.2%</u> | <u>36.5% ± 6.4%</u> | <u>35.9% ± 5.6%</u> | <u>39.7% ± 5.5%</u> |

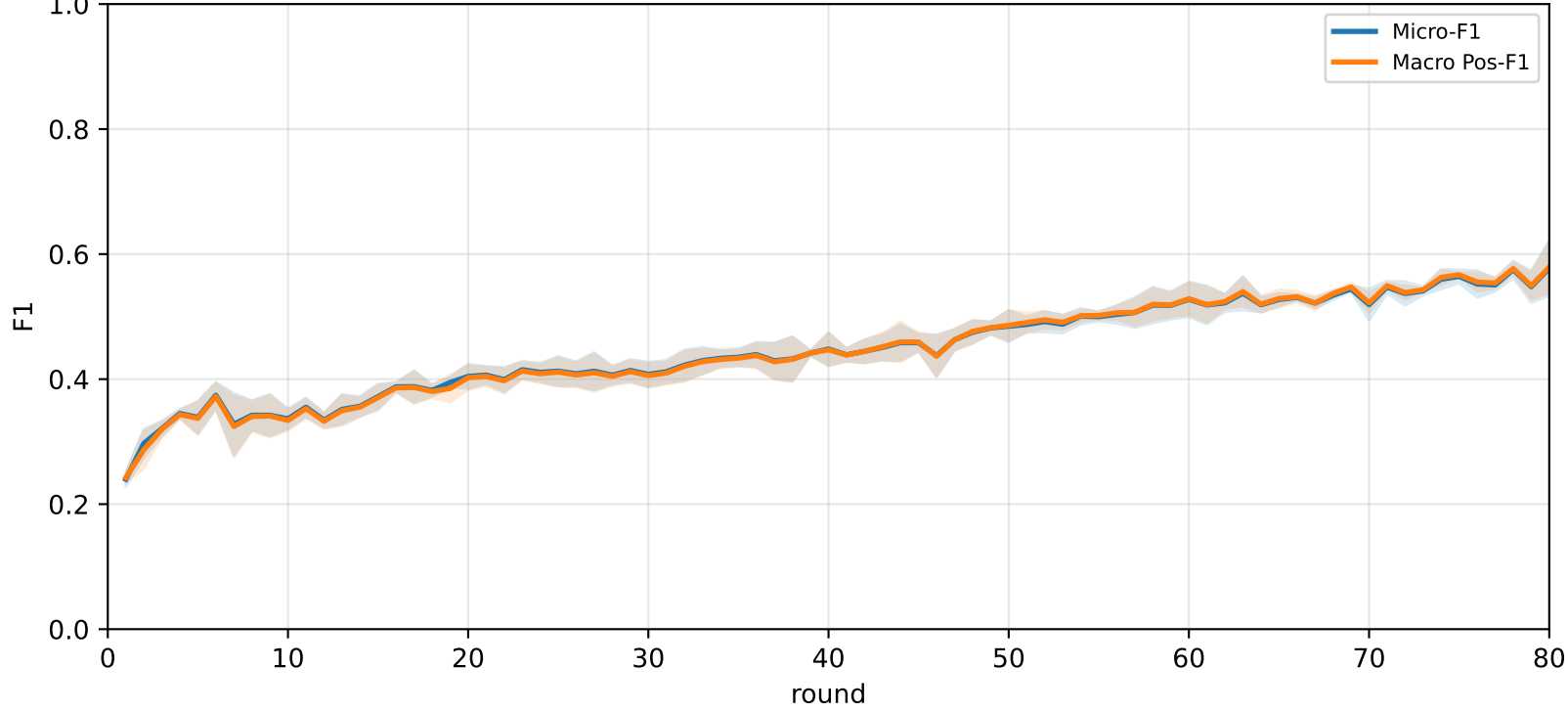
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



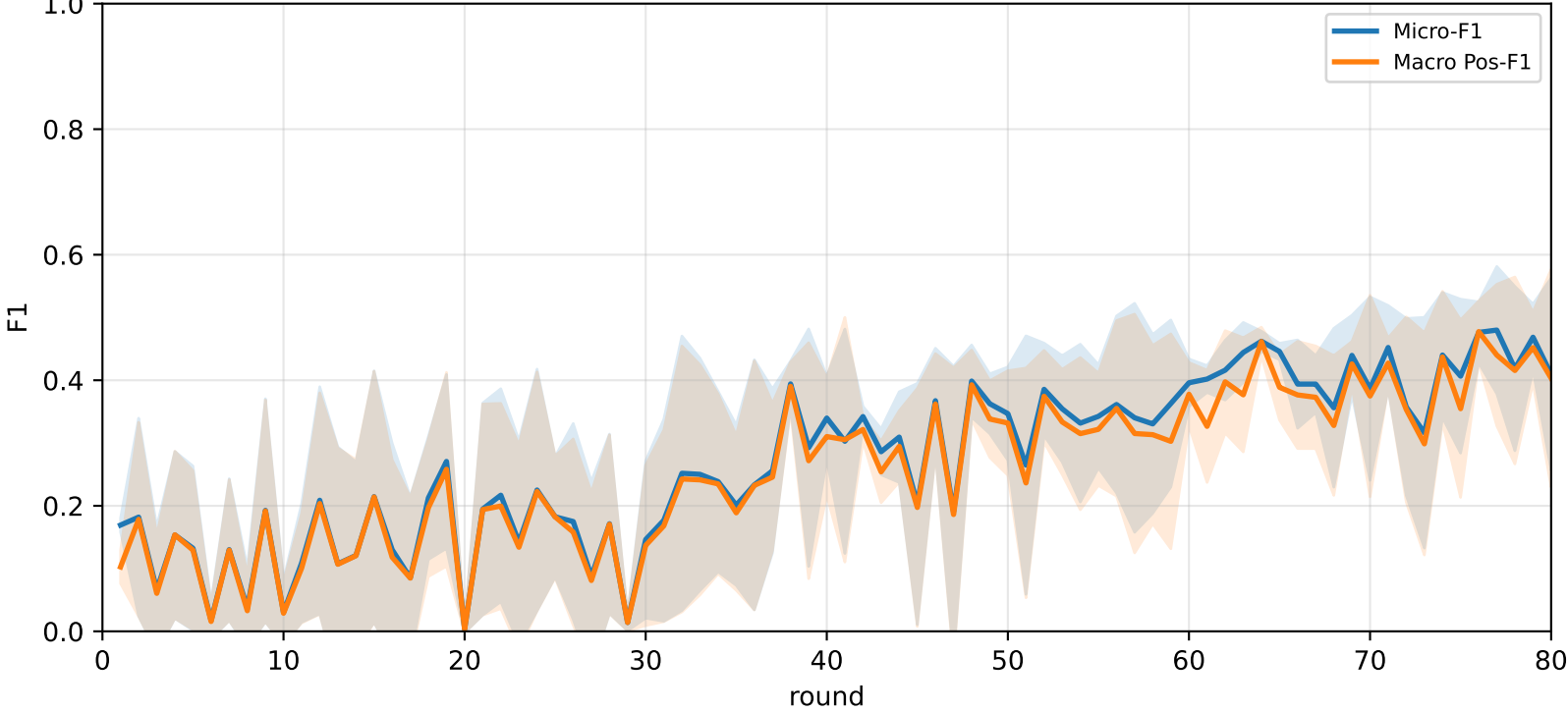
Pooled baseline micro / macro F1



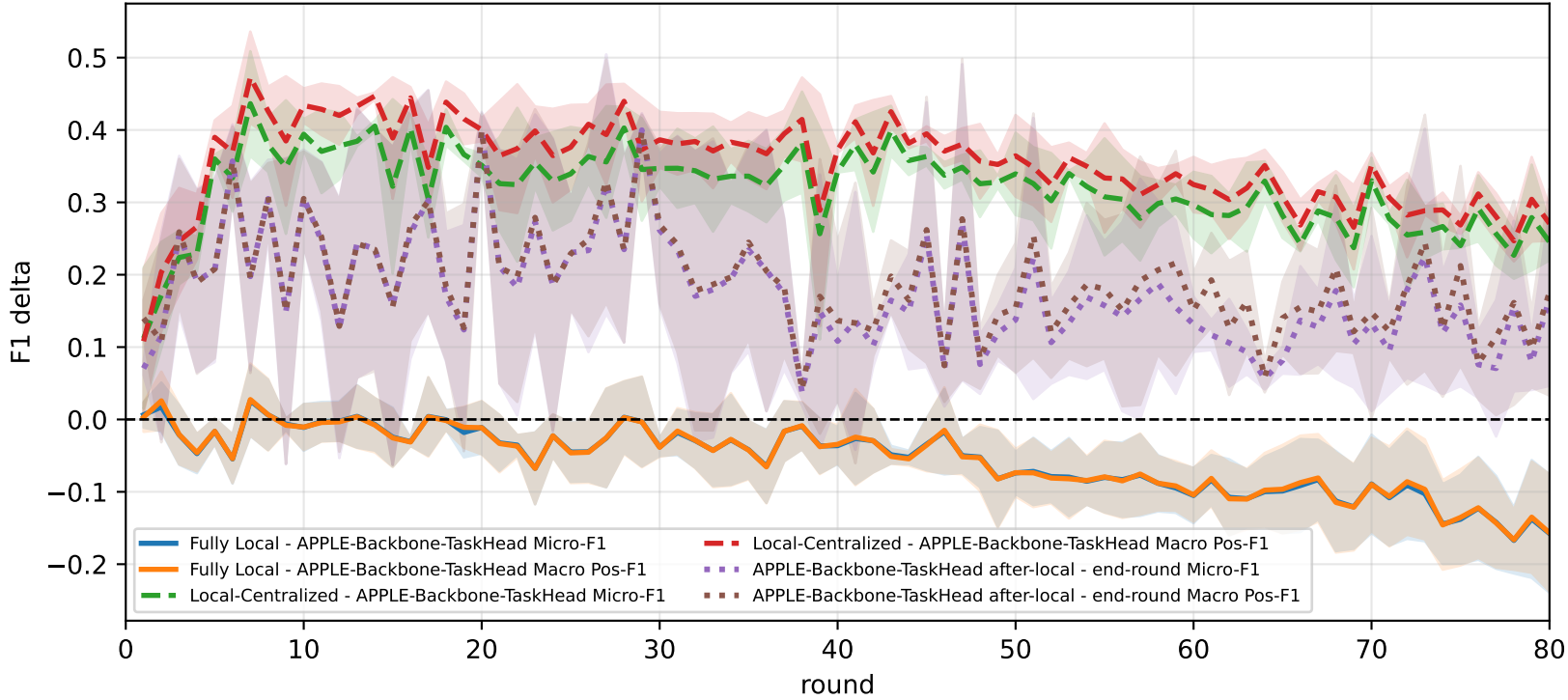
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap





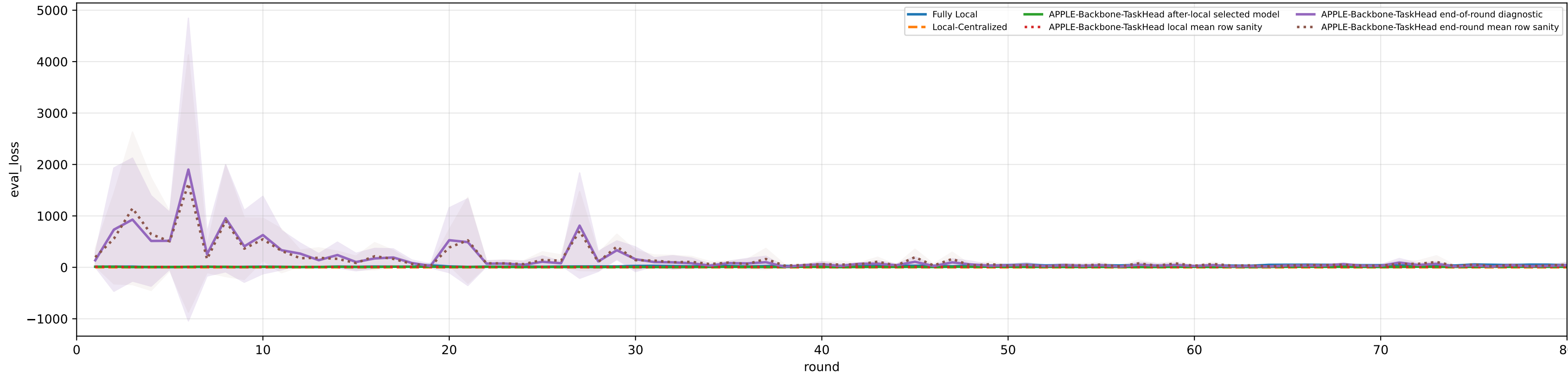
Controlled q-label heterogeneity: High | APPLE-Backbone-TaskHead diagnostics |  
family=q task\_cycle2 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |  
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

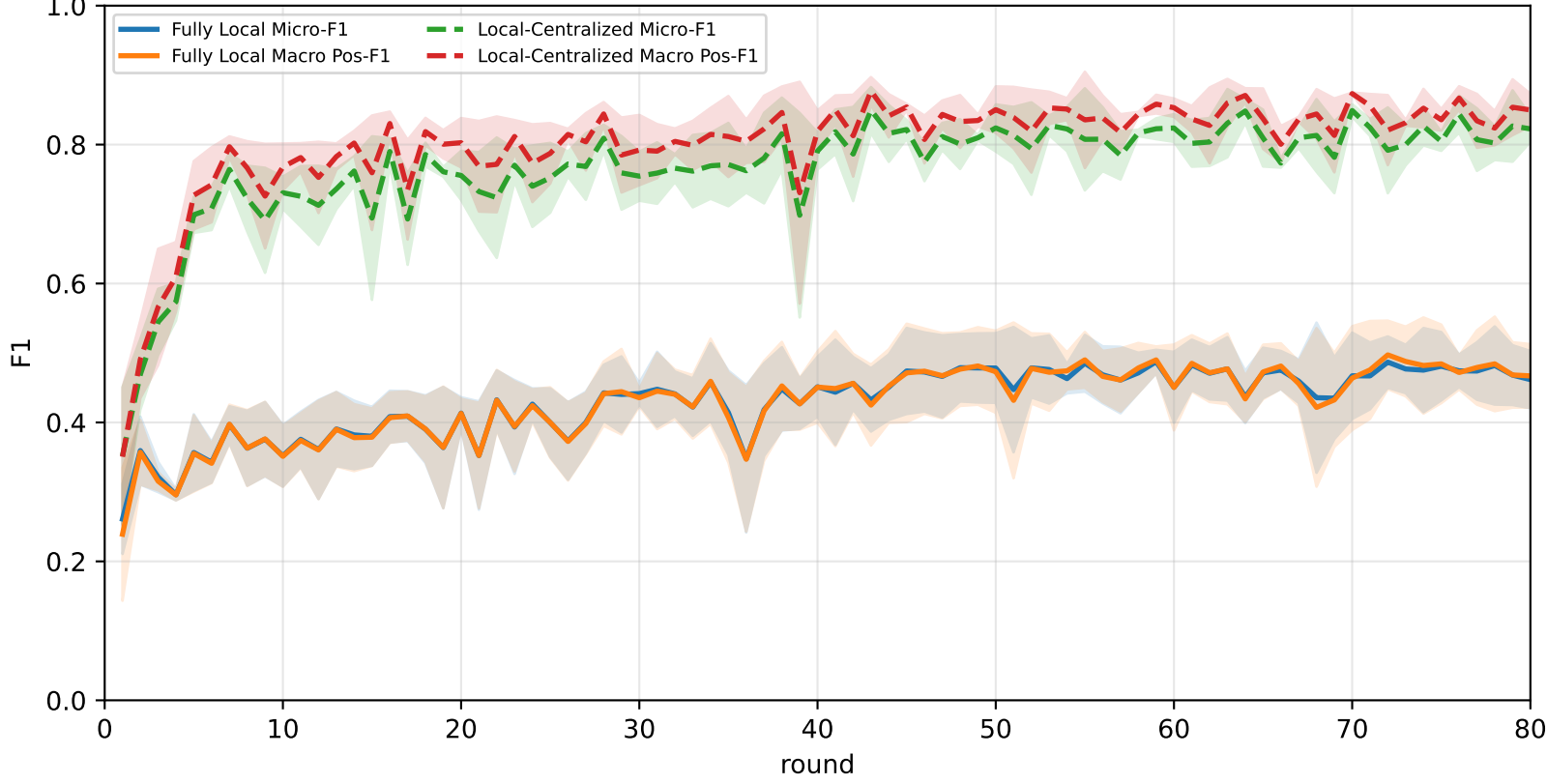
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 28.6% ± 0.7%        | 28.6% ± 0.7%        | 23.4% ± 1.7%        | 29.1% ± 1.2%        | 30.2% ± 0.8%        | 28.5% ± 1.4%        | 32.0% ± 1.3%        |
| Local-Centralized       | 84.9% ± 1.3%        | 86.7% ± 0.8%        | 96.8% ± 1.6%        | 95.1% ± 1.4%        | 86.6% ± 2.5%        | 81.4% ± 1.3%        | 73.7% ± 5.8%        |
| APPLE-Backbone-TaskHead | <u>36.6% ± 7.3%</u> | <u>36.6% ± 7.3%</u> | <u>31.2% ± 8.1%</u> | <u>37.9% ± 8.5%</u> | <u>36.7% ± 6.4%</u> | <u>36.4% ± 6.6%</u> | <u>40.8% ± 6.9%</u> |

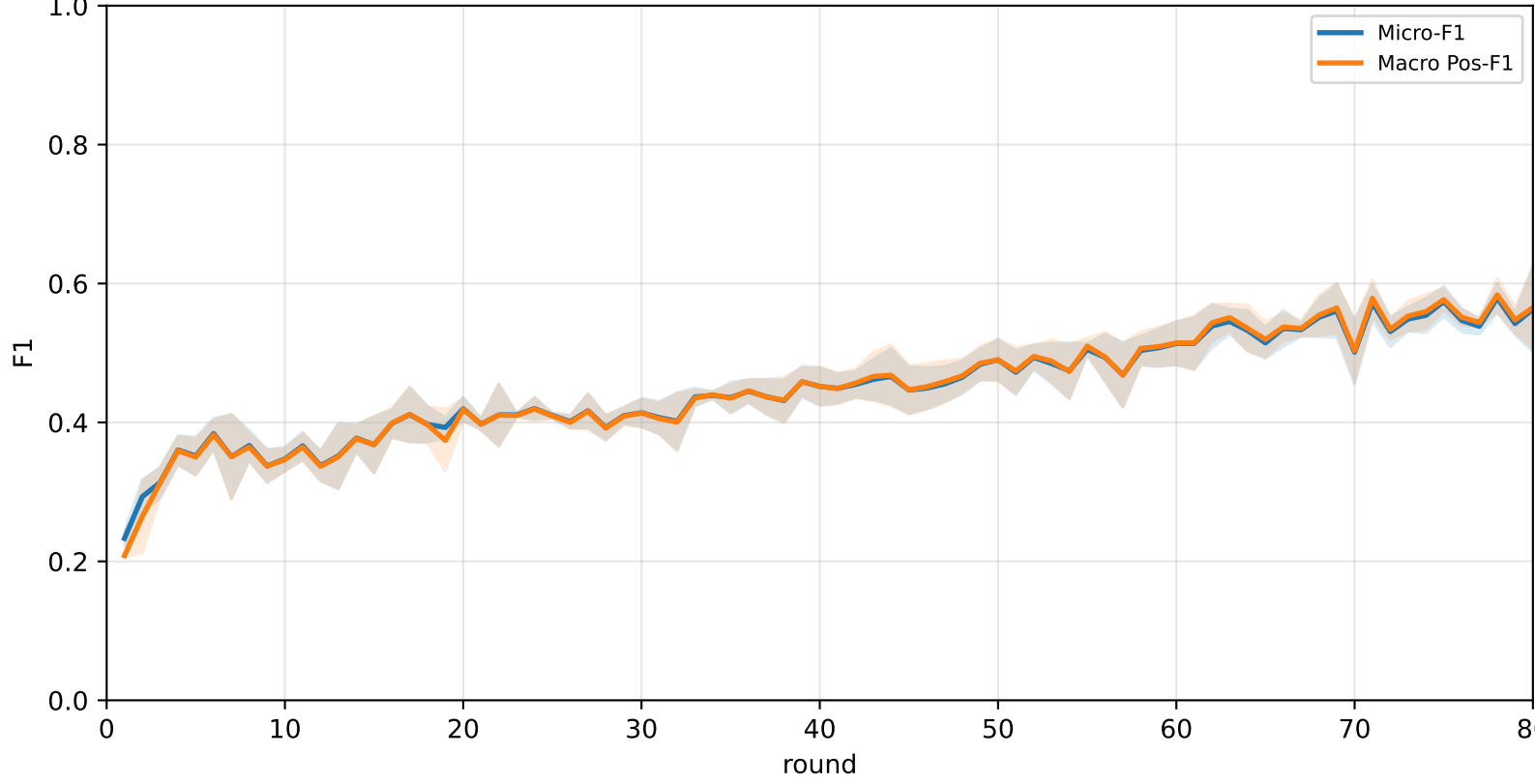
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



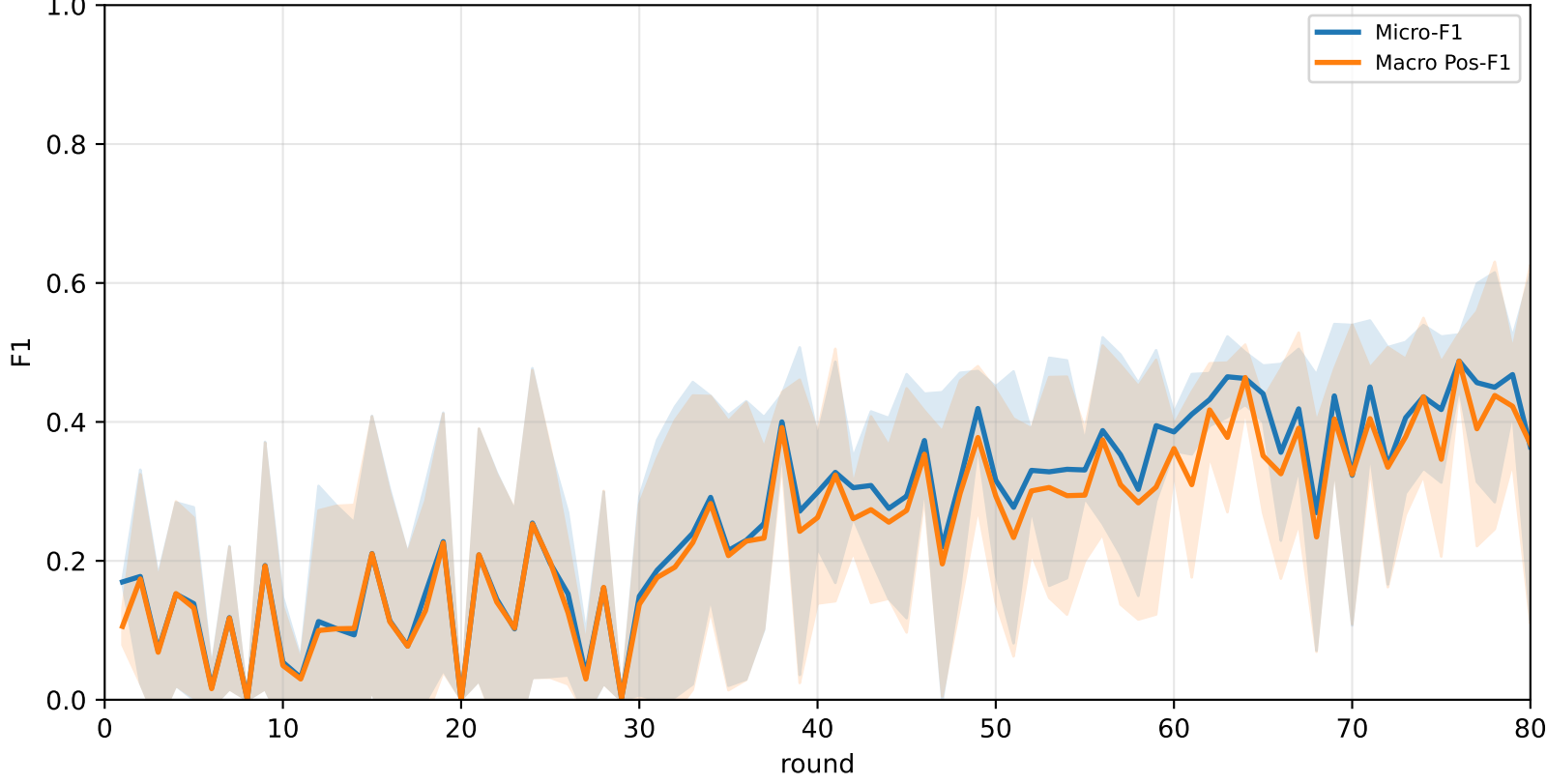
Pooled baseline micro / macro F1



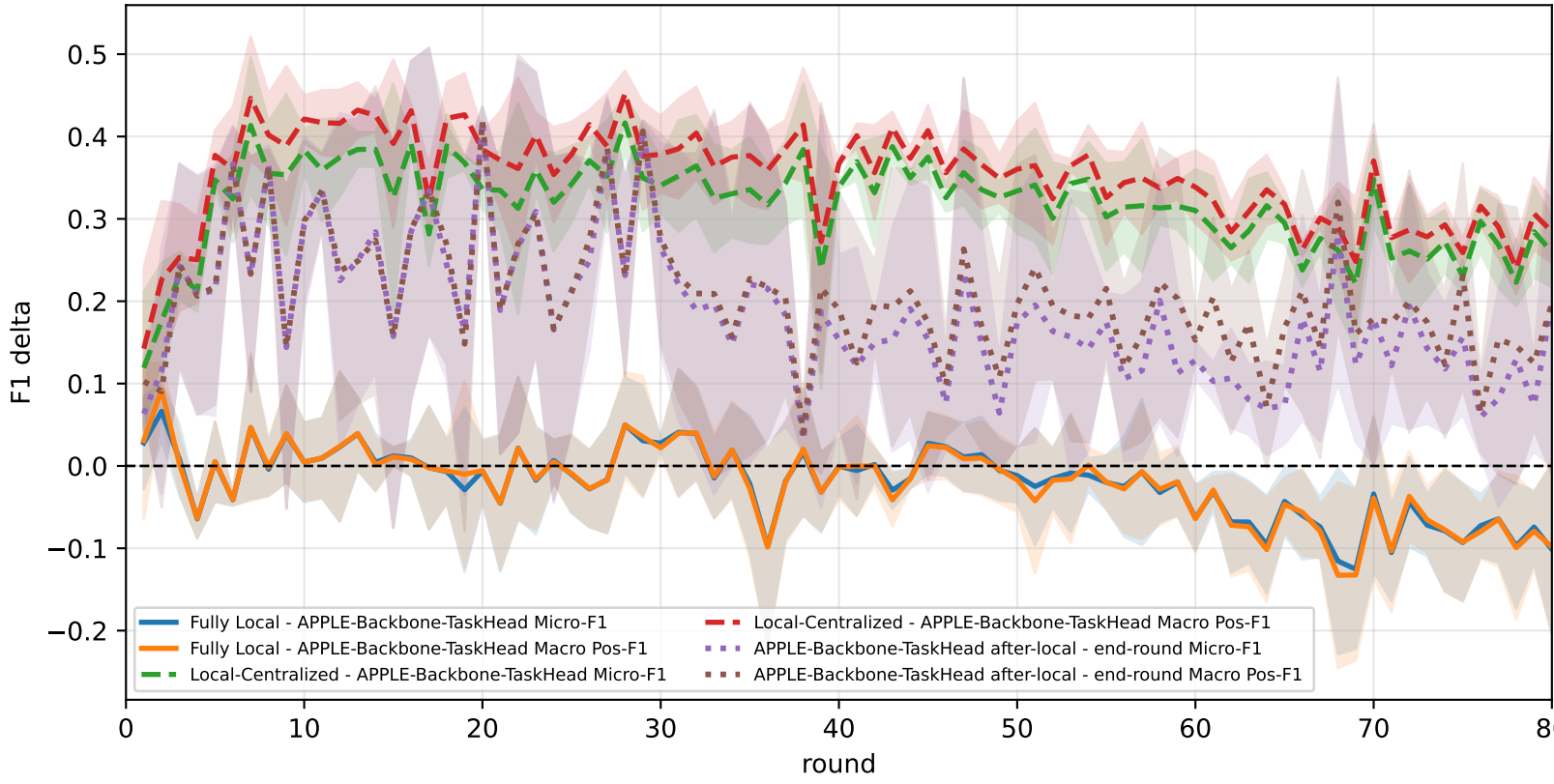
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap





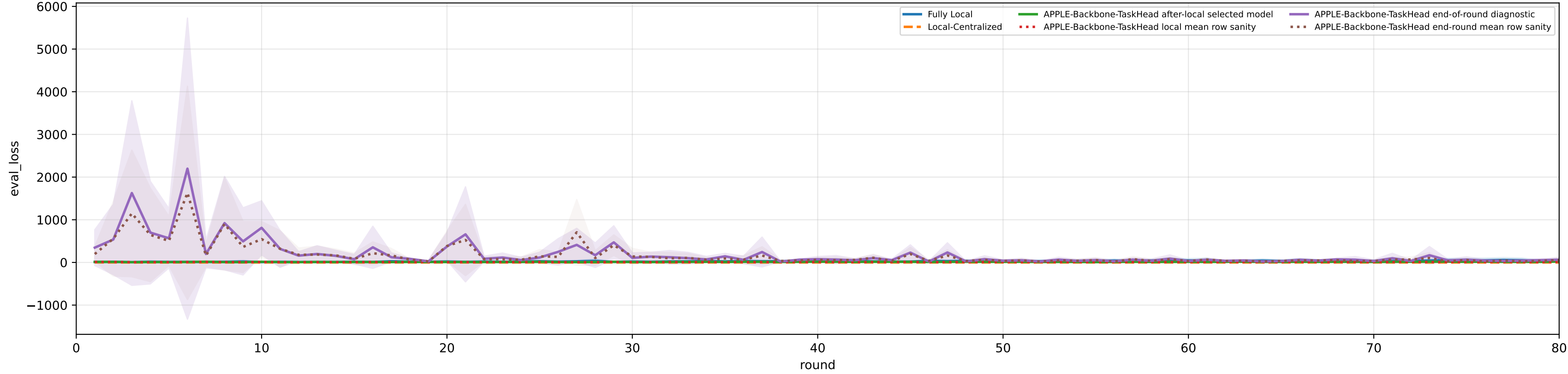
Controlled q-label heterogeneity: High | APPLE-Backbone-TaskHead diagnostics |  
family=q task\_cycle3 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |  
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

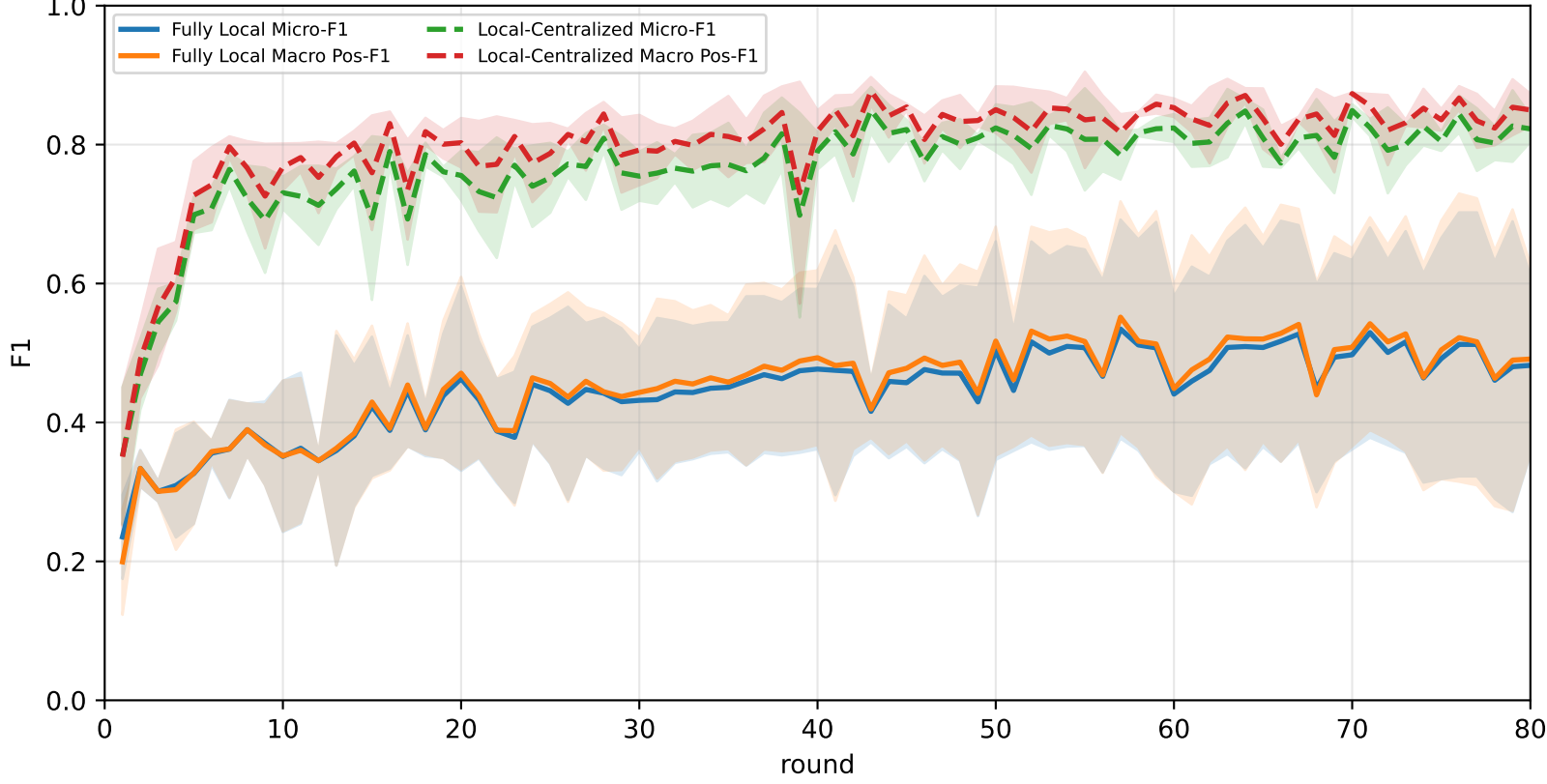
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 26.5% ± 4.7%        | 26.6% ± 4.7%        | 22.3% ± 4.2%        | 27.0% ± 7.0%        | 26.9% ± 5.3%        | 27.0% ± 4.1%        | 29.5% ± 3.2%        |
| Local-Centralized       | 84.9% ± 1.3%        | 86.7% ± 0.8%        | 96.8% ± 1.6%        | 95.1% ± 1.4%        | 86.6% ± 2.5%        | 81.4% ± 1.3%        | 73.7% ± 5.8%        |
| APPLE-Backbone-TaskHead | <u>36.0% ± 7.0%</u> | <u>36.0% ± 6.9%</u> | <u>30.4% ± 7.7%</u> | <u>36.1% ± 7.7%</u> | <u>36.8% ± 8.1%</u> | <u>36.4% ± 6.6%</u> | <u>40.1% ± 4.8%</u> |

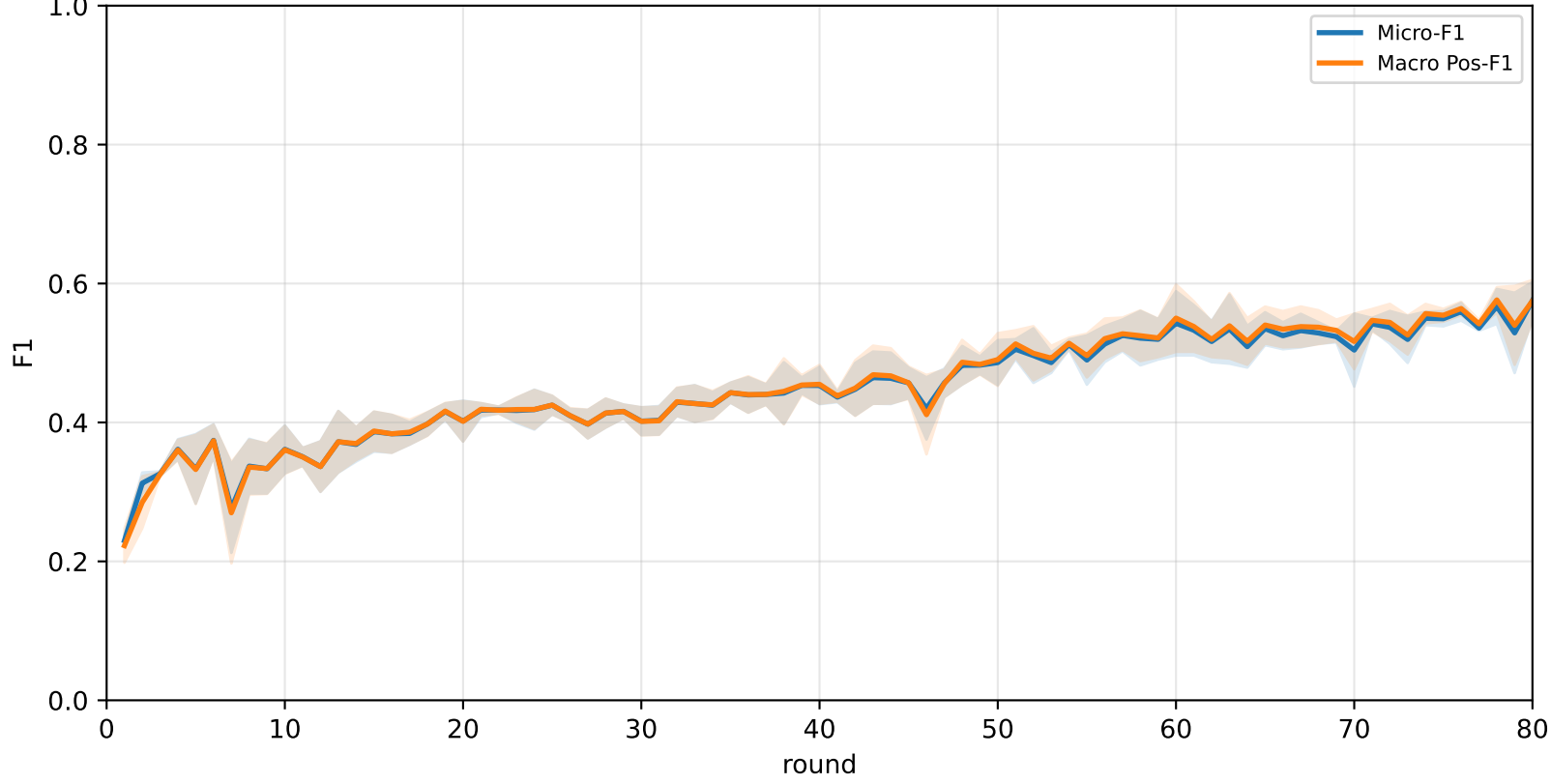
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



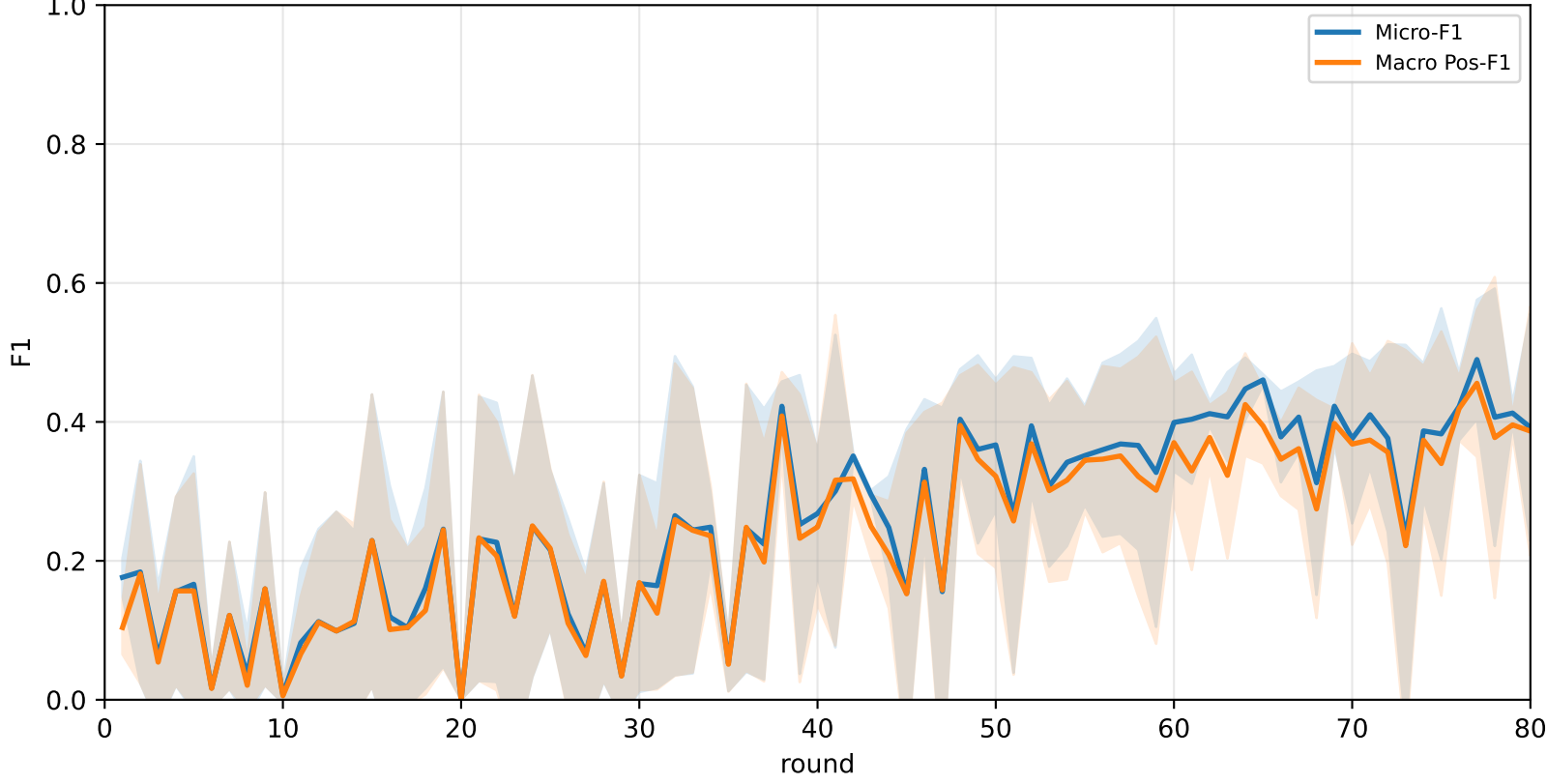
Pooled baseline micro / macro F1



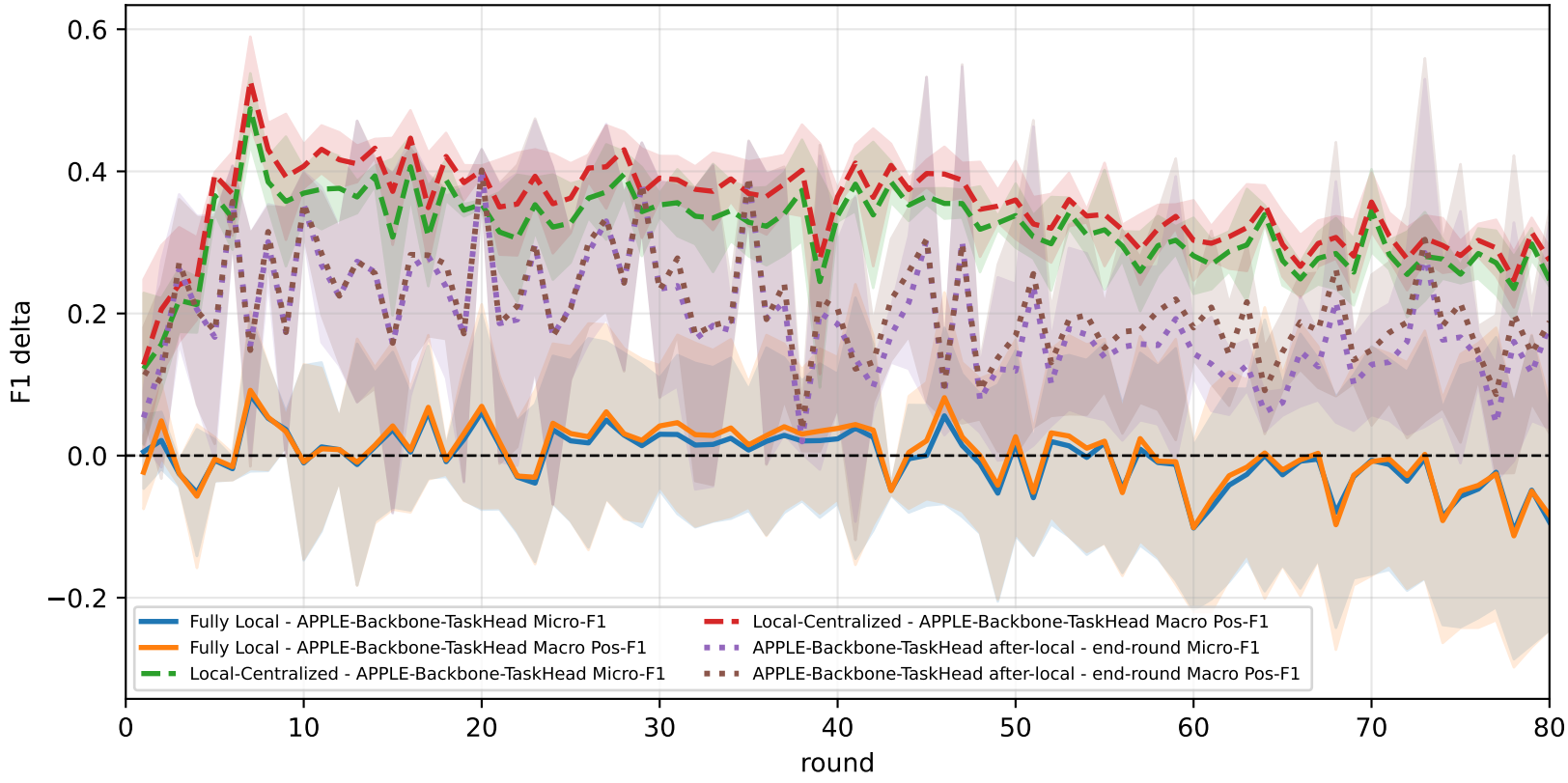
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap



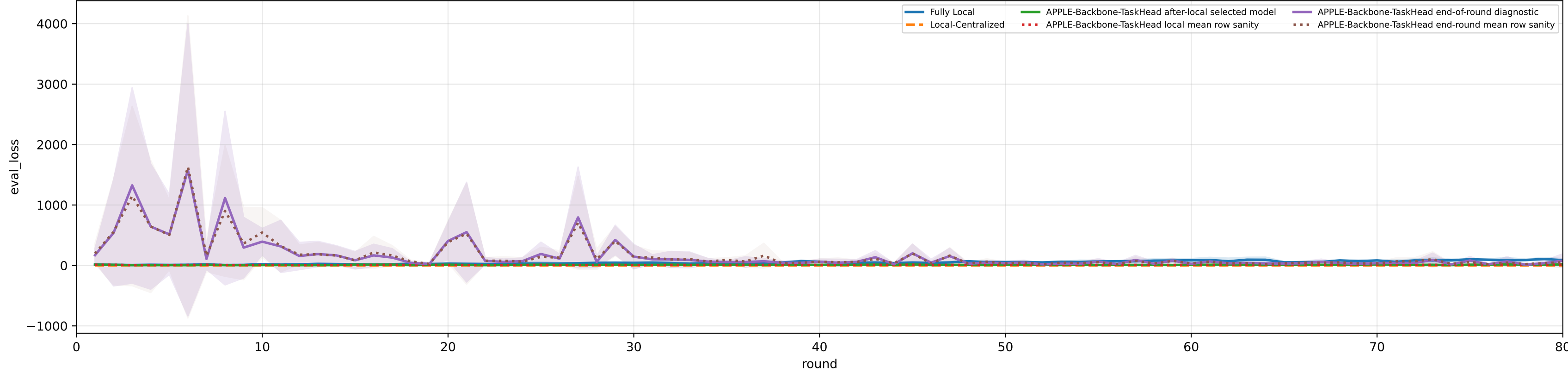
Controlled q-label heterogeneity: High | APPLE-Backbone-TaskHead diagnostics |  
family=q task\_cycle4 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |  
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

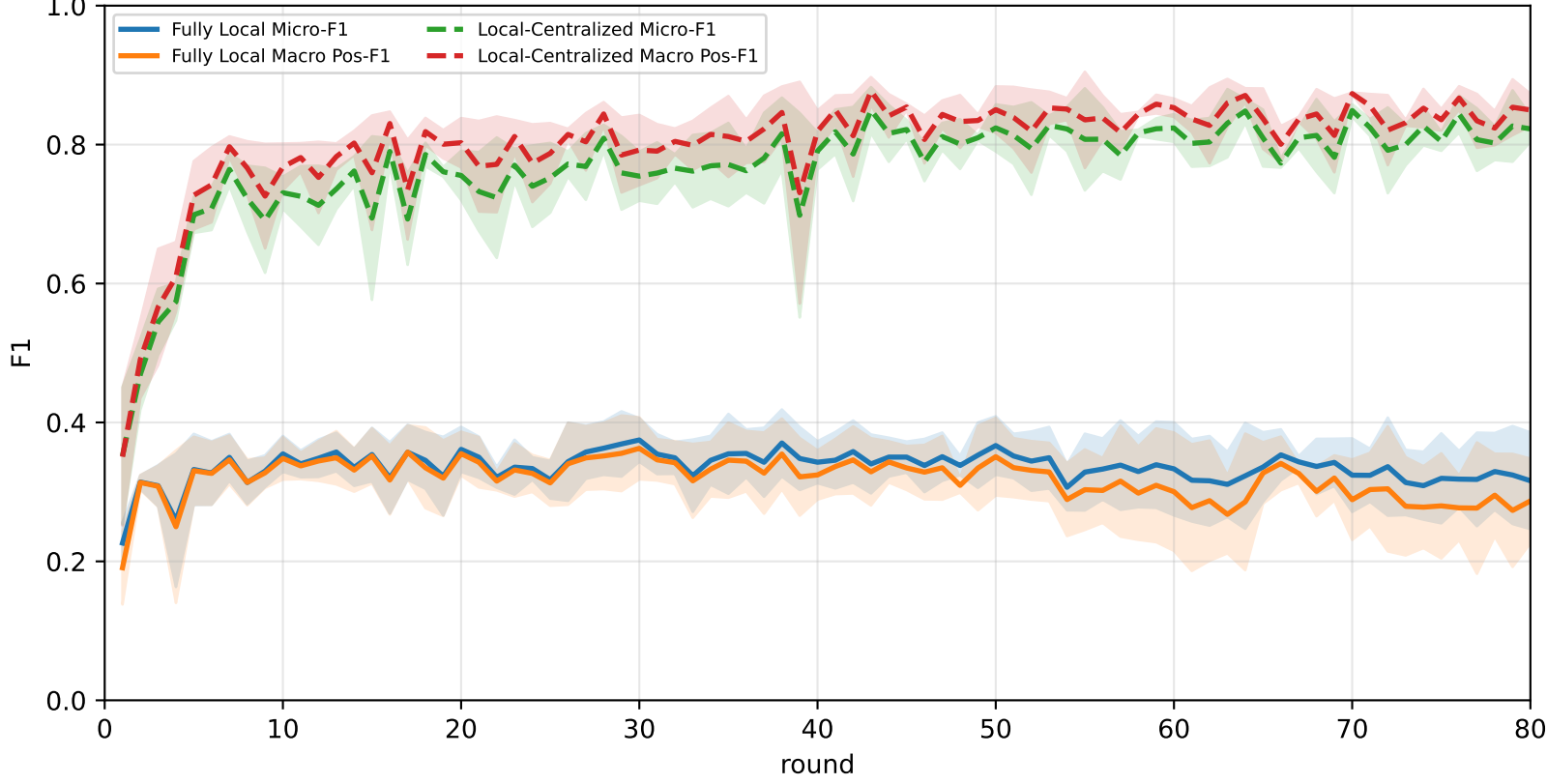
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 27.5% ± 1.5%        | 27.5% ± 1.4%        | 23.1% ± 1.8%        | 28.2% ± 0.7%        | 26.4% ± 1.8%        | 28.3% ± 1.7%        | 31.5% ± 2.1%        |
| Local-Centralized       | 84.9% ± 1.3%        | 86.7% ± 0.8%        | 96.8% ± 1.6%        | 95.1% ± 1.4%        | 86.6% ± 2.5%        | 81.4% ± 1.3%        | 73.7% ± 5.8%        |
| APPLE-Backbone-TaskHead | <u>35.9% ± 6.0%</u> | <u>35.8% ± 6.0%</u> | <u>31.1% ± 6.2%</u> | <u>36.8% ± 7.2%</u> | <u>35.7% ± 5.7%</u> | <u>36.4% ± 5.8%</u> | <u>39.1% ± 5.3%</u> |

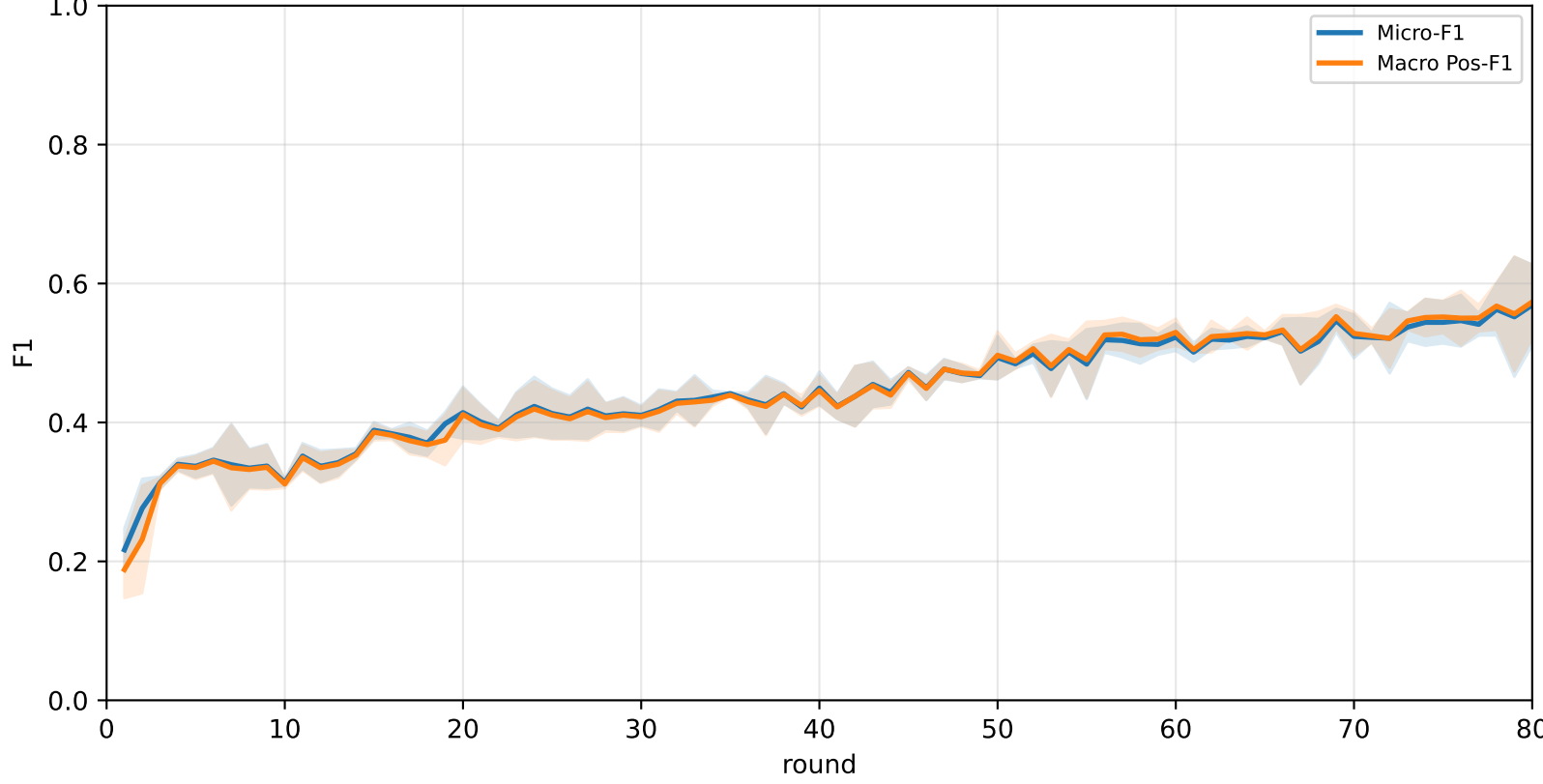
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



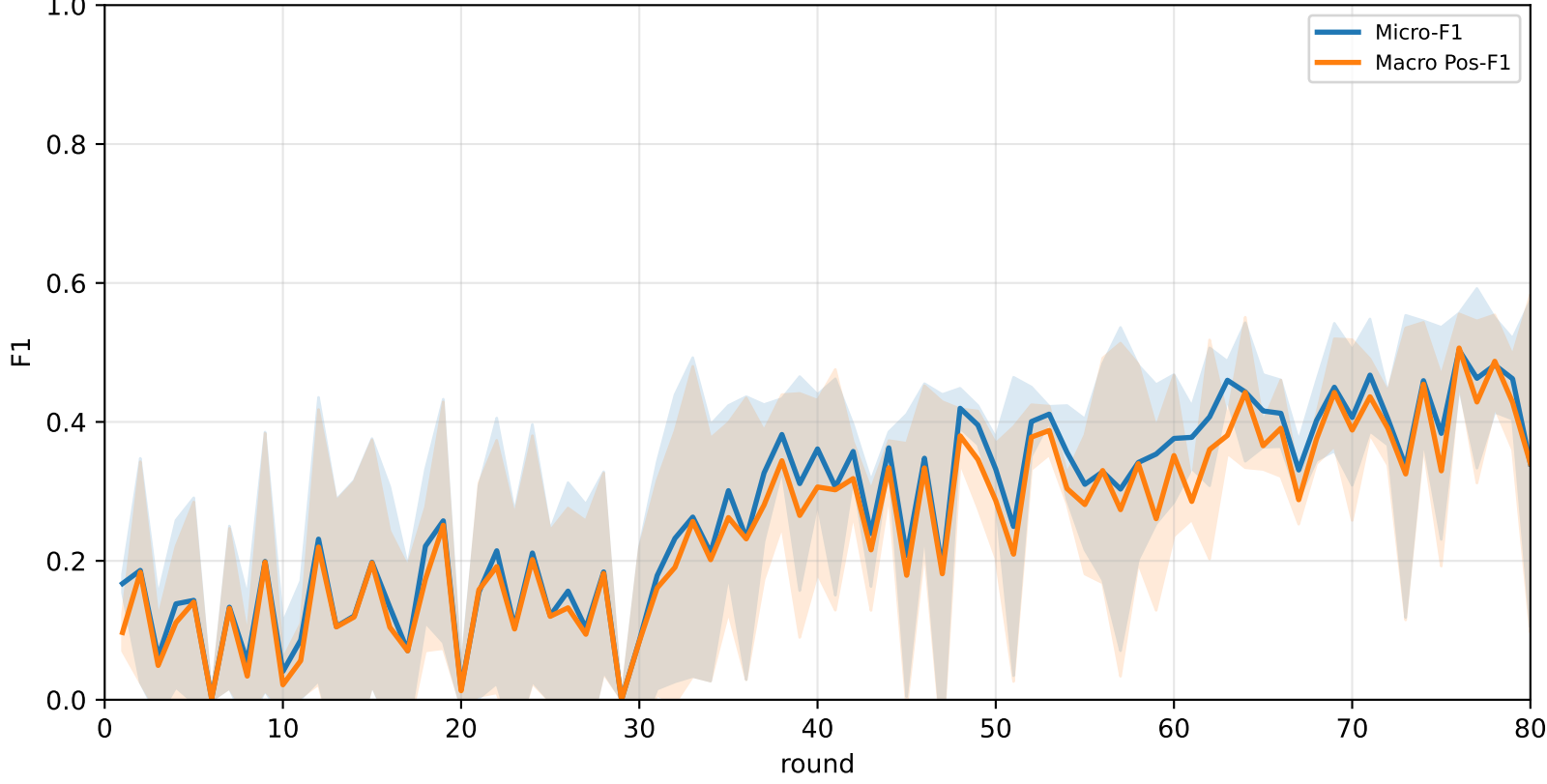
Pooled baseline micro / macro F1



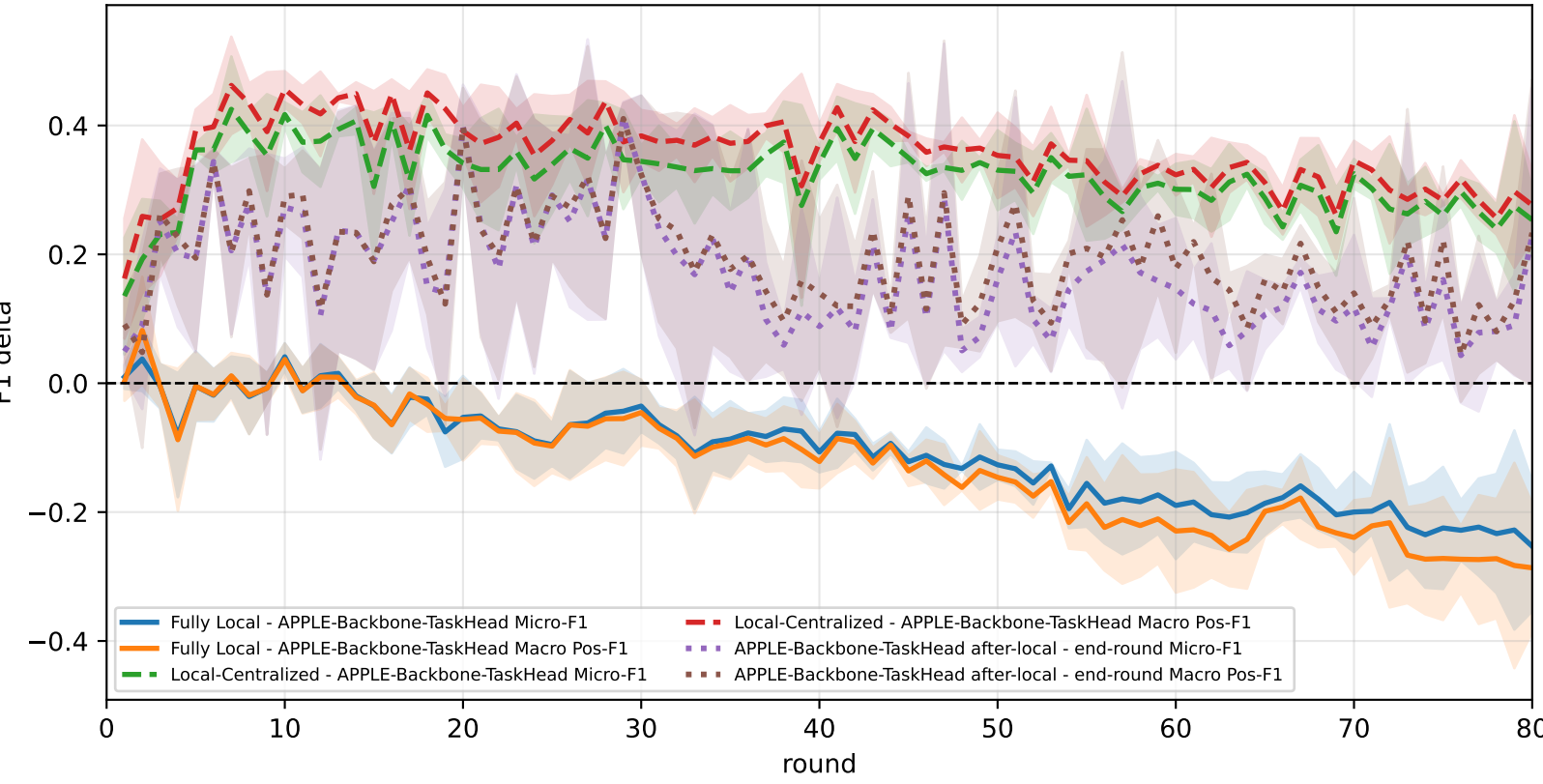
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap





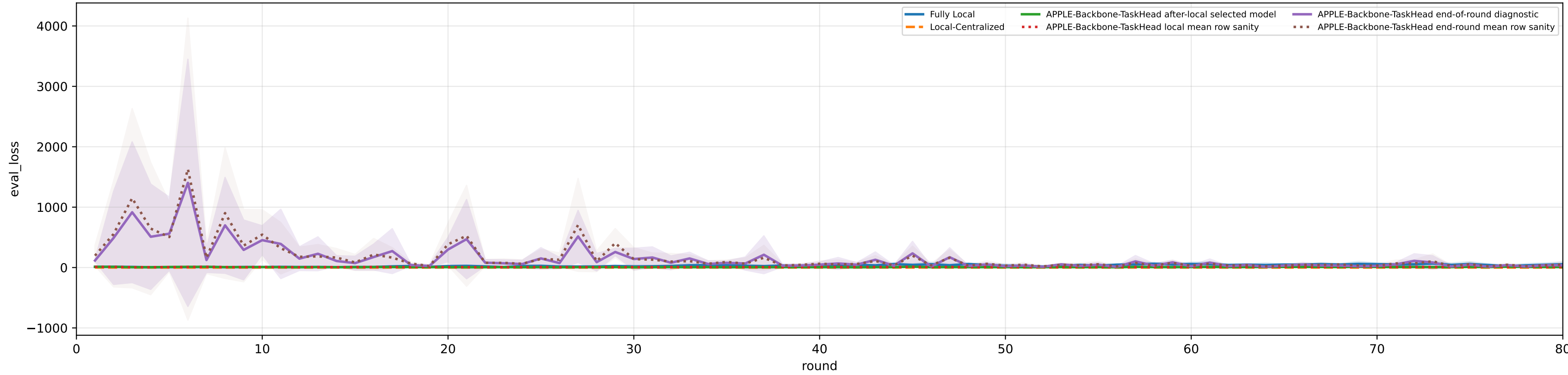
Controlled q-label heterogeneity: High | APPLE-Backbone-TaskHead diagnostics |  
family=q task\_cycle5 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |  
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% |  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

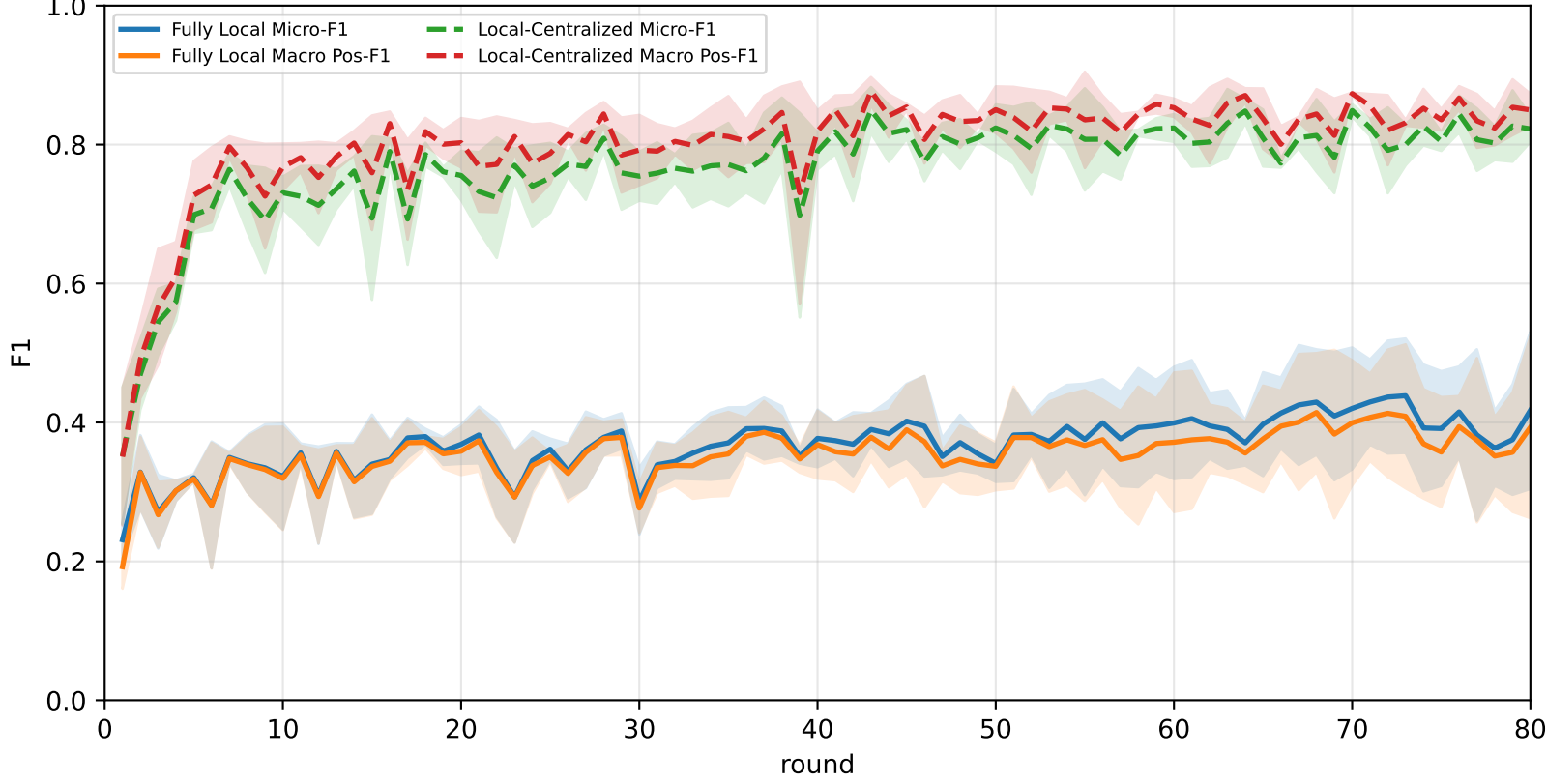
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 24.2% ± 1.1%        | 24.2% ± 1.1%        | 20.1% ± 0.7%        | 25.3% ± 1.1%        | 24.5% ± 1.6%        | 22.9% ± 1.3%        | 28.3% ± 1.1%        |
| Local-Centralized       | 84.9% ± 1.3%        | 86.7% ± 0.8%        | 96.8% ± 1.6%        | 95.1% ± 1.4%        | 86.6% ± 2.5%        | 81.4% ± 1.3%        | 73.7% ± 5.8%        |
| APPLE-Backbone-TaskHead | <u>35.1% ± 5.2%</u> | <u>35.0% ± 5.3%</u> | <u>29.9% ± 6.1%</u> | <u>35.7% ± 6.1%</u> | <u>35.7% ± 5.0%</u> | <u>34.3% ± 3.8%</u> | <u>39.4% ± 5.3%</u> |

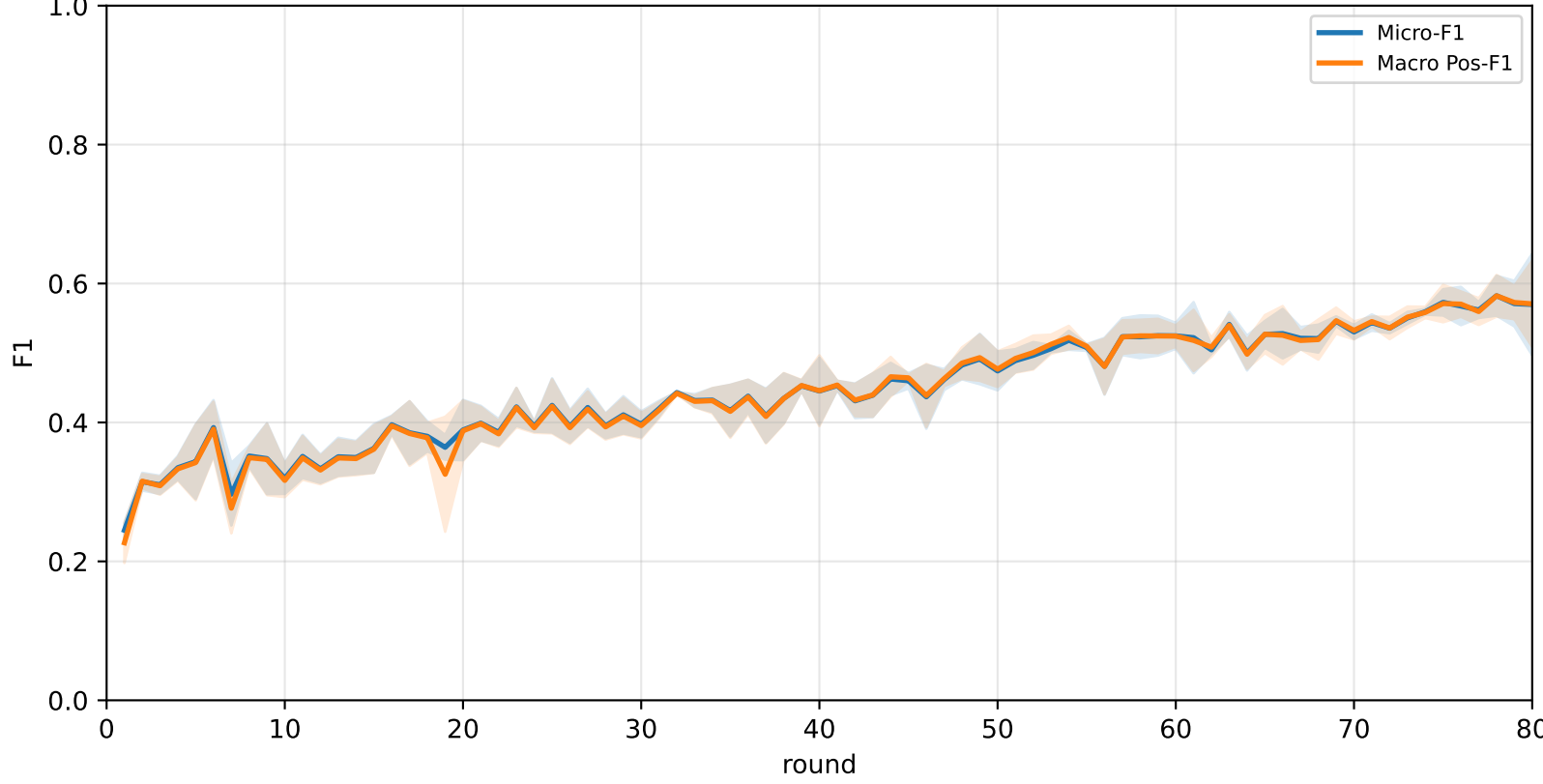
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



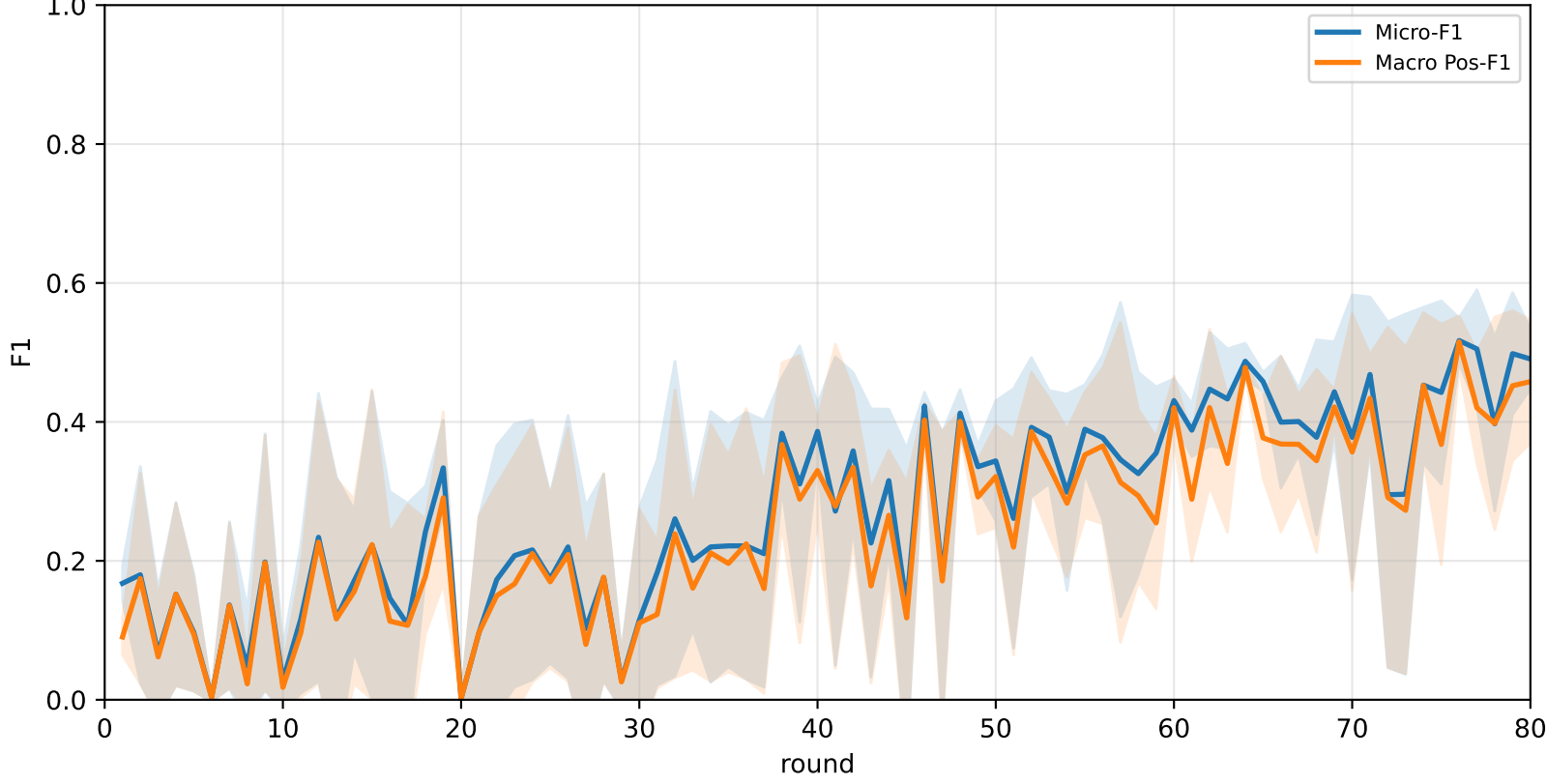
Pooled baseline micro / macro F1



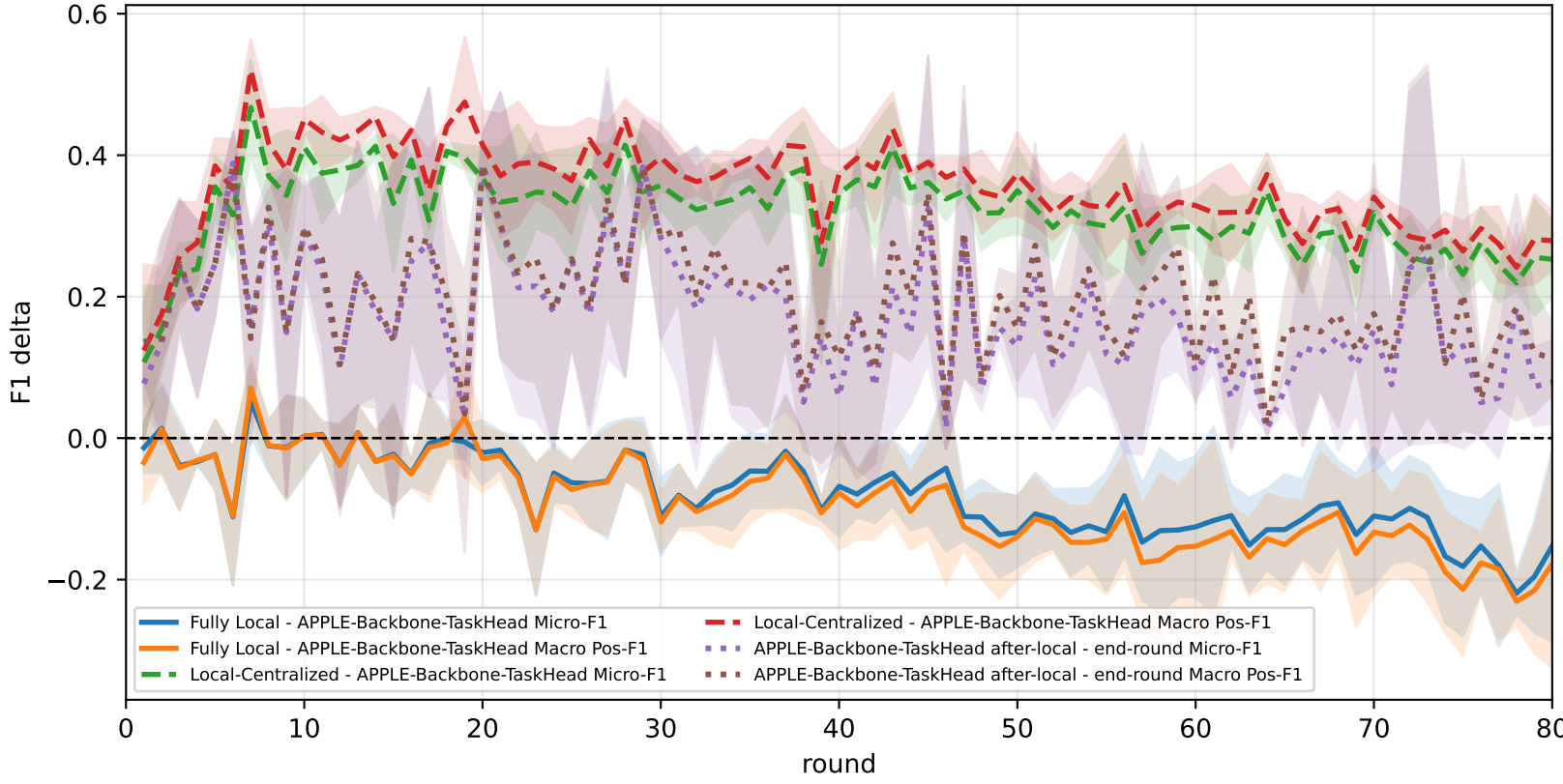
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap



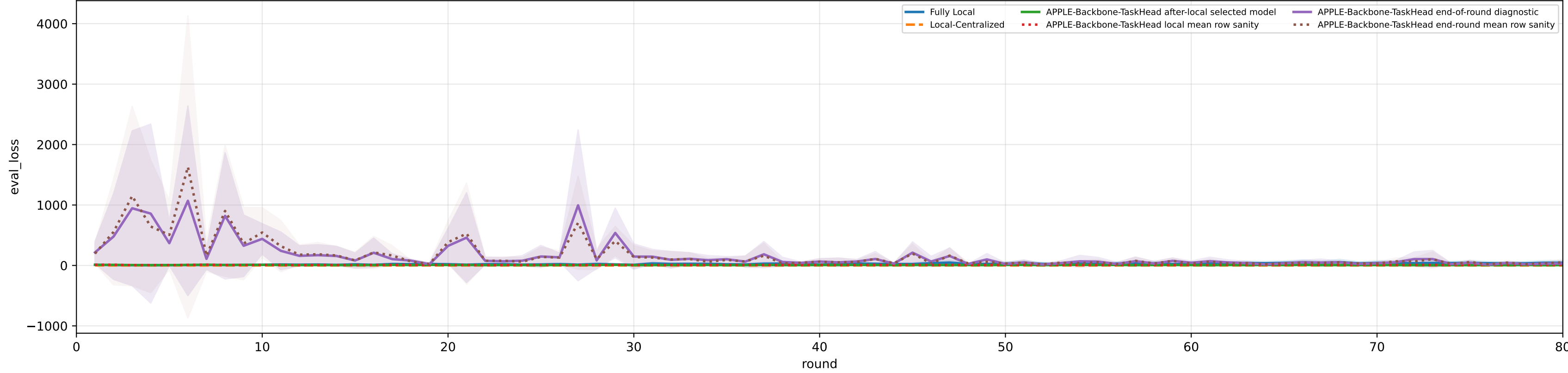
Controlled q-label heterogeneity: High | APPLE-Backbone-TaskHead diagnostics |  
family=q task cycle6 | n\_clients=1 |  
model=mcwauto\_layers6\_lr0.001\_wd0.0001\_do0.1\_hd64\_egoTrue\_bs64\_headmulti

family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |  
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

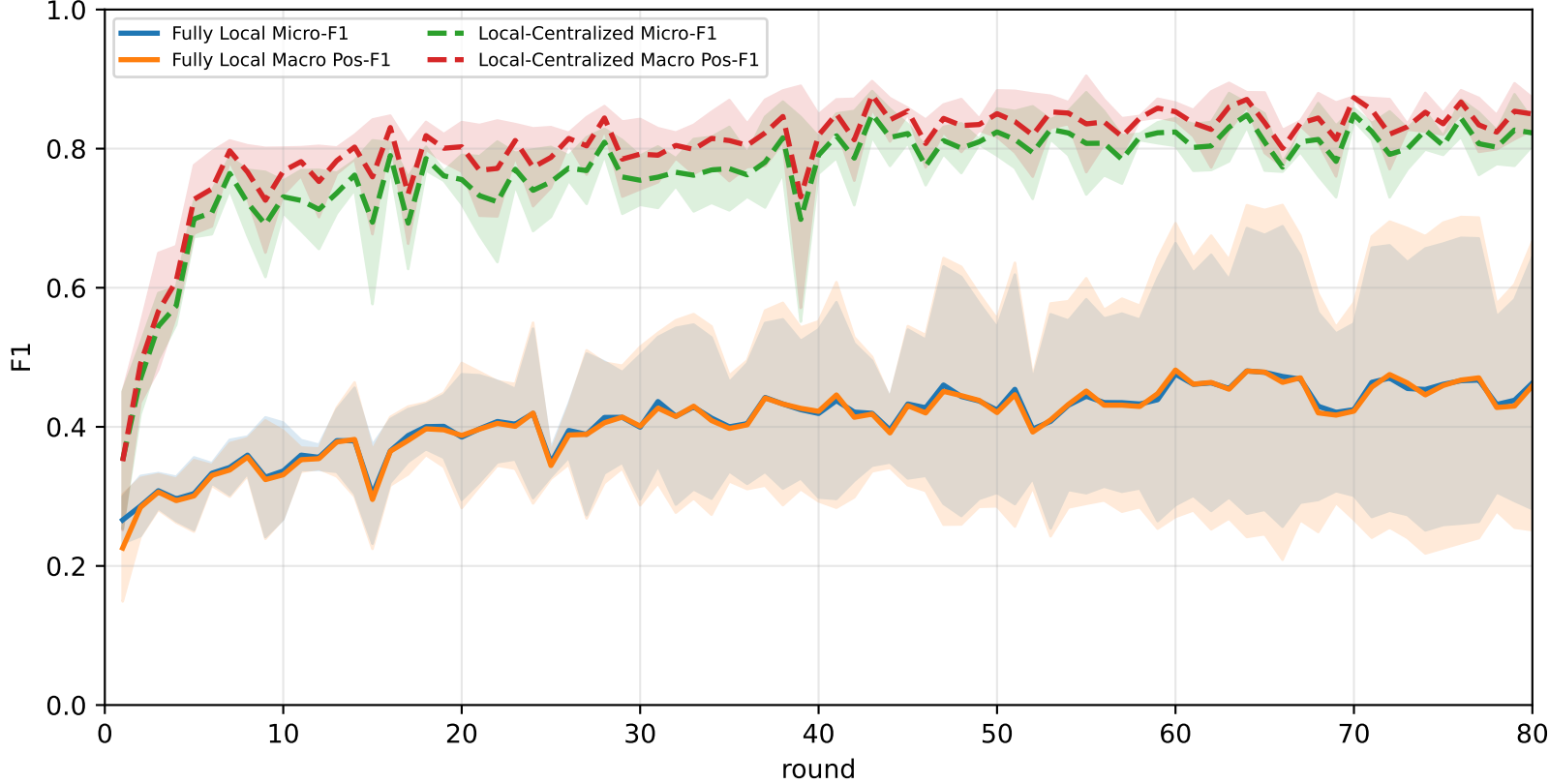
Best test performance (Fully Local vs Local-Centralized vs APPLE-Backbone-TaskHead, rounds=80)

| Algorithm               | Micro-F1            | Macro Pos-F1        | cycle2 pos-F1       | cycle3 pos-F1       | cycle4 pos-F1       | cycle5 pos-F1       | cycle6 pos-F1       |
|-------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Fully Local             | 31.2% ± 8.8%        | 31.1% ± 8.9%        | 26.7% ± 8.1%        | 32.1% ± 9.5%        | 33.6% ± 11.2%       | 31.3% ± 9.2%        | 32.1% ± 6.8%        |
| Local-Centralized       | 84.9% ± 1.3%        | 86.7% ± 0.8%        | 96.8% ± 1.6%        | 95.1% ± 1.4%        | 86.6% ± 2.5%        | 81.4% ± 1.3%        | 73.7% ± 5.8%        |
| APPLE-Backbone-TaskHead | <u>36.6% ± 7.0%</u> | <u>36.5% ± 7.0%</u> | <u>31.3% ± 8.8%</u> | <u>37.2% ± 7.0%</u> | <u>37.7% ± 7.3%</u> | <u>36.6% ± 5.8%</u> | <u>39.5% ± 5.8%</u> |

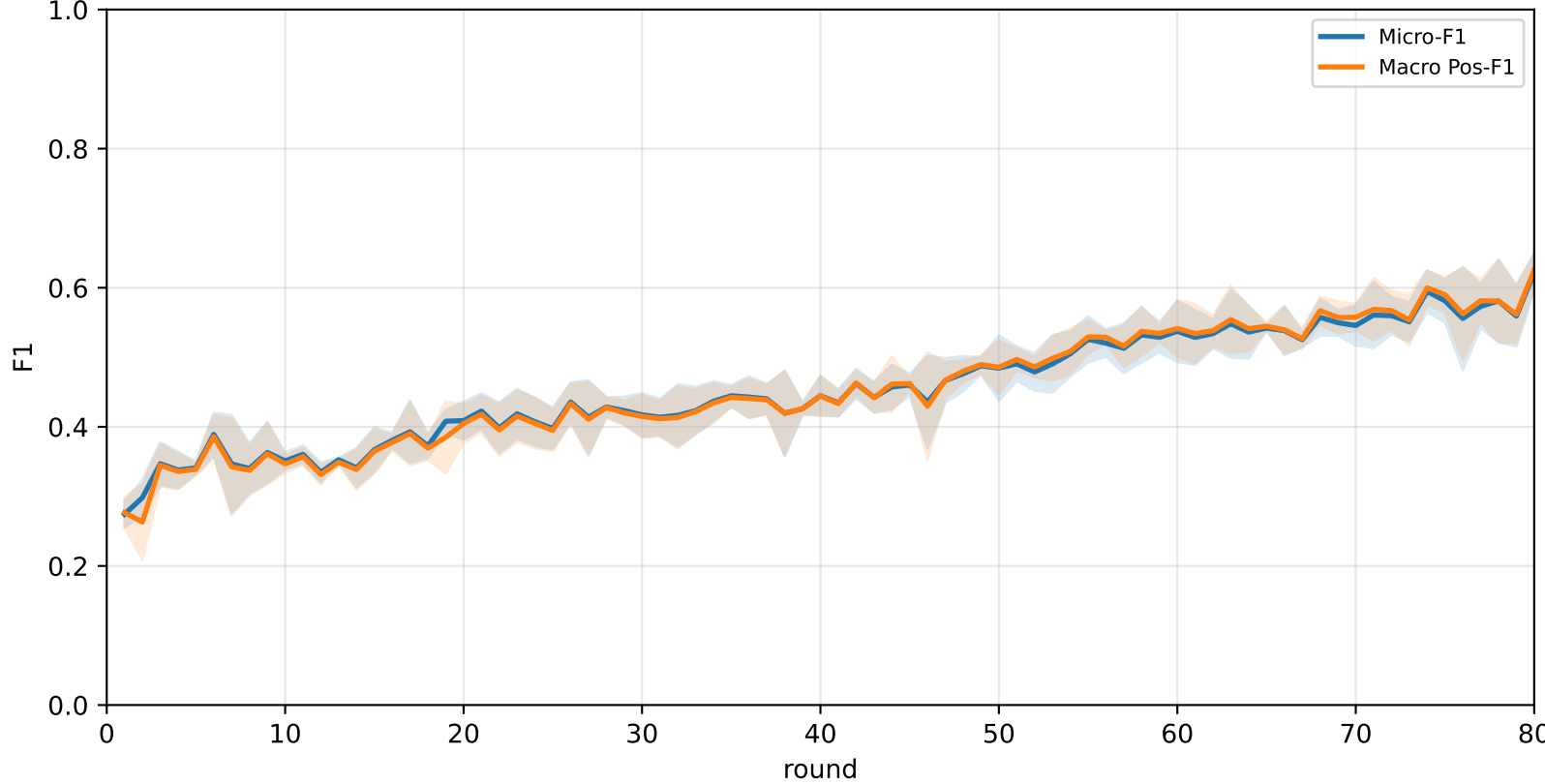
Validation loss: baselines vs selected APPLE-Backbone-TaskHead and end-of-round mismatch (rounds=80)



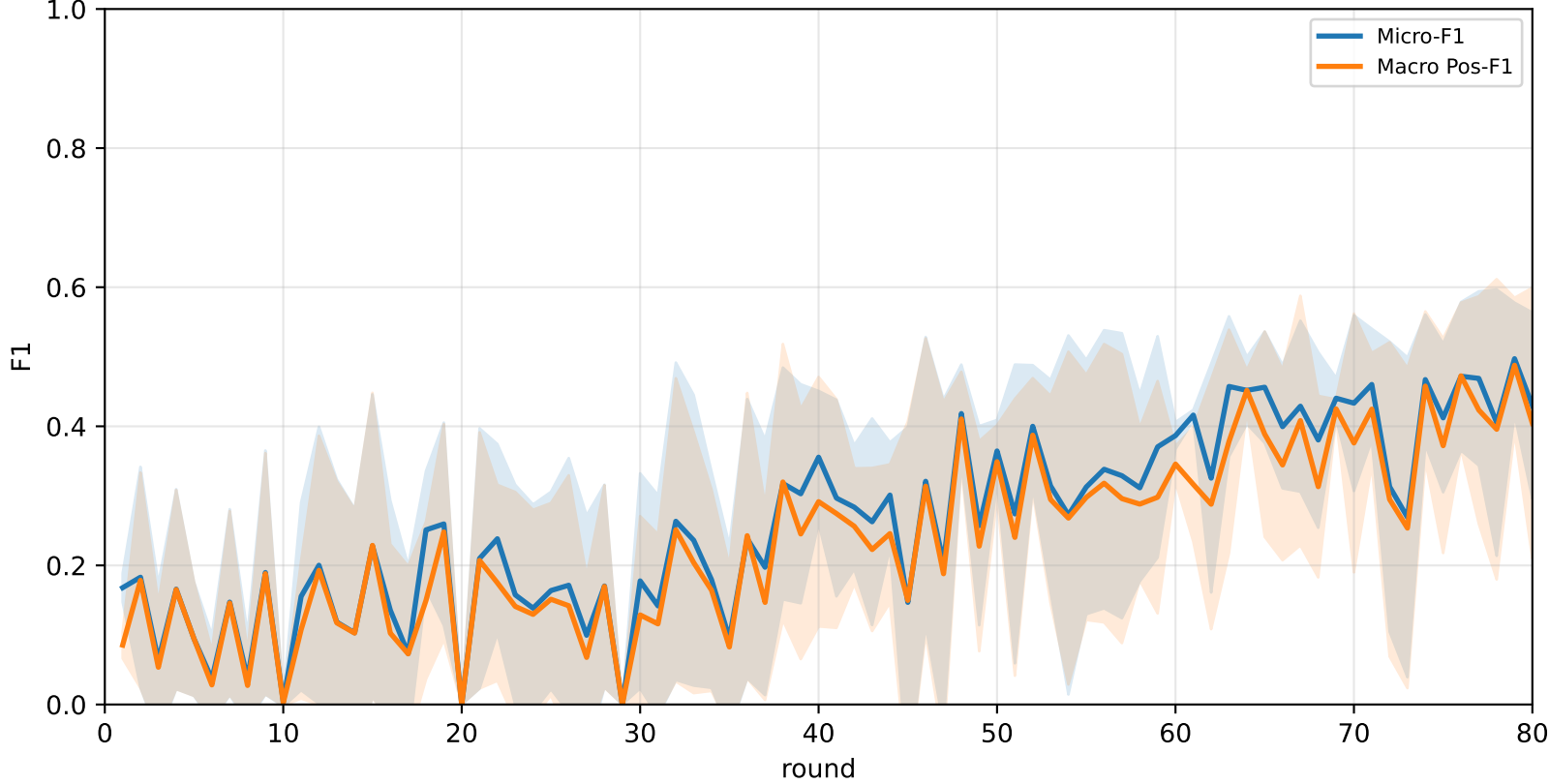
Pooled baseline micro / macro F1



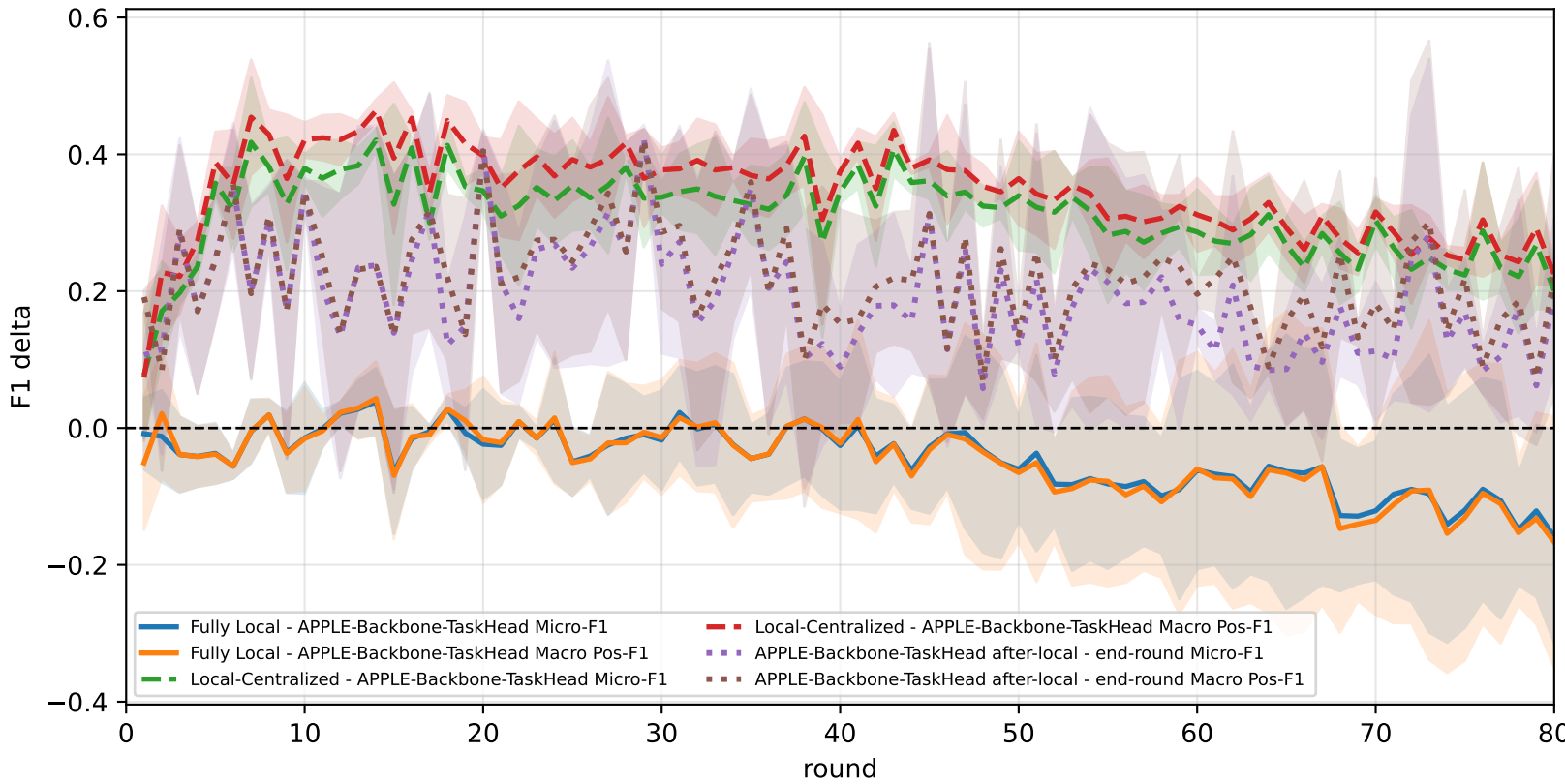
Pooled APPLE-Backbone-TaskHead micro / macro F1 (after-local selected model)



Pooled APPLE-Backbone-TaskHead micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-Backbone-TaskHead mismatch gap

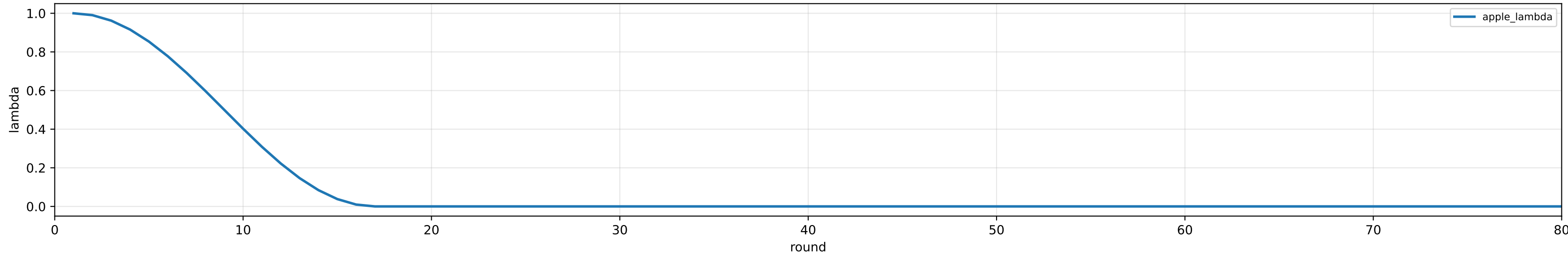




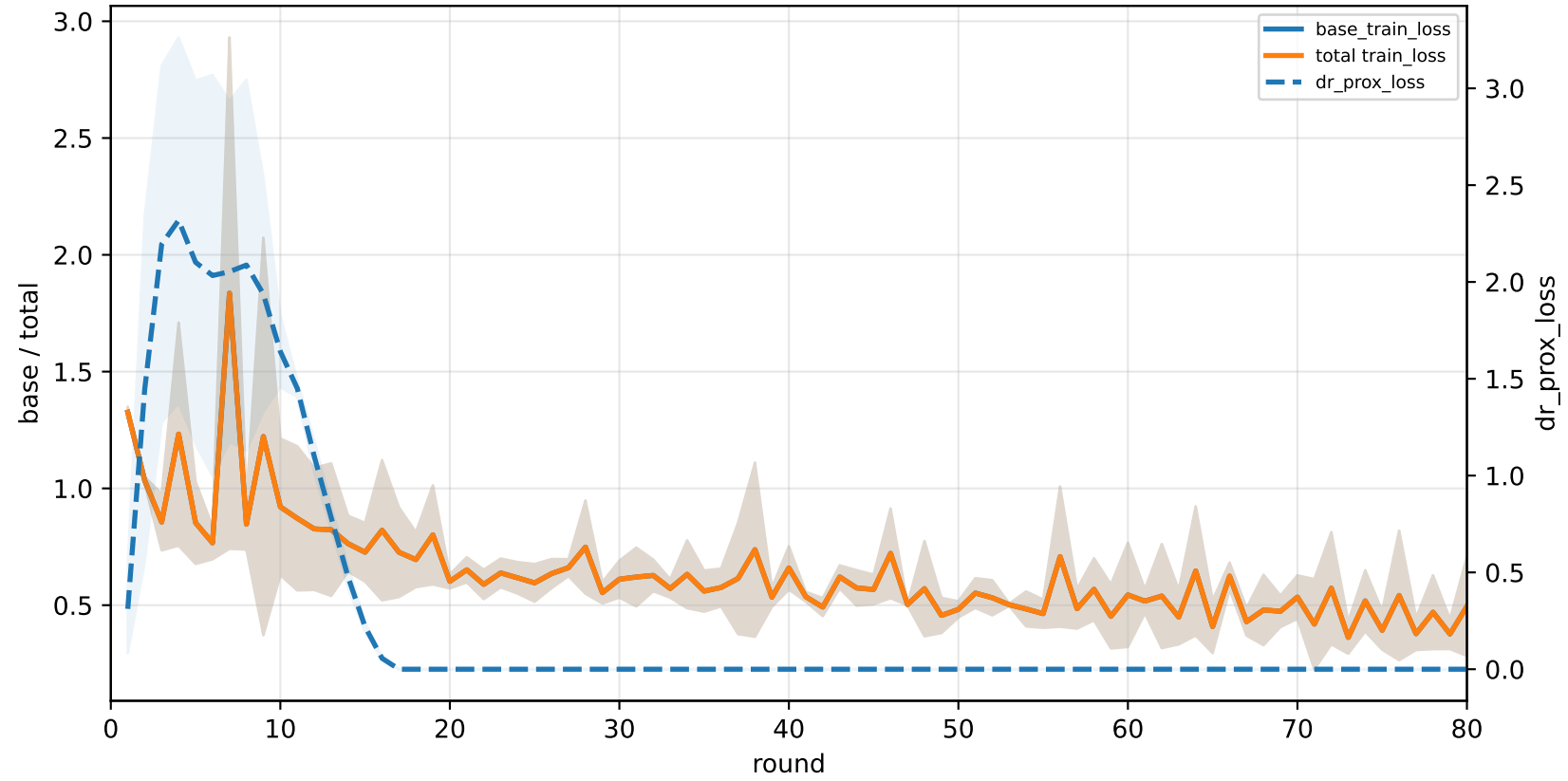
APPLE-Backbone-TaskHead training-loss decomposition | High heterogeneity

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

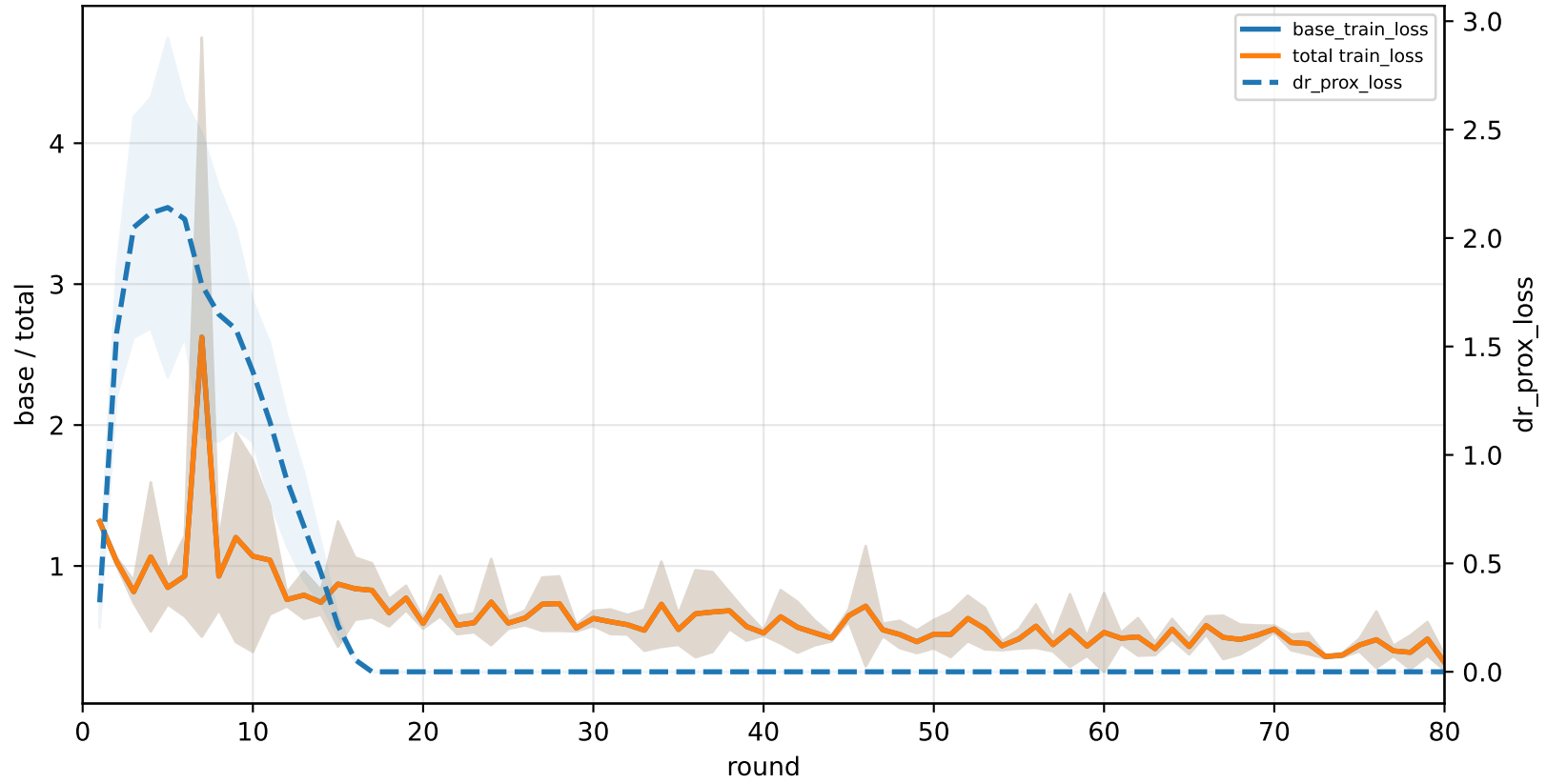
APPLE lambda scheduler | apple\_mu=0.001



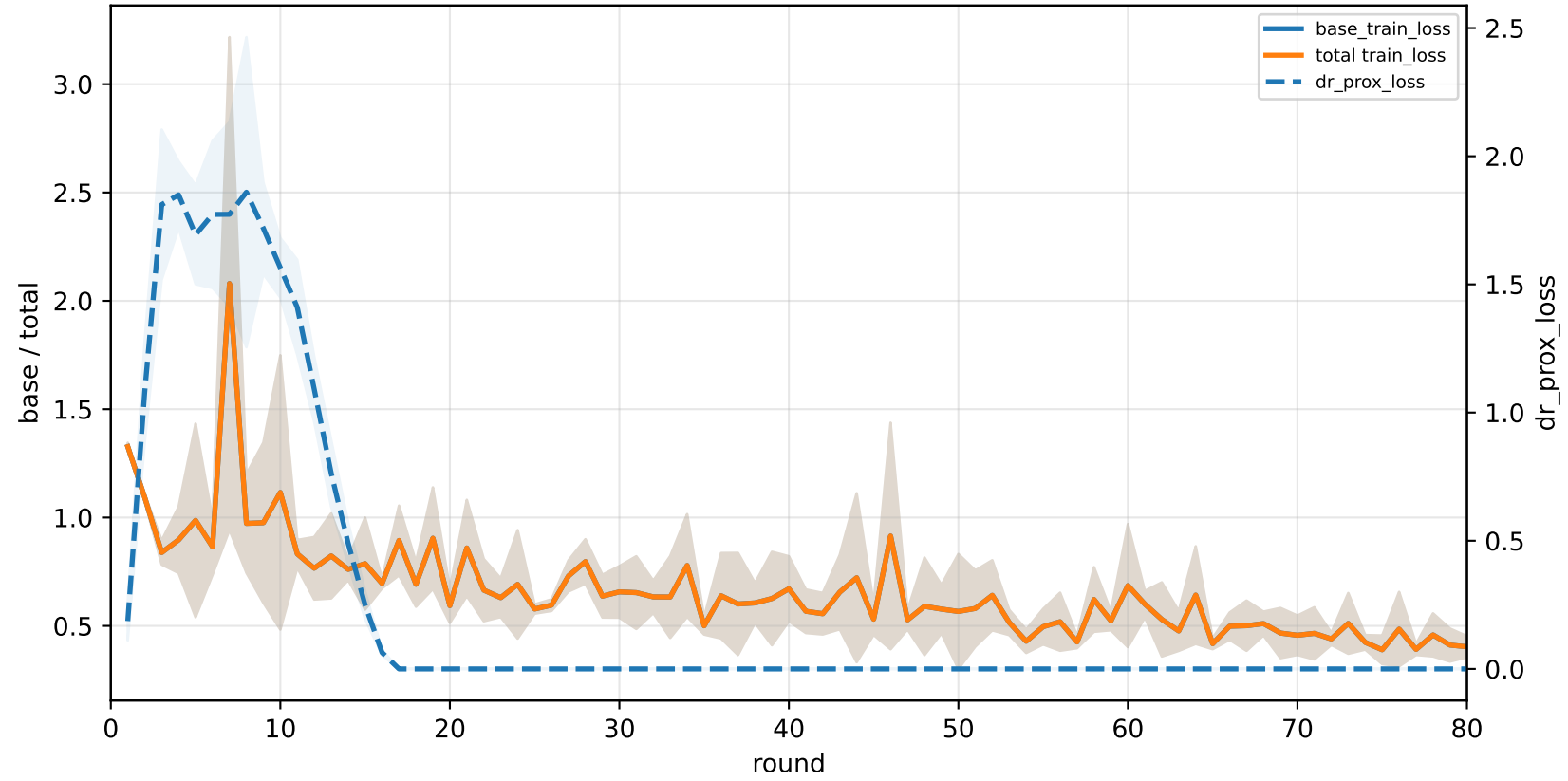
C2-specialist | graph\_id=3000010



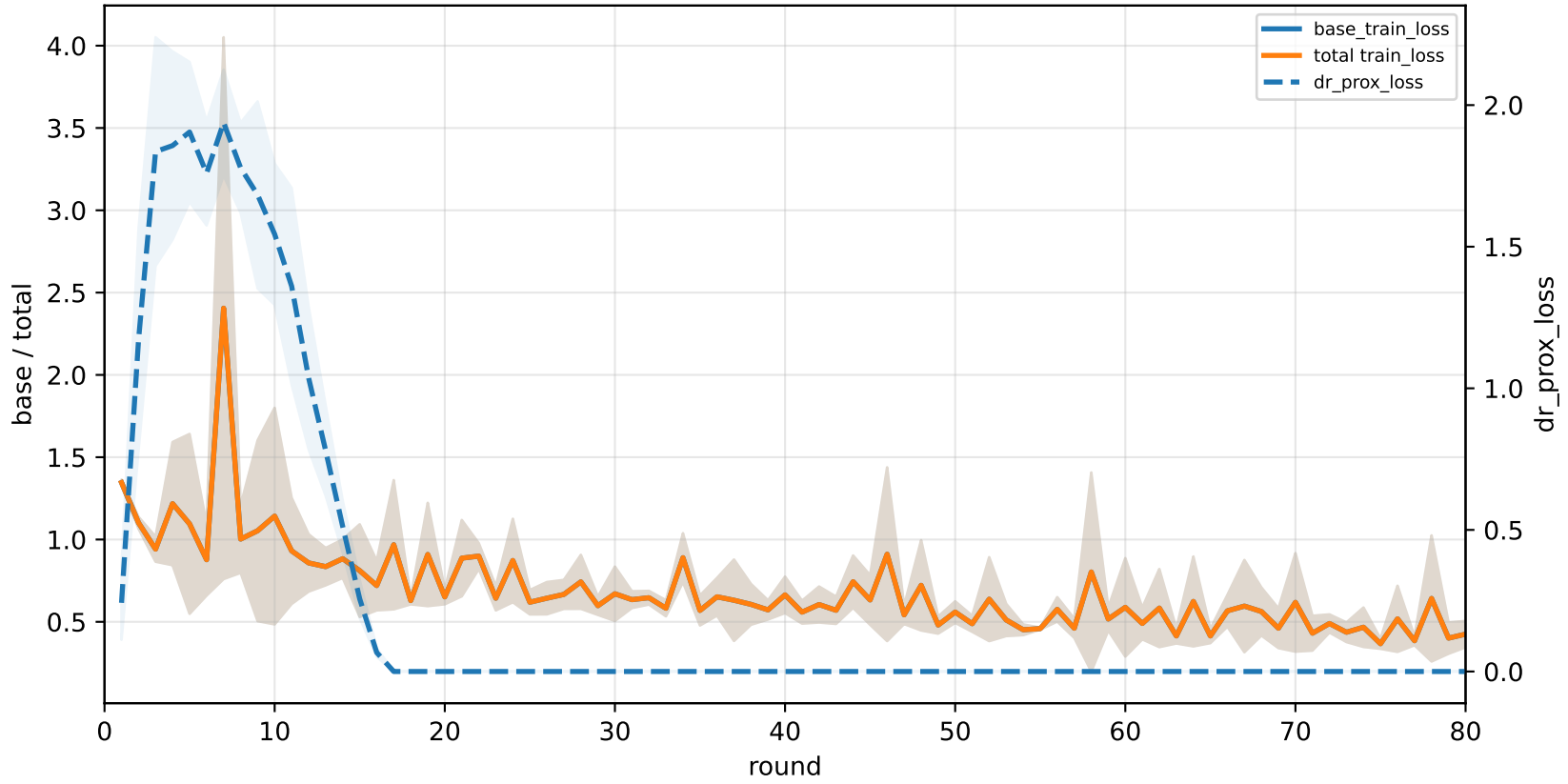
C3-specialist | graph\_id=3000011



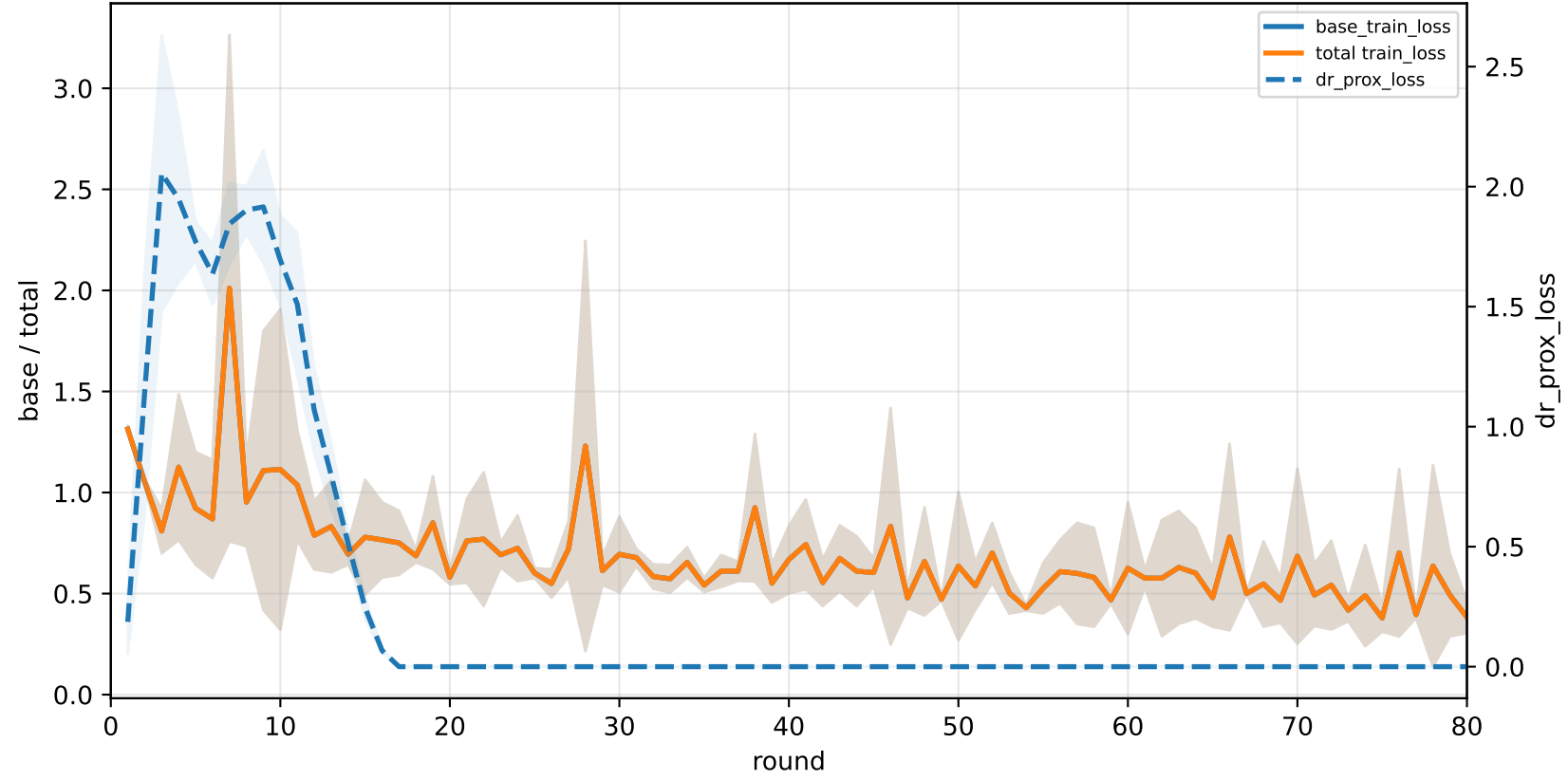
C4-specialist | graph\_id=3000012



C5-specialist | graph\_id=3000013



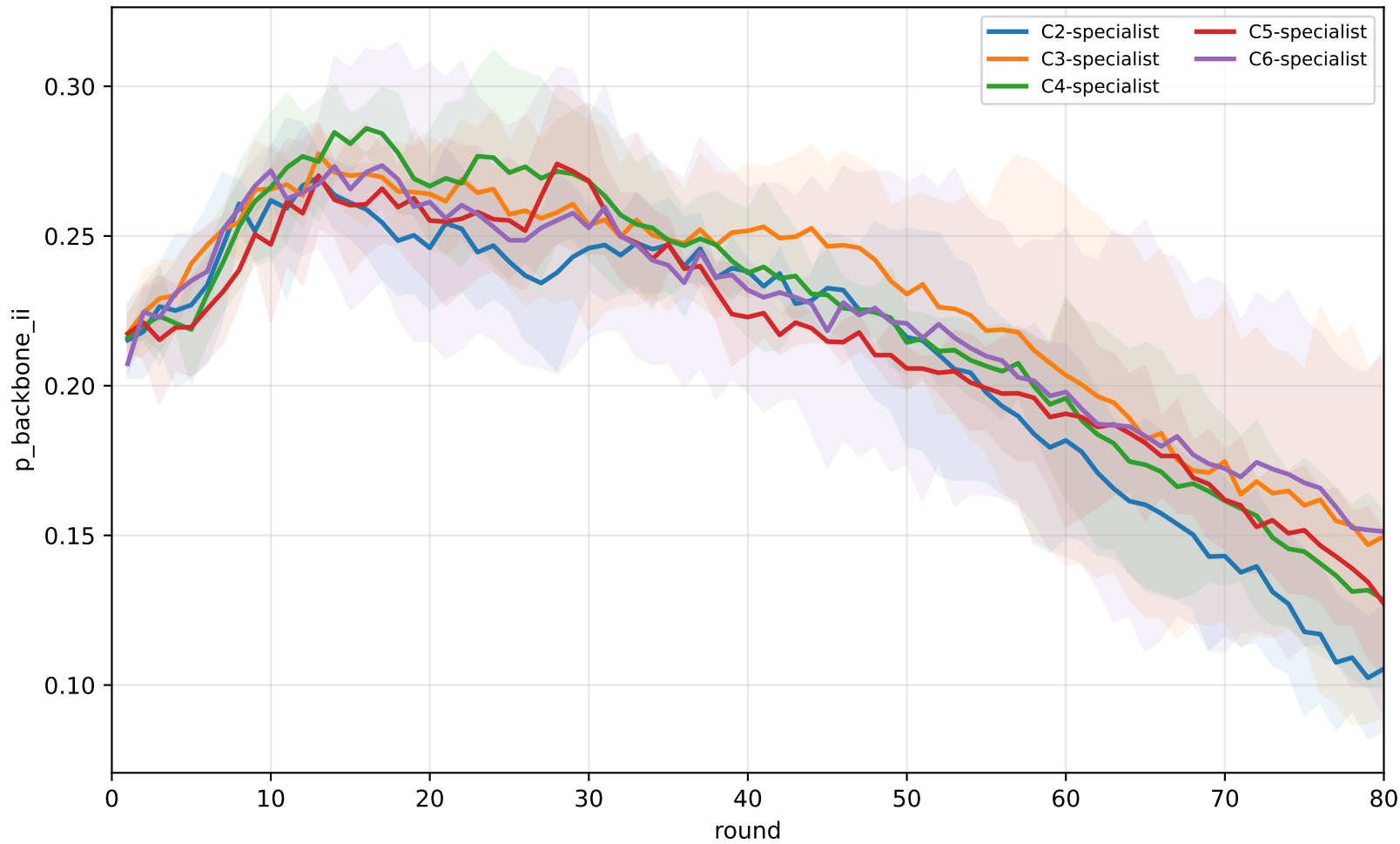
C6-specialist | graph\_id=3000014



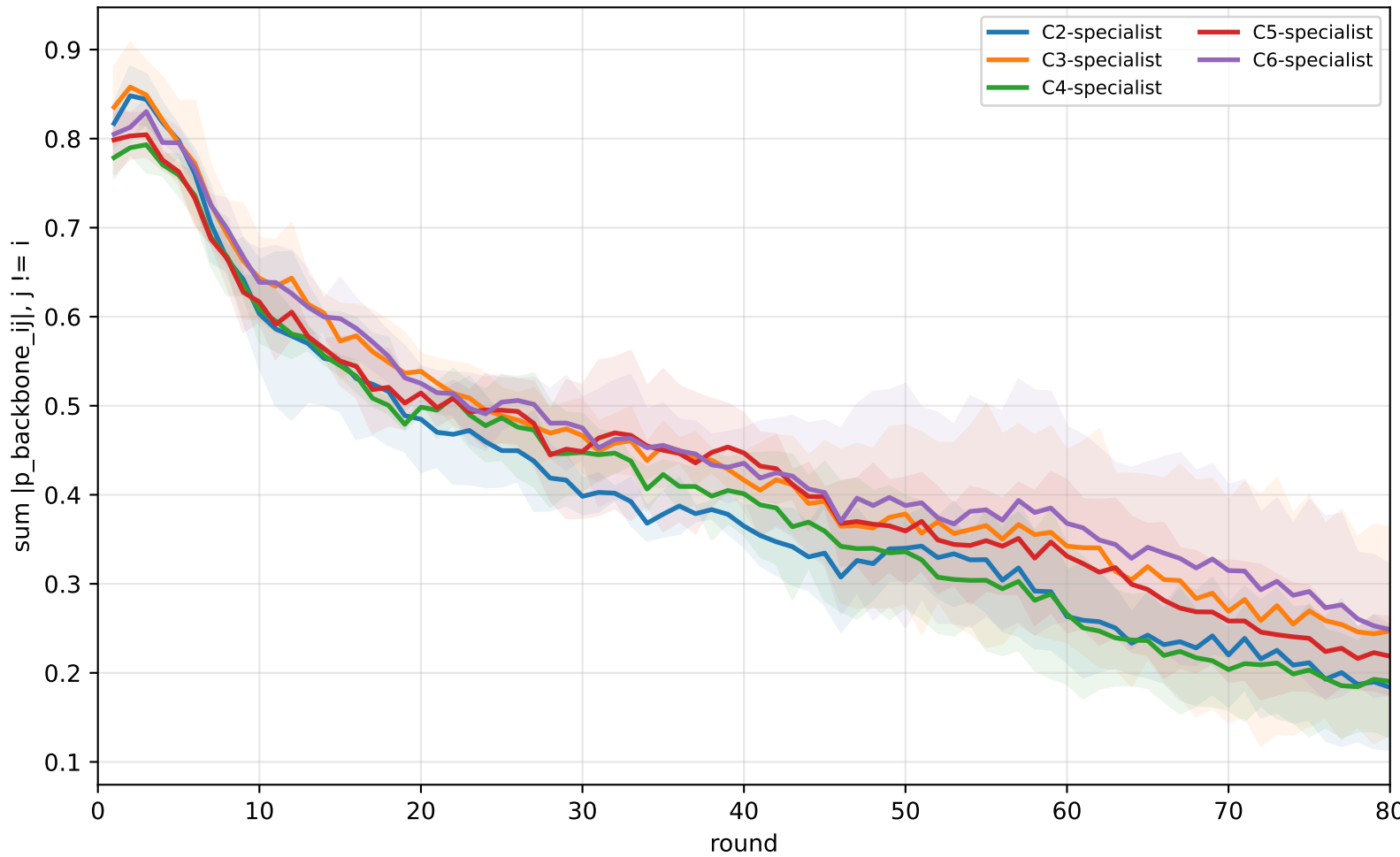
APPLE-Backbone-TaskHead backbone DR behavior | High heterogeneity

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

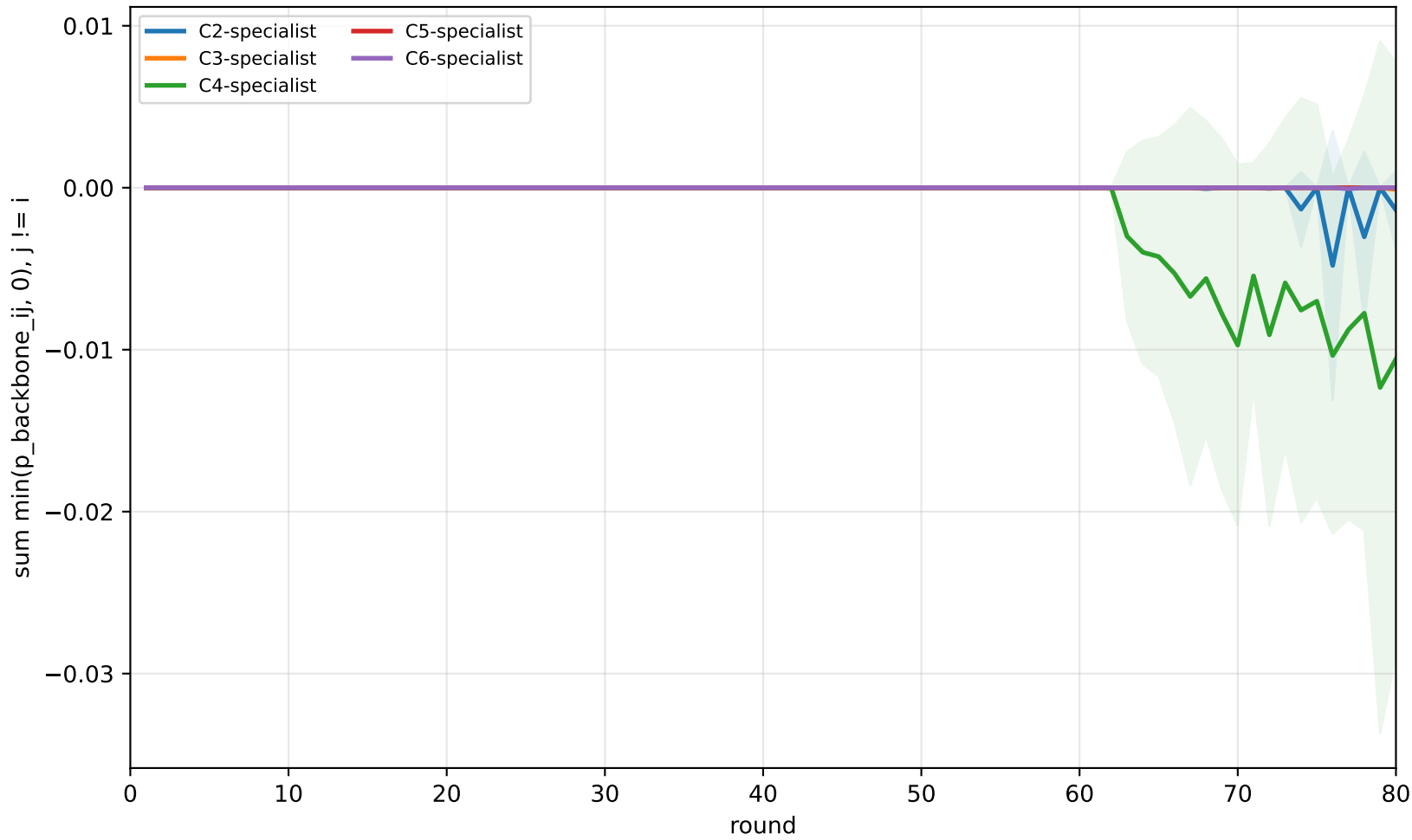
Backbone DR self-weight  $p_{\text{backbone\_ij}}$  over rounds



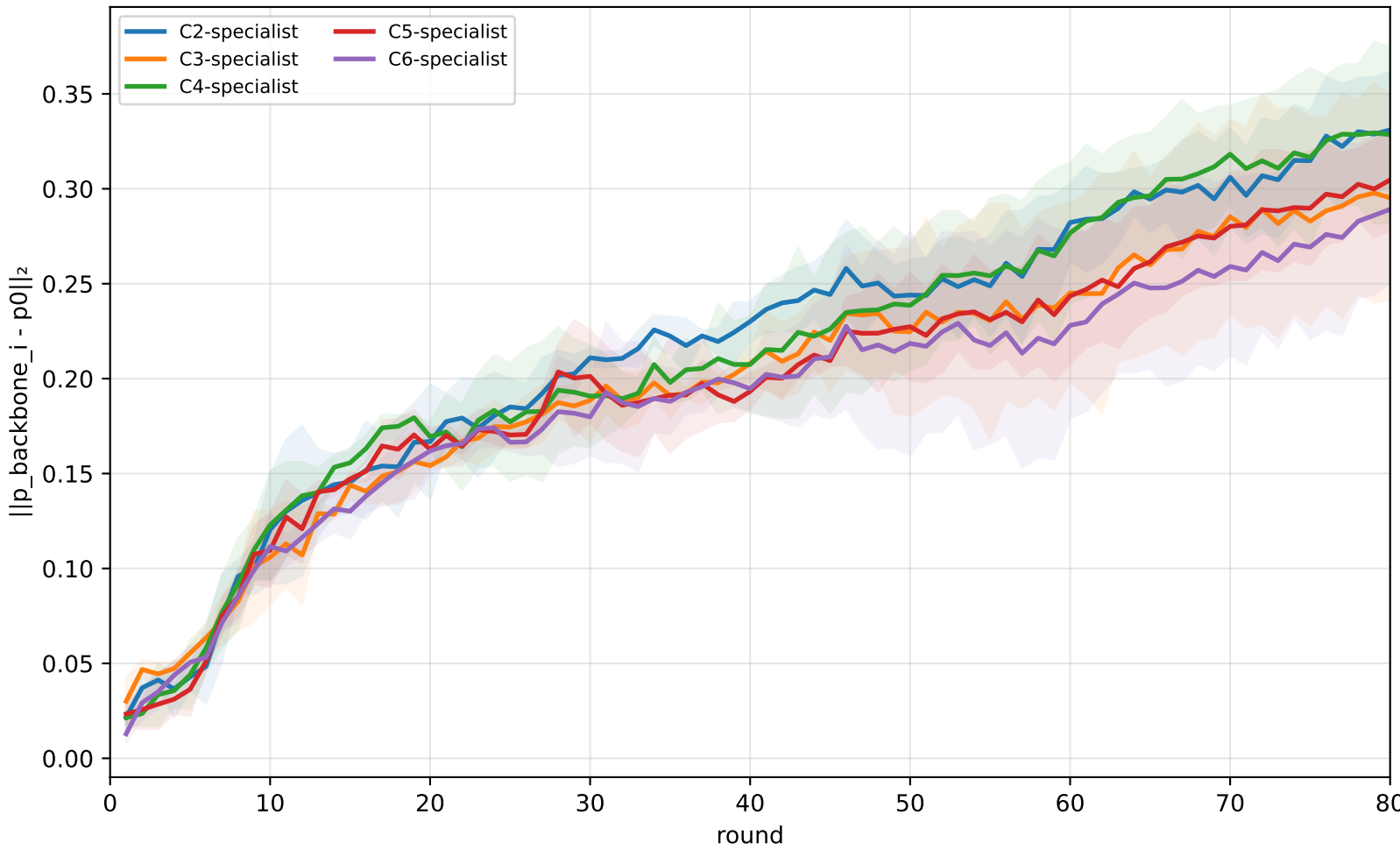
Backbone off-diagonal absolute mass over rounds



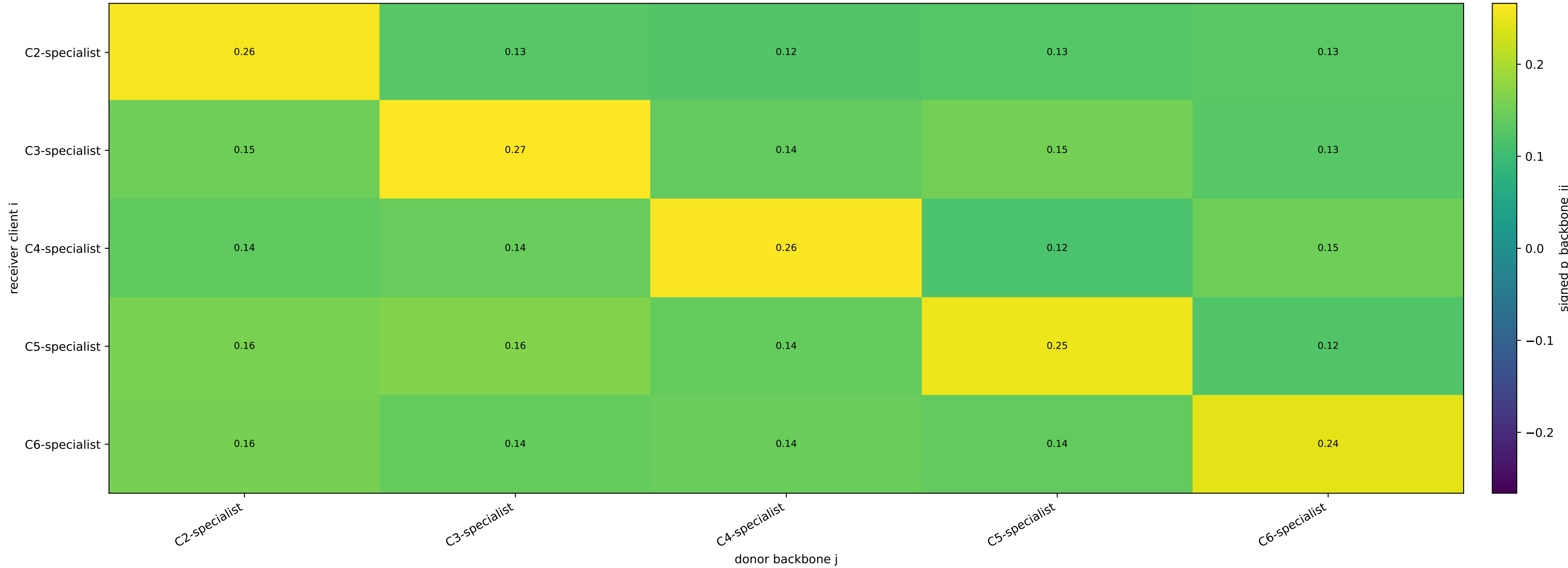
Backbone off-diagonal negative mass over rounds



Backbone DR movement away from  $p_0$



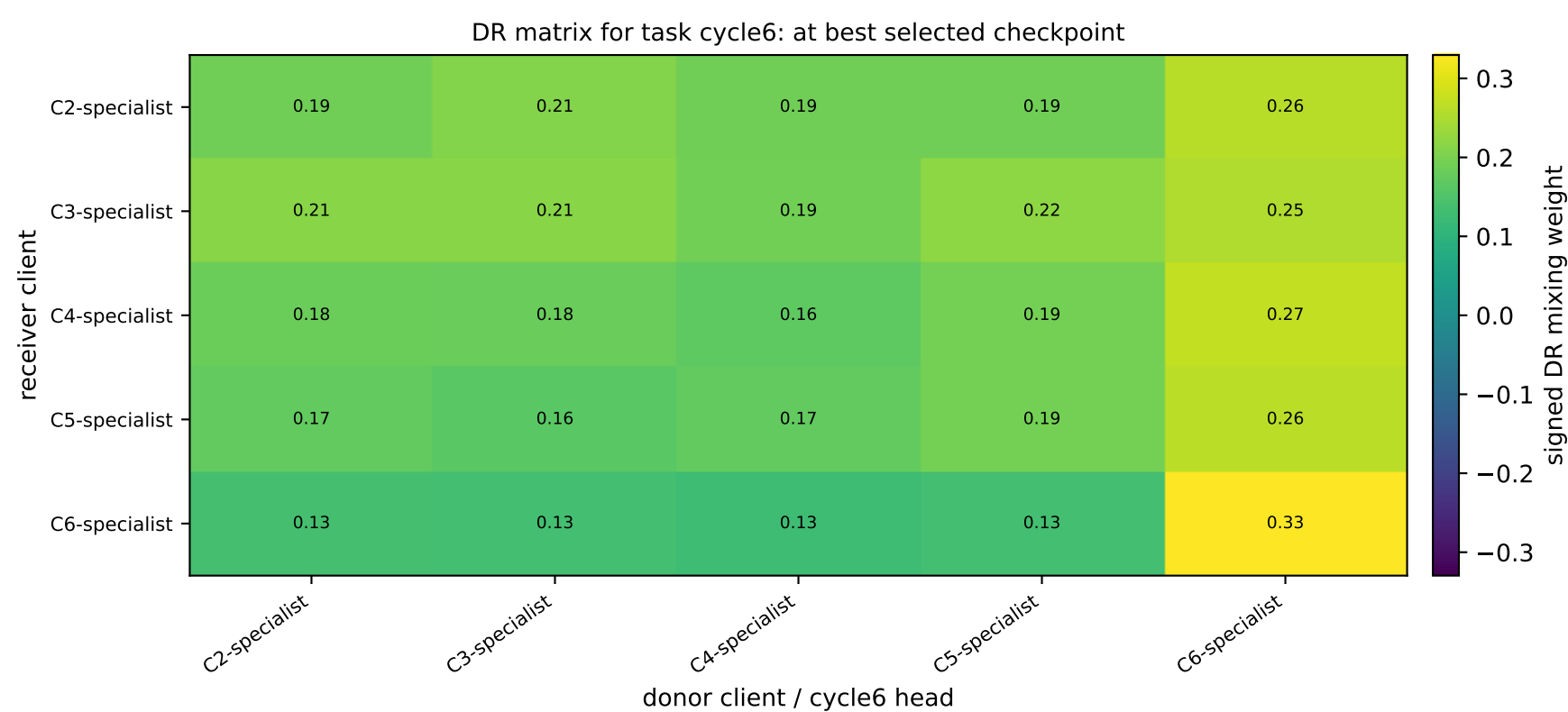
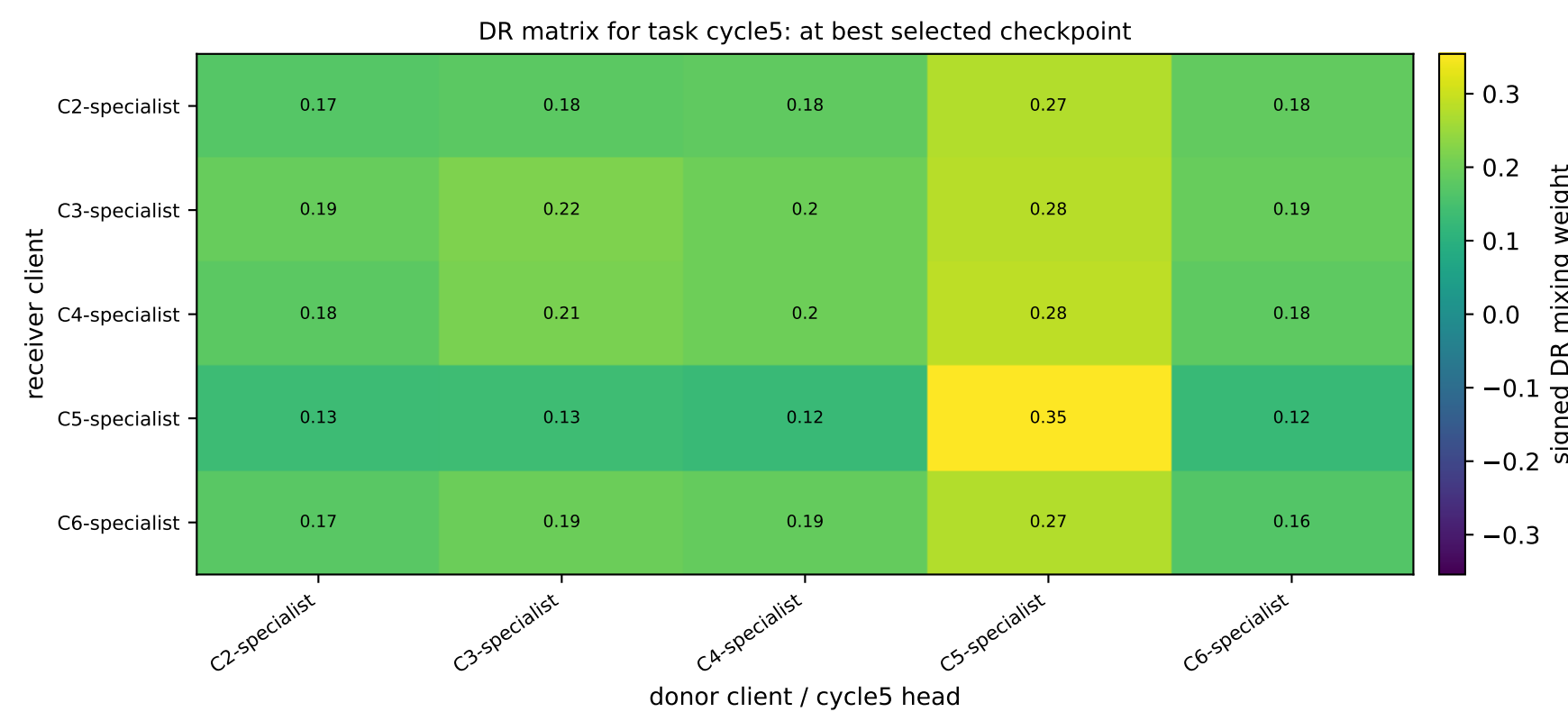
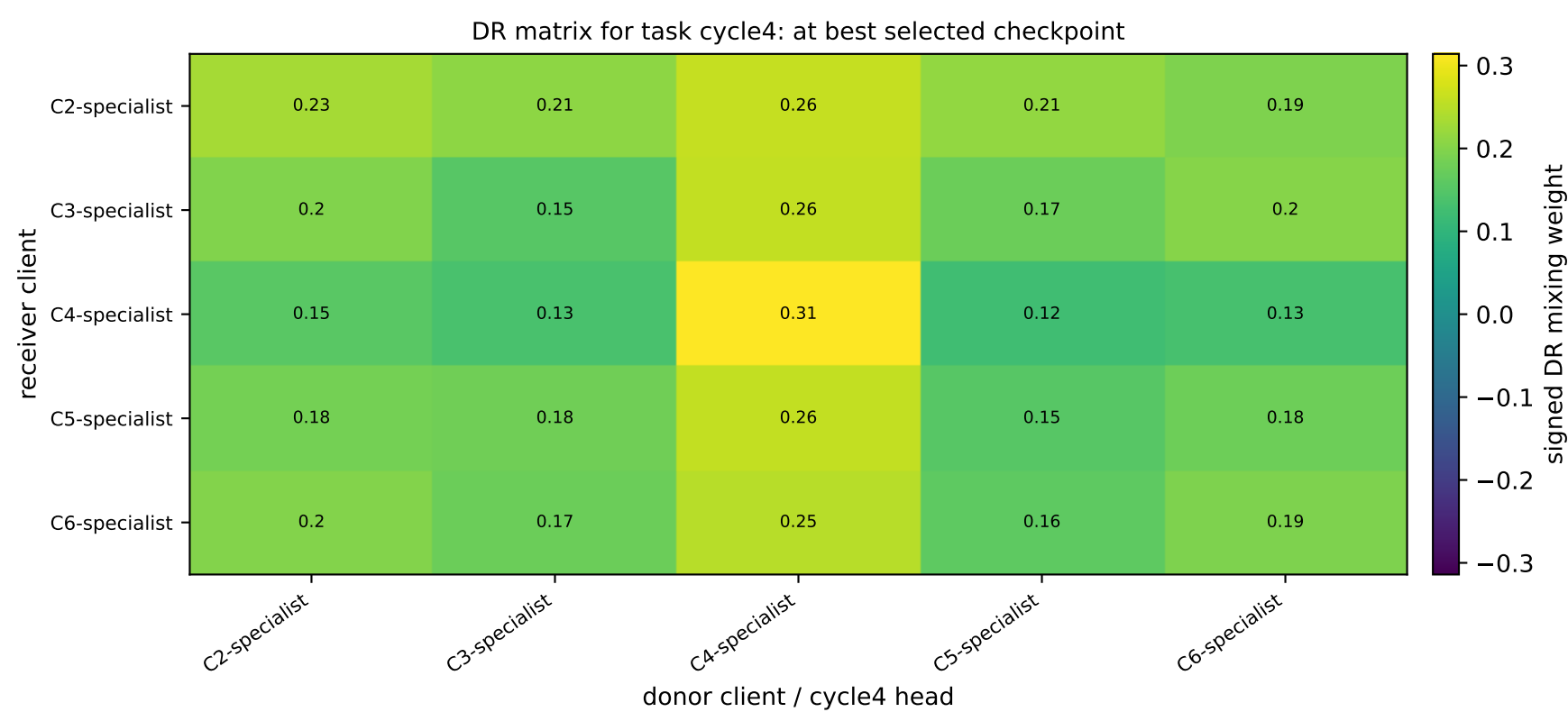
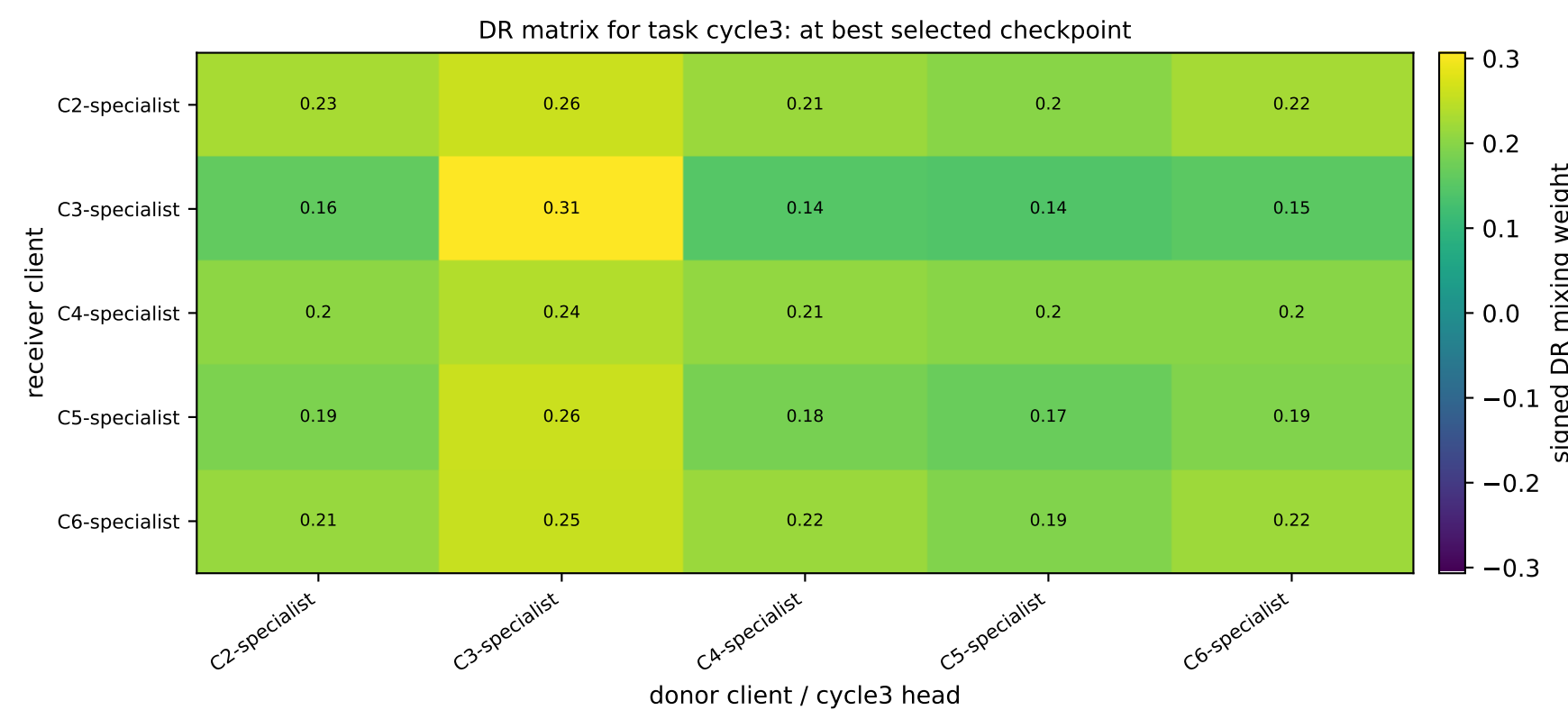
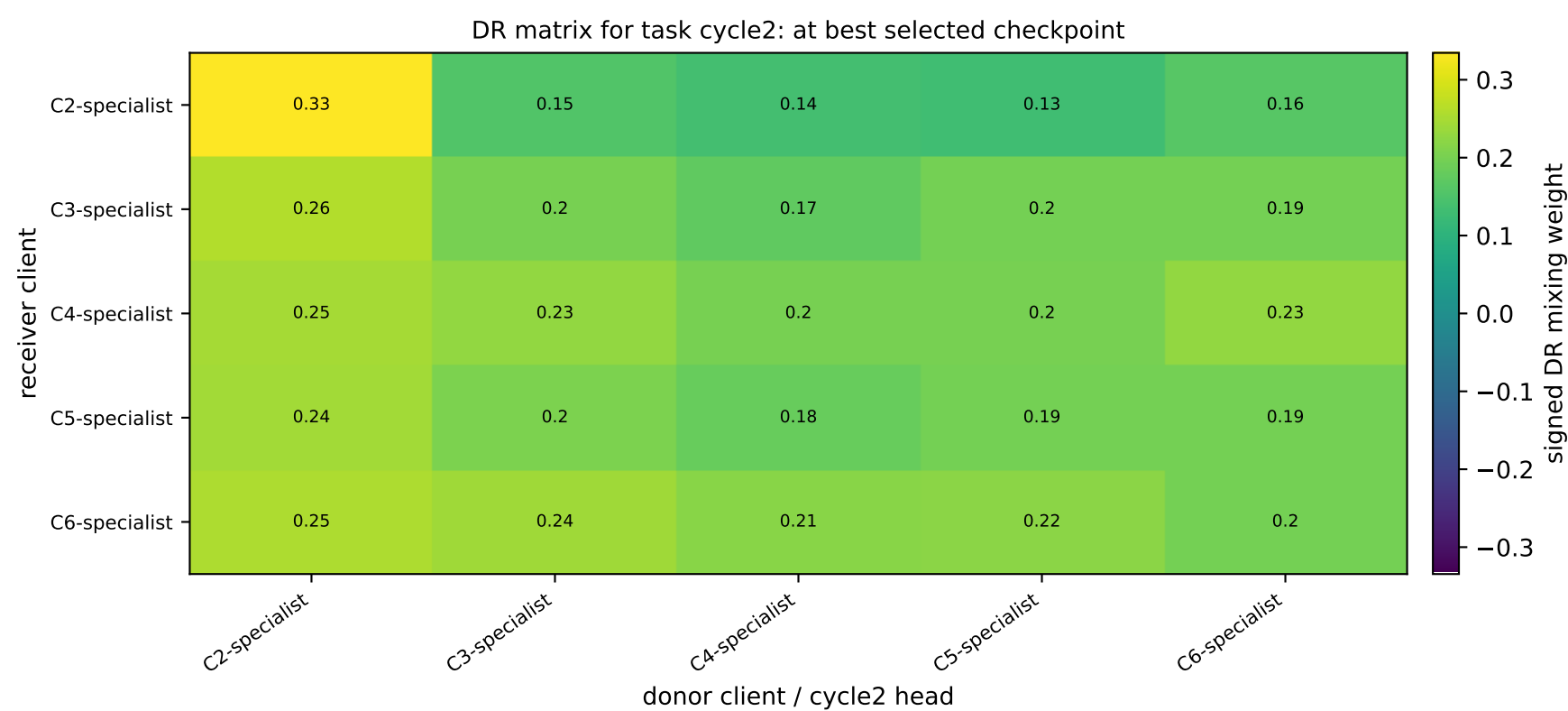
Selected-best-round backbone DR coefficients, mean over seeds





# APPLE-Backbone-TaskHead task-wise DR matrices: receiver clients × donor task-heads | High

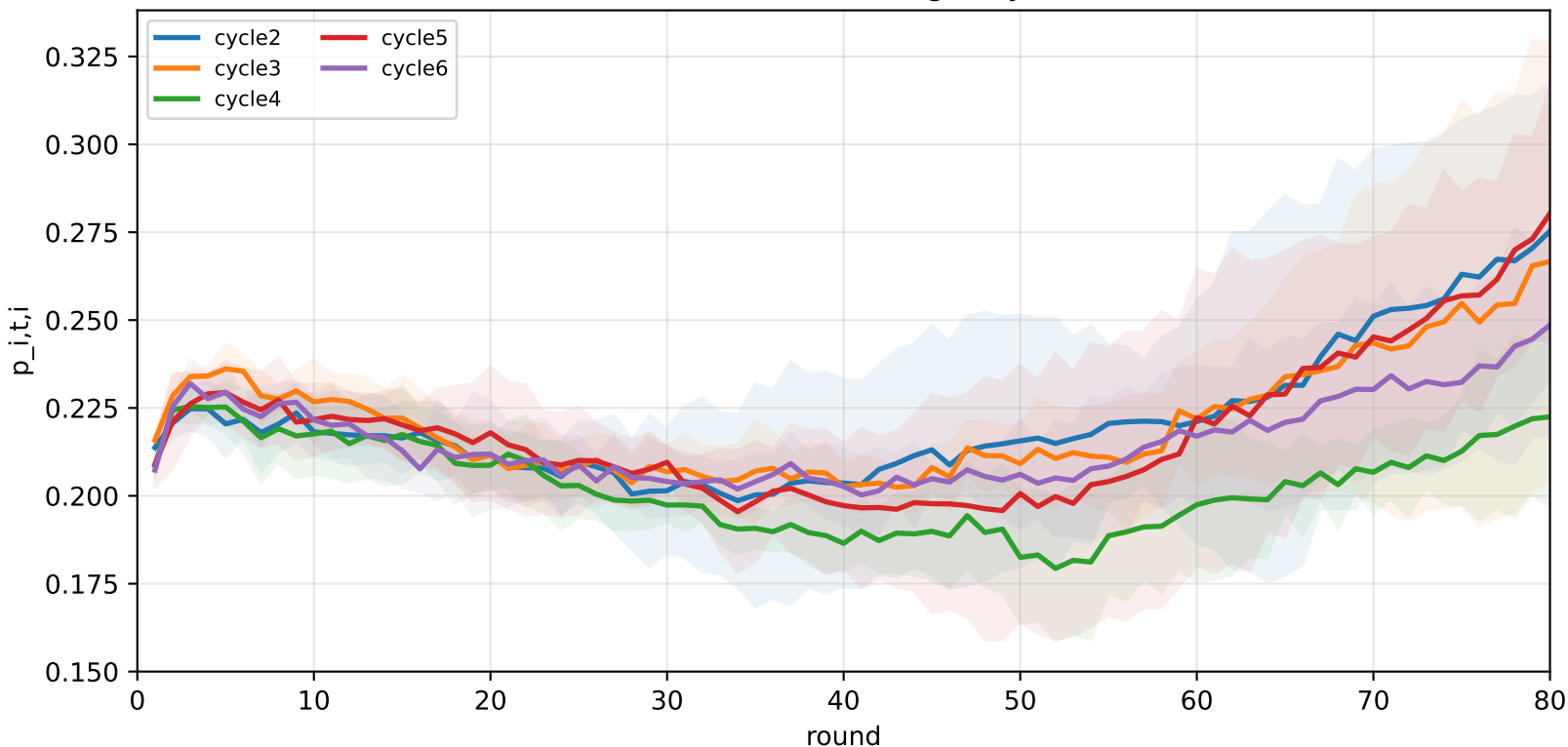
family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399



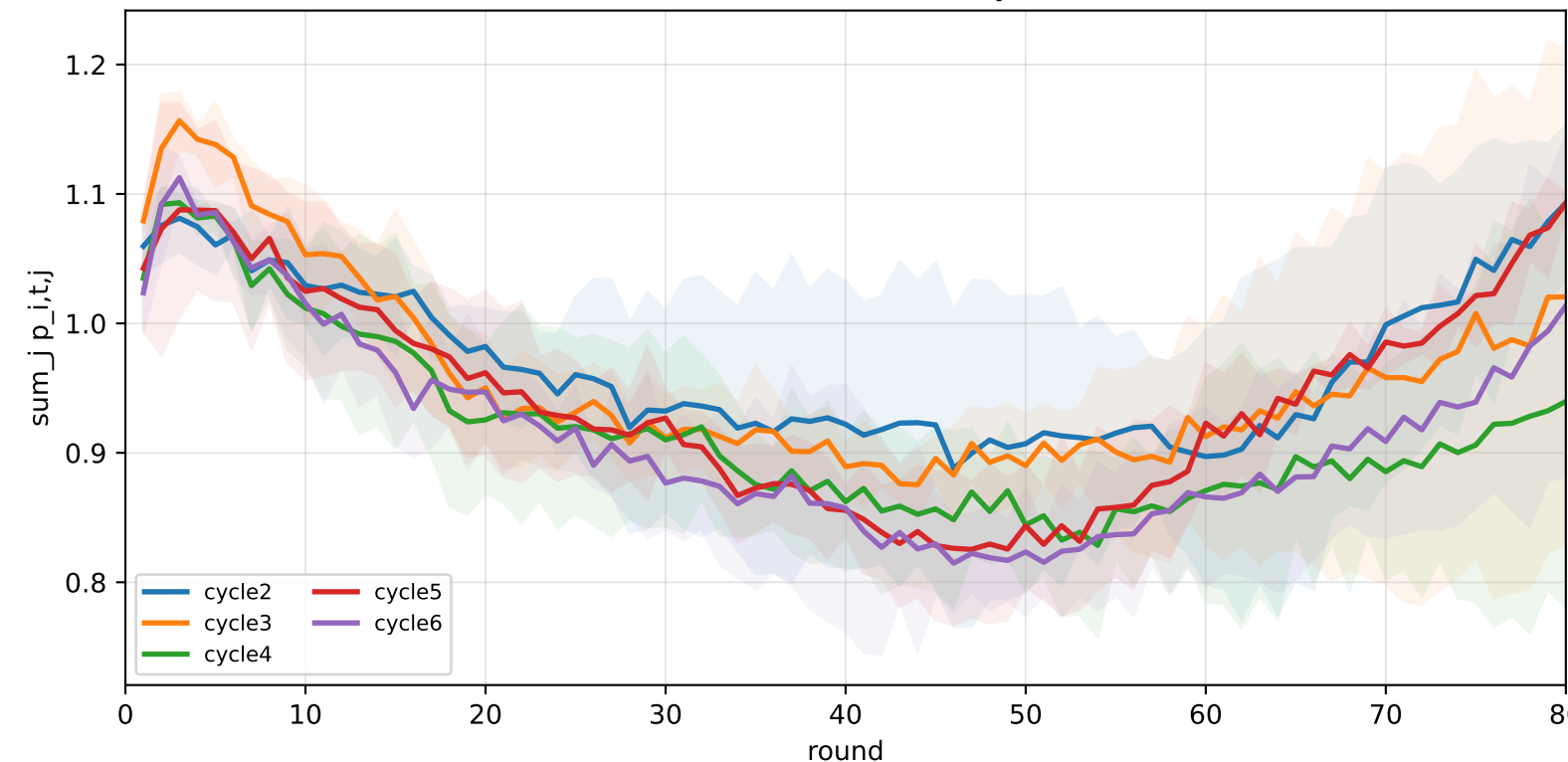
# APPLE-Backbone-TaskHead task-wise DR dynamics | High

family=q\_task\_cycle2 | n\_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
 family=q\_task\_cycle3 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
 family=q\_task\_cycle4 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
 family=q\_task\_cycle5 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%  
 family=q\_task\_cycle6 | n\_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%  
 heterogeneity=high | gamma=0.80 | task\_jsd\_mean=0.399 | task\_jsd\_median=0.399 | task\_jsd\_max=0.403 | family\_centroid\_jsd\_mean=0.399

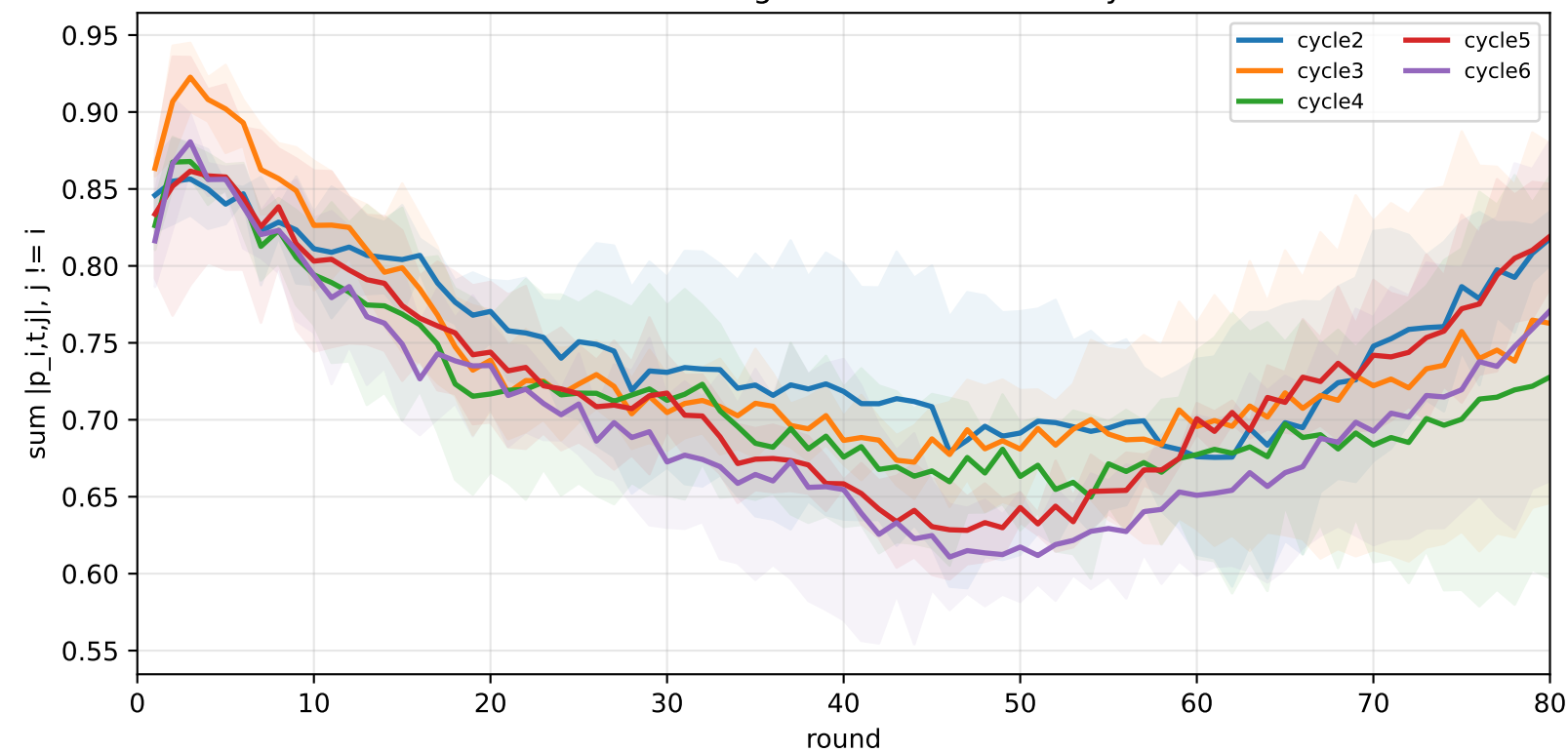
Task-head DR self-weight by task



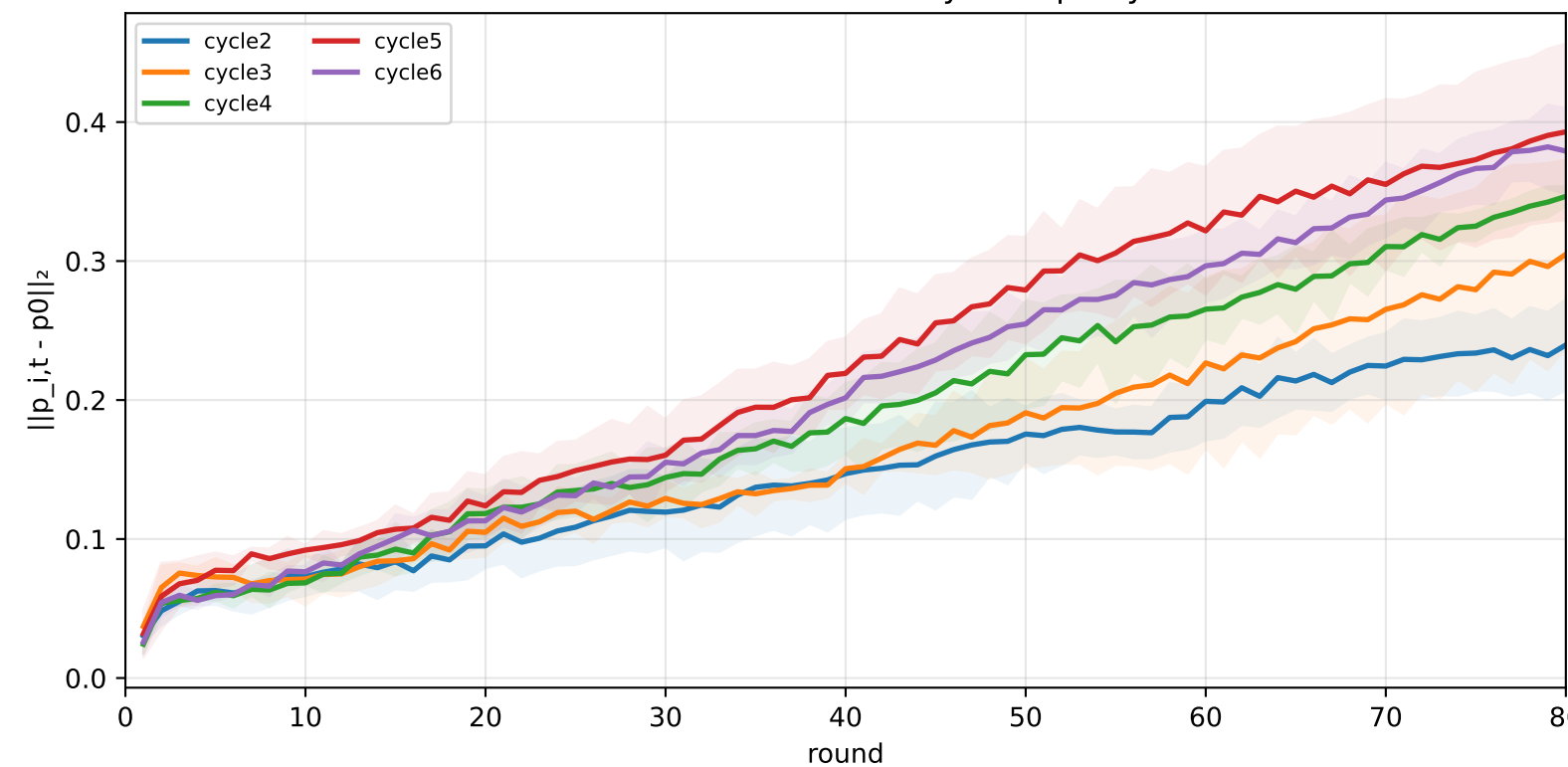
Task-head DR row sum by task



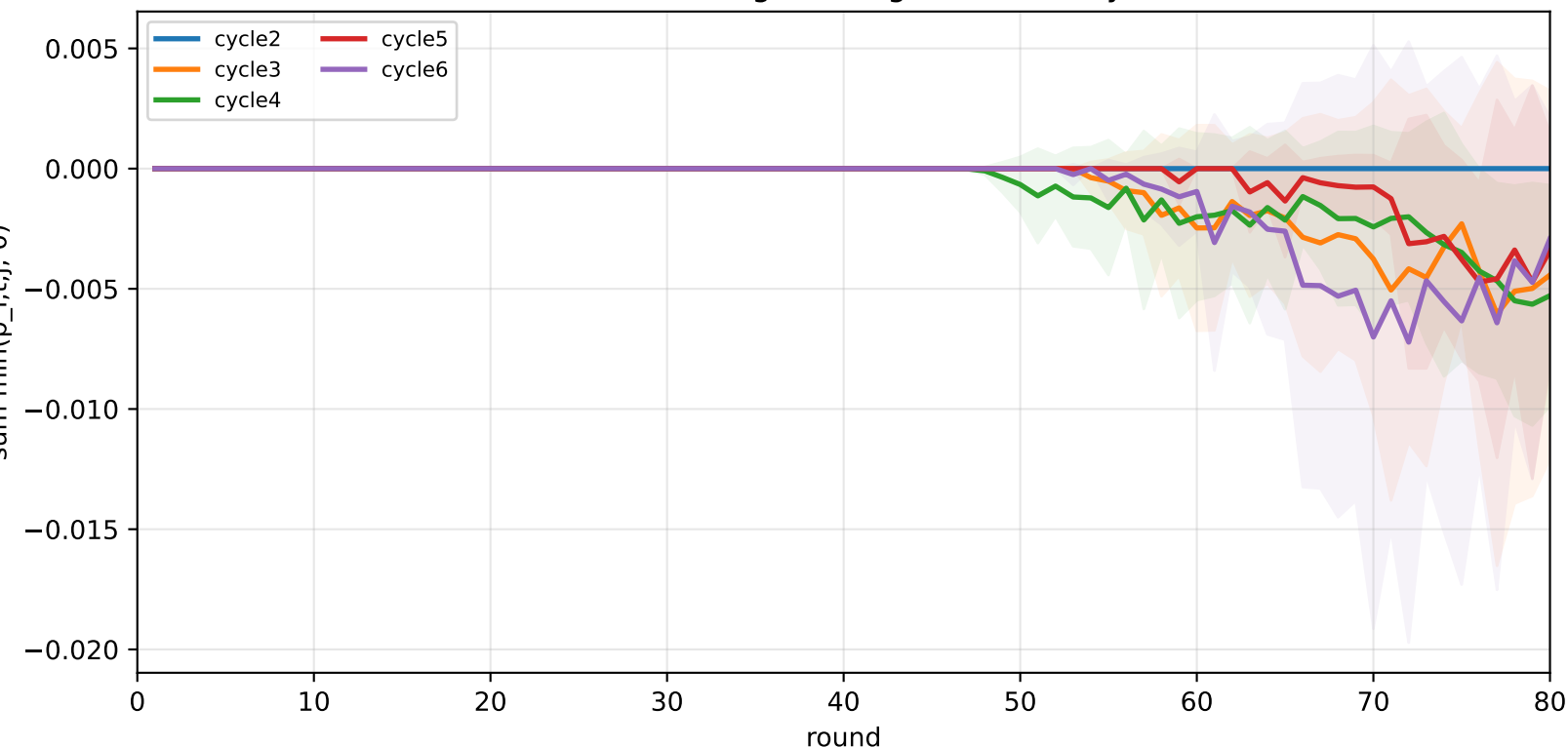
Task-head off-diagonal absolute mass by task



Task-head DR movement away from p0 by task



Task-head off-diagonal negative mass by task



Task-head DR alignment

